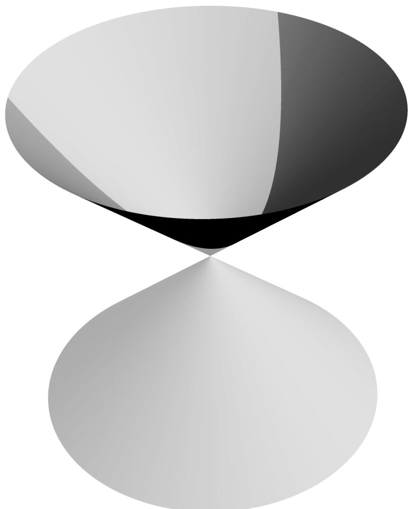
# Geometry I

Draft 2026



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# CHAPTER 1

## Affine space

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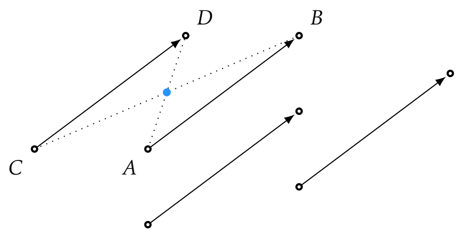
## 1.1 Affine space

### 1.1.1 The vector space structure of geometric vectors

We denote by $\mathbb{E}$ the Euclidean space. It describes the Platonic realm of perfect shapes which resonate with our intuition. To explore this world, the ancient Greeks used primitives and axioms in the form of Euclid's postulates [10]. Eventually, it was noticed that a more rigorous description is needed. A revision of the foundations of Euclidean geometry was proposed by Hilbert in his *Foundations of Geometry* [14]. An exploration of the axiomatics of $\mathbb{E}$ can be found in [16]. For now, we assume that $\mathbb{E}$ is a set of *points* governed by the geometric intuition which you have acquired so far in school.

A fundamental notion underlying our current way of understanding Euclidean geometry is the notion of *vector*. To define vectors we use the *equipollence* relation which we introduce next. Notice that a straight line segment $[AB]$, between the point $A$ and the point $B$, has an *orientation* implicit in the notation: we mention first $A$ and then $B$. To take this orientation into account we call $[AB]$ the *oriented line segment from $A$ to $B$*. Now, two oriented segments $[AB]$ and $[CD]$ are called *equipollent*, and we write $[AB] \sim [CD]$, if the segments $[AD]$ and $[BC]$ have the same midpoints.




For noncollinear points, this is equivalent to $ABDC$ being a parallelogram. Moreover, the equipollence relation is an equivalence relation. The equivalence classes of the equipollence relation are called *vectors*. The vector containing the oriented segment $[AB]$ is traditionally denoted by $\overrightarrow{AB}$. We also use the notation $B - A$ for this vector. Formally, we have

$$B - A = \overrightarrow{AB} := \Big\{ \text{ordered segments } [XY] \text{ such that } [XY] \sim [AB] \Big\}.$$

We say that $[AB]$ is a *representative* of the vector $\overrightarrow{AB}$. If $\overrightarrow{AB} = \overrightarrow{CD}$ we say that $[AB]$ and $[CD]$ *define the same vector*. The set of all vectors defined with points in $\mathbb{E}$ is the set of equivalence classes:

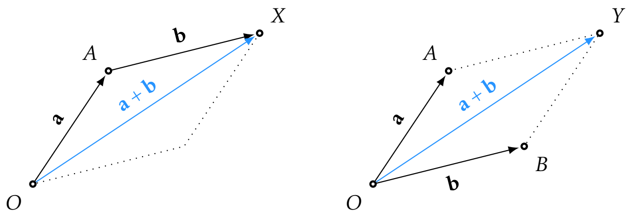
$$\mathbb{V} = \Big\{ \overrightarrow{AB} : \text{where } A \text{ and } B \text{ are points in } \mathbb{E} \Big\} = \mathbb{E} \times \mathbb{E} \big/ \sim .$$

The vector $\overrightarrow{AA}$ is called the *zero vector*, and we denote it by $\overrightarrow{0}$ or simply by $0$ when there is no risk of confusion. Since all representatives of a vector $\overrightarrow{AB}$ have the same length $|AB|$ this will also be the *length* of the vector $\overrightarrow{AB}$, and we denote it by $|\overrightarrow{AB}|$, equivalently $|B - A|$. This defines a length map $|\square| : \mathbb{V} \to \mathbb{R}_{\geq 0}$. The vector $\overrightarrow{BA}$ is called the *opposite of the vector* $\overrightarrow{AB}$ and is denoted by $-\overrightarrow{AB}$. This defines a map $-\square : \mathbb{V} \to \mathbb{V}$. Notice that $-(B - A) = A - B$ and that $|\overrightarrow{0}| = |\overrightarrow{AA}| = |A - A| = 0$.

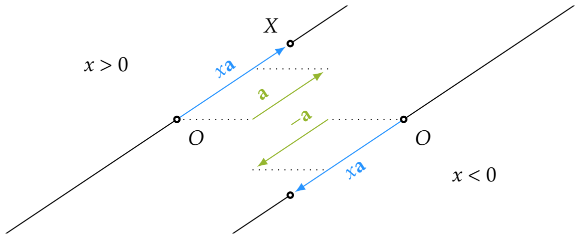
**Theorem 1.1.** The set of vectors $\mathbb{V}$ with vector addition and scalar multiplication is a vector space.

This theorem can be derived from any complete set of axioms describing Euclidean geometry. Here we simply recall the construction of the two operations.

**Addition of vectors:** Consider two vectors $\mathbf{a}$ and $\mathbf{b}$. If we fix a point $O$, there is a unique point $A$ such that $\mathbf{a} = \overrightarrow{OA}$ and for the point $A$ there exists a unique point $X$ such that $\mathbf{b} = \overrightarrow{AX}$. The *sum of* $\mathbf{a}$ *and* $\mathbf{b}$ is by definition the vector $\overrightarrow{OX}$, and we denote the sum by $\mathbf{a} + \mathbf{b}$. This defines a map $\square + \square : \mathbb{V} \times \mathbb{V} \to \mathbb{V}$. Equivalently, for a fixed point $O$ there are unique points $A$ and $B$ such that $\mathbf{a} = \overrightarrow{OA}$ and $\mathbf{b} = \overrightarrow{OB}$ and for the points $O, A$ and $B$ there is a *unique* point $Y$ such that $OAYB$ is a parallelogram. It follows that $X = Y$ and therefore $\mathbf{a} + \mathbf{b} = \overrightarrow{OY} = \overrightarrow{OX}$.



(a) Addition of vectors.



(b) Multiplication of vectors with scalars.

**Multiplication with scalars:** Consider a non-zero vector $\mathbf{a} = \overrightarrow{OA}$ and a scalar $x \in \mathbb{R}$. If $x > 0$, there is a unique point $X$ on the ray from $O$ to $A$ such that $|OX| = x \cdot |OA|$. The *multiplication of the vector* $\mathbf{a}$ *with the scalar* $x$, denoted $x \cdot \mathbf{a}$ (or simply $x\mathbf{a}$), defines a map $\square \cdot \square : \mathbb{R} \times \mathbb{V} \to \mathbb{V}$, and is given by

$$x \cdot \mathbf{a} = \begin{cases} \overrightarrow{OX} & \text{for } \mathbf{a} \neq \overrightarrow{0},\ x > 0 \text{ and } X \text{ as above,} \\ -(|x|\mathbf{a}) & \text{for } \mathbf{a} \neq \overrightarrow{0},\ x < 0, \\ \overrightarrow{0} & \text{for } \mathbf{a} = \overrightarrow{0} \text{ or } x = 0. \end{cases}$$

### 1.1.2 Vectors acting on points

Above we have described geometric vectors and their vector space structure. Points in $\mathbb{E}$ and vectors in $\mathbb{V}$ interact in two ways. Firstly, for any fixed point $O$, there is a bijection $\mathbb{E} \leftrightarrow \mathbb{V}$ which allows us to identify points with vectors. Indeed, for any point $A$ there is a unique vector from $O$ to $A$. The vector $\overrightarrow{OA}$ is called the *position vector of $A$ relative to $O$* or simply the *position vector of $A$* if it is clear from the context what $O$ is. It is customary to denote this vector by $\mathbf{r}_O(A)$ (the $\mathbf{r}$ stands for *radius vector*). Secondly, we may use vectors to 'move' points. Namely, for a vector $\mathbf{a}$ and a point $A$, there is a unique point $B$ such that $\mathbf{a} = \overrightarrow{AB}$. We may think about this as saying that the vector $\mathbf{a}$ is moving the point $A$ to the point $B$. In technical terms we say that *vectors act on points*. This interaction between points and vectors can be encapsulated in a data structure called *affine space*.

**Definition 1.2.** An affine space $\mathbb{A}$ is a set of points to which we attach a vector space $\mathbb{V}$ and two maps

$$\square - \square : \mathbb{A} \times \mathbb{A} \to \mathbb{V} \quad \text{and} \quad \square + \square : \mathbb{V} \times \mathbb{A} \to \mathbb{A}$$

subject to the following axioms:

(AS1) For any point $P$ we have $\overrightarrow{0} + P = P$

(AS2) For any point $P$ and any vectors $\mathbf{v}, \mathbf{w}$ we have $(\mathbf{v} + \mathbf{w}) + P = \mathbf{v} + (\mathbf{w} + P)$

(AS3) For any two points $A$ and $B$ we have $(A - B) + B = A$

(AS4) For any point $P$ and any vector $\mathbf{v}$ we have $(\mathbf{v} + P) - P = \mathbf{v}$

The first map produces a vector from two points. It overlaps with the notation $A - B$ for the vector $\overrightarrow{AB}$ mentioned in the previous section. The second map is called *translation* since it takes a vector $\mathbf{v}$ and a point $A$, and it produces the point $B$ such that $\mathbf{v} = \overrightarrow{AB}$.

The vector space $\mathbb{V}$ attached to the affine space $\mathbb{A}$ is denoted by $\mathrm{D}(\mathbb{A})$ and is called the *direction space of* $\mathbb{A}$. The dimension of $\mathbb{A}$ is by definition the dimension of $\mathrm{D}(\mathbb{A})$ and is denoted by $\dim \mathbb{A}$. It is customary to fix the dimension and indicate it with an upper index, e.g., we write $\mathbb{A}^n$ for an $n$-dimensional affine space.

**Example 1.3.** The term *affine* was coined by Leonhard Euler to refer to certain arguments and properties of certain geometric shapes which do not change when stretching or contracting the Euclidean space. We will make this more precise in a later chapter. An affine space is a first data structure that captures a considerable part of Euclidean geometry. One can deduce from Hilbert's axioms that the Euclidean space $\mathbb{E}$ with set of vectors $\mathrm{D}(\mathbb{E}) = \mathbb{V}$, defined in the previous section, is an affine space of dimension at least 3. However, when we want to talk about a particular dimension, we write it as an upper index, e.g., we write $\mathbb{E}^2$ for the Euclidean plane, $\mathbb{E}^3$ for the Euclidean space. The reframing of the geometric canvas as an affine space unlocks the full power of linear algebra. From Hilbert's axioms one can deduce in particular the following facts.

**Definition 1.4.** Let $\mathbb{A}$ be an affine space. A subset $S \subseteq \mathbb{A}$ is called an *affine subspace* if there exists a point $O \in S$ and a vector subspace $\mathbb{U} \subseteq \mathbb{V}$ such that

$$S = \mathbb{U} + O = \{\mathbf{u} + O : \mathbf{u} \in \mathbb{U}\}.$$

In this case, the subspace $\mathbb{U}$ is unique and is denoted by $\mathrm{D}(S)$. It is called the *direction subspace* of $S$. The dimension of $S$ is defined as $\dim(S) = \dim(\mathrm{D}(S))$.

- A *point* is an affine subspace of dimension 0.
- A *line* is an affine subspace of dimension 1.
- A *plane* is an affine subspace of dimension 2.
- A *hyperplane* is an affine subspace of dimension $\dim(\mathbb{A}) - 1$.

For the Euclidean space $\mathbb{E}$, these properties can be deduced from Hilbert's axioms.

**Example 1.5.** One can deduce from the axioms of an affine space that for any fixed point $O$ there is a bijection between vectors and points given by $\mathbf{v} \mapsto \mathbf{v} + O$. This allows one to identify, element-wise, vectors with points. Vector spaces and affine spaces are in fact in bijection since for any (real) vector space $\mathbb{V}$ one can construct an affine space $\mathbb{A}$ such that $\mathrm{D}(\mathbb{A}) = \mathbb{V}$. Indeed, the maps giving the affine space structure are subtraction and addition in the vector space. In this set-up we view the elements of $\mathbb{V}$ in two distinct ways: as points and as vectors which act on points. A vector space has an origin, the zero vector, but on a line or in a plane all points are equal. Marcel Berger [4, Chapter 2] summarizes this relationship nicely by stating that "an affine space is nothing more than a vector space whose origin we try to forget about, by adding translations to the linear maps."

**Remark.** The mathematical library *Mathlib* [23] written in Lean [22] defines affine spaces as above in order to treat Euclidean geometry (see [24]).

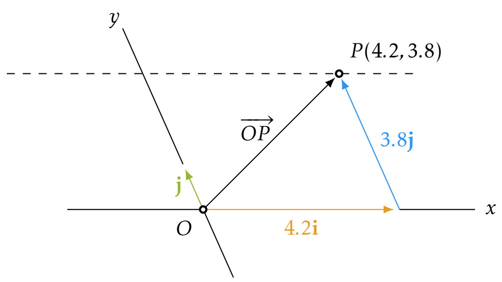
## 1.2 Cartesian coordinates

Coordinates are tuples of numbers associated with points in a given object or space. In this chapter, we focus on Cartesian frames, named after René Descartes. While modern views on the extent of Descartes' revolution vary, the utility of coordinate geometry in bridging geometry and algebra is undeniable [15, Section 6.8]. For other types of coordinate systems, see Appendix B.

### 1.2.1 Frames in dimension 2

In the Euclidean plane $\mathbb{E}^2$, fix two lines $\ell_1$ and $\ell_2$ which intersect at a unique point $O$. We can describe any point $P$ in $\mathbb{E}^2$ as follows: By Definition 1.4, $\mathrm{D}(\ell_1)$ and $\mathrm{D}(\ell_2)$ are 1-dimensional, linearly independent vector subspaces of the 2-dimensional vector space $\mathbb{V}^2 = \mathrm{D}(\mathbb{E}^2)$. If we select non-zero vectors $\mathbf{i} \in \mathrm{D}(\ell_1)$ and $\mathbf{j} \in \mathrm{D}(\ell_2)$, then $(\mathbf{i},\mathbf{j})$ is a basis of $\mathbb{V}^2$. Consequently, there exist *unique* scalars $x_P, y_P \in \mathbb{R}$ such that

$$\mathbf{r}_O(P) = \overrightarrow{OP} = x_P\mathbf{i} + y_P\mathbf{j}. \tag{1.1}$$




This establishes a bijection $\mathbb{E}^2 \leftrightarrow \mathbb{R}^2$ between points in the plane and ordered pairs of real numbers. This bijection arises as the composition of two bijections: 1. the bijection $\mathbf{r}_O : \mathbb{E}^2 \to \mathbb{V}^2$, which assigns to each point $P$ its position vector $\overrightarrow{OP}$, and 2. the decomposition of vectors $\mathbf{r}_O(P)$ with respect to a basis, establishing the identification $\mathbb{V}^2 \cong \mathbb{R}^2$. Thus, the bijection $\mathbb{E}^2 \leftrightarrow \mathbb{R}^2$ is the composition of a bijection $\mathbb{E}^2 \leftrightarrow \mathbb{V}^2$ and a bijection $\mathbb{V}^2 \leftrightarrow \mathbb{R}^2$, which depend on two choices: 1. selecting a point $O$ for the map $\mathbf{r}_O$, and 2. choosing two vectors $\mathbf{i}$ and $\mathbf{j}$ that form a basis of $\mathbb{V}^2$.

**Definition 1.6.** A *frame in* $\mathbb{E}^2$ is a pair $\mathcal{K} = (O,\mathcal{B})$, where $O$ is a point in $\mathbb{E}^2$ and $\mathcal{B} = (\mathbf{i},\mathbf{j})$ is a basis of $\mathbb{V}^2$. A frame is also referred to as a *Cartesian coordinate system*, or a *Cartesian frame*. We use the shorter term for convenience. Given a frame $\mathcal{K}$, the unique pair $(x_P, y_P)$ in (1.1) is called the *coordinates of* $P$ *with respect to the frame* $\mathcal{K}$. In other words, the coordinates of $P$ are the components of the position vector $\mathbf{r}_O(P) = \overrightarrow{OP}$ of $P$ relative to $O$ in the basis $\mathcal{B}$. We write $P_{\mathcal{K}}(x_P, y_P)$ when we want to indicate the coordinates. If it is clear from the context what $\mathcal{K}$ is, we omit the subscript $\mathcal{K}$ and simply write $P(x_P, y_P)$. By convention, the first coordinate is typically denoted $x$, and the second $y$. The line $\ell_1$ is called the $x$-axis, denoted $Ox$, while the line $\ell_2$ is the $y$-axis, denoted $Oy$. The point $O$ is referred to as the *origin of the frame* $\mathcal{K}$.

For computational purposes, points $P_{\mathcal{K}}(x_P, y_P)$ and vectors $\mathbf{a} = a_x\mathbf{i} + a_y\mathbf{j}$ are identified with column matrices, using their coordinates and components, respectively:

$$[P]_{\mathcal{K}} = \begin{bmatrix} x_P \\ y_P \end{bmatrix} \in \mathbb{R}^2 \quad \text{and} \quad [\mathbf{a}]_{\mathcal{B}} = \begin{bmatrix} a_x \\ a_y \end{bmatrix} \in \mathbb{R}^2.$$

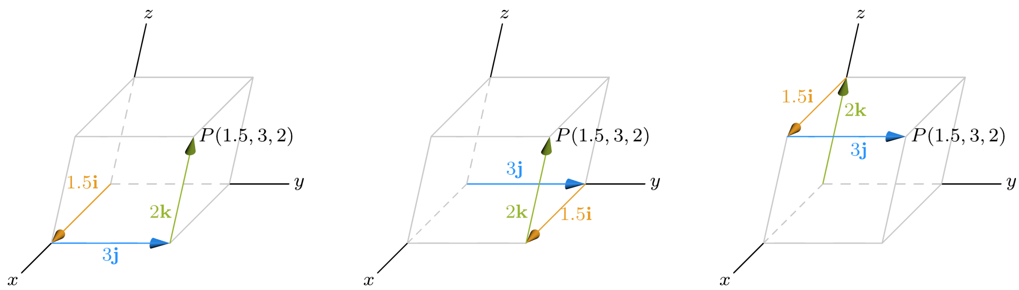
Here, the subscript $\mathcal{K}$ indicates the frame relative to which $P$ has the indicated coordinates, while the subscript $\mathcal{B}$ indicates the basis in which the components of the vector $\mathbf{a}$ are expressed.

### 1.2.2 Frames in dimension 3 and $n$

The concepts introduced for dimension 2 generalize naturally to dimension 3 and, more generally, to any dimension $n$.

In $\mathbb{E}^3$, we fix three non-coplanar lines intersecting at a point $O$. These lines determine three linearly independent directions in $\mathbb{V}^3 = \mathrm{D}(\mathbb{E}^3)$. Choosing basis vectors $\mathbf{i}, \mathbf{j}, \mathbf{k}$ along these directions gives us a basis $\mathcal{B} = (\mathbf{i}, \mathbf{j}, \mathbf{k})$. Any point $P \in \mathbb{E}^3$ is uniquely determined by the decomposition of its position vector:

$$\overrightarrow{OP} = x_P\mathbf{i} + y_P\mathbf{j} + z_P\mathbf{k}. \tag{1.2}$$




**Definition 1.7.** A *frame in* $\mathbb{A}^n$ (and specifically in $\mathbb{E}^3$ for $n = 3$) is a pair $\mathcal{K} = (O, \mathcal{B})$, where $O$ is a point in the affine space and $\mathcal{B} = (\mathbf{i}_1, \ldots, \mathbf{i}_n)$ is a basis of the direction space. The *coordinates* of a point $P$ with respect to $\mathcal{K}$ are the unique scalars $(x_1, \ldots, x_n)$ such that

$$\overrightarrow{OP} = x_1\mathbf{i}_1 + \cdots + x_n\mathbf{i}_n.$$

We write $P(x_1, \ldots, x_n)$. For $n = 3$, we typically use the notation $P(x, y, z)$ and the basis $(\mathbf{i}, \mathbf{j}, \mathbf{k})$. The lines determined by the basis vectors passing through $O$ are the *coordinate axes* $(Ox, Oy, Oz)$ and the planes determined by pairs of axes are *coordinate planes* $(Oxy, Oyz, Ozx)$.

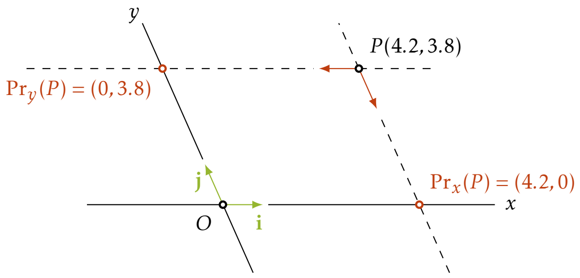
Points and vectors are identified with column matrices of coordinates and components respectively:

$$[P]_{\mathcal{K}} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^n \quad \text{and} \quad [\mathbf{a}]_{\mathcal{B}} = \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} \in \mathbb{R}^n.$$

## 1.3 Cartesian coordinates as projections

We explore projections extensively in Chapter 5. Here we simply observe that the coordinates can be interpreted as projections onto coordinate axes.

In $\mathbb{E}^2$, a point $P(x_P, y_P)$ satisfies $\overrightarrow{OP} = x_P\mathbf{i} + y_P\mathbf{j}$. This is equivalent to the existence of unique points $X \in Ox$ and $Y \in Oy$ such that $OXPY$ is a parallelogram.




**Definition 1.8.** With the above notation we may define the following maps

$$\mathrm{Pr}_x : \mathbb{E}^2 \to Ox, \quad \mathrm{Pr}_x(P) = X(x_P, 0) \quad \text{and} \quad \mathrm{Pr}_y : \mathbb{E}^2 \to Oy, \quad \mathrm{Pr}_y(P) = Y(0, y_P).$$

The map $\mathrm{Pr}_x$ is called *the projection on $Ox$ along $Oy$*, and $\mathrm{Pr}_y$ is called *the projection on $Oy$ along $Ox$*. For vectors $\mathbf{a} = a_x\mathbf{i} + a_y\mathbf{j}$, we have similar maps defined as follows

$$\mathrm{pr}_x : \mathbb{V}^2 \to \mathbb{R}, \quad \mathrm{pr}_x(\mathbf{a}) = a_x \quad \text{and} \quad \mathrm{pr}_y : \mathbb{V}^2 \to \mathbb{R}, \quad \mathrm{pr}_y(\mathbf{a}) = a_y.$$

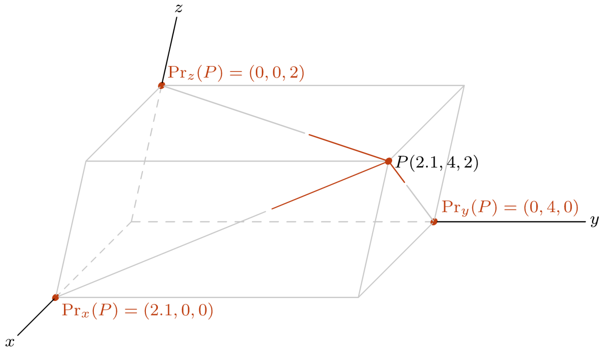
The map $\mathrm{pr}_x$ is called *the projection on the first component*, and $\mathrm{pr}_y$ is called *the projection on the second component*. From the definitions we immediately deduce the following identities

$$\overrightarrow{O\mathrm{Pr}_x(P)} = \mathrm{pr}_x(\overrightarrow{OP})\mathbf{i}, \quad \overrightarrow{O\mathrm{Pr}_y(P)} = \mathrm{pr}_y(\overrightarrow{OP})\mathbf{j}, \quad \overrightarrow{OP} = \mathrm{pr}_x(\overrightarrow{OP})\mathbf{i} + \mathrm{pr}_y(\overrightarrow{OP})\mathbf{j} \quad \text{and}$$

$$[P]_{\mathcal{K}} = \begin{bmatrix} x_P \\ y_P \end{bmatrix} = \begin{bmatrix} \mathrm{pr}_x(\overrightarrow{OP}) \\ \mathrm{pr}_y(\overrightarrow{OP}) \end{bmatrix} = [\overrightarrow{OP}]_{\mathcal{B}}.$$

**Remark.** If $\mathbf{i}$ and $\mathbf{j}$ are unit vectors, then for a point $P$ as above, we have $|OX| = x_P, |OY| = y_P$. In other words, the coordinates of $P$ are the lengths of the sides of the parallelogram $OXPY$.

In $\mathbb{E}^3$, for $P(x_P, y_P, z_P)$, there exist unique points $X \in Ox, Y \in Oy, Z \in Oz$ such that $O, X, Y, Z, P$ are vertices of a parallelepiped.




**Definition 1.9.** For vectors $\mathbf{a} = a_1\mathbf{i}_1 + \cdots + a_n\mathbf{i}_n$ in $\mathrm{D}(\mathbb{A}^n)$ we have *projection maps* $\mathrm{pr}_1, \ldots, \mathrm{pr}_n : \mathrm{D}(\mathbb{A}^n) \to \mathbb{R}$ defined by $\mathrm{pr}_k(\mathbf{a}) = a_k$. For points $P$ in $\mathbb{A}^n$, we define projections onto the coordinate axes $\mathrm{Pr}_k : \mathbb{A}^n \to Ox_k$ such that $\overrightarrow{O\mathrm{Pr}_k(P)} = \mathrm{pr}_k(\overrightarrow{OP})\mathbf{i}_k$.

In $\mathbb{E}^2$, $\mathrm{Pr}_x$ is the projection on $Ox$ along $Oy$. In $\mathbb{E}^3$, $\mathrm{Pr}_x$ is the projection on $Ox$ along the plane $Oyz$.

From the definitions we immediately deduce:

$$[P]_{\mathcal{K}} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} \mathrm{pr}_1(\overrightarrow{OP}) \\ \vdots \\ \mathrm{pr}_n(\overrightarrow{OP}) \end{bmatrix} = [\overrightarrow{OP}]_{\mathcal{B}}.$$

## 1.4 Changing Cartesian frames

A frame $\mathcal{K}$ allows us to identify the set of points in $\mathbb{A}^n$ with $\mathbb{R}^n$. However, an $n$-tuple of numbers has no geometric meaning in the absence of a frame. Moreover, with respect to different frames, points have different coordinates. The process of translating from one Cartesian frame to another is made precise with the following theorem. Recall from linear algebra the concept of change of basis matrix (see Appendix A).

**Theorem 1.10.** Let $\mathcal{K} = (O, \mathcal{B})$ and $\mathcal{K}' = (O', \mathcal{B}')$ be two frames in $\mathbb{A}^n$. For any point $P \in \mathbb{A}^n$ we have

$$[P]_{\mathcal{K}'} = \mathrm{M}_{\mathcal{B}',\mathcal{B}} \cdot ([P]_{\mathcal{K}} - [O']_{\mathcal{K}}) = \mathrm{M}^{-1}_{\mathcal{B},\mathcal{B}'} \cdot ([P]_{\mathcal{K}} - [O']_{\mathcal{K}}) = \mathrm{M}_{\mathcal{B}',\mathcal{B}} \cdot [P]_{\mathcal{K}} + [O]_{\mathcal{K}'}. \tag{1.3}$$

*Proof.* Since the coordinates of a point are the components of its position vector, Equation (1.3) is equivalent to:

$$[\overrightarrow{O'P}]_{\mathcal{B}'} = \mathrm{M}_{\mathcal{B}',\mathcal{B}} \cdot ([\overrightarrow{OP}]_{\mathcal{B}} - [\overrightarrow{OO'}]_{\mathcal{B}}) = \mathrm{M}^{-1}_{\mathcal{B},\mathcal{B}'} \cdot ([\overrightarrow{OP}]_{\mathcal{B}} - [\overrightarrow{OO'}]_{\mathcal{B}}) = \mathrm{M}_{\mathcal{B}',\mathcal{B}} \cdot [\overrightarrow{OP}]_{\mathcal{B}} + [\overrightarrow{O'O}]_{\mathcal{B}'}. \tag{1.4}$$

Since $[\overrightarrow{O'P}]_{\mathcal{B}} = [\overrightarrow{OP}]_{\mathcal{B}} - [\overrightarrow{OO'}]_{\mathcal{B}}$ and since $\mathrm{M}_{\mathcal{B}',\mathcal{B}}$ is the change of basis matrix from $\mathcal{B}$ to $\mathcal{B}'$, the first equality follows. The second equality follows from the property that $\mathrm{M}_{\mathcal{B}',\mathcal{B}} = \mathrm{M}^{-1}_{\mathcal{B},\mathcal{B}'}$ (see Appendix A). The last equality is obtained by opening the parentheses and noticing that

$$\mathrm{M}^{-1}_{\mathcal{B},\mathcal{B}'}[\overrightarrow{OO'}]_{\mathcal{B}} = [\overrightarrow{OO'}]_{\mathcal{B}'} = -[\overrightarrow{O'O}]_{\mathcal{B}'}.$$

$\square$

Let us flesh out the steps needed to translate coordinates from one Cartesian frame to another, by expanding Theorem 1.10 using Appendix A. Let $\mathcal{K} = (O,\mathcal{B})$ and $\mathcal{K}' = (O',\mathcal{B}')$ be two frames in $\mathbb{A}^n$ with $\mathcal{B} = (\mathbf{i}_1,\dots,\mathbf{i}_n)$ and $\mathcal{B}' = (\mathbf{i}'_1,\dots,\mathbf{i}'_n)$. Suppose we know $\mathcal{K}'$ in terms of $\mathcal{K}$, i.e., the coordinates of $O'$ relative to $\mathcal{K}$ and the components of $\mathbf{i}'_1,\dots,\mathbf{i}'_n$ with respect to $\mathcal{B}$ are given. If the coordinates of a point $P$ are given relative to $\mathcal{K}$, we can find the coordinates of $P$ relative to $\mathcal{K}'$ with the following steps:

1. Construct the change of basis matrix $\mathrm{M}_{\mathcal{B},\mathcal{B}'}$ by placing $[\mathbf{i}'_1]_{\mathcal{B}},\dots,[\mathbf{i}'_n]_{\mathcal{B}}$ in the columns of the matrix.

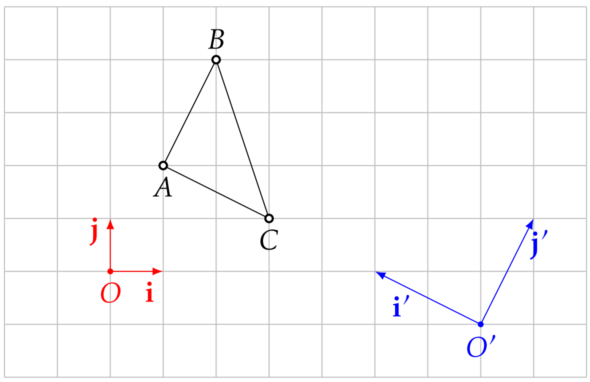
2. Calculate $\mathrm{M}_{\mathcal{B}',\mathcal{B}}$ by inverting $\mathrm{M}_{\mathcal{B},\mathcal{B}'}$.

3. Calculate $[P]_{\mathcal{K}'} = \mathrm{M}_{\mathcal{B}',\mathcal{B}}\cdot([P]_{\mathcal{K}} - [O']_{\mathcal{K}})$ since $[P]_{\mathcal{K}}$ and $[O']_{\mathcal{K}}$ are given.

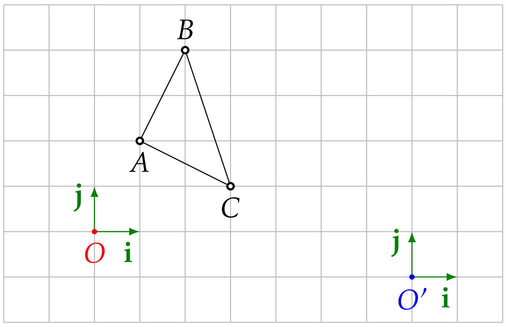
**Example 1.11** (In dimension 2)**.** Let $\mathcal{K} = (O,\mathcal{B})$ and $\mathcal{K}' = (O',\mathcal{B}')$ be two frames in $\mathbb{E}^2$, with $\mathcal{B} = (\mathbf{i},\mathbf{j})$ and $\mathcal{B}' = (\mathbf{i}',\mathbf{j}')$. Assume that $O'$, $\mathbf{i}'$ and $\mathbf{j}'$ are known relative to $\mathcal{K}$:

$$[O']_{\mathcal{K}} = \begin{bmatrix} 7 \\ -1 \end{bmatrix}, \quad [\mathbf{i}']_{\mathcal{B}} = \begin{bmatrix} -2 \\ 1 \end{bmatrix}, \quad [\mathbf{j}']_{\mathcal{B}} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}. \tag{1.5}$$

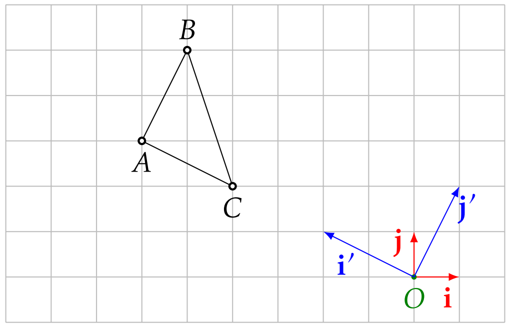
How can we translate the coordinates of points from $\mathcal{K}$ to $\mathcal{K}'$?




We can achieve this in two steps: (a) first, we change the origin, i.e., we go from $(O,\mathcal{B})$ to $(O',\mathcal{B})$, and (b) we change the direction of the coordinate axes, i.e., we go from $(O',\mathcal{B})$ to $(O',\mathcal{B}')$. The first step is simply a translation, while the second corresponds to the usual base change from linear algebra.




(a) Change the origin.




(b) Change the direction of the axes.

In the first step, when going from $\mathcal{K} = (O, \mathcal{B})$ to $\tilde{\mathcal{K}} = (O', \mathcal{B})$, we are looking for the components of the position vector of $A$ relative to $O'$ with respect to $\mathcal{B}$.

$$\overrightarrow{O'A} = \overrightarrow{OA} - \overrightarrow{OO'} \quad \text{and therefore} \quad [\,\overrightarrow{O'A}\,]_{\mathcal{B}} = [\,\overrightarrow{OA}\,]_{\mathcal{B}} - [\,\overrightarrow{OO'}\,]_{\mathcal{B}}.$$

In the second step, when going from $\tilde{\mathcal{K}} = (O', \mathcal{B})$ to $\mathcal{K}' = (O', \mathcal{B}')$, we are looking for the components of the position vector of $A$ relative to $O'$ with respect to $\mathcal{B}'$. From linear algebra, we know that this is done with the change of basis matrix (see Appendix A). Let $\mathrm{M}_{\mathcal{B}',\mathcal{B}}$ be the change of basis matrix from the basis $\mathcal{B}$ to the basis $\mathcal{B}'$. Then,

$$[\,\overrightarrow{OA}\,]_{\mathcal{B}'} = \mathrm{M}_{\mathcal{B}',\mathcal{B}}[\,\overrightarrow{OA}\,]_{\mathcal{B}}.$$

Thus, composing the two operations, (a) and (b), we obtain

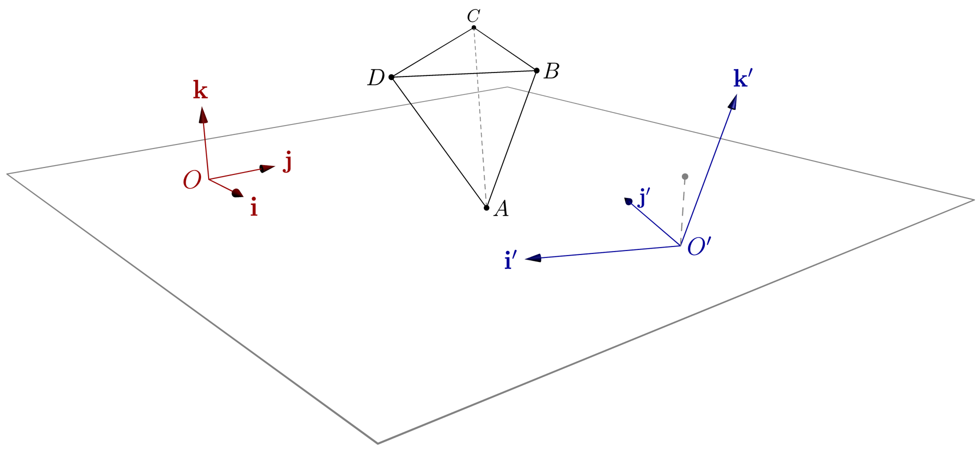
$$[A]_{\mathcal{K}'} = [\,\overrightarrow{O'A}\,]_{\mathcal{B}'} = \mathrm{M}_{\mathcal{B}',\mathcal{B}} \cdot [\,\overrightarrow{O'A}\,]_{\mathcal{B}} = \mathrm{M}_{\mathcal{B}',\mathcal{B}} \cdot \Big( [\,\overrightarrow{OA}\,]_{\mathcal{B}} - [\,\overrightarrow{OO'}\,]_{\mathcal{B}} \Big) \tag{1.6}$$

Now, suppose that $A$ has coordinates $(1,2)$ with respect to the frame $\mathcal{K}$. Since $[\,\overrightarrow{OO'}\,]_{\mathcal{B}} = [O']_{\mathcal{K}}$, by (1.5), we already know the terms in the parentheses on the right-hand side of (1.6). Furthermore, from our assumptions (1.5), it is easy to write down the matrix $\mathrm{M}_{\mathcal{B},\mathcal{B}'}$ and then $\mathrm{M}_{\mathcal{B}',\mathcal{B}} = \mathrm{M}^{-1}_{\mathcal{B},\mathcal{B}'}$. Thus

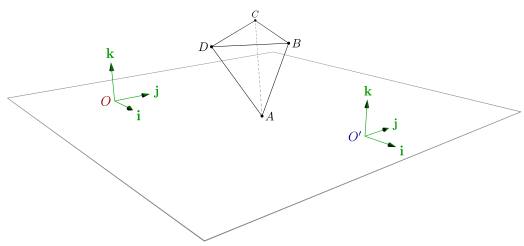
$$[A]_{\mathcal{K}'} = \begin{bmatrix} -2 & 1 \\ 1 & 2 \end{bmatrix}^{-1} \cdot \left( \begin{bmatrix} 1 \\ 2 \end{bmatrix} - \begin{bmatrix} 7 \\ -1 \end{bmatrix} \right) = \frac{1}{-5} \begin{bmatrix} 2 & -1 \\ -1 & -2 \end{bmatrix} \cdot \left( \begin{bmatrix} 1 \\ 2 \end{bmatrix} - \begin{bmatrix} 7 \\ -1 \end{bmatrix} \right) = \begin{bmatrix} 3 \\ 0 \end{bmatrix}.$$

**Example 1.12.** Let $\mathcal{K} = (O, \mathcal{B})$ and $\mathcal{K}' = (O', \mathcal{B}')$ be two frames in $\mathbb{E}^3$ with $\mathcal{B} = (\mathbf{i}, \mathbf{j}, \mathbf{k})$ and $\mathcal{B}' = (\mathbf{i}', \mathbf{j}', \mathbf{k}')$. Assume that $O'$, $\mathbf{i}', \mathbf{j}'$ and $\mathbf{k}'$ are known relative to $\mathcal{K}$:

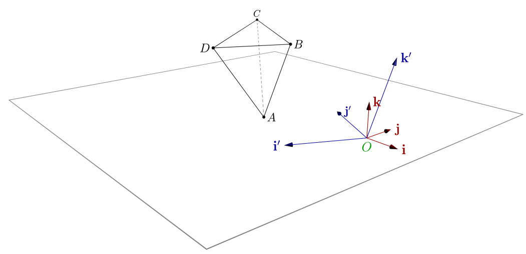
$$[O']_{\mathcal{K}} = \begin{bmatrix} 4 \\ 5 \\ -1 \end{bmatrix}, \quad \mathbf{i}' = -\mathbf{i} - 2\mathbf{j} = \begin{bmatrix} -1 \\ -2 \\ 0 \end{bmatrix}, \quad \mathbf{j}' = -2\mathbf{i} + \mathbf{j} = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{k}' = \mathbf{j} + 2\mathbf{k} = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}.$$




In any dimension, the coordinates with respect to one frame can be obtained from the coordinates with respect to another frame in two steps.




(a) Change the origin.



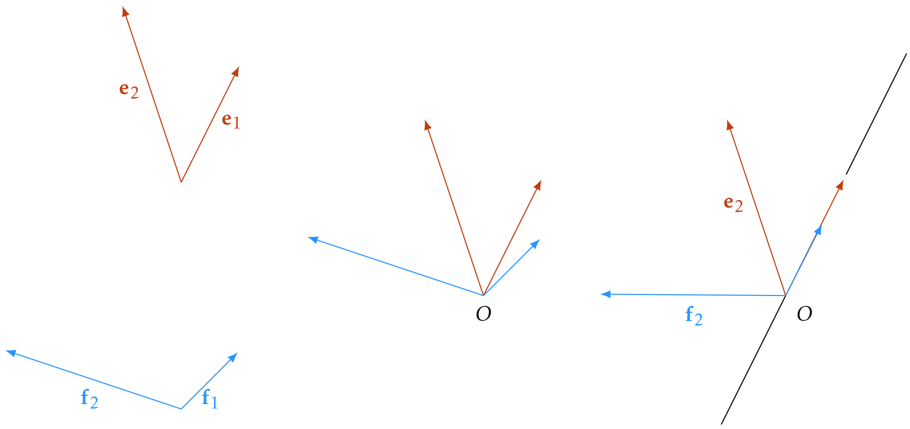

(b) Change the direction of the axes.

Let $B$ be the point with coordinates $(1,5,1)$ relative to $\mathcal{K}$. The argument used for dimension 2 (Example 1.5) literally translates to our 3-dimensional setting and we have

$$[B]_{\mathcal{K}'} = \mathrm{M}^{-1}_{\mathcal{B},\mathcal{B}'} \cdot ([B]_{\mathcal{K}} - [O']_{\mathcal{K}}) = \begin{bmatrix} -1 & -2 & 0 \\ -2 & 1 & 1 \\ 0 & 0 & 2 \end{bmatrix}^{-1} \left( \begin{bmatrix} 1 \\ 5 \\ 1 \end{bmatrix} - \begin{bmatrix} 4 \\ 5 \\ -1 \end{bmatrix} \right) = \frac{1}{10} \begin{bmatrix} -2 & -4 & 2 \\ -4 & 2 & -1 \\ 0 & 0 & 5 \end{bmatrix} \left( \begin{bmatrix} 1 \\ 5 \\ 1 \end{bmatrix} - \begin{bmatrix} 4 \\ 5 \\ -1 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

## 1.5 Orientation

Consider the bases $\mathcal{E} = (\mathbf{e}_1, \mathbf{e}_2)$ and $\mathcal{F} = (\mathbf{f}_1, \mathbf{f}_2)$ of $\mathbb{V}^2$. Represent all four vectors at a common point $O$ and rotate the basis $\mathcal{F}$ such that the vector $\mathbf{f}_1$ points in the same direction as the vector $\mathbf{e}_1$, i.e., such that $\mathbf{f}_1 = \lambda \mathbf{e}_1$ for some positive scalar $\lambda$.




The line passing through $O$ with direction vector $\mathbf{e}_1$ (and also $\mathbf{f}_1$) separates the plane $\mathbb{E}^2$ into two half-planes. Considering the positioning of the second vectors, two things can happen: $\mathbf{e}_2$ and $\mathbf{f}_2$ point toward the same half-plane or they point toward different half-planes.

**Proposition 1.13.** Let $\mathcal{E}$ and $\mathcal{F}$ be as above.

1. If $\det(\mathrm{M}_{\mathcal{E},\mathcal{F}}) > 0$ then $\mathbf{e}_2$ and $\mathbf{f}'_2$ point to the same half-plane.

2. If $\det(\mathrm{M}_{\mathcal{E},\mathcal{F}}) < 0$ then $\mathbf{e}_2$ and $\mathbf{f}'_2$ point to different half-planes.

*Sketch of the proof.* In the above process, we changed the basis $\mathcal{F} = (\mathbf{f}_1, \mathbf{f}_2)$ to the basis $\mathcal{F}' = (\mathbf{f}'_1, \mathbf{f}'_2)$ with a rotation. So, the change of basis matrix $\mathrm{M}_{\mathcal{F}',\mathcal{F}}$ is a $2\times 2$ rotation matrix which has determinant equal to 1 (for more on rotations, see Chapter 6). Now, considering the coordinates of the vectors in $\mathcal{F}'$ with respect to $\mathcal{E}$, we have $\mathbf{f}'_1 = (\lambda, 0)$ and $\mathbf{f}'_2 = (a, b)$. Thus, we notice that the vectors $\mathbf{e}_1$ and $\mathbf{f}_1$ point into the same half-plane if $b > 0$. If we now calculate the above determinant, we obtain

$$\det(\mathrm{M}_{\mathcal{E},\mathcal{F}}) = \det(\mathrm{M}_{\mathcal{E},\mathcal{F}'} \cdot \mathrm{M}_{\mathcal{F}',\mathcal{F}}) = \det(\mathrm{M}_{\mathcal{E},\mathcal{F}'}) \cdot \underbrace{\det(\mathrm{M}_{\mathcal{F}',\mathcal{F}})}_{=1} = \begin{vmatrix} \lambda & a \\ 0 & b \end{vmatrix} = \lambda \cdot b$$

and, since $\lambda$ is positive, the proof is finished. □

**Definition 1.14.** In the first case of Proposition 1.13 we say that $\mathcal{E}$ and $\mathcal{F}$ have the *same orientation* and in the second case we say that $\mathcal{E}$ and $\mathcal{F}$ have *opposite orientation*.

Why is this relevant? Besides the fact that it gives a geometric interpretation of the sign of the determinant $\det(\mathrm{M}_{\mathcal{E},\mathcal{F}})$, it also allows us to understand some signs which appear in calculations of areas (see Chapter 4). Moreover, the trigonometry that we know to hold true in $\mathbb{E}^2$ implicitly builds on the notion of oriented Euclidean plane $\mathbb{E}^2$. Mathematically, the distinction in orientation is only a matter of keeping track of the signs of some determinants. However, in relation to the physical world this distinction is more concrete.

**Definition 1.15.** Let $(\mathbf{i},\mathbf{j})$ be a basis of $\mathbb{V}^2$ represented in a common point $O \in \mathbb{E}^2$ such that $\mathbf{i} = \overrightarrow{OX}$ and $\mathbf{j} = \overrightarrow{OY}$. Rotate the plane such that $\mathbf{i}$ points downward. If $Y$ is in the right half-plane determined by the line $OX$, then we say that the basis $(\mathbf{i},\mathbf{j})$ is *right oriented*. If $Y$ lies in the left half-plane, we say that the basis $(\mathbf{i},\mathbf{j})$ is *left oriented*. A coordinate system $(O,\mathbf{i},\mathbf{j})$ is *left* or *right oriented* if the basis $(\mathbf{i},\mathbf{j})$ is left or right oriented, respectively.

Fixing an orientation in the Euclidean plane $\mathbb{E}^2$ is equivalent to choosing a coordinate system $\mathcal{K} = (O,\mathcal{B})$ and calling it right oriented. Then, all other bases of $\mathbb{V}^2$ either have the same orientation as $\mathcal{B}$, in which case they are also called right oriented, or they have opposite orientation, in which case they are called left oriented. When it comes to a concrete configuration of points, on a sheet of paper for instance, such a choice can be made with the right-hand rule. Once we have such a choice, $\mathbb{E}^2$ is called oriented. In other words, the *oriented plane* $\mathbb{E}^2$ is the usual Euclidean plane together with a choice of which of the two opposite classes of bases contains the “preferred” bases.

**Definition 1.16.** Two bases $\mathcal{E}$ and $\mathcal{F}$ of the vector space $\mathrm{D}(\mathbb{E}^n)$ are said to have the *same orientation* if $\det(\mathrm{M}_{\mathcal{E},\mathcal{F}}) > 0$. They have *opposite orientation* if $\det(\mathrm{M}_{\mathcal{E},\mathcal{F}}) < 0$. Two frames $\mathcal{K}$ and $\mathcal{K}'$ have the *same orientation* if their bases have the same orientation, and they have *opposite orientation* otherwise.

We say that the Euclidean space $\mathbb{E}^n$ is *oriented* if there is a choice of a frame $\mathcal{K} = (O,\mathcal{B})$ which is called *right oriented*. Then, all other frames of $\mathbb{E}^n$ with the same orientation as $\mathcal{K}$ are also *right oriented* and all other bases with opposite orientation are called *left oriented*.

**Remark.** Unless otherwise stated, whenever we consider a frame $\mathcal{K} = (O,\mathcal{B})$ of $\mathbb{E}^n$, we will assume that it is right oriented and that $\mathbb{E}^n$ is therefore an oriented Euclidean space.

# CHAPTER 2

## Affine subspaces

**Contents**

## 2.1 Lines in $\mathbb{A}^2$

The Euclidean plane $\mathbb{E}^2$ is a 2-dimensional real affine space. In order to emphasize the fact that we treat $\mathbb{E}^2$ as an affine space only, we denote it by $\mathbb{A}^2$. In this setting, a line in $\mathbb{A}^2$ is an affine subspace of dimension 1, i.e., $\mathrm{D}(S)$ is a 1-dimensional vector subspace of $\mathbb{V}^2 = \mathrm{D}(\mathbb{A}^2)$ (see Definition 1.4). In terms of the map $\mathbf{r}_Q : \mathbb{A}^2 \to \mathbb{V}^2$, which identifies points $P$ with position vectors relative to a fixed point $Q \in \mathbb{A}^2$, the subset $S \subseteq \mathbb{A}^2$ is a line if and only if for a point $Q \in S$ we have

$$\mathbf{r}_Q(S) = \left\{ \overrightarrow{QP} : P \in S \right\} \quad \text{is a 1-dimensional vector subspace of } \mathbb{V}^2.$$

It is not difficult to see that if the above description holds for one point $Q \in S$, it holds for any point $Q \in S$. The nonzero elements of the direction space $\mathrm{D}(S)$ of the line $S$ are called *direction vectors*.

### 2.1.1 Parametric equations

For any two distinct points $P, Q$ on a line $S$, the vector $\overrightarrow{QP}$ is a direction vector of $S$. Since $\mathrm{D}(S)$ is 1-dimensional, all direction vectors are linearly dependent, i.e., $\mathbf{v}$ is a direction vector for $S$ if and only if it is linearly dependent on $\overrightarrow{QP}$. So, for any direction vector $\mathbf{v}$ of $S$ there is a *unique* scalar $t \in \mathbb{R}$ such that

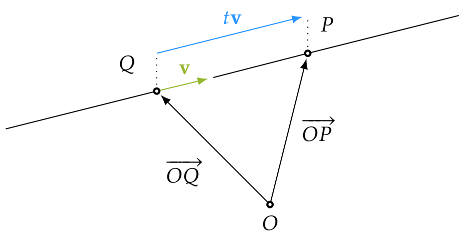
$$\overrightarrow{QP} = t\mathbf{v}.$$

Now, if we fix $Q$ and let $P$ vary on the line, then $t$ varies in $\mathbb{R}$. Since $\mathbf{r}_Q : \mathbb{A}^2 \to \mathbb{V}^2$ is a bijection, the line $S$ can be described as

$$S = \left\{ P \in \mathbb{A}^2 : \overrightarrow{QP} = t\mathbf{v} \text{ for some } t \in \mathbb{R} \right\}.$$

In this description, the point $Q \in S$ is arbitrary but fixed. If we want to emphasize that the description depends on fixing $Q$, we refer to this point as the *base point (of this description)* of the line. Moreover, for any point $O \in \mathbb{A}^2$ we may split the vector $\overrightarrow{QP}$ in the equation $\overrightarrow{QP} = t\mathbf{v}$ to obtain

$$\overrightarrow{OP} = \overrightarrow{OQ} + t\mathbf{v}. \tag{2.1}$$




So, we can describe the line $S$ as the set of points $P$ in $\mathbb{A}^2$ which satisfy Equation (2.1) for some $t \in \mathbb{R}$. This equation is called the *vector equation of the line $S$ relative to $O$, having base point $Q$ and direction vector* $\mathbf{v}$, or simply *a vector equation* of the line $S$. If $\mathbf{v} = \overrightarrow{QA}$, this equation also describes the segment $[QA]$ if we restrict the parameter to $t \in [0,1]$. Moreover, for $t > 0$ and $t < 0$ we obtain two rays emanating from $Q$.

Notice that a vector equation depends on the choice of the base point $Q$ and on the choice of the direction vector $\mathbf{v}$. In particular, a line does not have a unique vector equation. Notice also that the vector equation does not depend on the frame. In the above description $O$ can be any point in $\mathbb{A}^2$.

Now fix a frame $\mathcal{K} = (O, \mathcal{B})$. If we write Equation (2.1) in coordinates relative to $\mathcal{K}$, we obtain

$$S : \begin{cases} x = x_Q + tv_x \\ y = y_Q + tv_y \end{cases} \quad \text{or, in matrix form} \quad S : \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x_Q \\ y_Q \end{bmatrix} + t \begin{bmatrix} v_x \\ v_y \end{bmatrix} \tag{2.2}$$

where $Q = Q(x_Q, y_Q)$, $\mathbf{v} = \mathbf{v}(v_x, v_y)$ and where $t$ is the *parameter*; for different values of $t$ we obtain different points $(x, y)$ on the line. The two equations in the system (2.2) are called *parametric equations* of the line $S$. Traditionally, they are written in the form of a system of equations as indicated on the left. Writing them as one equation, as indicated on the right, is closer to the computational perspective where we identify points with column matrices. Clearly, the two ways of writing such parametric equations are equivalent.

### 2.1.2 Cartesian equations

It is possible to eliminate the parameter $t$ in (2.2). By expressing $t$ in both equations and setting the two expressions equal, we obtain

$$\frac{x - x_Q}{v_x} = \frac{y - y_Q}{v_y}. \tag{2.3}$$

We refer to Equation (2.3) as *a symmetric equation* of the line $S$. It could happen that $v_x$ or $v_y$ is zero. In that case, translate back to the parametric equations to understand what happens.

**Example 2.1.** The line with symmetric equation

$$\frac{x-3}{2} = \frac{y-5}{0} \quad \text{has parametric equations} \quad \begin{cases} x = 3 + 2 \cdot t \\ y = 5 + 0 \cdot t \end{cases}.$$

Thus, it is the line parallel to $Ox$ described by the equation $y = 5$.

We have just described a line with a linear equation relative to the frame $\mathcal{K}$. The converse is also true.

**Proposition 2.2.** Every line in $\mathbb{A}^2$ can be described with a linear equation in two variables

$$ax + by + c = 0 \tag{2.4}$$

relative to a fixed frame, and any linear equation in two variables relative to a fixed frame describes a line if the constants $a$ and $b$ are not both zero. Moreover, if Equation (2.4) describes the line $\ell$ relative to a frame $\mathcal{K} = (O, \mathcal{B})$, then the direction space $\mathrm{D}(\ell)$ of the line is the 1-dimensional subspace of $\mathbb{V}^2$ which, relative to the basis $\mathcal{B}$, satisfies the equation

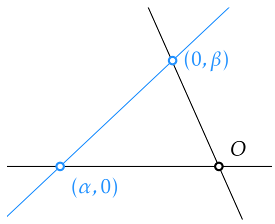
$$\mathrm{D}(\ell) : ax + by = 0.$$

*Proof.* This is a particular case of Theorem 2.6. $\square$

Equation (2.4) is called a *Cartesian equation* of the line which it describes. Notice that there are infinitely many Cartesian equations describing the same line, since we can multiply one equation by a nonzero constant. It is sometimes useful to rearrange the linear equation (2.4) in order to emphasize some geometric properties. For example, we can rearrange it in the form

$$\frac{x}{\alpha}+\frac{y}{\beta}=1 \quad \text{where} \quad \alpha=-\frac{c}{a} \quad \text{and} \quad \beta=-\frac{c}{b}.$$

In this form, we have the *equation of the line where we can read off the intersection points with the coordinate axes* since this line intersects $Ox$ in $(\alpha,0)$ and it intersects $Oy$ in $(0,\beta)$.




Alternatively, we may express the linear equation (2.4) with a determinant

$$\begin{vmatrix} x-x_Q & y-y_Q \\ v_x & v_y \end{vmatrix}=0$$

which says that a point $P$ belongs to the line if $\overrightarrow{QP}$ is linearly dependent on $\mathbf{v}$. In this form, we may describe the two half-planes which are separated by the line with the inequalities

$$\begin{vmatrix} x-x_Q & y-y_Q \\ v_x & v_y \end{vmatrix}<0 \quad \text{and} \quad \begin{vmatrix} x-x_Q & y-y_Q \\ v_x & v_y \end{vmatrix}>0$$

Indeed, any point $P$ in the plane either lies on the line or $(\overrightarrow{QP},\mathbf{v})$ is a left-oriented basis or $(\overrightarrow{QP},\mathbf{v})$ is a right-oriented basis.

### 2.1.3 Relative positions of two lines in $\mathbb{A}^2$

The tools of linear algebra readily apply to describe intersections of lines in $\mathbb{A}^2$. Assume that we have two lines

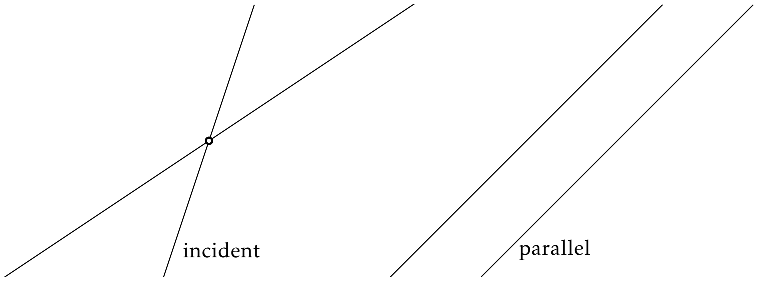
$$\ell_1: a_1x+b_1y+c_1=0 \quad \text{and} \quad \ell_2: a_2x+b_2y+c_2=0.$$

In order to determine if they intersect, one has to discuss the system:

$$\begin{cases} \ell_1: a_1x+b_1y+c_1 & = & 0 \\ \ell_2: a_2x+b_2y+c_2 & = & 0 \end{cases}. \tag{2.5}$$

Discussing this system is basic linear algebra (see, for example, [9, Section 3.6]). In the plane, the situation is very simple:

- two lines intersect in a unique point, the coordinates of which are the solution to (2.5); or
- they don't intersect and (2.5) doesn't have solutions, in which case the lines are parallel; or
- the system (2.5) has infinitely many solutions, in which case $\ell_1 = \ell_2$.




## 2.2 Planes in $\mathbb{A}^3$

The usual Euclidean space $\mathbb{E}^3$ is a 3-dimensional real affine space. In order to emphasize the fact that we treat $\mathbb{E}^3$ as an affine space only, we denote it by $\mathbb{A}^3$. A plane in $\mathbb{A}^3$ is a set of points $S$ which is an affine subspace of dimension 2 (see Definition 1.4). Considering the bijection $\mathbf{r}_Q : \mathbb{A}^3 \to \mathbb{V}^3$ which identifies points with position vectors relative to a fixed point $Q \in \mathbb{A}^3$, the subset $S$ is a plane if and only if for any $Q \in S$

$$\mathbf{r}_Q(S) = \{\overrightarrow{QP} : P \in S\} \quad \text{is a 2-dimensional vector subspace of } \mathbb{V}^3.$$

It is not difficult to see that the above description does not depend on the point $Q \in S$. If $S$ is a plane, the direction space $\mathrm{D}(S) = \mathbf{r}_Q(S)$ is a 2-dimensional vector subspace of $\mathbb{V}^3$.

### 2.2.1 Parametric equations

Since $\mathrm{D}(S)$ is a 2-dimensional vector space, any basis will contain two vectors. Let $(\mathbf{v}, \mathbf{w})$ be a basis of $\mathrm{D}(S)$. Then, for any two points $P, Q \in S$, the vector $\overrightarrow{QP}$ is a linear combination of the basis vectors, i.e., there exist unique scalars $s, t \in \mathbb{R}$ such that

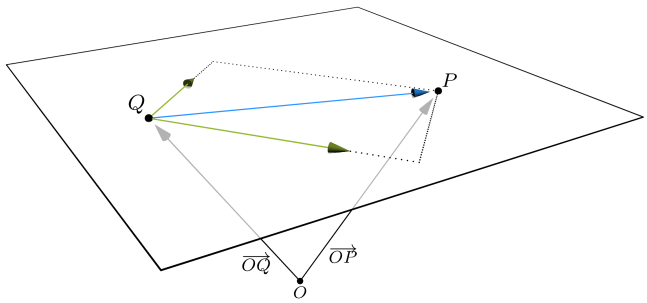
$$\overrightarrow{QP} = s\mathbf{v} + t\mathbf{w}.$$

Now, if we fix $Q$ and let $P$ vary in the plane $S$, then $s$ and $t$ vary in $\mathbb{R}$. Since $\mathbf{r}_Q : \mathbb{A}^3 \to \mathbb{V}^3$ is a bijection, the plane $S$ can be described as

$$S = \left\{P \in \mathbb{A}^3 : \overrightarrow{QP} = s\mathbf{v} + t\mathbf{w} \text{ for some } s, t \in \mathbb{R}\right\}.$$

In this description, the point $Q \in S$ is arbitrary but fixed. If we want to emphasize that the description depends on fixing $Q$, we refer to this point as the *base point (of this description)*. Moreover, for any point $O \in \mathbb{A}^3$ we may split the vector $\overrightarrow{QP}$ in the equation $\overrightarrow{QP} = s\mathbf{v} + t\mathbf{w}$ to obtain

$$\overrightarrow{OP} = \overrightarrow{OQ} + s\mathbf{v} + t\mathbf{w}. \tag{2.6}$$




So, we can describe the plane $S$ as the set of points $P$ in $\mathbb{A}^3$ which satisfy Equation (2.6) for some $s,t \in \mathbb{R}$. This equation is called the *vector equation of the plane $S$ relative to $O$, having base point $Q$ and direction vectors* $\mathbf{v}$ *and* $\mathbf{w}$, or simply *a vector equation* of the plane $S$. If $\mathbf{v} = \overrightarrow{QA}$, $\mathbf{w} = \overrightarrow{QC}$, and $\mathbf{v} + \mathbf{w} = \overrightarrow{QB}$, this equation also describes the interior of the parallelogram $QABC$ if we restrict the parameters $s,t$ to $(0,1)$. The interior of the triangle $QAC$ is obtained if we further impose the condition that $s+t<1$. Moreover, for $t>0$ and $t<0$ we obtain two half-planes separated by the line $QA$.

Notice that a vector equation depends on the choice of the base point $Q$ and on the choice of the vectors $\mathbf{v}$ and $\mathbf{w}$. In analogy with the case of the line in $\mathbb{A}^2$, we may call such vectors *direction vectors* for the plane $S$. In particular, a plane does not have a unique vector equation. Notice also that the vector equation does not depend on the frame. In the above description $O$ can be any point in $\mathbb{A}^3$.

Now fix a frame $\mathcal{K} = (O, \mathcal{B})$. If we write Equation (2.6) in coordinates relative to $\mathcal{K}$, we obtain

$$S : \begin{cases} x = x_Q + sv_x + tw_x \\ y = y_Q + sv_y + tw_y \\ z = z_Q + sv_z + tw_z \end{cases} \quad \text{or, in matrix form} \quad S : \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x_Q \\ y_Q \\ z_Q \end{bmatrix} + s \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix} + t \begin{bmatrix} w_x \\ w_y \\ w_z \end{bmatrix} \tag{2.7}$$

where $Q = Q(x_Q, y_Q, z_Q)$, $\mathbf{v} = \mathbf{v}(v_x, v_y, v_z)$, and $\mathbf{w} = \mathbf{w}(w_x, w_y, w_z)$. The values $s$ and $t$ are called *parameters*, and for different parameters we obtain different points $(x,y,z)$ in the plane $S$. The three equations in the system (2.7) are called *parametric equations* for the plane $S$.

### 2.2.2 Cartesian equations

Eliminating the parameters $s,t$ in (2.7) yields a linear equation in $x,y,z$. A more direct approach is to observe that a point $P(x,y,z)$ lies in the plane $S$ if and only if the vector $\overrightarrow{QP}$ is linearly dependent on the vectors $\mathbf{v}$ and $\mathbf{w}$. This is equivalent to:

$$\begin{vmatrix} x - x_Q & y - y_Q & z - z_Q \\ v_x & v_y & v_z \\ w_x & w_y & w_z \end{vmatrix} = 0. \tag{2.8}$$

In particular, four points $Q$, $E$, $F$, and $G$ are coplanar if and only if the vectors $\overrightarrow{QE}$, $\overrightarrow{QF}$, and $\overrightarrow{QG}$ are linearly dependent, which is equivalent to

$$\begin{vmatrix} x_E - x_Q & y_E - y_Q & z_E - z_Q \\ x_F - x_Q & y_F - y_Q & z_F - z_Q \\ x_G - x_Q & y_G - y_Q & z_G - z_Q \end{vmatrix} = 0. \tag{2.9}$$

Notice that Equation (2.9) is just a restatement of the fact that four points $Q$, $E$, $F$, and $G$ are coplanar if and only if the vectors $\overrightarrow{QE}$, $\overrightarrow{QF}$, and $\overrightarrow{QG}$ are linearly dependent. Notice also that if we replace the equals sign in (2.8) with inequalities, we describe the two half-spaces separated by this plane since, for a point $P$ not in the given plane, $(\overrightarrow{QP}, \mathbf{v}, \mathbf{w})$ is a basis which is either left- or right-oriented.

We have just described a plane with a linear equation (Equation (2.8)) relative to the frame $\mathcal{K}$. The converse is also true.

**Proposition 2.3.** Every plane in $\mathbb{A}^3$ can be described with a linear equation in three variables

$$ax + by + cz + d = 0 \tag{2.10}$$

relative to a fixed frame, and any linear equation relative to a fixed frame in three variables describes a plane if the constants $a, b,$and $c$ are not all zero. Moreover, if Equation (2.10) describes the plane $\pi$ relative to a frame $\mathcal{K} = (O, \mathcal{B})$, then the direction space $\mathrm{D}(\pi)$ of the plane is the 2-dimensional subspace of $\mathbb{V}^3$ which, relative to the basis $\mathcal{B}$, satisfies the equation

$$\mathrm{D}(\pi) : ax + by + cz = 0.$$

*Proof.* This is a particular case of Theorem 2.6. $\square$

Equation (2.10) is called a *Cartesian equation* of the plane it describes. Notice that there are infinitely many Cartesian equations describing the same plane, since we can multiply one equation by a nonzero constant. Here again, it may be useful to rearrange the linear equation (2.10) in order to emphasize some geometric properties. For example, we can rearrange it in the form

$$\frac{x}{\alpha} + \frac{y}{\beta} + \frac{z}{\gamma} = 1 \quad \text{where} \quad \alpha = -\frac{d}{a}, \quad \beta = -\frac{d}{b} \quad \text{and} \quad \gamma = -\frac{d}{c}.$$

In this form, we have the *equation of the plane where we can read off the intersection points with the coordinate axes* since the plane intersects $Ox$ in $(\alpha, 0, 0)$, it intersects $Oy$ in $(0, \beta, 0)$, and it intersects $Oz$ in $(0, 0, \gamma)$.

### 2.2.3 Relative positions of two planes in $\mathbb{A}^3$

In order to describe intersections of planes, we make use of linear algebra. Assume that we have two planes

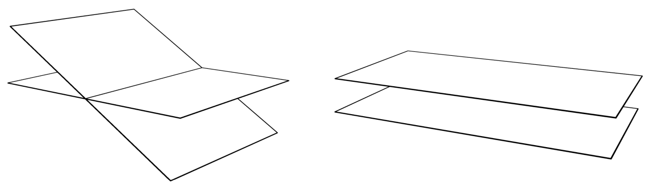
$$\pi_1 : a_1x + b_1y + c_1z + d_1 = 0 \quad \text{and} \quad \pi_2 : a_2x + b_2y + c_2z + d_2 = 0.$$

We determine if they intersect or not by discussing the system:

$$\begin{cases} \pi_1 : a_1x + b_1y + c_1z + d_1 &= 0 \\ \pi_2 : a_2x + b_2y + c_2z + d_2 &= 0 \end{cases} \tag{2.11}$$

Discussing this system is basic linear algebra (see, for example, [9, Section 3.6]). Here again, the situation is very simple. Let $M$ be the matrix of the system and $\widetilde{M}$ the extended matrix of the system. Then we have:

- two planes either intersect in a line (the coordinates of the points on the line will be solutions to (2.11)); this happens if the rank of $M$ and the rank of $\widetilde{M}$ equal 2; or
- they don't intersect and (2.11) doesn't have solutions, in which case the planes are parallel; this happens if the rank of $M$ is strictly less than the rank of $\widetilde{M}$; or
- the solutions to system (2.11) depend on two parameters, in which case $\pi_1 = \pi_2$; this happens if the rank of $M$ and the rank of $\widetilde{M}$ are equal to 1.



## 2.3 Lines in $\mathbb{A}^3$

Here again, we treat the usual Euclidean space $\mathbb{E}^3$ as a 3-dimensional affine space and denote it by $\mathbb{A}^3$. As in the case of $\mathbb{A}^2$, by Definition 1.4, a line in $\mathbb{A}^3$ is an affine subspace of dimension 1. Hence, the subset $S$ is a line if for any $Q \in S$ we have that

$$\mathbf{r}_Q(S) = \left\{\overrightarrow{QP} : P \in S\right\} \quad \text{is a 1-dimensional vector subspace of } \mathbb{V}^3.$$

### 2.3.1 Parametric equations

The vector description of a line in $\mathbb{A}^3$ is identical to that in $\mathbb{A}^2$. Given a base point $Q$ and a direction vector $\mathbf{v}$, the line $S$ is the set of points $P$ satisfying:

$$\overrightarrow{OP} = \overrightarrow{OQ} + t\mathbf{v}, \quad t \in \mathbb{R}. \tag{2.12}$$

Fixing a frame $\mathcal{K} = (O, \mathcal{B})$, we write Equation (2.12) in coordinates to obtain:

$$\begin{cases} x = x_Q + tv_x \\ y = y_Q + tv_y \\ z = z_Q + tv_z \end{cases} \quad \text{or, in matrix form} \quad \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x_Q \\ y_Q \\ z_Q \end{bmatrix} + t \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix} \tag{2.13}$$

where $Q = Q(x_Q, y_Q, z_Q)$, $\mathbf{v} = \mathbf{v}(v_x, v_y, v_z)$ relative to $\mathcal{K}$, and where $t$ is the *parameter* yielding different points $(x, y, z)$ on the line. The three equations in the system (2.13) are called *parametric equations* for the line $S$.

### 2.3.2 Cartesian equations

It is possible to eliminate the parameter $t$ in (2.13) in order to obtain

$$\frac{x-x_Q}{v_x}=\frac{y-y_Q}{v_y}=\frac{z-z_Q}{v_z}. \tag{2.14}$$

We refer to Equation (2.14) as *symmetric equations* of the line $S$. It could happen that $v_x$, $v_y$, or $v_z$ is zero. In that case, translate back to the parametric equations to understand what happens.

We have just described a line with two linear equations (Equations (2.14)) relative to the frame $\mathcal{K}$. The converse is also true.

**Proposition 2.4.** Every line in $\mathbb{A}^3$ can be described with two linear equations in three variables

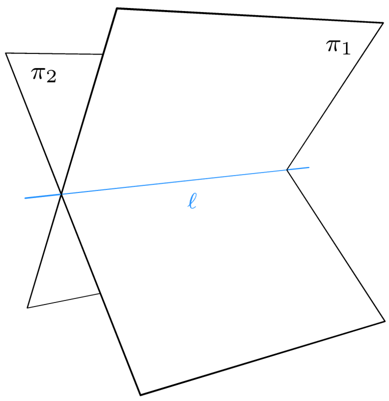
$$\begin{cases} a_1x+b_1y+c_1z+d_1 &= 0\\ a_2x+b_2y+c_2z+d_2 &= 0\end{cases} \tag{2.15}$$

relative to a fixed frame, and any compatible system of two linear equations of rank 2 in three variables relative to a fixed frame describes a line. Moreover, if the system of equations (2.15) describes the line $\ell$ relative to a frame $\mathcal{K}=(O,\mathcal{B})$, then the direction space $\mathrm{D}(\ell)$ of the line is the 1-dimensional subspace of $\mathbb{V}^3$ which, relative to the basis $\mathcal{B}$, satisfies the equations

$$\mathrm{D}(\ell):\begin{cases} a_1x+b_1y+c_1z &= 0\\ a_2x+b_2y+c_2z &= 0\end{cases}. \tag{2.16}$$

*Proof.* This is a particular case of Theorem 2.6. □

The system of equations (2.15) is called a system of *Cartesian equations* of the line which it describes. Notice that it describes a line as an intersection of two planes.




### 2.3.3 Relative positions of two lines in $\mathbb{A}^3$

Again, the intersections of lines can be determined with linear algebra. Assume we have two lines

$$\ell_1:\begin{cases} a_1x+b_1y+c_1z+d_1 &= 0\\ a_2x+b_2y+c_2z+d_2 &= 0\end{cases} \quad\text{and}\quad \ell_2:\begin{cases} a_3x+b_3y+c_3z+d_3 &= 0\\ a_4x+b_4y+c_4z+d_4 &= 0\end{cases}.$$

One way to determine if they intersect is to discuss the system:

$$\begin{cases} a_1x + b_1y + c_1z + d_1 &= 0 \\ a_2x + b_2y + c_2z + d_2 &= 0 \\ a_3x + b_3y + c_3z + d_3 &= 0 \\ a_4x + b_4y + c_4z + d_4 &= 0 \end{cases}. \tag{2.17}$$

Discussing this system is basic linear algebra (see, for example, [9, Section 3.6]). It is somewhat easier to discuss the relative positions of lines in $\mathbb{A}^3$ via their parametric equations:

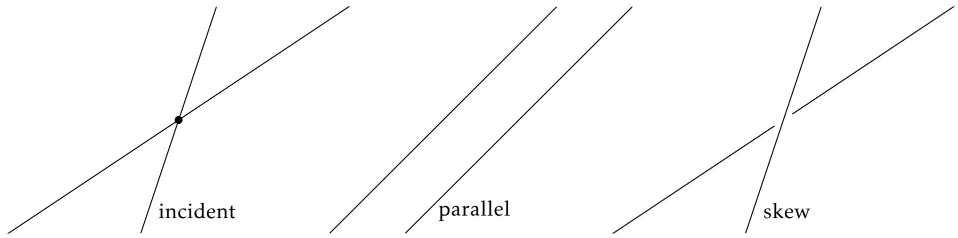
$$\ell_1 : \begin{cases} x = x_1 + tv_x \\ y = y_1 + tv_y \\ z = z_1 + tv_z \end{cases} \quad \text{and} \quad \ell_2 : \begin{cases} x = x_2 + tu_x \\ y = y_2 + tu_y \\ z = z_2 + tu_z \end{cases}.$$

We have the following cases:

- if the direction vectors $\mathbf{v}(v_1, v_2, v_3)$ and $\mathbf{u}(u_1, u_2, u_3)$ are proportional, then the two lines are parallel;
- if they are parallel and have a point in common, then the two lines are equal;
- if they are not parallel, then they are coplanar (they lie in the same plane) if

$$\begin{vmatrix} x_1 - x_2 & y_1 - y_2 & z_1 - z_2 \\ v_x & v_y & v_z \\ u_x & u_y & u_z \end{vmatrix} = 0,$$

  in which case they intersect in exactly one point;
- if they are not parallel and they don't intersect, then we say that the two lines $\ell_1$ and $\ell_2$ are *skew* relative to each other.




### 2.3.4 Relative positions of a line and a plane in $\mathbb{A}^3$

Consider the plane

$$\pi : ax + by + cz + d = 0$$

and the line

$$\ell : \begin{cases} x = x_0 + tv_x \\ y = y_0 + tv_y \\ z = z_0 + tv_z \end{cases}.$$

In order to see if they intersect, we check to see which points in $\ell$ satisfy the equation of $\pi$:

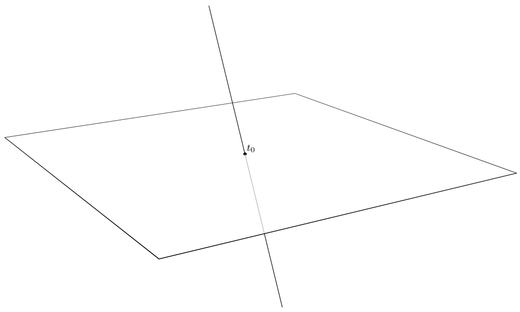
$$a(x_0 + tv_x) + b(y_0 + tv_y) + c(z_0 + tv_z) + d = 0 \quad \Leftrightarrow \quad (av_x + bv_y + cv_z)t + ax_0 + by_0 + cz_0 + d = 0. \tag{2.18}$$

The possibilities are:

- $av_x + bv_y + cv_z = 0$ and $ax_0 + by_0 + cz_0 + d \neq 0$, in which case Equation (2.18) has no solution, i.e., the plane and the line don't intersect (they are parallel); or
- $av_x + bv_y + cv_z = 0$ and $ax_0 + by_0 + cz_0 + d = 0$, in which case any $t \in \mathbb{R}$ is a solution to Equation (2.18), i.e., the line is contained in the plane (in particular, they are parallel); or
- $av_x + bv_y + cv_z \neq 0$, in which case Equation (2.18) has the unique solution

$$t_0 = -\frac{ax_0 + by_0 + cz_0 + d}{av_x + bv_y + cv_z}.$$

Hence, the point corresponding to the parameter $t_0$ is the intersection point $\ell \cap \pi$.




## 2.4 Affine subspaces of $\mathbb{A}^n$

Recall Definition 1.4 for affine subspaces. Two affine subspaces $S_1, S_2 \subseteq \mathbb{A}^n$ are *parallel* ($S_1 \| S_2$) if $\mathrm{D}(S_1) \subseteq \mathrm{D}(S_2)$ or $\mathrm{D}(S_2) \subseteq \mathrm{D}(S_1)$.

Affine subspaces are affine spaces in their own right, inheriting the structure from $\mathbb{A}^n$, analogous to vector subspaces.

**Proposition 2.5.** An affine subspace of $\mathbb{A}^n$ is an affine space with the affine structure inherited from $\mathbb{A}^n$.

*Proof.* The proof is a simple matter of unpacking the definition of affine spaces (Definition 1.2). The space $\mathbb{A}^n$ is a triple $(\mathbb{P}, \mathbb{V}, t)$, where $\mathbb{P}$ is the set of points, $\mathbb{V}$ is an $n$-dimensional vector space, and $t : \mathbb{V} \times \mathbb{P} \to \mathbb{P}$ is the translation map. If $S$ is an affine subspace, it is in particular a subset of $\mathbb{P}$. The direction space $\mathrm{D}(S)$ consists of vectors in $\mathbb{V}$ which can be represented by points in $S$, thus $\mathrm{D}(S)$ is a vector subspace of $\mathbb{V}$. The inherited affine structure is obtained by restricting $\square - \square : S \times S \to \mathrm{D}(S)$ and $\square + \square : \mathrm{D}(S) \times S \to S$. Since these maps satisfy the axioms of an affine space, so do their restrictions. Hence the triple $(S, \mathrm{D}(S), t')$ is an affine space. $\square$

Fixing a point $O \in \mathbb{A}^n$, a point $Q \in S$, and a basis $(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_d)$ of $\mathrm{D}(S)$, it follows from the definition that $S$ has the following description:

$$S = \left\{P \in \mathbb{A}^n : \overrightarrow{OP} = \overrightarrow{OQ} + t_1\mathbf{v}_1 + t_2\mathbf{v}_2 + \cdots + t_d\mathbf{v}_d \text{ for some } t_1, \dots, t_d \in \mathbb{R}\right\}. \tag{2.19}$$

The equation in (2.19) is called the *vector equation of the affine subspace $S$ relative to $O$, having base point $Q$ and direction vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_d$*, or simply *a vector equation* of $S$.

Fixing a frame with origin $O$ and translating the equation in (2.19) into coordinates, one obtains *parametric equations* of the affine space $S$.

$$S : \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} q_1 \\ q_2 \\ \vdots \\ q_n \end{bmatrix} + t_1 \begin{bmatrix} v_{1,1} \\ v_{1,2} \\ \vdots \\ v_{1,n} \end{bmatrix} + \cdots + t_d \begin{bmatrix} v_{d,1} \\ v_{d,2} \\ \vdots \\ v_{d,n} \end{bmatrix} \qquad t_1, \dots, t_d \in \mathbb{R}. \tag{2.20}$$

Another way of representing an affine subspace is by *Cartesian equations* (Equations (2.21)) as follows.

**Theorem 2.6.** Fix a frame $\mathcal{K} = (O, \mathcal{B})$ in the affine space $\mathbb{A}^n$. Let

$$\begin{cases} a_{11}x_1 + \cdots + a_{1n}x_n &= b_1 \\ & \vdots \\ a_{t1}x_1 + \cdots + a_{tn}x_n &= b_t \end{cases} \tag{2.21}$$

be a system of linear equations in the unknowns $x_1, \dots, x_n$. The set $S$ of points of $\mathbb{A}^n$ whose coordinates are solutions to (2.21), if there are any, is an affine space of dimension $d = n - r$, where $r$ is the rank of the matrix of coefficients of the system. The direction space $\mathrm{D}(S)$ is the vector subspace of $\mathbb{V}^n$ whose equations relative to $\mathcal{B}$ are given by the associated homogeneous system

$$\mathrm{D}(S) : \begin{cases} a_{11}x_1 + \cdots + a_{1n}x_n &= 0 \\ & \vdots \\ a_{t1}x_1 + \cdots + a_{tn}x_n &= 0 \end{cases}. \tag{2.22}$$

Conversely, for every affine subspace $S$ of $\mathbb{A}^n$ of dimension $d$ there is a system of $n - d$ linear equations in $n$ unknowns whose solutions correspond precisely to the coordinates of the points in $S$.

*Proof.* We follow the proof in [20, Section 8]. Denote by $\mathbb{W}$ the vector subspace defined by the homogeneous system (2.22). First, we show that the set of solutions to (2.21) is an affine subspace of $\mathbb{A}^n$ with the indicated properties. By assumption, we only consider the cases where $S \neq \emptyset$. Thus, we may fix a point $Q(q_1, \dots, q_n) \in S$. Then, for any point $P(p_1, \dots, p_n)$ belonging to $S$, we have

$$a_{j1}(p_1 - q_1) + \cdots + a_{jn}(p_n - q_n) = \underbrace{(a_{j1}p_1 + \cdots + a_{jn}p_n)}_{b_j} - \underbrace{(a_{j1}q_1 + \cdots + a_{jn}q_n)}_{b_j} = 0$$

for each $j = 1,\dots,t$. Thus, $\overrightarrow{QP} \in \mathbb{W}$. This shows that $S$ is contained in the affine subspace $T$ passing through $Q$ and parallel to $\mathbb{W}$. Conversely, if $R(r_1,\dots,r_n) \in T$, then $\overrightarrow{QR} \in \mathbb{W}$ and so the components $(r_1 - q_1,\dots,r_n - q_n)$ of $\overrightarrow{QR}$ are solutions to (2.22). Thus,

$$0 = a_{j1}(r_1 - q_1) + \cdots + a_{jn}(r_n - q_n) = a_{j1}r_1 + \cdots + a_{jn}r_n - \underbrace{(a_{j1}q_1 + \cdots + a_{jn}q_n)}_{b_j}$$

for each $j = 1,\dots,t$. That is, $R \in S$. Thus $S = T$, hence $S$ is an affine subspace. Moreover, $\dim(S) = \dim(\mathbb{W}) = n$ minus the rank of the matrix of coefficients of (2.22).

Next, we show that an affine subspace $S \subseteq \mathbb{A}^n$ has a description by a linear system of the form (2.21). Let $S$ be any affine subspace of $\mathbb{A}^n$ with direction space $\mathbb{W}$ of dimension $s$. Being an $s$-dimensional subspace of $\mathbb{V}$, $\mathbb{W}$ can be described by a homogeneous system with $n-s$ equations

$$\mathbb{W}: \begin{cases} a_{11}x_1 + \cdots + a_{1n}x_n &= 0 \\ & \vdots \\ a_{n-s,1}x_1 + \cdots + a_{n-s,n}x_n &= 0 \end{cases}.$$

Fix a point $Q \in S$. The points $P(p_1,\dots,p_n)$ of $S$ are characterized by the condition that $\overrightarrow{QP} \in \mathbb{W}$, i.e.,

$$a_{j1}(p_1 - q_1) + \cdots + a_{jn}(p_n - q_n) = 0$$

for each $j = 1,\dots,n-s$, or, equivalently,

$$a_{j1}p_1 + \cdots + a_{jn}p_n = b_j$$

where we have put $b_j = a_{j1}q_1 + \cdots + a_{jn}q_n$. Thus, the points $P(p_1,\dots,p_n) \in S$ are precisely those points in $\mathbb{A}^n$ whose coordinates satisfy the equations

$$\begin{cases} a_{11}x_1 + \cdots + a_{1n}x_n &= b_1 \\ & \vdots \\ a_{n-s,1}x_1 + \cdots + a_{n-s,n}x_n &= b_{n-s} \end{cases}.$$

□

### 2.4.1 Hyperplanes

Recall that affine subspaces in $\mathbb{A}^n$ which have dimension $n-1$ are called *hyperplanes* (see Definition 1.4). Let $H$ be a hyperplane and let $(\mathbf{v}_1,\dots,\mathbf{v}_{n-1})$ be a basis of $\mathrm{D}(H)$. With respect to a frame $\mathcal{K} = (O,\mathbf{e}_1,\dots,\mathbf{e}_n)$ of $\mathbb{A}^n$, parametric equations of $H$ are of the form

$$H: \begin{cases} x_1 &= q_1 + t_1 v_{1,1} + \cdots + t_{n-1}v_{n-1,1} \\ x_2 &= q_2 + t_1 v_{1,2} + \cdots + t_{n-1}v_{n-1,2} \\ \vdots & \\ x_n &= q_n + t_1 v_{1,n} + \cdots + t_{n-1}v_{n-1,n} \end{cases} \quad \text{or} \quad H: \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} q_1 \\ q_2 \\ \vdots \\ q_n \end{bmatrix} + t_1 \begin{bmatrix} v_{1,1} \\ v_{1,2} \\ \vdots \\ v_{1,n} \end{bmatrix} + \cdots + t_{n-1} \begin{bmatrix} v_{n-1,1} \\ v_{n-1,2} \\ \vdots \\ v_{n-1,n} \end{bmatrix} \tag{2.23}$$

where $\mathbf{v}_i = \mathbf{v}_i(v_{i,1},\dots,v_{i,n})$, where $Q = Q(q_1,\dots,q_n)$ is a point in $H$, and where $t_i \in \mathbb{R}$ for each $i \in \{1,\dots,n-1\}$. These parametric equations express the fact that a point $P$ belongs to $H$ if and only if the vector $\overrightarrow{QP}$ is a linear combination of the basis vectors $\mathbf{v}_1,\dots,\mathbf{v}_{n-1}$, i.e., if and only if the vectors $\overrightarrow{QP},\mathbf{v}_1,\dots,\mathbf{v}_{n-1}$ are linearly dependent. We can reformulate this as follows. A point $P(x_1,\dots,x_n)$ belongs to the hyperplane $H$ if and only if

$$\begin{vmatrix} x_1-q_1 & x_2-q_2 & \dots & x_n-q_n \\ v_{1,1} & v_{1,2} & \dots & v_{1,n} \\ \vdots & \vdots & & \vdots \\ v_{n-1,1} & v_{n-1,2} & \dots & v_{n-1,n} \end{vmatrix} = 0. \tag{2.24}$$

This is a *Cartesian equation* of the hyperplane $H$.

### 2.4.2 Lines

A line in $\mathbb{A}^n$ is a 1-dimensional affine subspace (see Definition 1.4). If $\ell$ is such a line, then the vectors which can be represented by points in $\ell$ are linearly dependent. Any such nonzero vector $\mathbf{v}$ is called a direction vector of $\ell$. Thus, $\ell$ can be described as

$$\ell = \left\{P \in \mathbb{A}^n : \overrightarrow{OP} = \overrightarrow{OQ} + t\mathbf{v} \text{ for some } t \in \mathbb{R}\right\}$$

for any point $O \in \mathbb{A}^n$ and any point $Q \in \ell$. The image that goes with this description is the same as the one in dimension 2, but here we interpret it in the $n$-dimensional space $\mathbb{A}^n$. In coordinates, relative to a given frame $\mathcal{K}$ of $\mathbb{A}^n$, we obtain parametric equations for $\ell$. They are of the form:

$$\ell : \begin{cases} x_1 &= q_1 + tv_1 \\ x_2 &= q_2 + tv_2 \\ &\vdots \\ x_n &= q_n + tv_n \end{cases} \quad \text{or, in matrix notation,} \quad \ell : \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} q_1 \\ q_2 \\ \vdots \\ q_n \end{bmatrix} + t \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$$

where $Q = Q(q_1,\dots,q_n)$ and $\mathbf{v} = \mathbf{v}(v_1,\dots,v_n)$ relative to $\mathcal{K}$. Here again, it is possible to eliminate the parameter $t$ in order to obtain *symmetric equations* of the line $\ell$:

$$\ell : \frac{x_1-q_1}{v_1} = \frac{x_2-q_2}{v_2} = \dots = \frac{x_n-q_n}{v_n}.$$

This is in fact a system of $(n-1)$ linear equations which we can rearrange to look like this:

$$\ell : \begin{cases} a_{1,1}x_1 + \dots + a_{1,n}x_n &= b_1 \\ &\vdots \\ a_{n-1,1}x_1 + \dots + a_{n-1,n}x_n &= b_{n-1} \end{cases}.$$

Notice that the rank of this system is $n-1$ since $\dim(\ell) = 1$. Moreover, notice that each linear equation in the above system describes a hyperplane. So, this is saying that a line in $\mathbb{A}^n$ can be described by the intersection of $n-1$ hyperplanes.

### 2.4.3 Relative positions

Let $S$ and $T$ be two affine subspaces of $\mathbb{A}^n$. If $S \parallel T$, then they are disjoint or one is included in the other (see Proposition 2.7 below). Notice that if $\dim(S) = \dim(T)$, then $S$ and $T$ are parallel if and only if $\mathrm{D}(S) = \mathrm{D}(T)$. In particular, if $S$ and $T$ are lines, they are parallel if they have the same direction, i.e., any two of their direction vectors are proportional. Notice also that two hyperplanes are parallel if the coefficients of the unknowns in their equations are proportional.

**Proposition 2.7.** Let $S$ and $T$ be parallel affine subspaces of $\mathbb{A}^n$ with $\dim(S) \leq \dim(T)$.

1) If $S$ and $T$ have a point in common, then $S \subseteq T$.

2) If $\dim(S) = \dim(T)$, and $S$ and $T$ have a point in common, then $S = T$.

*Proof.* Fix $Q \in S \cap T$. Since $S$ and $T$ are parallel, without loss of generality we may assume that $\mathrm{D}(S) \subseteq \mathrm{D}(T)$. Then, we obtain 1) from

$$S = \{P \in \mathbb{A}^n : \overrightarrow{QP} \in \mathrm{D}(S)\} \subseteq \{P \in \mathbb{A}^n : \overrightarrow{QP} \in \mathrm{D}(T)\} = T.$$

We obtain 2) by noticing that $\dim(S) = \dim(T)$ implies equality in the above equation. Indeed, by definition $\dim(S) = \dim(T)$ means that $\dim(\mathrm{D}(S)) = \dim(\mathrm{D}(T))$, but then $\mathrm{D}(S)$ is a vector subspace of $\mathrm{D}(T)$ of maximal dimension, hence $\mathrm{D}(S) = \mathrm{D}(T)$. □

As a consequence of Proposition 2.7, we obtain the following corollary which implies the 'parallel postulate' of Euclidean geometry (Axiom IV in [14]). The axioms of affine spaces therefore imply the validity of this postulate.

**Corollary 2.8.** If $S$ is an affine subspace of $\mathbb{A}^n$ and $Q$ is a point in $\mathbb{A}^n$, there is a unique affine subspace $T$ of $\mathbb{A}^n$ which contains $Q$, is parallel to $S$, and has the same dimension as $S$.

*Proof.* Fix a point $Q \in \mathbb{A}^n$ and a point $P \in S$. To see that $T$ exists, translate all points of $S$ by $\overrightarrow{QP}$, i.e., consider

$$T = \overrightarrow{QP} + S = \overrightarrow{QP} + \{P' \in \mathbb{A}^n : \overrightarrow{PP'} \in \mathrm{D}(S)\} = \{P' \in \mathbb{A}^n : \overrightarrow{QP'} \in \mathrm{D}(S)\}$$

where for the last equality we use $\overrightarrow{QP} + \overrightarrow{PP'} = \overrightarrow{QP'}$. Then $T$ is an affine subspace with $\mathrm{D}(T) = \mathrm{D}(S)$; in particular, it is parallel to $S$, and $\dim(T) = \dim(S)$. To see that $T$ is unique, assume that $T'$ is another affine subspace passing through $Q$ which is parallel to $S$, and of the same dimension. By point 2) of Proposition 2.7 we see that $T'$ has to equal $T$. □

**Definition 2.9.** If two affine subspaces $S$ and $T$ of $\mathbb{A}^n$ are not parallel, they are said to be *skew* if they do not meet, or *incident* if they have a point in common.

In order to determine the intersection $S \cap T$, suppose that the two subspaces are given by the Cartesian equations

$$S : \sum_{j=1}^{n} a_{ij}x_j = b_i \quad \text{for} \quad i = 1, \ldots, n-s \tag{2.25}$$

$$T : \sum_{j=1}^{n} c_{kj} x_j = d_k \quad \text{for} \quad k = 1, \ldots, n-t. \tag{2.26}$$

The intersection $S \cap T$ is the locus of points in $\mathbb{A}^n$ whose coordinates are simultaneously solutions to both (2.25) and (2.26), i.e., they are solutions to the system

$$S \cap T : \begin{cases} \sum_{j=1}^{n} a_{ij} x_j = b_i & \text{for} \quad i = 1, \ldots, n-s, \\ \sum_{j=1}^{n} c_{kj} x_j = d_k & \text{for} \quad k = 1, \ldots, n-t. \end{cases} \tag{2.27}$$

By Theorem 2.6, if the system (2.27) has a solution, then it describes an affine subspace. Thus, if $S \cap T$ is nonempty, it is an affine subspace of $\mathbb{A}^n$.

**Proposition 2.10.** If the intersection $S \cap T$ of two affine subspaces of $\mathbb{A}^n$ is nonempty, it is an affine subspace satisfying

$$\dim(S) + \dim(T) - \dim(\mathbb{A}^n) \leq \dim(S \cap T) \leq \min\Big\{\dim(S), \dim(T)\Big\}. \tag{2.28}$$

*Proof.* Let $s = \dim(S)$ and let $t = \dim(T)$. By Theorem 2.6 we have

$$\begin{cases} S : \sum_{j=1}^{n} a_{ij} x_j = 0 & \text{for } i = 1, \ldots, n-s \\ T : \sum_{j=1}^{n} c_{ij} x_j = 0 & \text{for } i = 1, \ldots, n-t \end{cases}$$

Since $S \cap T$ is nonempty, the above system is compatible, and the dimension of $S \cap T$ is $n - r$, where $r$ is the rank of the matrix of coefficients of this system. Notice that

$$r \leq n - s + n - t = 2n - (s + t)$$

Thus,

$$\dim(S \cap T) = n - r \geq n - [2n - (s + t)] = s + t - n = \dim(S) + \dim(T) - \dim(\mathbb{A}^n).$$

The last inequality is clear since $S \cap T \subseteq S$ and $S \cap T \subseteq T$ implies $\mathrm{D}(S \cap T) \subseteq \mathrm{D}(S)$ and $\mathrm{D}(S \cap T) \subseteq \mathrm{D}(T)$, and therefore $\dim(S \cap T) \leq \dim(S)$ and $\dim(S \cap T) \leq \dim(T)$. □

Notice that in the previous proposition, the second inequality is an equality if $S \subseteq T$ or $T \subseteq S$. What about the first inequality? When do we have equality there?

**Proposition 2.11.** Let $S$ and $T$ be two affine subspaces of $\mathbb{A}^n$. Then $\mathrm{D}(\mathbb{A}^n) = \mathrm{D}(S) + \mathrm{D}(T)$ if and only if $S \cap T \neq \emptyset$ *and*

$$\dim(S \cap T) = \dim(S) + \dim(T) - \dim(\mathbb{A}^n). \tag{2.29}$$

*Proof.* We use Grassmann's identity, which states that for any vector subspaces $\mathbb{W}$ and $\mathbb{U}$ we have

$$\dim(\mathbb{W}) + \dim(\mathbb{U}) = \dim(\mathbb{W} \cap \mathbb{U}) + \dim(\mathbb{W} + \mathbb{U}).$$

Let $s = \dim(S)$ and let $t = \dim(T)$. By Theorem 2.6 we have

$$\begin{cases} S : \sum_{j=1}^{n} a_{ij} x_j = b_i & \text{for } i = 1, \ldots, n-s \\ T : \sum_{j=1}^{n} c_{kj} x_j = d_k & \text{for } k = 1, \ldots, n-t \end{cases}$$

By convention, $\dim(\emptyset) = -\infty$; thus, (2.29) holds only if $S \cap T \neq \emptyset$. Assume that $S \cap T$ is nonempty. Then, the above system is compatible, and the dimension of $S \cap T$ is $n - r$, where $r$ denotes the rank of the matrix of coefficients of this system. Moreover, we notice that (2.29) is equivalent to

$$n - r = s + t - n \quad \Leftrightarrow \quad \dim(\mathrm{D}(S \cap T)) = \dim(\mathrm{D}(S)) + \dim(\mathrm{D}(T)) - \dim(\mathbb{V}^n)$$

and, since $S \cap T \neq \emptyset$, this is in turn equivalent to

$$\dim(\mathrm{D}(S) \cap \mathrm{D}(T)) = \dim(\mathrm{D}(S)) + \dim(\mathrm{D}(T)) - \dim(\mathbb{V}^n).$$

Rearranging the equation and using Grassmann's identity, the equation is equivalent to

$$\dim(\mathbb{V}^n) = \dim(\mathrm{D}(S)) + \dim(\mathrm{D}(T)) - \dim(\mathrm{D}(S) \cap \mathrm{D}(T)) = \dim(\mathrm{D}(S) + \mathrm{D}(T)).$$

At this point, we use the fact that the vector subspace $\mathrm{D}(S) + \mathrm{D}(T)$ of $\mathbb{V}^n$ has maximal dimension if and only if it equals the ambient space, i.e., if and only if $\mathrm{D}(S) + \mathrm{D}(T) = \mathbb{V}^n$. □

### 2.4.4 Changing the reference frame

Let $S$ be an affine subspace of $\mathbb{A}^n$ given with respect to the frame $\mathcal{K} = (O, \mathcal{B})$ via the parametric equations (2.20). Then, if $\mathcal{K}' = (O', \mathcal{B}')$ is another frame, by Theorem 1.10, the parametric equations with respect to $\mathcal{K}'$ are

$$S : \begin{bmatrix} x_1' \\ x_2' \\ \vdots \\ x_n' \end{bmatrix} = \mathrm{M}_{\mathcal{B}',\mathcal{B}} \cdot \begin{bmatrix} q_1 \\ q_2 \\ \vdots \\ q_n \end{bmatrix} + [O]_{\mathcal{K}'} + t_1 \cdot \mathrm{M}_{\mathcal{B}',\mathcal{B}} \cdot \begin{bmatrix} v_{1,1} \\ v_{1,2} \\ \vdots \\ v_{1,n} \end{bmatrix} + \cdots + t_d \cdot \mathrm{M}_{\mathcal{B}',\mathcal{B}} \cdot \begin{bmatrix} v_{d,1} \\ v_{d,2} \\ \vdots \\ v_{d,n} \end{bmatrix}.$$

In terms of Cartesian equations, notice that (2.21) can be written in the form

$$S : A \cdot \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = b.$$

Then, with respect to $\mathcal{K}'$, this system translates as follows:

$$S : \underbrace{\left(A \cdot \mathrm{M}_{\mathcal{B},\mathcal{B}'}\right)}_{=A'} \cdot \begin{bmatrix} x_1' \\ \vdots \\ x_n' \end{bmatrix} = \underbrace{b - A \cdot [O']_{\mathcal{K}}}_{=b'}.$$

# CHAPTER 3

# Euclidean space

## Contents

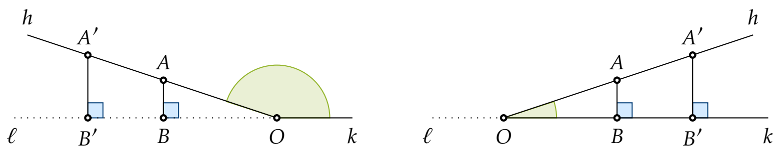
## 3.1 Angles

Based on Hilbert's Axioms [14], we define an angle $\measuredangle(h,k)$ as the information carried by two rays $h$ and $k$, emanating from a common point $O$. One can deduce from the Axioms that a nondegenerate angle defines a unique plane which the two rays separate into a convex and a concave subset. Visually, it is common to identify an angle with the convex subset they define. As a matter of notation, for points $A \in h$ and $B \in k$ we may write $\measuredangle AOB$ for the angle $\measuredangle(h,k)$. Notice that if three points $O, A, B$ are given, the symbols $\measuredangle AOB$ require that $A \neq O$ and $B \neq O$.

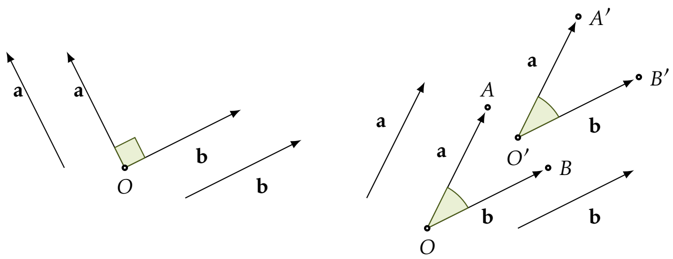
Two lines which intersect in exactly one point form four angles. The opposite angles are congruent, and the adjacent angles are called *supplementary*. A *right angle* is an angle which is congruent to its supplementary angle. If one of the angles of two intersecting lines is a right angle, then all of them are right angles, and we say that the lines are *orthogonal*, or that the lines are *perpendicular* to each other. An *acute angle* is an angle less than a right angle and an *obtuse angle* is an angle greater than a right angle. The sine and cosine are the standard numerical quantities attached to angles.

**Definition 3.1.** Let $\measuredangle(h,k)$ be an angle with the two rays emanating from the point $O$. Let $\ell$ be the line containing $k$ and choose $A \in h$ and $B \in \ell$ such that $AB$ is orthogonal to $\ell$. The *sine* and *cosine of the angle* $\measuredangle(h,k)$ are defined by

$$\sin\measuredangle(h,k) = \frac{|AB|}{|OA|} \quad \text{and} \quad \cos\measuredangle(h,k) = \begin{cases} 0 & \text{if the angle is a right angle;} \\ \frac{|OB|}{|OA|} & \text{if the angle is acute;} \\ -\frac{|OB|}{|OA|} & \text{if the angle is obtuse.} \end{cases}$$

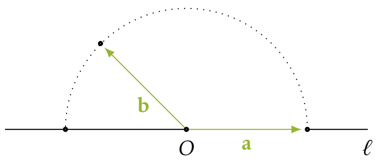



In this setup, one may deduce known properties from the Axioms. For example, one deduces that two angles are congruent if and only if their cosines are equal. Similar to how we abstracted the notion of oriented segments by introducing vectors, here too, we may treat angles up to congruence. This treatment is implicit in the notion of angle between two vectors.



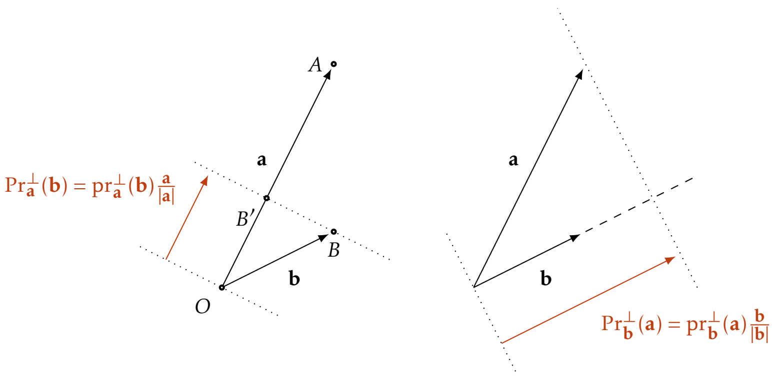
**Definition 3.2.** For two non-zero vectors $\mathbf{a} = \overrightarrow{OA}$ and $\mathbf{b} = \overrightarrow{OB}$, the *unoriented angle between* $\mathbf{a}$ *and* $\mathbf{b}$, denoted $\measuredangle(\mathbf{a},\mathbf{b})$, is the angle $\measuredangle AOB$ up to congruence. The values of sine and cosine do not change under congruence; thus, the *sine* $\sin\measuredangle(\mathbf{a},\mathbf{b})$ and *cosine* $\cos\measuredangle(\mathbf{a},\mathbf{b})$ *of the unoriented angle* $\measuredangle(\mathbf{a},\mathbf{b})$ are well-defined. If $\cos\measuredangle(\mathbf{a},\mathbf{b}) = 0$ we say that $\mathbf{a}$ *and* $\mathbf{b}$ *are orthogonal*, and we write $\mathbf{a}\perp\mathbf{b}$.

**Remark.** Let $\ell$ be a line in a plane $\pi$ and let $O$ be a point on $\ell$. Choose a side of $\pi$ relative to $\ell$ and consider the semicircle $\mathbb{S}^{\frac{1}{2}} = \Big\{ A : \text{ with } A \text{ on the given side of } \ell \text{ or on } \ell \text{ and } |OA| = 1 \Big\}$. One can deduce from the Axioms [14] that the set of unoriented angles is in bijection with $\mathbb{S}^{\frac{1}{2}}$.




**Definition 3.3.** For a vector $\mathbf{a}$ we have *orthogonal projection maps* $\mathrm{pr}^{\perp}_{\mathbf{a}}:\mathbb{V}\to\mathbb{R}$ and $\mathrm{Pr}^{\perp}_{\mathbf{a}}:\mathbb{V}\to\mathbb{V}$ defined as follows. For a vector $\mathbf{b}$ let $O,A,B$ be such that $\mathbf{a} = \overrightarrow{OA}$, $\mathbf{b} = \overrightarrow{OB}$. Drop a perpendicular $BB'$ on $OA$ with $B'\in OA$. Then

$$\mathrm{Pr}^{\perp}_{\mathbf{a}}(\mathbf{b}) = \overrightarrow{OB'} \quad \text{and} \quad \mathrm{Pr}^{\perp}_{\mathbf{a}}(\mathbf{b}) = \mathrm{pr}^{\perp}_{\mathbf{a}}(\mathbf{b})\frac{\mathbf{a}}{|\mathbf{a}|}.$$




**Proposition 3.4.** The orthogonal projection maps on vectors are linear maps. Moreover, for two vectors $\mathbf{a}$ and $\mathbf{b}$, we have

$$\cos\measuredangle(\mathbf{a},\mathbf{b}) = \frac{\mathrm{pr}^{\perp}_{\mathbf{a}}(\mathbf{b})}{|\mathbf{b}|} = \frac{\mathrm{pr}^{\perp}_{\mathbf{b}}(\mathbf{a})}{|\mathbf{a}|}.$$

*Proof.* It is enough to show that $\mathrm{pr}^{\perp}_{\mathbf{a}}$ is linear, since then

$$\mathrm{Pr}^{\perp}_{\mathbf{a}}(x\mathbf{b}+y\mathbf{c}) = \mathrm{pr}^{\perp}_{\mathbf{a}}(x\mathbf{b}+y\mathbf{c})\frac{\mathbf{a}}{|\mathbf{a}|} = x\,\mathrm{pr}^{\perp}_{\mathbf{a}}(\mathbf{b})\frac{\mathbf{a}}{|\mathbf{a}|} + y\,\mathrm{pr}^{\perp}_{\mathbf{a}}(\mathbf{c})\frac{\mathbf{a}}{|\mathbf{a}|} = x\,\mathrm{Pr}^{\perp}_{\mathbf{a}}(\mathbf{b}) + y\,\mathrm{Pr}^{\perp}_{\mathbf{a}}(\mathbf{c}).$$

Let $\mathbf{i}$ be the unit vector $\frac{\mathbf{a}}{|\mathbf{a}|}$ and let $\mathbf{j}$ be a vector orthogonal to $\mathbf{i}$. The linearity of $\mathrm{pr}^{\perp}_{\mathbf{a}}$ follows from the fact that it is a projection onto the coordinate $x$-axis of the frame having $O$ as origin and $(\mathbf{i},\mathbf{j})$ as basis. The last claim follows from the definition of cosine. $\square$

### 3.1.1 Orthonormal frames

**Definition 3.5.** A basis $\mathcal{B} = (\mathbf{e}_1, \dots, \mathbf{e}_n)$ of $\mathbb{V}^n$ is called *orthogonal* if the vectors $\mathbf{e}_i$ are mutually orthogonal, i.e., if $\mathbf{e}_i \perp \mathbf{e}_j$ for all $i, j \in \{1, \dots, n\}$ with $i \neq j$. The basis $\mathcal{B}$ is called *orthonormal* if it is orthogonal and all $\mathbf{e}_i$ are unit vectors. A coordinate system $\mathcal{K} = (O, \mathcal{B})$ is called *orthogonal* or *orthonormal* if the basis $\mathcal{B}$ is orthogonal or, respectively, orthonormal.

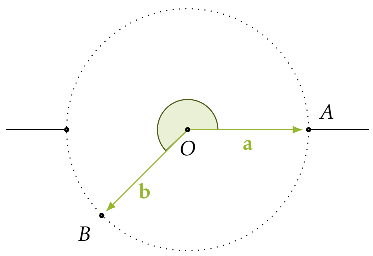
In dimension 2, the existence of orthonormal frames is a consequence of the existence of right angles. In general, the existence of these frames follows from the defining properties of the scalar product (Proposition 3.10) in the context of bilinear forms (see Corollary F.9) or constructively using the Gram-Schmidt algorithm (see Section 3.2.2).

Notice that by Proposition 3.4, for any vector $\mathbf{a}$, we have $\mathrm{pr}^{\perp}_{\mathbf{e}_i}(\mathbf{a}) = |\mathbf{a}| \cos\measuredangle(\mathbf{a}, \mathbf{e}_i)$. Thus, with respect to an orthonormal frame the coordinates of a point $P(x_1, \dots, x_n)$ and its position vector $\mathbf{a} = \overrightarrow{OP}$ are

$$[P]_{\mathcal{K}} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} |\mathbf{a}|\cos\measuredangle(\mathbf{a}, \mathbf{e}_1) \\ \vdots \\ |\mathbf{a}|\cos\measuredangle(\mathbf{a}, \mathbf{e}_n) \end{bmatrix} = |\mathbf{a}| \cdot \begin{bmatrix} \cos\measuredangle(\mathbf{a}, \mathbf{e}_1) \\ \vdots \\ \cos\measuredangle(\mathbf{a}, \mathbf{e}_n) \end{bmatrix} = [\,\overrightarrow{OP}\,]_{\mathcal{B}}.$$

### 3.1.2 Oriented angles

We have already noticed that the set of (unoriented) angles can be represented with a semicircle. If we are in dimension 2, i.e., when considering the Euclidean plane $\mathbb{E}^2$, we may consider the exterior of an angle to be an angle as well. This only works in dimension 2 because lines are hyperplanes here.




**Definition 3.6.** For an angle $\measuredangle(h, k)$ denote by $-\measuredangle(h, k)$ the *exterior* of the angle, i.e., the region in the plane which is *not between* the two rays $h$ and $k$. Assume that a right-oriented frame in $\mathbb{E}^2$ has been chosen. Let $h$ and $k$ be two rays emanating from the same point $O$ and let $A \in h$ and $B \in k$. The oriented angle defined by $h$ and $k$ is

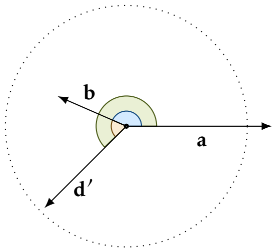
$$\measuredangle_{\mathrm{or}}(h, k) = \begin{cases} \measuredangle(h, k) & \text{if } (\overrightarrow{OA}, \overrightarrow{OB}) \text{ is right-oriented,} \\ -\measuredangle(h, k) & \text{otherwise.} \end{cases}$$

For two vectors $\mathbf{a} = \overrightarrow{OA}$ and $\mathbf{b} = \overrightarrow{OB}$, the oriented angle $\measuredangle_{\mathrm{or}}(\mathbf{a}, \mathbf{b})$ is the oriented angle defined by the rays $(OA$ and $(OB$. The orientation of $\measuredangle_{\mathrm{or}}(h, k)$ is the orientation of the basis $(\mathbf{a}, \mathbf{b})$.

**Remark.** There is a bijection between the set of oriented angles and the unit circle $\mathbb{S}^1$.

**Definition 3.7** (Counterclockwise sum of angles)**.** We define the sum of two oriented angles $\measuredangle_{\text{or}}(\mathbf{a},\mathbf{b})$ and $\measuredangle_{\text{or}}(\mathbf{c},\mathbf{d})$ as follows. There is a unique unit vector $\mathbf{d}'$ such that $\measuredangle_{\text{or}}(\mathbf{c},\mathbf{d}) = \measuredangle_{\text{or}}(\mathbf{b},\mathbf{d}')$ and we define

$$\measuredangle_{\text{or}}(\mathbf{a},\mathbf{b}) + \measuredangle_{\text{or}}(\mathbf{c},\mathbf{d}) = \measuredangle_{\text{or}}(\mathbf{a},\mathbf{d}')$$



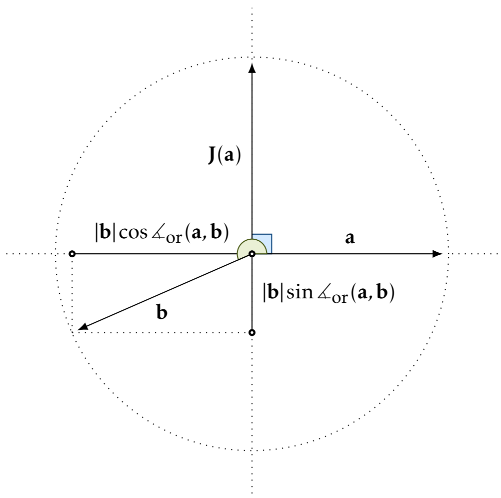

**Remark.** The set of oriented angles with addition is an abelian group.

**Definition 3.8.** For a vector $\mathbf{v} \in \mathbb{V}^2$ let $\mathbf{J}(\mathbf{v})$ be the unique vector in $\mathbb{V}^2$ satisfying the following properties

$$\mathbf{J}(\mathbf{v}) \perp \mathbf{v}, \quad |\mathbf{J}(\mathbf{v})| = |\mathbf{v}| \quad \text{and} \quad (\mathbf{v}, \mathbf{J}(\mathbf{v})) \text{ is a right-oriented basis of } \mathbb{V}^2.$$

The *sine of the oriented angle* $\measuredangle_{\text{or}}(\mathbf{a},\mathbf{b})$ is defined to be

$$\sin \measuredangle_{\text{or}}(\mathbf{a},\mathbf{b}) = \frac{\mathrm{pr}^{\perp}_{\mathbf{J}(\mathbf{a})}(\mathbf{b})}{|\mathbf{b}|} = -\frac{\mathrm{pr}^{\perp}_{\mathbf{J}(\mathbf{b})}(\mathbf{a})}{|\mathbf{a}|}. \tag{3.1}$$



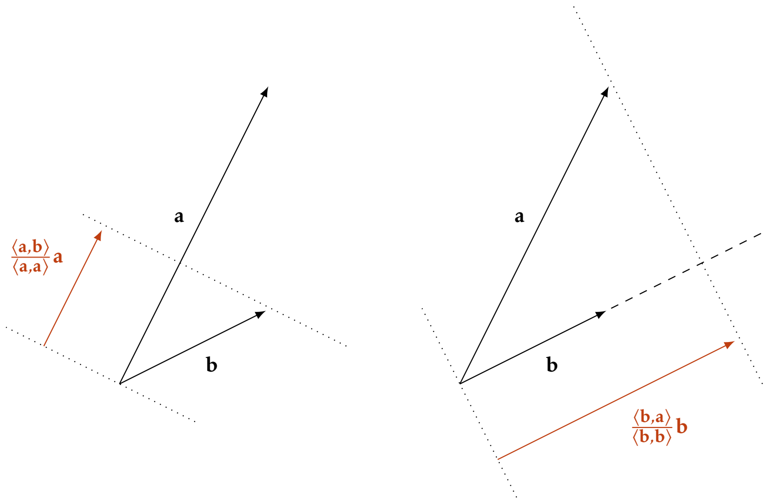
## 3.2 Scalar product

**Definition 3.9.** The *scalar product* (or *dot product*) of two vectors $\mathbf{a},\mathbf{b}\in\mathbb{V}$ is the real number

$$\langle \mathbf{a},\mathbf{b}\rangle = \begin{cases} 0, & \text{if one of the two vectors is zero;}\\ |\mathbf{a}|\cdot|\mathbf{b}|\cdot\cos\measuredangle(\mathbf{a},\mathbf{b}) & \text{if both vectors are non-zero.}\end{cases}$$

It is a map $\langle \_,\_\rangle:\mathbb{V}\times\mathbb{V}\to\mathbb{R}$. Notice that from the definitions, we have

$$\mathrm{Pr}^{\perp}_{\mathbf{a}}(\mathbf{b}) = \frac{\langle \mathbf{a},\mathbf{b}\rangle}{\langle \mathbf{a},\mathbf{a}\rangle}\mathbf{a} \quad \text{and} \quad \mathrm{Pr}^{\perp}_{\mathbf{b}}(\mathbf{a}) = \frac{\langle \mathbf{b},\mathbf{a}\rangle}{\langle \mathbf{b},\mathbf{b}\rangle}\mathbf{b}. \tag{3.2}$$




**Proposition 3.10.** The scalar product satisfies the following properties.

(SP1) It is *bilinear*, i.e., for all $a,b\in\mathbb{R}$ and all $\mathbf{v},\mathbf{w},\mathbf{u}\in\mathbb{V}^2$ we have

$$\langle a\mathbf{v}+b\mathbf{w},\mathbf{u}\rangle = a\langle \mathbf{v},\mathbf{u}\rangle + b\langle \mathbf{w},\mathbf{u}\rangle \quad \text{and} \quad \langle \mathbf{v},a\mathbf{w}+b\mathbf{u}\rangle = a\langle \mathbf{v},\mathbf{w}\rangle + b\langle \mathbf{v},\mathbf{u}\rangle.$$

(SP2) It is *symmetric*, i.e., for all $\mathbf{v},\mathbf{w}\in\mathbb{V}^2$ we have

$$\langle \mathbf{v},\mathbf{w}\rangle = \langle \mathbf{w},\mathbf{v}\rangle.$$

(SP3) It is *positive definite*, i.e., for all $\mathbf{v}\in\mathbb{V}^2$

$$\mathbf{v}\neq 0 \quad\Rightarrow\quad \langle \mathbf{v},\mathbf{v}\rangle > 0.$$

(SP4) It recognizes right angles and unit lengths, i.e., for all non-zero vectors $\mathbf{v},\mathbf{w}\in\mathbb{V}^2$

$$\mathbf{v}\perp\mathbf{w} \quad\Leftrightarrow\quad \langle \mathbf{v},\mathbf{w}\rangle = 0 \qquad \text{and} \qquad |\mathbf{v}| = 1 \quad\Leftrightarrow\quad \langle \mathbf{v},\mathbf{v}\rangle = 1.$$

*Proof.* Since $\cos\measuredangle(\mathbf{a},\mathbf{b}) = \cos\measuredangle(\mathbf{b},\mathbf{a})$, the scalar product is symmetric. Since $\cos\measuredangle(\mathbf{a},\mathbf{a}) = 1$, the scalar product is positive definite. In order to show bilinearity, notice that by symmetry, it is enough to show that the scalar product is linear in the first argument. For any non-zero vectors $\mathbf{a},\mathbf{b}$, by Proposition 3.4, we have

$$\langle \mathbf{a},\mathbf{b}\rangle = |\mathbf{a}|\cdot|\mathbf{b}|\cdot\cos\measuredangle(\mathbf{a},\mathbf{b}) = |\mathbf{a}|\cdot|\mathbf{b}|\cdot\frac{\mathrm{pr}^{\perp}_{\mathbf{b}}(\mathbf{a})}{|\mathbf{a}|} = |\mathbf{b}|\cdot\mathrm{pr}^{\perp}_{\mathbf{b}}(\mathbf{a})$$

which is linear in $\mathbf{a}$ since $\mathbf{b}$ is fixed and since $\mathrm{pr}^{\perp}_{\mathbf{b}}$ is a linear map. The last property follows directly from the definition. □

**Proposition 3.11.** Let $\mathcal{K} = (O,\mathcal{B})$ be an orthonormal frame. For two vectors $\mathbf{a}(a_1,\dots,a_n)$ and $\mathbf{b}(b_1,\dots,b_n)$, we have

$$\langle \mathbf{a},\mathbf{b}\rangle = a_1b_1+\cdots+a_nb_n \tag{3.3}$$

and therefore

$$|\mathbf{a}| = \sqrt{a_1^2+a_2^2+\cdots+a_n^2},$$

$$\cos\measuredangle(\mathbf{a},\mathbf{b}) = \frac{a_1b_1+a_2b_2+\cdots+a_nb_n}{\sqrt{a_1^2+a_2^2+\cdots+a_n^2}\sqrt{b_1^2+b_2^2+\cdots+b_n^2}},$$

$$\mathbf{a}\perp\mathbf{b} \Leftrightarrow a_1b_1+a_2b_2+\cdots+a_nb_n = 0.$$

### 3.2.1 The Euclidean space $\mathbb{R}^n$

The Euclidean space $\mathbb{E}^n$ can be viewed as an affine space $\mathbb{A}^n$ equipped with a metric structure derived from the scalar product. While Hilbert's Axioms provide the synthetic foundation, the analytic approach identifies $\mathbb{E}^n$ with $\mathbb{R}^n$ endowed with the standard dot product. This identification relies on the existence of a unique positive definite symmetric bilinear form recognizing right angles and unit lengths (Corollary F.10).

**Definition 3.12.** The *n-dimensional Euclidean space* $\mathbb{E}^n$ is the pair $(\mathbb{A}^n,\langle\_,\_\rangle)$ where $\mathbb{A}^n$ is the $n$-dimensional real affine space and where $\langle\_,\_\rangle$ is the unique positive definite symmetric bilinear form on $\mathrm{D}(\mathbb{A}^n)$ which recognizes right angles and unit lengths, i.e., with respect to an orthonormal basis $\mathcal{B}$, it has the expression (3.3). Then, the *distance between two points* $P,Q\in\mathbb{E}^n$ is

$$d(P,Q) = |\overrightarrow{QP}| = \sqrt{\langle\overrightarrow{QP},\overrightarrow{QP}\rangle} = \sqrt{(p_1-q_1)^2+(p_2-q_2)^2+\cdots+(p_n-q_n)^2}. \tag{3.4}$$

where the coordinates of $P(p_1,\dots,p_n)$ and $Q(q_1,\dots,q_n)$ are relative to an orthonormal frame $\mathcal{K}$.

**Remark** (The Euclidean space $\mathbb{R}^n$)**.** Choosing an orthonormal frame, we may identify both $\mathbb{A}^n$ and $\mathbb{V}^n$ with $\mathbb{R}^n$. Since computationally there is no difference, it is more economical to say that the Euclidean space is $\mathbb{R}^n$ with the standard basis an orthonormal basis. This is the starting point of any Analysis course, and it is the advantage that Newton and Leibniz in particular saw in Descartes' work.

**Remark.** The advantages of Definition 3.12 are conceptual. For example, by Proposition 2.5, a $d$-dimensional affine subspace $S$ of $\mathbb{A}^n$ is itself an affine space, which can be identified with $\mathbb{A}^d$; we write $S \cong \mathbb{A}^d$. It is easy to see that a scalar product on $\mathbb{A}^n$ defined with the properties in Proposition 3.10 restricts to a scalar product on $S \cong \mathbb{A}^d \subseteq \mathbb{A}^n$. Thus, an inclusion $S \cong \mathbb{A}^d \subseteq \mathbb{A}^n$ automatically translates to an inclusion $S \cong \mathbb{E}^d \subseteq \mathbb{E}^n$. Formally, this can be stated as follows: An affine subspace of $\mathbb{E}^n$ is a Euclidean space with the scalar product inherited from $\mathbb{E}^n$. In particular, all the results which we know to hold true for $\mathbb{E}^2$ or $\mathbb{E}^3$ will hold true when we consider 2-dimensional or 3-dimensional subspaces of the Euclidean space $\mathbb{E}^n$.

### 3.2.2 Gram-Schmidt algorithm

Clearly, not all coordinate systems are orthonormal, so what do we do if we have to deal with a non-orthonormal reference frame $\mathcal{K}$? The familiar formulas in Proposition 3.11 no longer hold true. We have two options: 1. We deal with the scalar product in the given reference frame $\mathcal{K}$; or 2. We find an orthonormal reference frame $\mathcal{K}'$ starting from $\mathcal{K}$ and translate everything to $\mathcal{K}'$. For the first option, one makes use of the Gram matrix (see Appendix F). We discuss the second option in this section.

Fix a basis $\mathcal{B} = (\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n)$ of $\mathbb{V}^n$. We want to construct an orthonormal basis $\mathcal{B}'$ starting from $\mathcal{B}$. Recall from (3.2) that the orthogonal projection of $\mathbf{e}_i$ onto $\mathbf{e}_j$ is $\frac{\langle \mathbf{e}_j, \mathbf{e}_i\rangle}{\langle \mathbf{e}_j, \mathbf{e}_j\rangle}\mathbf{e}_j$. We construct $\mathcal{B}'$ in two steps:

1. Construct an orthogonal basis $\mathbf{e}'_1, \mathbf{e}'_2, \dots, \mathbf{e}'_n$ as follows

$$\begin{aligned}
\mathbf{e}'_1 &= \mathbf{e}_1 && = \mathbf{e}_1\\
\mathbf{e}'_2 &= \mathbf{e}_2 - \tfrac{\langle \mathbf{e}'_1, \mathbf{e}_2\rangle}{\langle \mathbf{e}'_1, \mathbf{e}'_1\rangle}\mathbf{e}'_1 && = \mathbf{e}_2 - \mathrm{Pr}^{\perp}_{\mathbf{e}'_1}(\mathbf{e}_2)\\
\mathbf{e}'_3 &= \mathbf{e}_3 - \tfrac{\langle \mathbf{e}'_1, \mathbf{e}_3\rangle}{\langle \mathbf{e}'_1, \mathbf{e}'_1\rangle}\mathbf{e}'_1 - \tfrac{\langle \mathbf{e}'_2, \mathbf{e}_3\rangle}{\langle \mathbf{e}'_2, \mathbf{e}'_2\rangle}\mathbf{e}'_2 && = \mathbf{e}_3 - \mathrm{Pr}^{\perp}_{\mathbf{e}'_1}(\mathbf{e}_3) - \mathrm{Pr}^{\perp}_{\mathbf{e}'_2}(\mathbf{e}_3)\\
&\vdots
\end{aligned}$$

2. Normalize the vectors to obtain the basis

$$\mathcal{B}' = \left\{ \frac{\mathbf{e}'_1}{|\mathbf{e}'_1|}, \dots, \frac{\mathbf{e}'_n}{|\mathbf{e}'_n|} \right\}.$$

This process of obtaining an orthonormal basis from a given basis is called the *Gram-Schmidt process*. It can be used in an infinite-dimensional vector space, hence the name *process*. If the vector space is finite-dimensional, the process terminates, and we may call it an *algorithm*.

**Proposition 3.13.** The basis $\mathcal{B}'$ obtained from the basis $\mathcal{B}$ with the Gram-Schmidt algorithm is an orthonormal basis.

*Proof.* A more general statement and proof are given in [20, Theorem 17.4]. The vectors $\mathbf{e}'_1, \mathbf{e}'_2, \dots, \mathbf{e}'_k, \dots$ are constructed by induction on $k$. For each $k$, we consider the vector subspace $\mathbb{V}_k$ generated by $\mathcal{B}_k = (\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_k)$ and show that $\mathcal{B}'_k = (\mathbf{e}'_1, \mathbf{e}'_2, \dots, \mathbf{e}'_k)$ is an orthogonal basis for $\mathbb{V}_k$.

If $k = 1$, then $\mathbf{e}'_1 = \mathbf{e}_1$, and the claim is obvious. Assume that the claim holds for $k$. By definition, we have

$$\mathbf{e}'_{k+1} = \mathbf{e}_{k+1} - \underbrace{\sum_{i=1}^{k} \frac{\langle \mathbf{e}'_i, \mathbf{e}_{k+1}\rangle}{\langle \mathbf{e}'_i, \mathbf{e}'_i\rangle}\mathbf{e}'_i}_{\mathbf{v}}.$$

Since the vectors $\mathbf{e}'_i$ form a basis, $\mathbf{e}'_{k+1} \neq 0 \Leftrightarrow \mathbf{v} \neq \mathbf{e}_{k+1}$; otherwise, $\mathbf{e}_{k+1}$ would be a linear combination of $\mathcal{B}'_k$ and therefore of $\mathcal{B}_k$ (since both are bases of $\mathbb{V}_k$). It follows that $\mathcal{B}'_{k+1}$ is a basis of $\mathbb{V}_{k+1}$. Moreover, for each $i = 1, \dots, k$, we have

$$\begin{aligned}
\langle \mathbf{e}'_{k+1}, \mathbf{e}'_i\rangle &= \langle \mathbf{e}_{k+1} - \textstyle\sum_{j=1}^{k} \frac{\langle \mathbf{e}'_j, \mathbf{e}_{k+1}\rangle}{\langle \mathbf{e}'_j, \mathbf{e}'_j\rangle}\mathbf{e}'_j, \mathbf{e}'_i\rangle \\
&= \langle \mathbf{e}_{k+1}, \mathbf{e}'_i\rangle - \tfrac{\langle \mathbf{e}'_i, \mathbf{e}_{k+1}\rangle}{\langle \mathbf{e}'_i, \mathbf{e}'_i\rangle}\langle \mathbf{e}'_i, \mathbf{e}'_i\rangle \\
&= \langle \mathbf{e}_{k+1}, \mathbf{e}'_i\rangle - \langle \mathbf{e}_{k+1}, \mathbf{e}'_i\rangle = 0
\end{aligned}$$

since, by the inductive hypothesis, $\langle \mathbf{e}'_i, \mathbf{e}'_j\rangle = 0$ for all $1 \leq i, j \leq k$. Thus, $\mathcal{B}'_{k+1}$ is an orthogonal basis, and by renormalizing it, we obtain an orthonormal basis for $\mathbb{V}_{k+1}$. □

### 3.2.3 Normal vectors

Recall that, with respect to a frame $\mathcal{K} = (O, \mathcal{B})$, a hyperplane $\mathcal{H}$ is given by a linear equation

$$\mathcal{H} : a_1x_1 + a_2x_2 + \cdots + a_nx_n = b. \tag{3.5}$$

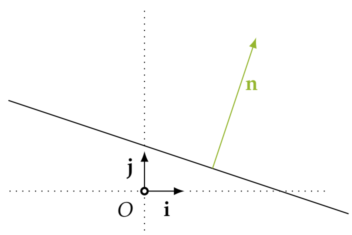
Assume now that $\mathcal{K}$ is orthonormal, i.e., that $\mathcal{B} = (\mathbf{e}_1, \dots, \mathbf{e}_n)$ is orthonormal. Fix a point $Q(q_1, \dots, q_n)$ in $\mathcal{H}$. Since $Q$ lies in $\mathcal{H}$, it satisfies the equation (3.5), i.e., we have $b = a_1q_1 + a_2q_2 + \cdots + a_nq_n$. Having expressed the constant $b$ in terms of $Q$, any other point $P(p_1, \dots, p_n)$ is a solution to (3.5) if and only if

$$a_1(p_1 - q_1) + a_2(p_2 - q_2) + \cdots + a_n(p_n - q_n) = 0.$$

Therefore, if we denote by $\mathbf{n}$ the vector with components $(a_1, \dots, a_n)$, then

$$P \in \mathcal{H} \quad \text{if and only if} \quad \langle \mathbf{n}, \overrightarrow{QP}\rangle = 0, \quad \text{equivalently, if and only if} \quad \mathbf{n} \perp \overrightarrow{QP}.$$

In other words, the vector $\mathbf{n}$ of coefficients in (3.5) is orthogonal to any vector parallel to $\mathcal{H}$, and we say that it is *orthogonal to* $\mathcal{H}$.



**Definition 3.14.** Let $\mathcal{H}$ be a hyperplane in $\mathbb{E}^n$. A vector $\mathbf{v}$ is called a *normal vector* of $\mathcal{H}$ if it is orthogonal to $\mathcal{H}$.

**Example 3.15.** Hyperplanes in $\mathbb{E}^2$ are lines. The line with equation $\ell : x + 3y - 3 = 0$, relative to some orthonormal coordinate system, admits $\mathbf{v}(1,3)$ as a normal vector.

**Example 3.16.** Hyperplanes in $\mathbb{E}^3$ are planes. The plane with equation

$$\pi : 2x - y + \frac{1}{3}z + 7 = 0,$$

relative to some orthonormal coordinate system, admits $\mathbf{v}(6,-3,1)$ as a normal vector.

**Proposition 3.17** (Hesse normal form)**.** Let $\mathcal{K} = (O, \mathcal{B})$ be an orthonormal frame with $\mathcal{B} = (\mathbf{e}_1, \ldots, \mathbf{e}_n)$. Any hyperplane $\mathcal{H}$ has, up to sign, a unique normal vector $\mathbf{n}$ of length 1. The components of $\mathbf{n}$ are $(\cos(\theta_1), \cos(\theta_2), \ldots, \cos(\theta_n))$ where $\theta_i$ is the angle $\measuredangle(\mathbf{n}, \mathbf{e}_i)$. Consequently, relative to $\mathcal{K}$ any hyperplane $\mathcal{H}$ has a unique equation of the form

$$\mathcal{H} : \cos(\theta_1)x_1 + \cos(\theta_2)x_2 + \cdots + \cos(\theta_n)x_n = c \tag{3.6}$$

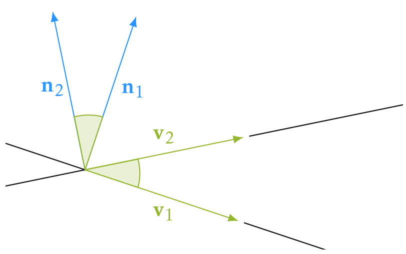
for some $c \geq 0$. Moreover, the distance $d(O, \mathcal{H})$ from the origin to the hyperplane equals $c$.

*Proof.* Let $\mathcal{H}$ be a hyperplane with equation (3.5). A normal vector of $\mathcal{H}$ is $\mathbf{n}(a_1, \ldots, a_n)$. By the discussion in Section 3.1.1, the normalized vector $\tilde{\mathbf{n}} = \mathbf{n}/|\mathbf{n}|$ has components $(\cos(\theta_1), \ldots, \cos(\theta_n))$ for some angles $\theta_1, \ldots, \theta_n \in [0, \pi)$. Thus, with $c = b/|\mathbf{n}|$, the equation of $\mathcal{H}$ has the form (3.6). If $c < 0$, replace each $\theta_i$ by $\pi - \theta_i$ to change the sign of $c$ (this is equivalent to changing the sign of $\tilde{\mathbf{n}}$). The last claim follows from the distance formula (3.8) deduced below in Proposition 3.20. □

### 3.2.4 Angles between lines and hyperplanes

Let $\ell_1$ and $\ell_2$ be two lines in $\mathbb{E}^2$. They define two angles; if $\mathbf{v}_1$ is a direction vector for $\ell_1$ and if $\mathbf{v}_2$ is a direction vector for $\ell_2$, then the two angles described by $\ell_1$ and $\ell_2$ are $\measuredangle(\mathbf{v}_1, \mathbf{v}_2)$ and $\measuredangle(-\mathbf{v}_1, \mathbf{v}_2)$. They are supplementary angles, so if you know one of them, you know the other one. We may calculate this with the scalar product, since

$$\cos \measuredangle(\mathbf{v}_1, \mathbf{v}_2) = \frac{\langle \mathbf{v}_1, \mathbf{v}_2 \rangle}{|\mathbf{v}_1| \cdot |\mathbf{v}_2|}.$$



Notice also that the two angles can be described with normal vectors: if $\mathbf{n}_1$ and $\mathbf{n}_2$ are normal vectors for $\ell_1$ and $\ell_2$, respectively, then the two angles between $\ell_1$ and $\ell_2$ are $\measuredangle(\mathbf{n}_1, \mathbf{n}_2)$ and $\measuredangle(-\mathbf{n}_1, \mathbf{n}_2)$. So, if these vectors are known, we may calculate

$$\cos\measuredangle(\mathbf{n}_1, \mathbf{n}_2) = \frac{\langle \mathbf{n}_1, \mathbf{n}_2 \rangle}{|\mathbf{n}_1| \cdot |\mathbf{n}_2|}.$$

On the other hand, if we know a direction vector $\mathbf{v}_1$ for the first line and a normal vector $\mathbf{n}_2$ for the second line, then the acute angle between $\ell_1$ and $\ell_2$ is

$$\frac{\pi}{2} - \arccos(\left|\frac{\langle \mathbf{v}_1, \mathbf{n}_2 \rangle}{|\mathbf{v}_1| \cdot |\mathbf{n}_2|}\right|) \in [0, \frac{\pi}{2}).$$

This generalizes in three ways. In $\mathbb{E}^n$, consider two lines $\ell_1$ and $\ell_2$ with direction vectors $\mathbf{v}_1$ and $\mathbf{v}_2$, respectively, as well as two hyperplanes $\mathcal{H}_1$ and $\mathcal{H}_2$ with normal vectors $\mathbf{n}_1$ and $\mathbf{n}_2$, respectively.

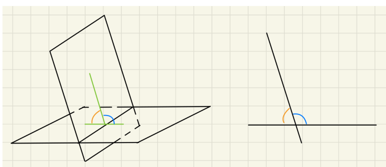
1. $\ell_1$ and $\ell_2$ define two supplementary angles: $\measuredangle(\mathbf{v}_1, \mathbf{v}_2)$ and $\measuredangle(-\mathbf{v}_1, \mathbf{v}_2)$, which can be calculated with
$$\cos\measuredangle(\mathbf{v}_1, \mathbf{v}_2) = \frac{\langle \mathbf{v}_1, \mathbf{v}_2 \rangle}{|\mathbf{v}_1| \cdot |\mathbf{v}_2|}.$$

2. $\mathcal{H}_1$ and $\mathcal{H}_2$ define two supplementary angles: $\measuredangle(\mathbf{n}_1, \mathbf{n}_2)$ and $\measuredangle(-\mathbf{n}_1, \mathbf{n}_2)$, which can be calculated with
$$\cos\measuredangle(\mathbf{n}_1, \mathbf{n}_2) = \frac{\langle \mathbf{n}_1, \mathbf{n}_2 \rangle}{|\mathbf{n}_1| \cdot |\mathbf{n}_2|}.$$

3. $\ell_1$ and $\mathcal{H}_1$ define two supplementary angles: if $\cos\measuredangle(\mathbf{v}_1, \mathbf{n}_1) \geq 0$, then $\measuredangle(\mathbf{v}_1, \mathbf{n}_1)$ is acute and the acute angle between $\ell_1$ and $\mathcal{H}_1$ is
$$\frac{\pi}{2} - \arccos\left(\frac{\langle \mathbf{v}_1, \mathbf{n}_1 \rangle}{|\mathbf{v}_1| \cdot |\mathbf{n}_1|}\right).$$

   Otherwise, if $\cos\measuredangle(\mathbf{v}_1, \mathbf{n}_1) < 0$, replace $\mathbf{n}_1$ with the normal vector $-\mathbf{n}_1$ of $\mathcal{H}$.

The angles between (hyper)planes in $\mathbb{E}^3$ are referred to as *dihedral angles*. Two planes, $\pi_1$ and $\pi_2$, define four dihedral angles, which are the four regions in which the two planes divide $\mathbb{E}^3$. More precisely, let $\ell$ be the line $\pi_1 \cap \pi_2$; choose a plane $\pi$ orthogonal to $\ell$ and consider the lines $\ell_1 = \pi \cap \pi_1$ and $\ell_2 = \pi \cap \pi_2$. The angles between $\ell_1$ and $\ell_2$ do not depend on the choice of $\pi$; i.e., if we choose a different plane orthogonal to $\ell$, we obtain congruent angles. So, up to congruence, we have two angles, and they can be calculated using normal vectors as indicated above.



## 3.3 Distance

**Definition 3.18.** The *distance between two points* $P,Q$ in $\mathbb{E}^n$, denoted $d(P,Q)$, is the length of the segment $[PQ]$, and we calculate it with (3.4). The *distance between two sets of points* $S_1$ *and* $S_2$ is

$$d(S_1,S_2) := \inf\Big\{d(P,Q) : P \in S_1 \text{ and } Q \in S_2\Big\}. \tag{3.7}$$

### 3.3.1 Distance from a point to a hyperplane

**Proposition 3.19.** Consider a hyperplane $\mathcal{H}$ and a point $P$ not in $\mathcal{H}$. Drop a perpendicular line from $P$ to $\mathcal{H}$, and let $P'$ be the point in which the line intersects the hyperplane. Then

$$d(P,\mathcal{H}) = |PP'|.$$

*Proof.* For any other point $Q$ in $\mathcal{H}$, distinct from $P'$, we have a right-angled triangle $PP'Q$. Since the hypotenuse is larger than the catheti, we have $d(P,\mathcal{H}) \leq |PP'| < |PQ|$ for any point $Q \in \mathcal{H}$. Thus $d(P,\mathcal{H}) \leq |PP'|$. □

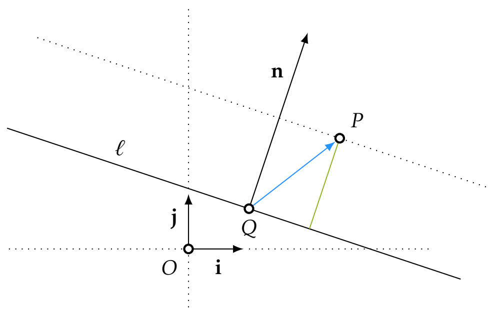
**Proposition 3.20.** Let $\mathcal{K}$ be a frame of $\mathbb{E}^n$. Consider the point $P(p_1,\dots,p_n)$ and a hyperplane $\mathcal{H}$ : $a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$. Then

$$d(P,\mathcal{H}) = \frac{|a_1p_1 + a_2p_2 + \cdots + a_np_n - b|}{\sqrt{a_1^2 + a_2^2 + \cdots + a_n^2}}. \tag{3.8}$$

*Proof.* Drop a perpendicular line from $P$ to $\mathcal{H}$, intersecting $\mathcal{H}$ at $P'$. By Proposition 3.19, we have $d(P,\mathcal{H}) = |PP'|$. Now let $Q$ be a point in $\mathcal{H}$, distinct from $P'$, and consider the normal vector $\mathbf{n} = \mathbf{n}(a_1,\dots,a_n)$. Then $|PP'|$ is the length of the orthogonal projection of $\overrightarrow{QP}$ onto $\mathbf{n}$. To see this, let $N$ be such that $\mathbf{n} = \overrightarrow{QN}$ and look at the quadrilateral $QP'PN$. We have

$$d(P,\mathcal{H}) = |\frac{\langle \mathbf{n}, \overrightarrow{QP}\rangle}{|\mathbf{n}|^2}\mathbf{n}| = \frac{|\langle \mathbf{n}, \overrightarrow{QP}\rangle|}{|\mathbf{n}|} = \frac{|\langle \mathbf{n}, \overrightarrow{OP} - \overrightarrow{OQ}\rangle|}{|\mathbf{n}|} = \frac{|\langle \mathbf{n}, \overrightarrow{OP}\rangle - b|}{|\mathbf{n}|}.$$

If we write this explicitly in coordinates, we obtain the claim. □



**Proposition 3.21.** Let $S$ be an affine subspace of $\mathbb{E}^n$ parallel to a hyperplane $\mathcal{H}$. Then

$$d(S,\mathcal{H}) = d(P,\mathcal{H})$$

for any $P$ in $S$.

*Proof.* Notice that a point is parallel to any affine subspace (this follows from Definition 1.4). We can generalize the distance formula (3.8) to other affine subspaces parallel to a hyperplane. Consider an affine subspace $S$ of $\mathbb{E}^n$ which is parallel to $\mathcal{H}$. Take two points $A$ and $B$ and from each of them drop a perpendicular to $\mathcal{H}$ which intersects the hyperplane at $M$ and $N$, respectively. Since the sides $MA$ and $NB$ are parallel to $\mathbf{n}$, the quadrilateral $ABNM$ is planar (lies in a plane). But then $AB$ has to be parallel to $MN$; otherwise, the two lines intersect at a point which would lie in $S \cap \mathcal{H}$, contradicting $S \| \mathcal{H}$. Since we have right angles, $ABNM$ is in fact a rectangle. This shows that $d(A,\mathcal{H}) = d(B,\mathcal{H})$. Thus, the distance from $S$ to $\mathcal{H}$ is the distance from any point $A \in S$ to $\mathcal{H}$

$$d(S,\mathcal{H}) = d(A,\mathcal{H}) \quad \forall A \in S.$$

□

### 3.3.2 Loci of points equidistant from affine subspaces

**Definition 3.22.** Let $S$ be a set of points in $\mathbb{E}^n$. For $c \in \mathbb{R}$, the set of points at distance $c$ from $S$ is

$$\mathcal{L}(S,c) = \Big\{P \in \mathbb{E}^n : d(P,S) = c\Big\}.$$

**Proposition 3.23.** Let $S$ be an affine subspace of $\mathbb{E}^n$ and let $c > 0$ be a constant. Table 3.1 classifies the possible shapes of $\mathcal{L}(S,c)$.

| n | a point | a line | a plane |
|---|---|---|---|
| 1 | two points | - | - |
| 2 | circle | two lines | - |
| 3 | sphere | circular cylinder | two planes |

Table 3.1: Loci of points at constant distance from an affine subspace.

*Proof.* We notice first that if $c = 0$, then $\mathcal{L}(S,c)$ is the set $S$ itself by definition (3.7). Moreover, if $c < 0$, then $\mathcal{L}(S,c)$ is empty since distances are positive. The non-trivial cases appear when $c > 0$. If $S$ consists of a single point, in dimension $n = 1$, then $\mathcal{L}(S,c)$ consists of two points, namely the endpoints of a segment with midpoint $S$. We can discuss the diagonal entries in Table 3.1 together by considering hyperplanes, since, in dimension 1, these are points; in dimension 2, these are lines; and, in dimension 3, these are planes. Let the hyperplane $S$ be given by the equation $a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$. By (3.8), a point $Q(q_1,\ldots,q_n)$ belongs to $\mathcal{L}(S,c)$ if and only if $d(Q,S) = c$, which means

$$d(P,Q) = c \quad \Leftrightarrow \quad |a_1q_1 + a_2q_2 + \cdots + a_nq_n| = c'$$

where $c' = c \cdot \sqrt{a_1^2 + a_2^2 + \cdots + a_n^2}$. Thus, $\mathcal{L}(S,c)$ is the union of two hyperplanes with equations

$$a_1q_1 + a_2q_2 + \cdots + a_nq_n = c' \quad \text{and} \quad a_1q_1 + a_2q_2 + \cdots + a_nq_n = -c'.$$

Consider the case where $S$ is a line in $\mathbb{E}^3$. We notice that in dimension 3, we don't yet have a formula for the distance between a point and a line. Such a formula will be deduced in Section 4.2.2 and can be used to deduce equations for $\mathcal{L}(S,c)$ if $S$ is an arbitrary line. However, we have other options as well; for instance, we can 'scan' the three-dimensional space with planes passing through $S$ and notice that in each such plane we obtain two lines at equal distance from $S$. Yet another option, since we are only interested in the possible shapes of $\mathcal{L}(S,c)$, is to make a good choice of a frame. Choose the frame $\mathcal{K}$ such that $S$ is the $z$-axis. An arbitrary point in $S$ is $P(0,0,t)$ with $t \in \mathbb{R}$. Then, for a point $Q(x_Q, y_Q, z_Q)$, we have

$$d(P,Q) = c \quad \Leftrightarrow \quad x_Q^2 + y_Q^2 = c^2$$

Thus, this locus of points has equation $\mathcal{L}(S,c) : x^2 + y^2 = c^2$, which describes a cylinder with axis $S$.

Lastly, in any dimension, the set of points at distance $c$ from a point $Q(x_Q, y_Q, z_Q)$ is the (hyper)sphere of radius $c$ centered at $Q$, since

$$d(P,Q) = c \quad \Leftrightarrow \quad (p_1 - q_1)^2 + (p_2 - q_2)^2 + \cdots + (p_n - q_n)^2 = c^2.$$

□

**Definition 3.24.** If $S'$ is another set, the *locus of points equidistant from $S$ and $S'$* is

$$\mathcal{L}(S,S') = \Big\{P \in \mathbb{E}^n : d(P,S) = d(P,S')\Big\}.$$

**Proposition 3.25.** Let $S$ and $S'$ be two affine subspaces of $\mathbb{E}^n$. Table 3.2 classifies the possible shapes of $\mathcal{L}(S,S')$.

| n | 2 points | point + line | point + plane | 2 lines | line + plane | 2 planes |
|---|---|---|---|---|---|---|
| 1 | point | - | - | - | - | - |
| 2 | line | parabola | - | line/lines | - | - |
| 3 | plane | parabolic cylinder | paraboloid | planes or hyperbolic paraboloid | cone or parabolic cylinder | plane or two planes |

Table 3.2: Loci of points equidistant from two distinct affine subspaces.

*Proof.* **Two points:** For fixed points $A, B$, the condition $d(P,A) = d(P,B)$ yields

$$(a_1 - p_1)^2 + \cdots + (a_n - p_n)^2 = (b_1 - p_1)^2 + \cdots + (b_n - p_n)^2.$$

Simplifying gives a linear equation defining the *perpendicular bisecting hyperplane* of $[AB]$.

**Point and Hyperplane:** Choose coordinates such that $Q(0,\dots,0,1)$ and $\mathcal{H} : x_n + 1 = 0$. The condition $d(P,\mathcal{H}) = d(P,Q)$ becomes

$$|p_n + 1| = \sqrt{p_1^2 + \cdots + p_{n-1}^2 + (p_n - 1)^2} \quad \Leftrightarrow \quad x_1^2 + \cdots + x_{n-1}^2 = 4x_n,$$

which describes a paraboloid (or parabola in $\mathbb{E}^2$).

**Two Hyperplanes:** For normalized equations $\mathbf{n}\cdot\mathbf{x} + a = 0$ and $\mathbf{n}'\cdot\mathbf{x} + b = 0$, equidistance implies $|\mathbf{n}\cdot\mathbf{x} + a| = |\mathbf{n}'\cdot\mathbf{x} + b|$. This yields two linear equations representing the angle bisecting hyperplanes (or a single parallel hyperplane if $\mathbf{n} = \mathbf{n}'$).

**Point and Line in $\mathbb{E}^3$:** Choosing a frame where $Q(1,0,0)$ and $S'$ is the $y$-axis shifted to $(-1,0,0)$, the condition $d(P,Q) = d(P,S')$ leads to $y_P^2 = 4x_P$, a parabolic cylinder.

**Two Lines in $\mathbb{E}^3$:** If the lines intersect (e.g., $x$ and $y$ axes), we obtain $(x_P - y_P)(x_P + y_P) = 0$, describing two orthogonal planes. If the lines are skew (e.g., $x$-axis and a line through $(0,1,0)$ with direction $(\lambda,0,1)$), the condition yields

$$y_P^2 + z_P^2 = (y_P - 1)^2 + \frac{(x_P - \lambda z_P)^2}{\lambda^2 + 1}$$

which corresponds to a hyperbolic paraboloid. Finally, for a line and a plane, one obtains an elliptic cone or a parabolic cylinder. □

### 3.3.3 Convergence

Once we have a clear notion of distance, the notion of proximity to a point also becomes clear. Being close to a point $P$ means lying in a ball of radius $\varepsilon$ centered at the point $P$, and you may adjust $\varepsilon$ at will. The set of open balls centered at all points defines a topology on $\mathbb{E}^n$, called the *standard topology*. Recall Chapter 7 of your Analysis course [17]. All the results from your analysis course hold true for $\mathbb{E}^n$ by fixing an orthonormal frame, which identifies $\mathbb{E}^n$ with $\mathbb{R}^n$.

# CHAPTER 4

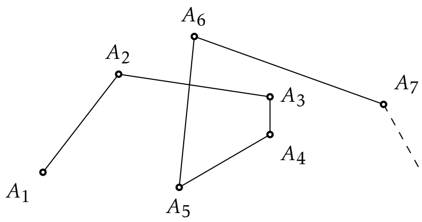
# Area and volume

## Contents

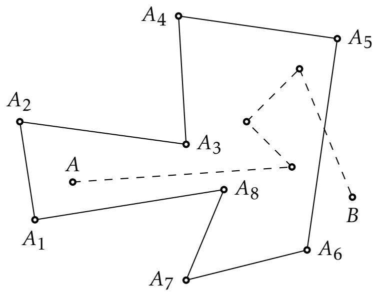
## 4.1 Area

### 4.1.1 Area of polygons

**Definition 4.1.** A *polygonal segment* $A_1A_2\dots A_n$ is a union of segments $[A_1A_2],\dots,[A_{n-1}A_n]$, such that consecutive segments $[A_iA_{i+1}]$ and $[A_{i+1}A_{i+2}]$ intersect only in their shared *vertex* $A_{i+1}$. If all the vertices lie in the same plane, the polygonal segment is said to be *planar*. The polygonal segment $A_1A_2\dots A_n$ is said to *intersect itself* at a point $P$ if there exist indices $i$ and $j$ with $i+1<j$, such that $[A_iA_{i+1}]$ intersects $[A_jA_{j+1}]$ at $P$.

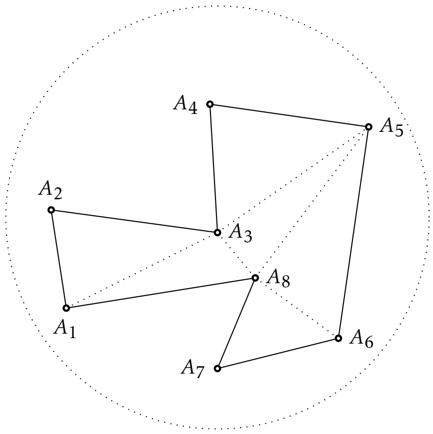



A *polygon* is an $n$-tuple of points $(A_1,A_2,\dots,A_n)$, denoted $A_1A_2\dots A_n$, such that the polygonal segment $A_1A_2\dots A_nA_1$ intersects itself only at the vertex $A_1$. The segments $[A_iA_{i+1}]$ are called the *sides of the polygon*. Two polygons are said to be *congruent* if their corresponding sides and the angles between them are pairwise congruent. Two polygons $P_1=A_1A_2\dots A_n$ and $P_2=B_1B_2\dots B_m$ are said to be *similar* if their corresponding angles are congruent. In this case, the ratio of the lengths of corresponding sides is constant, and we denote this *ratio* by $P_1:P_2$.




We will only consider planar polygons. A planar polygon separates the points of the plane not lying on its sides into two regions (see [14, Theorem 9]): the *interior* and the *exterior*. These regions have the following property: if $A$ is a point of the interior (an *inner point*) and $B$ is a point of the exterior (an *exterior point*), then every polygonal segment lying in the plane of the polygon and connecting $A$ with $B$ must intersect the sides of the polygon at least once.

The interior $S$ of a (planar) polygon is *bounded* in the sense that, for any point $P$ in the plane, there is an $r>0$ such that all points in $S$ are at distance at most $r$ from $P$. Indeed, take $r$ to be the maximum of the distances from $P$ to all vertices. Our goal is to measure such bounded regions of the plane—namely, interiors of polygons. We do this by means of triangles, which are the simplest possible polygons. Any polygon can be subdivided into triangles (see for example [3, Theorem 3.1]), for instance with the so-called ear clipping method. Such a subdivision partitions the interior of a polygon into interiors of triangles, if we are willing to disregard segments. Given all this, we may measure the interior of polygons by comparing them with the interior of a unit square.




**Definition 4.2.** For brevity, we use the term *area of a polygon* to mean the *area of the interior of a polygon*. To define area, we consider two polygons to be *disjoint* if their interiors are disjoint. Denote by $\mathcal{P}$ the set of all finite unions of polygons. We define an *area* function $\text{Area} : \mathcal{P} \to \mathbb{R}_{\geq 0}$ through the following properties:

(A1) If $S$ is a square of side length 1, then $\text{Area}(S) = 1$.

(A2) If $S_1$ and $S_2$ are similar polygons with ratio $x$, then $\text{Area}(S_1) = x^2\,\text{Area}(S_2)$.

(A3) If $S_1$ and $S_2$ are disjoint polygons, then $\text{Area}(S_1 \cup S_2) = \text{Area}(S_1) + \text{Area}(S_2)$.

**Remark.** The above definition captures basic principles of the intuitive notion of area. We note that in property (A2), the term 'similar polygons' can be replaced with 'congruent polygons', resulting in a weaker assumption. Furthermore, in property (A3) the finite union can be extended to a countable union, in which case the sum is replaced by a countable sum (see Section 4.3.3). Our definition of area allows us to deduce the following well-known facts.

**Proposition 4.3.** We have

1. The area of a rectangle of side lengths $a$ and $b$ is $ab$.

2. The area of a triangle $ABC$ with side length $a$ and corresponding height $h$ is $ah/2$.

3. The area of a parallelogram $ABCD$ is twice the area of the triangle $ABC$.

*Proof.* Denote by $S_a$ a square of side length $a$ and by $R_{a,b}$ a rectangle of side lengths $a$ and $b$. First, notice that since the ratio $S_a : S_1$ equals $a$, we have $\text{Area}(S_a) = a^2\,\text{Area}(S_1) = a^2$, by (A2) and (A1).

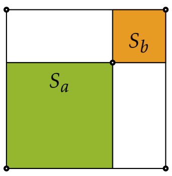
1. Place $S_a$ and $S_b$ in opposite corners of $S_{a+b}$. The complement of the two small squares in the big square consists of two disjoint rectangles which are congruent to $R_{a,b}$. Thus, by (A3) we have

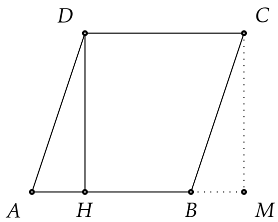
$$\mathrm{Area}(S_{a+b}) = \mathrm{Area}(S_a \cup R_{a,b} \cup R_{a,b} \cup S_b) = \mathrm{Area}(S_a) + \mathrm{Area}(S_b) + 2\,\mathrm{Area}(R_{a,b})$$

hence

$$(a+b)^2 = a^2 + b^2 + 2\,\mathrm{Area}(R_{a,b})$$

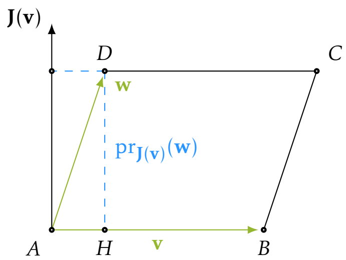
and therefore $\mathrm{Area}(R_{a,b}) = ab$.




For claim 3, it suffices to notice that a diagonal divides a parallelogram into two congruent triangles. For claim 2, we use claim 3 and check the known area formula for a parallelogram $ABCD$, i.e., $\mathrm{Area}(ABCD) = ah$ where $h$ is the height and $a$ the length of the corresponding side. Let $H$ be a point on $AB$ such that $h = |DH|$. The triangle $ADH$ can be moved to the opposite side of the parallelogram to form a rectangle with side lengths $a$ and $h$. Thus, by claim 1, $\mathrm{Area}(ABCD) = \mathrm{Area}(R_{a,h}) = ah$. □

Orthonormal frames offer an efficient method for calculating the area of a parallelogram, and therefore, the areas of triangles and polygons.




**Proposition 4.4.** Suppose the vertices $A(x_A, y_A)$, $B(x_B, y_B)$ and $D(x_D, y_D)$ of the parallelogram $ABCD$ are given with respect to an orthonormal frame. Then,

$$\mathrm{Area}(ABCD) = \left| \begin{vmatrix} x_A & y_A & 1 \\ x_B & y_B & 1 \\ x_D & y_D & 1 \end{vmatrix} \right| = \left| \begin{vmatrix} v_x & v_y \\ w_x & w_y \end{vmatrix} \right|$$

where $\mathbf{v}(v_x, v_y) = \overrightarrow{AB}$ and $\mathbf{w}(w_x, w_y) = \overrightarrow{AD}$.

*Proof.* Drop a perpendicular line from $D$ to $AB$ and denote by $H$ its intersection with $AB$. By Proposition 4.3, we have

$$\text{Area}(ABCD) = |\overrightarrow{AB}| \cdot |\overrightarrow{HD}| = |\mathbf{v}| \cdot |\text{pr}_{\mathbf{J}(\mathbf{v})}(\mathbf{w})| = |\mathbf{v}| \cdot |\frac{\langle \mathbf{J}(\mathbf{v}), \mathbf{w}\rangle}{|\mathbf{J}(\mathbf{v})|}| = |\langle \mathbf{J}(\mathbf{v}), \mathbf{w}\rangle| = |\begin{vmatrix} v_x & v_y \\ w_x & w_y \end{vmatrix}|.$$

Moreover, using properties of the determinants we obtain

$$\begin{vmatrix} v_x & v_y \\ w_x & w_y \end{vmatrix} = \begin{vmatrix} x_B - x_A & y_B - y_A \\ x_D - x_A & y_D - y_A \end{vmatrix} = \begin{vmatrix} x_A & y_A & 1 \\ x_B - x_A & y_B - y_A & 0 \\ x_D - x_A & y_D - y_A & 0 \end{vmatrix} = \begin{vmatrix} x_A & y_A & 1 \\ x_B & y_B & 1 \\ x_D & y_D & 1 \end{vmatrix}.$$

□

### 4.1.2 Oriented area

Fix an orientation in $\mathbb{E}^2$, i.e., fix a right-oriented basis $\mathcal{B}$. Proposition 4.4 shows that, up to sign, the area of a parallelogram $ABDC$ is given by the determinant of the base change matrix $M_{\mathcal{B},\mathcal{B}'}$, where $\mathcal{B}' = (\overrightarrow{AB}, \overrightarrow{AC})$. The value $\det(M_{\mathcal{B},\mathcal{B}'})$ does not depend on the orthonormal basis $\mathcal{B}$ in which the vectors are expressed. The proof of this claim is the same in any dimension (see Proposition 4.21).

**Definition 4.5.** With the above notation, the *box product of* $\mathbf{v}$ *and* $\mathbf{w}$ is $\det(M_{\mathcal{B},\mathcal{B}'})$, and we denote it by $[\mathbf{v},\mathbf{w}]$. The *oriented area of the parallelogram* $ABDC$ spanned by $\mathbf{v} = \overrightarrow{AB}$ and $\mathbf{w} = \overrightarrow{AC}$ is

$$\text{Area}_{\text{or}}(ABDC) = [\mathbf{v},\mathbf{w}].$$

Similarly, the oriented area of the triangle $ABC$ is $\text{Area}_{\text{or}}(ABC) = \text{Area}_{\text{or}}(ABDC)/2$.

**Proposition 4.6.** Let $\mathbf{v} = \overrightarrow{AB}$ and $\mathbf{w} = \overrightarrow{AC}$ be two vectors with $ABC$ a triangle.

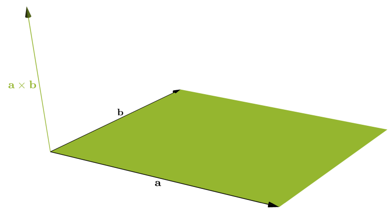
$$\sin \measuredangle_{\text{or}}(\mathbf{v},\mathbf{w}) = \frac{[\mathbf{v},\mathbf{w}]}{|\mathbf{v}| \cdot |\mathbf{w}|} = \frac{2 \cdot \text{Area}_{\text{or}}(ABC)}{|AB| \cdot |AC|}.$$

where $\mathcal{B}' = (\mathbf{v},\mathbf{w})$.

*Proof.* This is a direct consequence of the definition of the sine function for oriented angles and of the definition of oriented area of a triangle. □

## 4.2 Cross product

The cross product is the 3-dimensional analogue of the operator $J$ introduced in Section 3.1.2. Throughout this section, we consider $\mathbb{E}^3$. For two non-zero vectors $\mathbf{a}$ and $\mathbf{b}$, the orthogonal complements $\mathbb{V}^{\perp\mathbf{a}}$ and $\mathbb{V}^{\perp\mathbf{b}}$ are 2-dimensional vector subspaces of $\mathbb{V}^3$. If $\mathbf{a}$ and $\mathbf{b}$ are linearly independent, then $\mathbb{V}^{\perp\mathbf{a}} \cap \mathbb{V}^{\perp\mathbf{b}}$ is a 1-dimensional vector subspace of $\mathbb{V}^3$. In other words, up to scalar multiple, there is a unique vector $\mathbf{c}$ perpendicular to both $\mathbf{a}$ and $\mathbf{b}$. When prescribing the length of $\mathbf{c}$, we have exactly two options: $\pm\mathbf{c}$. For these options, $(\mathbf{a},\mathbf{b},\mathbf{c})$ is either left- or right-oriented. By fixing such an orientation, the choice of $\mathbf{c}$ becomes unique. This brings us to the following definition.




**Definition 4.7.** Let $\mathbf{a},\mathbf{b}\in\mathbb{V}^3$ be two vectors. The *cross product* (or *vector product*) of $\mathbf{a}$ and $\mathbf{b}$, denoted $\mathbf{a}\times\mathbf{b}$, is the vector defined by the following properties:

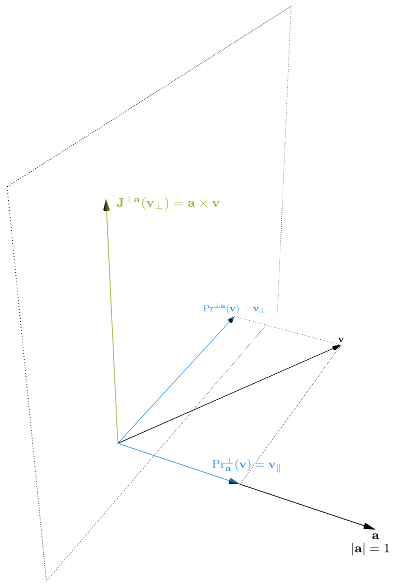
1. if $\mathbf{a}$ and $\mathbf{b}$ are parallel, then $\mathbf{a}\times\mathbf{b}=0$.
2. if $\mathbf{a}$ and $\mathbf{b}$ are not parallel, then
   (a) $|\mathbf{a}\times\mathbf{b}|$ equals the area of a parallelogram spanned by $\mathbf{a}$ and $\mathbf{b}$,
   (b) $\mathbf{a}\times\mathbf{b}\perp\mathbf{a}$ and $\mathbf{a}\times\mathbf{b}\perp\mathbf{b}$,
   (c) $(\mathbf{a},\mathbf{b},\mathbf{a}\times\mathbf{b})$ is a right-oriented basis of $\mathbb{V}^3$.

In particular, the vectors $\mathbf{a}$ and $\mathbf{b}$ are parallel if and only if $\mathbf{a}\times\mathbf{b}=0$.

The above geometric definition can be translated into a more accurate algebraic description as follows. Fix a non-zero vector $\mathbf{a}$. Consider the orthogonal complement of $\mathbf{a}$, i.e., $\mathbb{V}^{\perp\mathbf{a}}=\{\mathbf{v}\in\mathbb{V}^3:\langle\mathbf{v},\mathbf{a}\rangle=0\}$. It is a 2-dimensional vector subspace of $\mathbb{V}^3$. For any vector $\mathbf{v}\in\mathbb{V}^3$, we have a unique decomposition

$$\mathbf{v}=\mathbf{v}_{||}+\mathbf{v}_{\perp}\quad\text{with } \mathbf{v}_{||} \text{ parallel to } \mathbf{a} \text{ and } \mathbf{v}_{\perp} \text{ orthogonal to } \mathbf{a}.$$

This gives a projection map $\mathrm{Pr}^{\perp\mathbf{a}}:\mathbb{V}^3\to\mathbb{V}^{\perp\mathbf{a}}$ defined by $\mathrm{Pr}^{\perp\mathbf{a}}(\mathbf{v})=\mathbf{v}_{\perp}$. Let $\mathbf{J}^{\perp\mathbf{a}}(\mathbf{v}_{\perp})$ denote the unique vector in $\mathbb{V}^{\perp\mathbf{a}}$ of length $|\mathbf{v}_{\perp}|$ and such that $(\mathbf{a},\mathbf{v}_{\perp},\mathbf{J}^{\perp\mathbf{a}}(\mathbf{v}_{\perp}))$ is right-oriented.



**Proposition 4.8.** With the above notation, for any two non-zero vectors $\mathbf{a},\mathbf{b}\in\mathbb{V}^3$ we have

$$\mathbf{a}\times\mathbf{b}=|\mathbf{a}|\cdot\mathbf{J}^{\perp\mathbf{a}}(\mathrm{Pr}^{\perp\mathbf{a}}(\mathbf{b})).$$

*Proof.* By construction, we see that $\mathbf{J}^{\perp\mathbf{a}}(\mathrm{Pr}^{\perp\mathbf{a}}(\mathbf{b}))$ is orthogonal to both $\mathbf{a}$ and $\mathbf{b}$. Furthermore, by the definition of the operator $\mathbf{J}^{\perp\mathbf{a}}$, the vector $\mathbf{J}^{\perp\mathbf{a}}(\mathrm{Pr}^{\perp\mathbf{a}}(\mathbf{b}))$ is a positive scalar multiple of $\mathbf{a}\times\mathbf{b}$. Therefore, all we need to show is that the vector on the right-hand side has length $|\mathbf{a}\times\mathbf{b}|$. We have

$$|\mathbf{a}\times\mathbf{b}|=|\mathbf{a}|\cdot|\mathbf{b}|\cdot\sin\measuredangle(\mathbf{a},\mathbf{b})=|\mathbf{a}|\cdot|\mathrm{Pr}^{\perp\mathbf{a}}(\mathbf{b})|=|\mathbf{a}|\cdot|\mathbf{J}^{\perp\mathbf{a}}(\mathrm{Pr}^{\perp\mathbf{a}}(\mathbf{b}))|$$

where the last equality follows from the fact that $\mathbf{J}^{\perp\mathbf{a}}$ is a rotation by a right angle—in particular, it does not change the length of vectors. □

**Proposition 4.9.** The cross product $\square\times\square:\mathbb{V}^3\times\mathbb{V}^3\to\mathbb{V}^3$ satisfies the following properties.

(CP1) It is *bilinear*, i.e., for all $a,b\in\mathbb{R}$ and all $\mathbf{v},\mathbf{w},\mathbf{u}\in\mathbb{V}^3$ we have

$$(a\mathbf{v}+b\mathbf{w})\times\mathbf{u}=a(\mathbf{v}\times\mathbf{u})+b(\mathbf{w}\times\mathbf{u})\quad\text{and}\quad\mathbf{v}\times(a\mathbf{w}+b\mathbf{u})=a(\mathbf{v}\times\mathbf{w})+b(\mathbf{v}\times\mathbf{u}).$$

(CP2) It is *skew-symmetric*, i.e., for all $\mathbf{v},\mathbf{w}\in\mathbb{V}^3$ we have

$$\mathbf{v}\times\mathbf{w}=-\mathbf{w}\times\mathbf{v}.$$

*Proof.* Skew-symmetry follows from the orientation requirement in the definition of the cross product. Indeed, if $(\mathbf{a},\mathbf{b},\mathbf{v})$ is right-oriented then $(\mathbf{b},\mathbf{a},\mathbf{v})$ is left-oriented, hence $(\mathbf{b},\mathbf{a},-\mathbf{v})$ is right-oriented. To check bilinearity, we need to verify that the cross product is linear in each argument. If the first argument is fixed, the map $\mathbf{a}\times\square:\mathbb{V}^3\to\mathbb{V}^3$ is a composition of linear maps (by Proposition 4.8), so it is linear. For the second argument we may use the skew-symmetry of the cross product. □

Since the cross product is bilinear, its values are determined by the values on a basis. If $\mathcal{B}=(\mathbf{i},\mathbf{j},\mathbf{k})$ is a right-oriented orthonormal basis, one can check with the definition of the cross product that the values on the basis vectors are:

| $\times$ | $\mathbf{i}$ | $\mathbf{j}$ | $\mathbf{k}$ |
|---|---|---|---|
| $\mathbf{i}$ | $0$ | $\mathbf{k}$ | $-\mathbf{j}$ |
| $\mathbf{j}$ | $-\mathbf{k}$ | $0$ | $\mathbf{i}$ |
| $\mathbf{k}$ | $\mathbf{j}$ | $-\mathbf{i}$ | $0$ |

.

This table allows us to calculate the cross product for arbitrary vectors $\mathbf{a}(a_1,a_2,a_3)$ and $\mathbf{b}(b_1,b_2,b_3)$, with components relative to $\mathcal{B}$:

$$(a_1\mathbf{i}+a_2\mathbf{j}+a_3\mathbf{k})\times(b_1\mathbf{i}+b_2\mathbf{j}+b_3\mathbf{k})=(a_2b_3-a_3b_2)\mathbf{i}+(a_3b_1-a_1b_3)\mathbf{j}+(a_1b_2-a_2b_1)\mathbf{k}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\a_1&a_2&a_3\\b_1&b_2&b_3\end{vmatrix}.$$

We can therefore derive, for example, a formula for the area of a parallelogram $P$ spanned by $\mathbf{a}$ and $\mathbf{b}$:

$$\mathrm{Area}(P)=|\mathbf{a}\times\mathbf{b}|=\sqrt{\begin{vmatrix}a_2&a_3\\b_2&b_3\end{vmatrix}^2+\begin{vmatrix}a_1&a_3\\b_1&b_3\end{vmatrix}^2+\begin{vmatrix}a_1&a_2\\b_1&b_2\end{vmatrix}^2}.$$

### 4.2.1 Algebraic identities

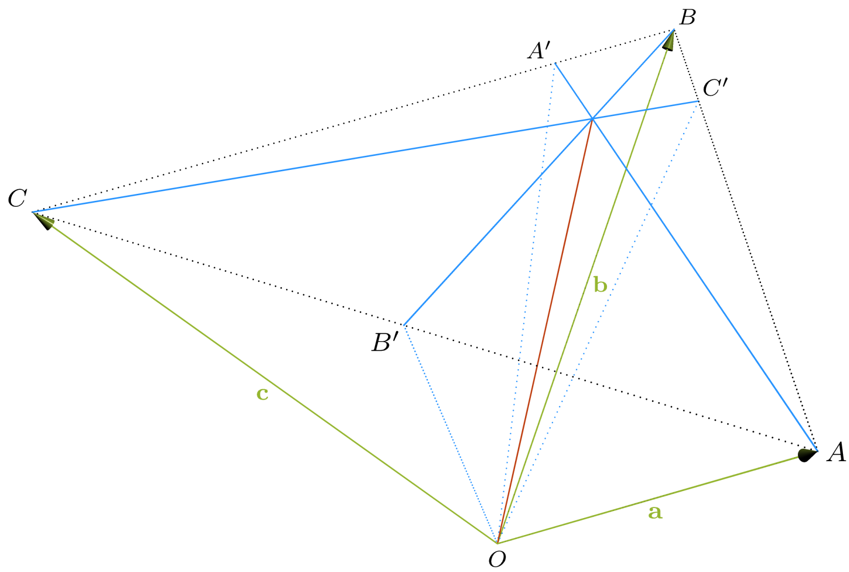
When calculating consecutive cross products, we notice that the cross product is not associative. For example, if $(\mathbf{i},\mathbf{j},\mathbf{k})$ is a right oriented orthonormal basis, then:

$$(\mathbf{i}\times\mathbf{j})\times\mathbf{j}=\mathbf{k}\times\mathbf{j}=-\mathbf{i}\quad\text{whereas}\quad\mathbf{i}\times(\mathbf{j}\times\mathbf{j})=0.$$

However, there is a rule which explains how iterated cross products behave when evaluated in different ways. This rule is the *Jacobi identity*:

$$(\mathbf{a}\times\mathbf{b})\times\mathbf{c}+(\mathbf{b}\times\mathbf{c})\times\mathbf{a}+(\mathbf{c}\times\mathbf{a})\times\mathbf{b}=0.\quad\forall\mathbf{a},\mathbf{b},\mathbf{c}\in\mathbb{V}^3.\tag{4.1}$$

One way to prove this identity is to write out the above expression in coordinates and check that the left-hand side simplifies to the zero vector. The geometric interpretation of the identity is as follows: given a tetrahedron $OABC$, the three planes passing through the edges adjacent to $O$ and orthogonal to the respective opposite faces intersect in a single line.




A short way to prove (4.1) is by using *the double cross formula*. This formula is of independent interest, as it provides an efficient way of calculating iterated cross products. Specifically, it replaces the calculation of a determinant with the calculation of two scalar products. Other beautiful identities, which can be derived with linear algebra only, can be found in the exercises and in [11, Chapter 4].

**Theorem 4.10** (Double Cross Formula)**.** For any vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ we have

$$(\mathbf{a}\times\mathbf{b})\times\mathbf{c}=\langle\mathbf{a},\mathbf{c}\rangle\cdot\mathbf{b}-\langle\mathbf{b},\mathbf{c}\rangle\cdot\mathbf{a}.\tag{4.2}$$

*Proof.* Notice that (4.2) can be checked directly in coordinates. A different proof is provided in [11, p.74] or in [5, p.70]. Our proof has three steps. First, note that we may assume all three vectors to be non-zero; otherwise, it is straightforward to check that both sides of the equality are zero.

(Step 1) It suffices to prove (4.2) in the special case where all three vectors are unit vectors. Indeed, under this assumption, using the linearity of the cross product (Proposition 4.9) and the linearity of the scalar product, we obtain:

$$(\mathbf{a}\times\mathbf{b})\times\mathbf{c}=|\mathbf{a}|\cdot|\mathbf{b}|\cdot|\mathbf{c}|\cdot\left[\left(\frac{\mathbf{a}}{|\mathbf{a}|}\times\frac{\mathbf{b}}{|\mathbf{b}|}\right)\times\frac{\mathbf{c}}{|\mathbf{c}|}\right]=|\mathbf{a}|\cdot|\mathbf{b}|\cdot|\mathbf{c}|\cdot\left[\langle\frac{\mathbf{a}}{|\mathbf{a}|},\frac{\mathbf{c}}{|\mathbf{c}|}\rangle\frac{\mathbf{b}}{|\mathbf{b}|}-\langle\frac{\mathbf{b}}{|\mathbf{b}|},\frac{\mathbf{c}}{|\mathbf{c}|}\rangle\frac{\mathbf{a}}{|\mathbf{a}|}\right]=\langle\mathbf{a},\mathbf{c}\rangle\cdot\mathbf{b}-\langle\mathbf{b},\mathbf{c}\rangle\cdot\mathbf{a}.$$

(Step 2) It suffices to prove (4.2) in the special cases where $\mathbf{c}=\mathbf{a}$ and $\mathbf{c}=\mathbf{b}$. Indeed, since $\mathbf{a}$ and $\mathbf{b}$ are non-parallel unit vectors, $(\mathbf{a},\mathbf{b},\mathbf{a}\times\mathbf{b})$ is a basis and $\mathbf{c}=\alpha\mathbf{a}+\beta\mathbf{b}+\gamma\mathbf{a}\times\mathbf{b}$ for some scalars $\alpha,\beta,\gamma\in\mathbb{R}$. Then, using the linearity of the cross product (Proposition 4.9) and the properties of the scalar product, we have:

$$\begin{aligned}(\mathbf{a}\times\mathbf{b})\times\mathbf{c}&=(\mathbf{a}\times\mathbf{b})\times(\alpha\mathbf{a}+\beta\mathbf{b}+\gamma\mathbf{a}\times\mathbf{b})\\&=\alpha(\mathbf{a}\times\mathbf{b})\times\mathbf{a}+\beta(\mathbf{a}\times\mathbf{b})\times\mathbf{b}\\&=\alpha\big(\langle\mathbf{a},\mathbf{a}\rangle\mathbf{b}-\langle\mathbf{b},\mathbf{a}\rangle\mathbf{a}\big)+\beta\big(\langle\mathbf{a},\mathbf{b}\rangle\mathbf{b}-\langle\mathbf{b},\mathbf{b}\rangle\mathbf{a}\big)\\&=\big(-\alpha\langle\mathbf{b},\mathbf{a}\rangle-\beta\langle\mathbf{b},\mathbf{b}\rangle\big)\mathbf{a}+\big(\alpha\langle\mathbf{a},\mathbf{a}\rangle+\beta\langle\mathbf{a},\mathbf{b}\rangle\big)\mathbf{b}\\&=\langle-\alpha\mathbf{a}-\beta\mathbf{b},\mathbf{b}\rangle\mathbf{a}+\langle\alpha\mathbf{a}+\beta\mathbf{b},\mathbf{a}\rangle\mathbf{b}\\&=\langle-\alpha\mathbf{a}-\beta\mathbf{b}-\gamma\mathbf{a}\times\mathbf{b},\mathbf{b}\rangle\mathbf{a}+\langle\alpha\mathbf{a}+\beta\mathbf{b}+\gamma\mathbf{a}\times\mathbf{b},\mathbf{a}\rangle\mathbf{b}\\&=\langle\mathbf{a},\mathbf{c}\rangle\mathbf{b}-\langle\mathbf{b},\mathbf{c}\rangle\mathbf{a}.\end{aligned}$$

(Step 3) It remains to show that (4.2) holds in the special cases where $\mathbf{c}=\mathbf{a}$ and $\mathbf{c}=\mathbf{b}$, assuming $\mathbf{a}$ and $\mathbf{b}$ are unit vectors. Using the skew-symmetry of the cross product, it is easy to see that the two cases are equivalent. By Step 1, we may also assume that all three vectors are unit vectors. In particular, $\langle\mathbf{a},\mathbf{a}\rangle=1$ and $\langle\mathbf{b},\mathbf{a}\rangle=\cos\theta$, where $\theta=\measuredangle(\mathbf{a},\mathbf{b})$. Thus, it suffices to prove the identity:

$$(\mathbf{a}\times\mathbf{b})\times\mathbf{a}=\mathbf{b}-\cos\theta\cdot\mathbf{a}.$$

Let $\mathbf{v}$ denote the left-hand side and $\mathbf{w}$ the right-hand side. Notice that $\mathcal{B}_{\mathbf{v}}=(\mathbf{a}\times\mathbf{b},\mathbf{a},\mathbf{v})$ is a right-oriented orthogonal basis. Since there is a unique vector of length $|\mathbf{v}|$ with this property, it suffices to show that $\mathcal{B}_{\mathbf{w}}=(\mathbf{a}\times\mathbf{b},\mathbf{a},\mathbf{w})$ is a right-oriented orthogonal basis and that $|\mathbf{v}|=|\mathbf{w}|$. The first property can be verified by calculating the determinant of the base-change matrix to the basis $\mathcal{B}=(\mathbf{a},\mathbf{b},\mathbf{a}\times\mathbf{b})$, which is also right-oriented:

$$\det(M_{\mathcal{B},\mathcal{B}_{\mathbf{w}}})=\begin{vmatrix}0&1&-\cos\theta\\0&0&1\\1&0&0\end{vmatrix}=1>0$$

Consequently, $\mathcal{B}_{\mathbf{w}}$ is right-oriented. Furthermore, because $\langle\mathbf{a},\mathbf{w}\rangle=\langle\mathbf{b},\mathbf{a}\rangle-\langle\mathbf{b},\mathbf{a}\rangle=0$, it follows that $(\mathbf{a}\times\mathbf{b},\mathbf{a},\mathbf{w})$ is an orthogonal basis (the remaining orthogonality requirements are straightforward to verify). It remains to check the length of $\mathbf{w}$. Since the vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ are unit vectors, and because $\measuredangle(\mathbf{a}\times\mathbf{b},\mathbf{a})$ is a right angle, we have

$$|\mathbf{v}|^2=|\mathbf{a}\times\mathbf{b}|^2=(\sin\theta)^2=1-2(\cos\theta)^2+(\cos\theta)^2=\langle\mathbf{b}-\langle\mathbf{b},\mathbf{a}\rangle\cdot\mathbf{a},\mathbf{b}-\langle\mathbf{b},\mathbf{a}\rangle\cdot\mathbf{a}\rangle=|\mathbf{w}|^2.$$

Thus, $\mathcal{B}_{\mathbf{w}}$ equals $\mathcal{B}_{\mathbf{v}}$, hence $\mathbf{v}=\mathbf{w}$. This completes the proof of Step 3 and of the theorem. □

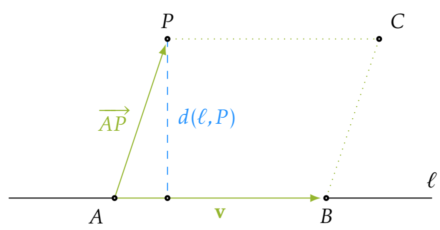
### 4.2.2 Distance between a point and a line

A line $\ell$ in dimension 2 is a hyperplane, and for any point $P$ of $\mathbb{E}^2$ we have an efficient way of calculating $d(\ell,P)$ with respect to an orthonormal frame of $\mathbb{E}^2$.

Now consider the 3-dimensional case. Let $\ell$ be a line in $\mathbb{E}^3$ and let $P$ be a point which does not lie on $\ell$. Suppose that $P$ has coordinates $(x_P, y_P, z_P)$ with respect to an orthonormal frame $\mathcal{K}$ of $\mathbb{E}^3$, and that $\ell$ is given by a point $A(x_A, y_A, z_A) \in \ell$ and a direction vector $\mathbf{v}(v_x, v_y, v_z)$ with respect to $\mathcal{K}$. There is a plane $\pi$ containing both $\ell$ and $P$. It is possible to calculate a frame $\mathcal{K}_\pi$ of $\pi$, extend it to a frame $\mathcal{K}'$ of $\mathbb{E}^3$, translate the given coordinates and components from $\mathcal{K}$ to $\mathcal{K}'$, and then do the calculations in $\pi$ (with respect to $\mathcal{K}_\pi$) where $\ell$ is a hyperplane.

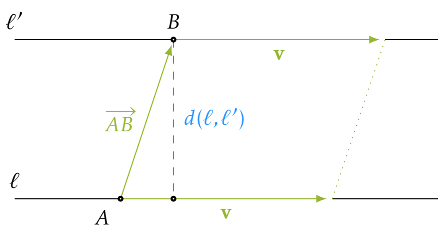
A much more efficient way to calculate $d(\ell,P)$ can be deduced using the cross product. Let $B$ be another point on $\ell$ such that $\overrightarrow{AB} = \mathbf{v}$ and consider the parallelogram spanned by $\overrightarrow{AB}$ and $\overrightarrow{AP}$. Then $d(\ell,P)$ is the height of the parallelogram corresponding to the side $[AB]$. Thus,

$$d(\ell,P) = \frac{|\overrightarrow{AP} \times \mathbf{v}|}{|\mathbf{v}|}.$$



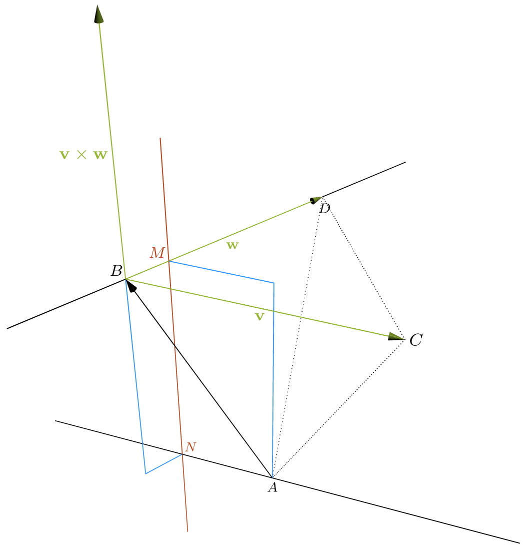

From this, we immediately deduce a formula for the distance between two parallel lines in $\mathbb{E}^3$. Let $\ell'$ be a line parallel to $\ell$ which passes through a point $B(x_B, y_B, z_B)$. We may apply Proposition 3.21 to conclude that $d(\ell,\ell') = d(\ell,Q)$ for any point $Q$ in $\ell'$. Thus,

$$d(\ell,\ell') = d(\ell,B) = \frac{|\overrightarrow{AB} \times \mathbf{v}|}{|\mathbf{v}|}.$$



### 4.2.3 Common perpendicular line of two skew lines

Consider two vectors $\mathbf{v}$ and $\mathbf{w}$. When proving Theorem 4.10, we reduced the claim to $(\mathbf{v}\times\mathbf{w})\times\mathbf{v}$. Let us give more insight into this expression by deducing equations for the common perpendicular line of two skew lines. Let $\ell$ and $\ell'$ be two skew lines in $\mathbb{E}^3$ passing respectively through $A$ and $B$. Let $\mathbf{v}$ be the direction vector for $\ell$ and let $\mathbf{w}$ be the direction vector for $\ell'$.




The *common perpendicular line* of $\ell$ and $\ell'$ is the unique line $d$ which intersects the two lines and is orthogonal to both of them, i.e., the line satisfying the following properties

- $d\perp\ell$ and $d\perp\ell'$, and
- $d\cap\ell\neq\emptyset$ and $d\cap\ell'\neq\emptyset$.

Since the two lines are skew relative to each other, the vectors $\mathbf{v}$ and $\mathbf{w}$ are linearly independent, hence the vector $\mathbf{v}\times\mathbf{w}$ is non-zero and perpendicular to both $\mathbf{v}$ and $\mathbf{w}$. Let $\pi$ be the plane passing through $A$ and parallel to $\mathbf{v}$ and $\mathbf{v}\times\mathbf{w}$. Notice that $\mathbf{n}=(\mathbf{v}\times\mathbf{w})\times\mathbf{v}$ is a normal vector for $\pi$ and that $\ell$ is included in $\pi$. Similarly, there is a plane $\pi'$ containing $\ell'$ and having normal vector $\mathbf{n}'=(\mathbf{v}\times\mathbf{w})\times\mathbf{w}$. The intersection of these two planes is a line $d$, and we claim that it is the common perpendicular of $\ell$ and $\ell'$. Since $d=\pi\cap\pi'$, a direction vector for $d$ is

$$\mathbf{n}\times\mathbf{n}'=\Big((\mathbf{v}\times\mathbf{w})\times\mathbf{v}\Big)\times\Big((\mathbf{v}\times\mathbf{w})\times\mathbf{w}\Big).$$

Denote $\mathbf{v}\times\mathbf{w}$ by $\mathbf{a}$ and notice that, by Theorem 4.10, we have

$$\mathbf{n}\times\mathbf{n}'=(\mathbf{a}\times\mathbf{v})\times(\mathbf{a}\times\mathbf{w})=\underbrace{\langle\mathbf{a},\mathbf{a}\times\mathbf{w}\rangle}_{=0}\mathbf{v}-\langle\mathbf{v},\mathbf{a}\times\mathbf{w}\rangle\mathbf{a}\quad(\text{proportional to }\mathbf{a}=\mathbf{v}\times\mathbf{w})$$

where $\langle\mathbf{a},\mathbf{a}\times\mathbf{w}\rangle=0$ since the two vectors are orthogonal. It follows that $\mathbf{a}$ is a direction vector for $d$. But $\mathbf{a}=\mathbf{v}\times\mathbf{w}$ is orthogonal to both $\mathbf{v}$ and $\mathbf{w}$, hence it is orthogonal to both $\ell$ and $\ell'$. Moreover, since $d$

and $\ell$ lie in the plane $\pi$ and have non-parallel direction vectors, they necessarily intersect. Similarly, we see that $d \cap \ell' \neq \emptyset$. This shows that $d$ is indeed the common perpendicular of $\ell$ and $\ell'$.

Writing the equations of $\pi$ and $\pi'$ in coordinates, we obtain

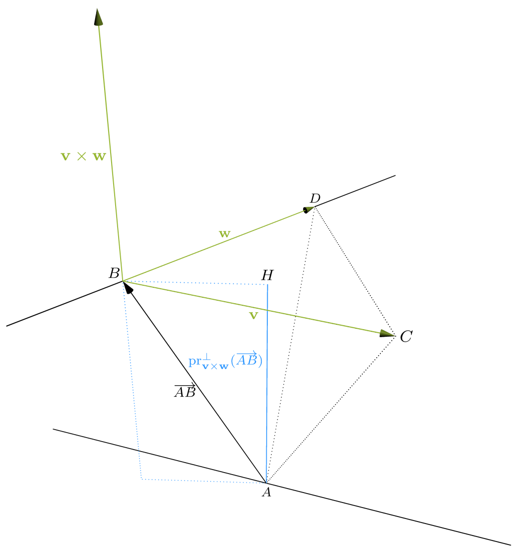
$$d = \pi \cap \pi' : \begin{cases} \pi : \begin{vmatrix} x - x_A & y - y_A & z - z_A \\ v_x & v_y & v_z \\ a_x & a_y & a_z \end{vmatrix} = 0 \\ \pi' : \begin{vmatrix} x - x_B & y - y_B & z - z_B \\ w_x & w_y & w_z \\ a_x & a_y & a_z \end{vmatrix} = 0 \end{cases}$$

where we used the coordinates of $A$ and $B$ with respect to an orthonormal frame $\mathcal{K} = (O, \mathcal{B})$ and the components of $\mathbf{v}$, $\mathbf{w}$ and $\mathbf{a}$ with respect to $\mathcal{B}$.

### 4.2.4 Distance between two skew lines

As in the previous section, let $\ell$ and $\ell'$ be two skew lines in $\mathbb{E}^3$ passing respectively through $A(x_A, y_A, z_A)$ and $B(x_B, y_B, z_B)$. Let $\mathbf{v}(v_x, v_y, v_z)$ be the direction vector for $\ell$ and let $\mathbf{w}(w_x, w_y, w_z)$ be the direction vector for $\ell'$. We assume that the coordinates of the points and the components of the vectors are known.

Let $MN$ be the common perpendicular line of $\ell$ and $\ell'$ with $M \in \ell$ and $N \in \ell'$. Consider the parallelepiped spanned by $\overrightarrow{MA}, \overrightarrow{NM}, \overrightarrow{NB}$; it is easy to show that $d(A, B) \geq d(M, N) = |MN|$. Hence, $d(\ell, \ell') = |MN|$. Moreover, we could calculate this distance by writing down equations for $MN$ (as in Section 4.2.3), intersecting with $\ell$ and $\ell'$ to determine the coordinates of $M$ and $N$, and then calculating $|MN|$. However, there is a shorter way of obtaining this distance.




Since the lines are skew, there is a unique plane $\pi$ containing $\ell'$ which is parallel to $\ell$. Let $P$ be a point in $\pi$. Considering the parallelepiped with vertices $M$, $N$ and $P$ we see that $d(\ell, \pi) = |MN|$.

Hence, $d(\ell,\ell') = d(\ell,\pi)$. Now, dropping a perpendicular from $A$ to $\pi$ which intersects $\pi$ at $H$, we see that $d(\ell,\pi) = |AH|$. Therefore,

$$d(\ell,\ell') = d(\ell,\pi) = |AH| = |\mathrm{Pr}^{\perp}_{\mathbf{v}\times\mathbf{w}}(\overrightarrow{AB})| = |\frac{\langle \mathbf{v}\times\mathbf{w}, \overrightarrow{AB}\rangle}{|\mathbf{v}\times\mathbf{w}|}|$$

Hence, if we let $\mathbf{a}(a_x,a_y,a_z) = \mathbf{v}\times\mathbf{w}$, we obtain

$$d(\ell,\ell') = |\frac{\langle \mathbf{v}\times\mathbf{w}, \overrightarrow{AB}\rangle}{|\mathbf{v}\times\mathbf{w}|}| = \frac{\left|a_x(x_B-x_A)+a_y(y_B-y_A)+a_z(z_B-z_A)\right|}{\sqrt{a_x^2+a_y^2+a_z^2}}.$$

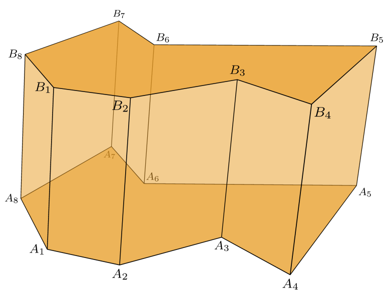
A further interpretation of this formula is the following. Let $P$ be a parallelepiped spanned by the three vectors $\mathbf{v},\mathbf{w},\overrightarrow{AB}$ and denote by $F$ a face spanned by $\mathbf{v}$ and $\mathbf{w}$. The distance between the two lines is the height of $P$ corresponding to the face $F$. Moreover, anticipating the discussion in Section 4.3, we have

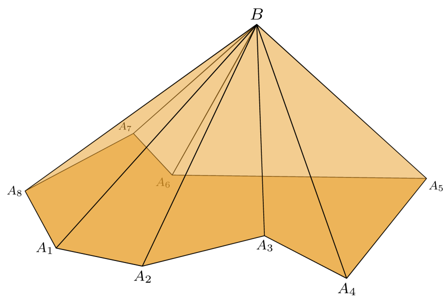
$$d(\ell,\ell') = \frac{\mathrm{Vol}(P)}{\mathrm{Area}(F)} = |\frac{[\mathbf{v},\mathbf{w},\overrightarrow{AB}]}{|\mathbf{v}\times\mathbf{w}|}|.$$

## 4.3 Volume

### 4.3.1 Volume of polyhedra

**Definition 4.11.** 'A polyhedron may be defined as a finite, connected set of plane polygons, such that every side of each polygon belongs also to just one other polygon, with the proviso that the polygons surrounding each vertex form a single circuit' [7]. Well-known examples are *prisms* having two congruent parallel faces, in particular triangular prisms and parallelepipeds, or *pyramids* having just one point, the *apex*, outside the plane of the polygonal *base*. The *height of a prism* is the distance between the two main faces, and the *height of a pyramid* is the distance from the apex to the base.




Triangulation of a polyhedron, or *tetrahedralization*, refers to the subdivision of a polyhedron into tetrahedral meshes. Similar to polygons in dimension 2, it is always possible to break a polyhedron into tetrahedra. Unlike the two-dimensional case, it is not always possible to do this without adding vertices. The smallest example of a polyhedron where this is not possible is the Schönhardt polyhedron. For a discussion of the various challenges and algorithms, see [21, Chapter 25].

For our purposes, the following non-standard definition suffices. The interior of a tetrahedron is a polyhedral set. If $ABCD$ and $ABCD'$ are two tetrahedra with disjoint interiors, then the union of the interiors of the two tetrahedra together with the interior of the triangle $ABC$ is a polyhedral set obtained by *gluing the two tetrahedra*. Any polyhedral set is obtained after a finite number of gluings of tetrahedra.

**Definition 4.12.** Let $\mathcal{P}$ denote the set of all polyhedral sets in $\mathbb{E}^3$. We define a *volume* function $\mathrm{Vol} : \mathcal{P} \to \mathbb{R}_{\geq 0}$ through the following properties:

(V1) If $S$ is a cuboid of side lengths 1, 1, and $x$, then $\mathrm{Vol}(S) = x$.

(V2) If $S_1$ and $S_2$ are similar polyhedral sets with ratio $x = S_1 : S_2$, then $\mathrm{Vol}(S_1) = x^3 \cdot \mathrm{Vol}(S_2)$.

(V3) If $S_1$ and $S_2$ are disjoint polyhedral sets, then $\mathrm{Vol}(S_1 \cup S_2) = \mathrm{Vol}(S_1) + \mathrm{Vol}(S_2)$.

**Proposition 4.13.** We have

1. The volume of a prism of height $h$ and base area $a$ is $ah$. In particular, the volume of a parallelepiped with a face of area $a$ and corresponding height $h$ is $ah$.

2. The volume of a pyramid of height $h$ and base area $a$ is $ah/3$. In particular, the volume of a tetrahedron $ABCD$ with $\mathrm{Area}(BCD) = a$ and $d(A, BCD) = h$ is $ah/3$.

3. The volume of a rectangular parallelepiped of side lengths $a$, $b$, and $c$ is $abc$.

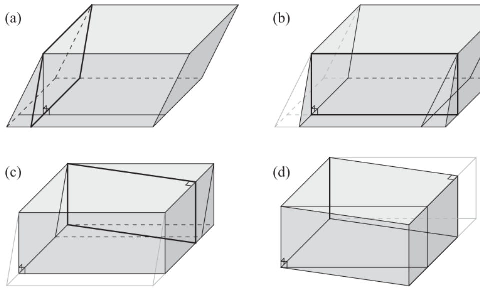
*Proof.* It is clear that 3 follows from 1. However, the proof needs to start from the definitions. We do this by first proving 3. Denote by $R_{a,b,c}$ a rectangular parallelepiped with side lengths $a$, $b$, and $c$. By (V1) and (V2), we have

$$\mathrm{Vol}(R_{a,a,1}) = a^3 \cdot \mathrm{Vol}(R_{1,1,\frac{1}{a}}) = a^3 \cdot \frac{1}{a} = a^2$$

for any $a > 0$. Now, as in the proof of Proposition 4.3, consider $R_{a,a,1}$ and $R_{b,b,1}$ in opposite corners of $R_{a+b,a+b,1}$. By (V3), we have $\mathrm{Vol}(R_{a+b,a+b,1}) = \mathrm{Vol}(R_{a,a,1}) + \mathrm{Vol}(R_{b,b,1}) + 2 \cdot \mathrm{Vol}(R_{a,b,1})$ hence $(a+b)^2 = a^2 + b^2 + 2 \cdot \mathrm{Vol}(R_{a,b,1})$ and therefore $\mathrm{Vol}(R_{a,b,1}) = ab$ for any $a, b > 0$. Then, for an arbitrary rectangular parallelepiped we have

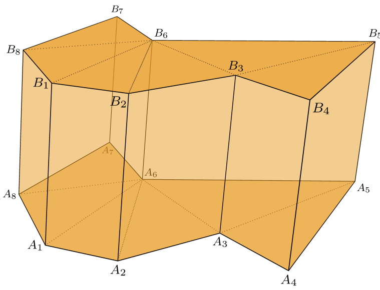
$$\mathrm{Vol}(R_{a,b,c}) = c^3 \cdot \mathrm{Vol}(R_{\frac{a}{c},\frac{b}{c},1}) = c^3 \cdot \frac{ab}{c^2} = abc.$$

Next, we use 3 to deduce the volume of an arbitrary parallelepiped. For this, we section and rearrange the parallelepiped as in the figure (this is Figure 4.1.2 from [1]) below.

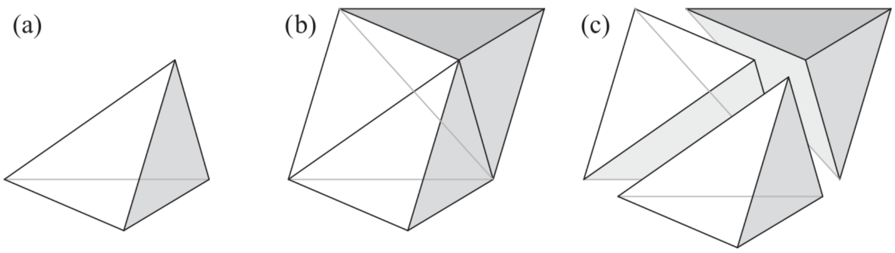



In (a), (b), and (c), the parallelepiped is transformed into a prism whose base is a parallelogram by cutting and pasting congruent triangular prisms. In particular, the height $h$ and the base area $a$ are unchanged. In the last step (d), the base is rearranged into a rectangle without changing its area —as in the proof of Proposition 4.3. The final result is a rectangular parallelepiped $R_{x,y,h}$ with volume $ah$ equal to the volume of the initial parallelepiped.

Now, slicing a parallelepiped along the diagonals of two opposite faces, we obtain a triangular prism with volume $ha$ where $a$ is half the area of the face which was cut. Thus, a triangular prism has the claimed volume. For an arbitrary prism, use a triangulation of the base polygon to deduce the claim.




For 2, we first consider tetrahedra and follow the argument in [1, §4.1]. Slice a rectangular prism as in the following figure[1]. The proof of Proposition 5 of Book XII of Euclid's *Elements* (see for example [10]) is correct for our settings and assumptions. It implies that two triangular pyramids (tetrahedra) with the same height and with congruent bases have the same volume [10, Volume 3, p.390].



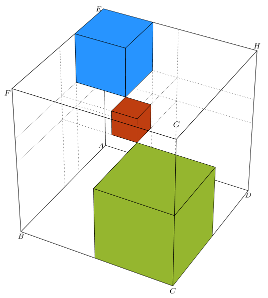

[1] Figure 4.1.4 from [1]

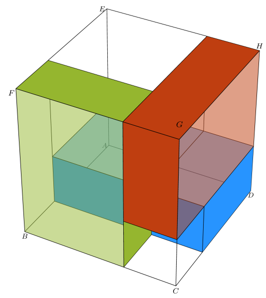
In the above slicing, the two tetrahedra with white bases have the same heights and the bottom and top tetrahedra are clearly congruent. Thus, if $T$ is any of the three tetrahedra of the triangular prism $P$, then

$$\mathrm{Vol}(T) = \frac{1}{3}\mathrm{Vol}(P) = \frac{ah}{3}$$

where $a$ is the area of a base triangle of $P$ and $h$ is the height of $P$. Then, for an arbitrary pyramid, the claim follows by considering a tetrahedralization. □

**Remark.** Similar to our definition of area, the above definition of volume tries to capture basic principles of the intuitive notion of volume. It is an oversimplified version of the so-called Lebesgue measure (see Section 4.3.3). In property (V1) we may replace the rectangular parallelepiped with a cube of side length 1. However, after weakening the assumption, it is not possible to cover an arbitrary rectangular parallelepiped allowing only the uniform scaling given in (V2). To see this difficulty, keep the notation in the proof of Proposition 4.13 and place three cubes $C_a$, $C_b$, and $C_c$ on the diagonal of a cube $C_{a+b+c}$.





The planes of the small cubes divide $C_{a+b+c}$ into cuboids, the sides of which can be checked to correspond to the algebraic formula for the cube of $a+b+c$. We have

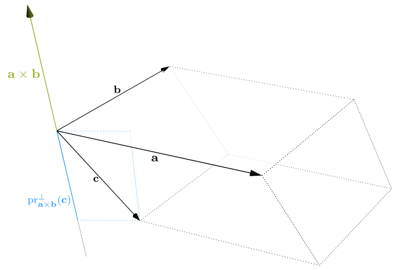
$$\mathrm{Vol}(C_{a+b+c}) = (a+b+c)^3 = a^3+b^3+c^3+3a^2b+3a^2c+3ab^2+3b^2c+3ac^2+3bc^2+6abc. \tag{4.3}$$

Assuming that the volume $\mathrm{Vol}(C_{x,y,y})$ of a cuboid with two equal adjacent sides is $xy^2$, it follows from (V2), (V3) and (4.3) that $\mathrm{Vol}(R_{a,b,c}) = abc$. Thus, it is necessary and sufficient to show that $\mathrm{Vol}(C_{x,y,y}) = xy^2$. For this, place two cubes $C_a$ and $C_b$ on the diagonal of $C_{a+b}$ and notice that the subdivision of the big cube corresponds to

$$\mathrm{Vol}(C_{a+b}) = (a+b)^3 = a^3+b^3+3ab(a+b).$$

From this equation and (V2), (V3) it follows that $\mathrm{Vol}(R_{a,b,(a+b)}) = ab(a+b)$ for any $a,b>0$. However, this does not yield a decomposition of a cube into smaller cubes and rectangles congruent to $R_{x,y,y}$ for arbitrary $x$ and $y$. Compare this to the two dimensional case.

Orthonormal frames offer an efficient way of calculating the volume of a parallelogram, and therefore volumes of tetrahedra and polyhedra provided that the coordinates of vertices are known.




**Proposition 4.14.** The volume of a parallelepiped $P$ spanned by the vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ is

$$\mathrm{Vol}(P) = |\langle \mathbf{a}\times\mathbf{b}, \mathbf{c}\rangle| = |\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}|$$

where the components of $\mathbf{a}(a_1, a_2, a_3)$, $\mathbf{b}(b_1, b_2, b_3)$, and $\mathbf{c}(c_1, c_2, c_3)$ are with respect to an orthonormal basis.

*Proof.* The area of the face spanned by $\mathbf{a}$ and $\mathbf{b}$ is $|\mathbf{a}\times\mathbf{b}|$ and the height corresponding to this face is the length of the projection of $\mathbf{c}$ onto $\mathbf{a}\times\mathbf{b}$. Thus

$$\mathrm{Vol}(P) = |\mathbf{a}\times\mathbf{b}|\cdot|\mathrm{pr}^{\perp}_{\mathbf{a}\times\mathbf{b}}(\mathbf{c})| = |\mathbf{a}\times\mathbf{b}|\cdot|\frac{\langle \mathbf{a}\times\mathbf{b}, \mathbf{c}\rangle}{\langle \mathbf{a}\times\mathbf{b}, \mathbf{a}\times\mathbf{b}\rangle}\mathbf{a}\times\mathbf{b}| = |\langle \mathbf{a}\times\mathbf{b}, \mathbf{c}\rangle|.$$

Moreover, with respect to an orthonormal basis, we have

$$\langle \mathbf{a}\times\mathbf{b}, \mathbf{c}\rangle = \langle \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}, c_1\mathbf{i} + c_2\mathbf{j} + c_3\mathbf{k}\rangle = \begin{vmatrix} c_1 & c_2 & c_3 \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}.$$

□

### 4.3.2 Oriented volume

**Definition 4.15.** The above proposition shows that, up to sign, the volume of a parallelepiped spanned by $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ is the determinant of the base change matrix $\mathrm{M}_{\mathcal{B},\mathcal{B}'}$ where $\mathcal{B}$ is a right-oriented basis and $\mathcal{B}' = (\mathbf{a}, \mathbf{b}, \mathbf{c})$. The value $\det(\mathrm{M}_{\mathcal{B},\mathcal{B}'})$ does not depend on the orthonormal basis $\mathcal{B}$ in which the vectors are expressed. The proof of this claim is the same in any dimension (see Proposition 4.21). We call this determinant the *box product of* $\mathbf{a}$, $\mathbf{b}$, *and* $\mathbf{c}$ and denote it by $[\mathbf{a}, \mathbf{b}, \mathbf{c}]$. We define the *oriented volume of a parallelepiped $P$ spanned by the vectors* $\mathbf{a}$, $\mathbf{b}$, *and* $\mathbf{c}$ to be

$$\mathrm{Vol}_{\mathrm{or}}(P) = [\mathbf{a}, \mathbf{b}, \mathbf{c}].$$

Similarly, the oriented volume of the tetrahedron $ABCD$ is $\mathrm{Vol}_{\mathrm{or}}(ABCD) = [\overrightarrow{AB}, \overrightarrow{AC}, \overrightarrow{AD}]/6$. Proposition 4.14 shows that mixing the cross product with the scalar product gives the box product. For this reason $\langle \mathbf{a}\times\mathbf{b}, \mathbf{c}\rangle = [\mathbf{a}, \mathbf{b}, \mathbf{c}]$ is sometimes called the *mixed product* of the vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$.

**Proposition 4.16.** The box product defines a map $[\_,\_,\_]:\mathbb{V}^3\times\mathbb{V}^3\times\mathbb{V}^3\to\mathbb{R}\quad(\mathbf{a},\mathbf{b},\mathbf{c})\mapsto[\mathbf{a},\mathbf{b},\mathbf{c}]$ which is linear in each argument. Moreover, for three vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$, we have

$$[\mathbf{a},\mathbf{b},\mathbf{c}]=[\mathbf{b},\mathbf{c},\mathbf{a}]=[\mathbf{c},\mathbf{a},\mathbf{b}]=-[\mathbf{b},\mathbf{a},\mathbf{c}]=-[\mathbf{a},\mathbf{c},\mathbf{b}]=-[\mathbf{c},\mathbf{b},\mathbf{a}].$$

The coplanarity (i.e., linear dependency) condition for $\mathbf{a},\mathbf{b},\mathbf{c}$ is

$$[\mathbf{a},\mathbf{b},\mathbf{c}]=0.$$

The sign of the box product determines the orientation of the basis $(\mathbf{a},\mathbf{b},\mathbf{c})$

$$[\mathbf{a},\mathbf{b},\mathbf{c}]\quad\begin{cases}>0 & \text{then } (\mathbf{a},\mathbf{b},\mathbf{c}) \text{ is right oriented}\\ =0 & \text{then } (\mathbf{a},\mathbf{b},\mathbf{c}) \text{ is not a basis}\\ <0 & \text{then } (\mathbf{a},\mathbf{b},\mathbf{c}) \text{ is left oriented}\end{cases}.$$

*Proof.* In dimension 3, linearity of the box product follows from the fact that it is the composition of linear maps: the cross product and the scalar product. You can also deduce the linearity of this map from the fact that with respect to an orthonormal basis, it is the determinant of a base change matrix. The other properties follow from this latter description and from known properties of the determinant. □

**Proposition 4.17** (Triple cross product formula)**.** For vectors $\mathbf{a},\mathbf{b},\mathbf{c},\mathbf{d}\in\mathbb{V}^3$ we have

$$(\mathbf{a}\times\mathbf{b})\times(\mathbf{c}\times\mathbf{d})=\mathbf{b}\cdot[\mathbf{a},\mathbf{c},\mathbf{d}]-\mathbf{a}\cdot[\mathbf{b},\mathbf{c},\mathbf{d}]=\mathbf{c}\cdot[\mathbf{a},\mathbf{b},\mathbf{d}]-\mathbf{d}\cdot[\mathbf{a},\mathbf{b},\mathbf{c}]$$

### 4.3.3 Hypervolume

The higher-dimensional analogues of polygons and polyhedra are *polytopes*. The simplest polytope in dimension $n$ is an $n$-simplex. If $n=2$, these are triangles, and if $n=3$, these are tetrahedra. In general, in dimension $n$, consider $n+1$ points $P_0,P_1,\dots,P_n$ which do not lie in a hyperplane. This is equivalent to asking for the vectors $\overrightarrow{P_0P_1},\dots,\overrightarrow{P_0P_n}$ to be linearly independent, i.e., to form a basis of $\mathbb{V}^n$. The convex hull of $P_0,P_1,\dots,P_n$ is an $n$-simplex. The $(n-1)$-faces of an $n$-simplex are $(n-1)$-simplices; in particular, the faces of a 4-simplex are tetrahedra. We may define polytopes as sets obtained by gluing $n$-simplices along their faces.

A hyperparallelepiped spanned by $\mathbf{v}_1,\dots,\mathbf{v}_n$ is the set of all points obtained from a given point $P$ with linear combinations of the given vectors, considering only coefficients in the interval $[0,1]$, concretely

$$\mathcal{H}=P+\{\alpha_1\mathbf{v}_1+\cdots+\alpha_n\mathbf{v}_n:\alpha_1,\dots,\alpha_n\in[0,1]\}.$$

A *hypercube* is a hyperparallelepiped with all 1-faces of equal length and all angles right angles.

**Definition 4.18.** Let $\mathcal{P}$ denote the set of all polytopes in $\mathbb{E}^3$. We define a *rational hypervolume* function $\mathrm{Vol}:\mathcal{P}\to\mathbb{R}_{\geq 0}$ through the following properties:

(V1) If $S$ is a hypercube of side length 1, then $\mathrm{Vol}(S)=1$.

(V2) If $S_1$ and $S_2$ are congruent polytopes, then $\mathrm{Vol}(S_1)=\mathrm{Vol}(S_2)$.

(V3) If $S_1$ and $S_2$ are disjoint polytopes, then $\mathrm{Vol}(S_1 \cup S_2) = \mathrm{Vol}(S_1) + \mathrm{Vol}(S_2)$.

This volume function is called rational because it does not allow us to measure all polytopes. For instance, it is not possible to use it to deduce the volume of a cube with side lengths being arbitrary irrational numbers.

**Definition 4.19.** We say that a set $S$ is *measurable* if it contains a sequence of strictly growing internal polytopes $I_i$ and if it is contained in a sequence of strictly shrinking exterior polytopes $E_i$ such that the volumes of these sets converge to a common value. Formally, the following properties need to hold

$$I_i \subseteq I_j \subseteq S \subseteq E_j \subseteq E_i \quad \forall i < j \quad \text{and} \quad \lim_{i\to\infty} \mathrm{Area}(I_i) = \lim_{i\to\infty} \mathrm{Area}(E_i)$$

If the limit exists, it is unique, and we denote it by $\mathrm{Vol}(S)$. Denoting by $\mathcal{M}$ the set of all measurable sets of $\mathbb{E}^n$, we extend the (hyper)volume function to $\mathrm{Vol} : \mathcal{M} \to \mathbb{R}_{\geq 0}$. This is a version of the Lebesgue measure (see, for example, [19, Chapter 11]).

From this definition, one may deduce that a cuboid with side lengths $a_1, \dots, a_n$ has volume $\prod a_i$. Or that if $P$ is a hyperparallelepiped in dimension $n$, and if $S$ is any $n$-simplex constructed from vertices of $P$, then

$$\mathrm{Vol}(S) = \frac{1}{n!} \mathrm{Vol}(P)$$

which is the higher-dimensional analogue of the volume formula for a tetrahedron.

As in the case of dimensions 2 and 3, the efficient way of calculating hypervolume is given by the box product, which allows us to calculate the hypervolume of hyperparallelepipeds, and therefore of $n$-simplices and of polytopes in general.

**Definition 4.20.** Let $\mathbf{v}_1, \dots, \mathbf{v}_n$ be vectors in $\mathbb{V}^n$ with components $\mathbf{v}_i(v_{i,1}, \dots, v_{i,n})$ relative to a right-oriented orthonormal basis $\mathcal{B}$ of $\mathbb{V}^n$. The *(n-fold) box product* of these vectors is

$$[\mathbf{v}_1, \dots, \mathbf{v}_n] = \begin{vmatrix} v_{1,1} & v_{2,1} & \dots & v_{n,1} \\ v_{1,2} & v_{2,2} & \dots & v_{n,2} \\ \vdots & \vdots & & \vdots \\ v_{1,n} & v_{2,n} & \dots & v_{n,n} \end{vmatrix}.$$

**Proposition 4.21.** The definition of the box product does not depend on the choice of the right-oriented orthonormal basis $\mathcal{B}$.

*Proof.* Let $\mathcal{V} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ be an $n$-tuple of $n$-dimensional vectors and let $\mathrm{M}_{\mathcal{B},\mathcal{V}}$ be the matrix whose columns are the components of the $\mathbf{v}_i$s. Notice that

$$[\mathbf{v}_1, \dots, \mathbf{v}_n] = \det(\mathrm{M}_{\mathcal{B},\mathcal{V}})$$

If the vectors are linearly dependent, then $\det(\mathrm{M}_{\mathcal{B},\mathcal{V}}) = 0$ for any basis $\mathcal{B}$. If the vectors are linearly independent, then $\mathcal{V}$ is a basis, and if $\mathcal{B}'$ is another right-oriented orthonormal basis, then

$$\det(\mathrm{M}_{\mathcal{B}',\mathcal{V}}) = \det(\mathrm{M}_{\mathcal{B}',\mathcal{B}} \mathrm{M}_{\mathcal{B},\mathcal{V}}) = \det(\mathrm{M}_{\mathcal{B}',\mathcal{B}}) \det(\mathrm{M}_{\mathcal{B},\mathcal{V}})$$

and it suffices to show that $\det(M_{\mathcal{B}',\mathcal{B}}) = 1$. Let $\mathbf{e}_i$ denote the vectors in $\mathcal{B}$. Then, since $\mathcal{B}'$ is orthonormal, we have

$$M^T_{\mathcal{B}',\mathcal{B}} M_{\mathcal{B}',\mathcal{B}} = \begin{bmatrix} \langle \mathbf{e}_1, \mathbf{e}_1 \rangle & \langle \mathbf{e}_1, \mathbf{e}_2 \rangle & \dots & \langle \mathbf{e}_1, \mathbf{e}_n \rangle \\ \langle \mathbf{e}_2, \mathbf{e}_1 \rangle & \langle \mathbf{e}_2, \mathbf{e}_2 \rangle & \dots & \langle \mathbf{e}_2, \mathbf{e}_n \rangle \\ \vdots & \vdots & \dots & \vdots \\ \langle \mathbf{e}_n, \mathbf{e}_1 \rangle & \langle \mathbf{e}_n, \mathbf{e}_2 \rangle & \dots & \langle \mathbf{e}_n, \mathbf{e}_n \rangle \end{bmatrix} = I_n$$

hence

$$1 = \det(I_n) = \det(M^T_{\mathcal{B}',\mathcal{B}} M_{\mathcal{B}',\mathcal{B}}) = \det(M^T_{\mathcal{B}',\mathcal{B}}) \det(M_{\mathcal{B}',\mathcal{B}}) = \det(M_{\mathcal{B}',\mathcal{B}})^2.$$

It follows that $\det(M_{\mathcal{B}',\mathcal{B}}) = \pm 1$, and since the two bases are both right-oriented, we deduce that $\det(M_{\mathcal{B}',\mathcal{B}}) = 1$, and the claim follows. □

**Definition 4.22.** Let $\mathcal{B}$ be a basis of $\mathbb{V}^n$. The *oriented volume* of the basis $\mathcal{B}$, denoted by $\mathrm{Vol}_{\mathrm{or}}(\mathcal{B})$, is the value of the box product of the vectors in $\mathcal{B}$. The *volume* of the basis $\mathcal{B}$ is the absolute value $|\mathrm{Vol}_{\mathrm{or}}(\mathcal{B})|$, and we denote it by $\mathrm{Vol}(\mathcal{B})$. If we are in dimension 2, we refer to these values as the *area* and *oriented area* of $\mathcal{B}$, and denote them by $\mathrm{Area}(\mathcal{B})$ and $\mathrm{Area}_{\mathrm{or}}(\mathcal{B})$, respectively.

One can also generalize the **J**-operator and the cross product to higher dimensions. For $n-1$ vectors $\mathbf{v}_1, \dots, \mathbf{v}_{n-1}$, the *wedge product* of these vectors, denoted by $\mathbf{v}_1 \wedge \dots \wedge \mathbf{v}_{n-1}$, is the unique vector $\mathbf{w}$ such that $|\mathbf{w}|$ is the hypervolume of the hyperparallelepiped spanned by $\mathbf{v}_1, \dots, \mathbf{v}_{n-1}$, the vector $\mathbf{w}$ is orthogonal to each $\mathbf{v}_i$, and $(\mathbf{v}_1, \dots, \mathbf{v}_{n-1}, \mathbf{w})$ is right-oriented.

CHAPTER 5

# Affine maps

## Contents



## 5.1 Properties of affine maps

**Definition 5.1.** A map $\phi : \mathbb{A}^n \to \mathbb{A}^m$ is an affine map if and only if there exist $A \in \mathrm{Mat}_{m\times n}(\mathbb{R})$ and $b \in \mathrm{Mat}_{m\times 1}(\mathbb{R})$ such that

$$[\phi(P)]_{\mathcal{K}'} = A\cdot[P]_{\mathcal{K}} + b \tag{5.1}$$

relative to some frame $\mathcal{K} = (O,\mathcal{B})$ of $\mathbb{A}^n$ and some frame $\mathcal{K}' = (O',\mathcal{B}')$ of $\mathbb{A}^m$. Such a map defines a linear map $\mathrm{lin}(\phi) : \mathrm{D}(\mathbb{A}^n) \to \mathrm{D}(\mathbb{A}^m)$ given by

$$[\mathrm{lin}(\phi)(\overrightarrow{PQ})]_{\mathcal{B}'} = A\cdot[\overrightarrow{PQ}]_{\mathcal{B}}. \tag{5.2}$$

This is called *the linear map associated to* $\phi$, or *the linear part of* $\phi$. To see that (5.2) indeed defines a map, one just needs to unpack the definition:

$$[\mathrm{lin}(\phi)(\overrightarrow{PQ})]_{\mathcal{B}'} = [\overrightarrow{\phi(P)\phi(Q)}]_{\mathcal{B}'} = (A\cdot[Q]_{\mathcal{K}}+b)-(A\cdot[P]_{\mathcal{K}}+b) = A\cdot([Q]_{\mathcal{K}}-[P]_{\mathcal{K}}) = A\cdot[\overrightarrow{PQ}]_{\mathcal{B}}.$$

We notice that both $A$ and $b$ depend on the choice of the frames $\mathcal{K}$ and $\mathcal{K}'$ and moreover $A$ is the matrix of $\operatorname{lin}(\phi)$ relative to the bases in $\mathcal{K}$ and $\mathcal{K}'$.

If $n = m$, then $\phi : \mathbb{A}^n \to \mathbb{A}^n$ is called an *affine endomorphism*. The set of all such endomorphisms is denoted by $\operatorname{End}_{\text{aff}}(\mathbb{A}^n)$. It is easy to see that the affine map $\phi$ is invertible if and only if the map $\operatorname{lin}(\phi)$ is invertible; equivalently, $\phi$ is invertible if and only if the matrix $A$ of the linear map $\operatorname{lin}(\phi)$ is invertible (see proof of Proposition 5.5). An invertible affine endomorphism $\phi$ is called an *affine automorphism* or *affine transformation*. The set of all affine transformations of $\mathbb{A}^n$ is denoted by $\operatorname{AGL}(\mathbb{A}^n)$.

Moreover, if in addition to $n = m$, $O = O'$ and $b = 0$ in (5.1) then $\phi$ can be viewed as a linear map from $\mathbb{V}^n$ to $\mathbb{V}^n$ since it is given by multiplication with the matrix $A$. The set of invertible linear maps $\mathbb{V}^n \to \mathbb{V}^n$ is denoted by $\operatorname{GL}(\mathbb{V}^n)$ and we have the following inclusion:

$$\operatorname{GL}(\mathbb{V}^n) \subseteq \operatorname{AGL}(\mathbb{A}^n).$$

**Example 5.2.** A homothety $\phi_{C,\lambda}$ of $\mathbb{E}^n$ with center $C$ is the map which rescales the space with a factor $\lambda$ along lines passing through the point $C$. With respect to a coordinate system $\mathcal{K} = (C, \mathcal{B})$ with origin in $C$ it has the form

$$[\phi_{C,\lambda}(P)]_{\mathcal{K}} = \lambda \cdot [P]_{\mathcal{K}}.$$

Notice that if $\lambda = 1$ this is the identity map, if $\lambda < 1$ this is a *contraction*, if $\lambda > 1$ it is an *expansion*.

**Example 5.3.** The various parallel projections and reflections described in the next sections are examples of affine maps, as are the isometries discussed in Chapter 6.

**Proposition 5.4.** Let $\phi : \mathbb{E}^n \to \mathbb{E}^m$ be an affine map. If a line $\ell$ is mapped onto a line $\ell'$ under $\phi$, then $\phi$ preserves the oriented ratio on $\ell$, i.e. if $A, B, C$ are points on $\ell$, then

$$\frac{\overrightarrow{AC}}{\overrightarrow{AB}} = \frac{\overrightarrow{\phi(A)\phi(C)}}{\overrightarrow{\phi(A)\phi(B)}}. \tag{5.3}$$

In particular, the ratios on $\ell$ are preserved.

*Proof.* The oriented ratio $\frac{\overrightarrow{AC}}{\overrightarrow{AB}}$ denotes a scalar $\lambda$ such that $\overrightarrow{AC} = \lambda \overrightarrow{AB}$. Since $\overrightarrow{\phi(A)\phi(C)} = \operatorname{lin}(\phi)(\overrightarrow{AC})$ and since $\operatorname{lin}(\phi)$ is a linear map we have

$$\overrightarrow{\phi(A)\phi(C)} = \operatorname{lin}(\phi)(\overrightarrow{AC}) = \operatorname{lin}(\phi)(\lambda \overrightarrow{AB}) = \lambda \operatorname{lin}(\phi)(\overrightarrow{AB}) = \lambda \overrightarrow{\phi(A)\phi(B)}$$

and the claim follows. □

**Proposition 5.5** (Characterization of affine transformations)**.** A map $\phi : \mathbb{E}^n \to \mathbb{E}^n$ is an affine transformation if and only if

1. $\phi$ is injective;
2. $\phi$ preserves lines;
3. $\phi$ preserves the oriented ratio on lines.

### 5.1.1 Homogeneous coordinates and homogeneous matrix

Conceptually, homogeneous coordinates are used to describe the affine space $\mathbb{A}^n$ inside the projective space $\mathbb{P}^n$. This is outside the scope of these notes. However, there is also a computational advantage to homogeneous coordinates, which is what we describe here.

**Definition 5.6.** Let $\mathcal{K}$ be a reference frame of $\mathbb{A}^n$. The *homogeneous coordinates* of the point $P(p_1,\dots,p_n)$ are $(p_1,\dots,p_n,1)$. Yes, they are the ordinary coordinates with an extra 1 at the end.

**Definition 5.7.** The *homogeneous matrix* of an affine map $\phi:\mathbb{A}^n\to\mathbb{A}^m$, defined with respect to some reference frames $\mathcal{K}$ and $\mathcal{K}'$ by $\phi(\mathbf{x})=A\mathbf{x}+b$, is

$$\hat{\mathrm{M}}_{\mathcal{K}',\mathcal{K}}(\phi)=\begin{bmatrix}A & b\\ 0 & 1\end{bmatrix}.$$

The utility of introducing these notions is the following: the composition of affine maps corresponds to the multiplication of homogeneous matrices. Indeed, let $\psi(\mathbf{x})=A'\mathbf{x}+b'$ be another affine map defined on $\mathbb{A}^m$. Then

$$\psi\circ\phi(\mathbf{x})=\psi(\phi(\mathbf{x}))=\psi(A\mathbf{x}+b)=A'(A\mathbf{x}+b)+b'=(A'\cdot A)\mathbf{x}+(A'\cdot b+b').$$

In terms of homogeneous matrices, we notice that

$$\hat{\mathrm{M}}_{\mathcal{K}'',\mathcal{K}}(\psi\circ\phi)=\begin{bmatrix}A'\cdot A & A'\cdot b+b'\\ 0 & 1\end{bmatrix}=\begin{bmatrix}A' & b'\\ 0 & 1\end{bmatrix}\cdot\begin{bmatrix}A & b\\ 0 & 1\end{bmatrix}=\hat{\mathrm{M}}_{\mathcal{K}'',\mathcal{K}'}(\psi)\cdot\hat{\mathrm{M}}_{\mathcal{K}',\mathcal{K}}(\phi)$$

and the homogeneous coordinates of the values of the map $\phi$ can be obtained through a matrix multiplication as well

$$\begin{bmatrix}\phi(\mathbf{x})\\ 1\end{bmatrix}=\begin{bmatrix}A & b\\ 0 & 1\end{bmatrix}\cdot\begin{bmatrix}\mathbf{x}\\ 1\end{bmatrix}.$$

Thus

$$\begin{bmatrix}\psi\circ\phi(\mathbf{x})\\ 1\end{bmatrix}=\begin{bmatrix}A' & b'\\ 0 & 1\end{bmatrix}\cdot\begin{bmatrix}A & b\\ 0 & 1\end{bmatrix}\cdot\begin{bmatrix}\mathbf{x}\\ 1\end{bmatrix}.$$

## 5.2 Projections and reflections

### 5.2.1 Tensor product

Tensor products are widely used throughout mathematics. Here we look at the particular case of tensor products of vectors in $\mathbb{R}^n$ relative to a basis $\mathcal{B}$. In this case, the tensor product of two vectors is a matrix, also called the *outer product* of the two vectors. We use this to give compact expressions for the projections and reflections discussed in the following subsections.

**Definition 5.8.** Let $\mathbf{v}(v_1,\dots,v_n)$ and $\mathbf{w}(w_1,\dots,w_n)$ be two vectors with components relative to a basis $\mathcal{B}$. The *tensor product* $\mathbf{v}\otimes_{\mathcal{B}}\mathbf{w}$ is the $n\times n$ matrix defined by $(\mathbf{v}\otimes_{\mathcal{B}}\mathbf{w})_{ij}=v_iw_j$. In other words

$$\mathbf{v}\otimes_{\mathcal{B}}\mathbf{w}=\mathbf{v}\cdot\mathbf{w}^T=\begin{bmatrix}v_1\\ \vdots\\ v_n\end{bmatrix}\cdot\begin{bmatrix}w_1\dots w_n\end{bmatrix}=\begin{bmatrix}v_1w_1 & \dots & v_1w_n\\ \vdots & & \vdots\\ v_nw_1 & \dots & v_nw_n\end{bmatrix}.$$

We write $\otimes$ instead of $\otimes_{\mathcal{B}}$ if it is clear from the context what $\mathcal{B}$ is.

**Proposition 5.9.** The map $\_\otimes_{\mathcal{B}}\_:\mathbb{R}^n\times\mathbb{R}^n\to\mathrm{Mat}_{n\times n}(\mathbb{R})$ given by $(\mathbf{v},\mathbf{w})\mapsto\mathbf{v}\otimes_{\mathcal{B}}\mathbf{w}$ has the following properties:

1. It is linear in both arguments.
2. $(\mathbf{v}\otimes_{\mathcal{B}}\mathbf{w})^T=\mathbf{w}\otimes_{\mathcal{B}}\mathbf{v}$.
3. If $\mathcal{B}$ is orthonormal, then $(\mathbf{u}\otimes_{\mathcal{B}}\mathbf{v})\cdot\mathbf{w}=\langle\mathbf{v},\mathbf{w}\rangle\mathbf{u}$ for any $\mathbf{u},\mathbf{v},\mathbf{w}\in\mathbb{V}^n$.

*Proof.* 1. Linearity of the tensor product follows from the linearity of matrix multiplication. To see this for the first argument, take two scalars $\alpha,\beta$, three vectors $\mathbf{u},\mathbf{v},\mathbf{w}$, and notice that

$$(\alpha\mathbf{u}+\beta\mathbf{v})\otimes\mathbf{w}=(\alpha\mathbf{u}+\beta\mathbf{v})\cdot\mathbf{w}^T=\alpha(\mathbf{u}\cdot\mathbf{w}^T)+\beta(\mathbf{v}\cdot\mathbf{w}^T)=\alpha\mathbf{u}\otimes\mathbf{w}+\beta\mathbf{v}\otimes\mathbf{w}.$$

2. For the second claim, we use properties of matrix transposition:

$$(\mathbf{v}\otimes_{\mathcal{B}}\mathbf{w})^T=(\mathbf{v}\cdot\mathbf{w}^T)^T=(\mathbf{w}^T)^T\cdot\mathbf{v}^T=\mathbf{w}\cdot\mathbf{v}^T=\mathbf{w}\otimes\mathbf{v}.$$

3. For the last point, we use the fact that $\langle\mathbf{v},\mathbf{w}\rangle=\mathbf{v}^T\cdot\mathbf{w}$, since we are in an orthonormal basis. Then

$$(\mathbf{u}\otimes_{\mathcal{B}}\mathbf{v})\cdot\mathbf{w}=(\mathbf{u}\cdot\mathbf{v}^T)\cdot\mathbf{w}=\mathbf{u}\cdot(\mathbf{v}^T\cdot\mathbf{w})=\mathbf{u}\cdot\langle\mathbf{v},\mathbf{w}\rangle=\langle\mathbf{v},\mathbf{w}\rangle\cdot\mathbf{u}.$$

□

### 5.2.2 Parallel projection on a hyperplane

**Example 5.10.** Fix a reference frame $\mathcal{K}$ of $\mathbb{A}^2$. We want to project onto the line (hyperplane)

$$\ell: x+y-1=0$$

in the direction of the vector $\mathbf{v}(-2,-1)$. How do we do this? We do it pointwise. Take an arbitrary point $P(x_P,y_P)$ and consider the line $\ell_P$ passing through $P$ in the direction of $\mathbf{v}$. It has parametric equations

$$\ell_P:\begin{cases}x=x_P-2t\\ y=y_P-t.\end{cases}$$

The projection of $P$ onto $\ell$ in the direction of $\mathbf{v}$ is the point $P'=\ell\cap\ell_P$.

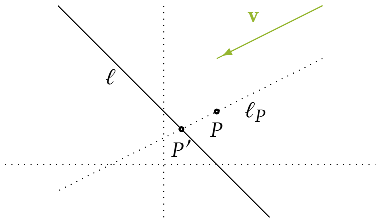



Figure 5.1: Projection of the point $P$ on the line $\ell$ in the direction of the vector $\mathbf{v}$.

To determine $P'$, we check which point on $\ell_P$ satisfies the equation of $\ell$

$$x_P - 2t + y_P - t - 1 = 0 \quad \Rightarrow \quad t = \frac{1}{3}(x_P + y_P - 1).$$

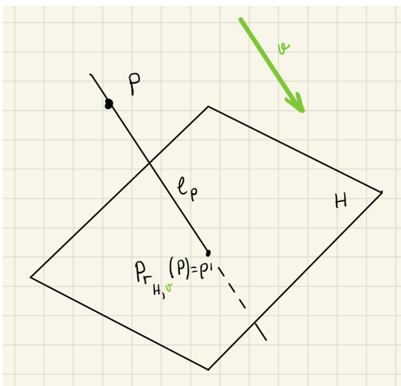
Thus, the projection of $P$ is

$$[P']_{\mathcal{K}} = \begin{bmatrix} \frac{1}{3}x_P - \frac{2}{3}y_p + \frac{2}{3} \\ -\frac{1}{3}x_P + \frac{2}{3}y_p + \frac{1}{3} \end{bmatrix} = \underbrace{\frac{1}{3}\begin{bmatrix} 1 & -2 \\ -1 & 2 \end{bmatrix}}_{A} \begin{bmatrix} x_P \\ y_P \end{bmatrix} + \underbrace{\frac{1}{3}\begin{bmatrix} 2 \\ 1 \end{bmatrix}}_{b}.$$

**Definition 5.11.** Let $H$ be a hyperplane, and let $\mathbf{v}$ be a vector in $\mathbb{V}^n$ that is not parallel to $H$. For any point $P \in \mathbb{E}^n$, there is a unique line $\ell_P$ passing through $P$ and having $\mathbf{v}$ as its direction vector. The line $\ell_P$ is not parallel to $H$; hence, it intersects $H$ in a unique point $P'$. We denote $P'$ by $\mathrm{Pr}_{H,\mathbf{v}}(P)$ and call it the *projection of the point P onto the hyperplane H parallel to* $\mathbf{v}$. This gives a map

$$\mathrm{Pr}_{H,\mathbf{v}} : \mathbb{A}^n \to \mathbb{A}^n$$

called the *projection onto the hyperplane H parallel to* $\mathbf{v}$.




Fix a reference frame $\mathcal{K} = (O, \mathcal{B})$ of $\mathbb{A}^n$. Consider the hyperplane

$$H : a_1x_1 + \cdots + a_nx_n + a_{n+1} = 0 \tag{5.4}$$

and a line $\ell_P$ passing through the point $P(p_1,\dots,p_n)$ and having $\mathbf{v}(v_1,\dots,v_n)$ as direction vector:

$$\ell_P = \{P + t\mathbf{v} : t \in \mathbb{R}\}. \tag{5.5}$$

The intersection $\ell_P \cap H$ can be described as follows:

$$P + t\mathbf{v} \in \ell_P \cap H \quad \Leftrightarrow \quad a_1(p_1+tv_1)+\cdots+a_n(p_n+tv_n)+a_{n+1} = 0.$$

So, the intersection point $\mathrm{Pr}_{H,\mathbf{v}}(P) = P'$ is:

$$P' = P - \frac{a_1p_1+\cdots+a_np_n+a_{n+1}}{a_1v_1+\cdots+a_nv_n}\mathbf{v} = P - \frac{\mathbf{a}^T\cdot P + a_{n+1}}{\mathbf{a}^T\cdot\mathbf{v}}\mathbf{v}, \tag{5.6}$$

where $\mathbf{a} = \mathbf{a}(a_1,\dots,a_n)$, and where, in the second equality, we use the convention that points and vectors are identified with column matrices of their coordinates and components, respectively. Hence, if we denote by $p'_1,\dots,p'_n$ the coordinates of the projected point $\mathrm{Pr}_{H,\mathbf{v}}(P)$, then

$$\begin{cases} p'_1 = p_1 + \mu v_1 \\ \qquad\vdots \\ p'_n = p_n + \mu v_n \end{cases} \qquad \text{where} \quad \mu = -\frac{a_1p_1+\cdots+a_np_n+a_{n+1}}{a_1v_1+\cdots+a_nv_n}.$$

Since $\mathbf{a}^T\cdot P\cdot\mathbf{v} = \mathbf{v}\cdot\mathbf{a}^T\cdot P$, in matrix form we have:

$$\mathrm{Pr}_{H,\mathbf{v}}(P) = \left(I_n - \frac{\mathbf{v}\cdot\mathbf{a}^T}{\mathbf{v}^T\cdot\mathbf{a}}\right)\cdot P - \frac{a_{n+1}}{\mathbf{v}^T\cdot\mathbf{a}}\mathbf{v}$$

where $I_n$ is the $n\times n$ identity matrix. In particular, if $\mathcal{B}$ is orthonormal, the linear part of this map is

$$\mathrm{M}_{\mathcal{B}}(\mathrm{lin}(\mathrm{Pr}_{H,\mathbf{v}})) = \left(I_n - \frac{\mathbf{v}\otimes\mathbf{a}}{\langle\mathbf{v},\mathbf{a}\rangle}\right).$$

Parallel projections on hyperplanes are affine maps. Obviously, they are not bijective, so

$$\mathrm{Pr}_{H,\mathbf{v}} \in \mathrm{End}_{\mathrm{aff}}(\mathbb{A}^n) \quad \text{but} \quad \mathrm{Pr}_{H,\mathbf{v}} \notin \mathrm{AGL}(\mathbb{A}^n).$$

**Definition 5.12.** The *orthogonal projection* $\mathrm{Pr}_H^{\perp}$ onto the hyperplane $H \subseteq \mathbb{E}^n$ is the projection onto $H$ in the direction of a vector that is orthogonal to $H$, i.e.,

$$\mathrm{Pr}_H^{\perp} = \mathrm{Pr}_{H,\mathbf{v}}$$

where $\mathbf{v}$ is a normal vector of $H$. With the above notation we see that

$$\mathrm{Pr}_H^{\perp}(P) = \left(I_n - \frac{\mathbf{a}\otimes\mathbf{a}}{|\mathbf{a}|^2}\right)\cdot P - \frac{a_{n+1}}{|\mathbf{a}|^2}\mathbf{a}$$

since we may choose $\mathbf{v} = \mathbf{a}$.

### 5.2.3 Parallel reflection in a hyperplane

**Example 5.13.** We consider again the setup in Example 5.10. But this time, we want to reflect in the line (hyperplane)

$$\ell : x + y - 1 = 0$$

in the direction of the vector $\mathbf{v}(-2,-1)$. We do it pointwise. Take an arbitrary point $P(x_P, y_P)$ and consider the line $\ell_P$ passing through $P$ in the direction of $\mathbf{v}$. It has parametric equations

$$\ell_P : \begin{cases} x = x_P - 2t \\ y = y_P - t. \end{cases}$$

The reflection of $P$ in $\ell$ in the direction of $\mathbf{v}$ is the point $P'$ such that $\mathrm{Pr}_{\ell,\mathbf{v}}(P) = \ell \cap \ell_P$ is the midpoint of the segment $[PP']$. Thus, identifying points with column matrices of their coordinates relative to $\mathcal{K}$ we have

$$\frac{P + P'}{2} = \mathrm{Pr}_{\ell,\mathbf{v}}(P) \quad \Rightarrow \quad P' = 2\,\mathrm{Pr}_{\ell,\mathbf{v}}(P) - P.$$

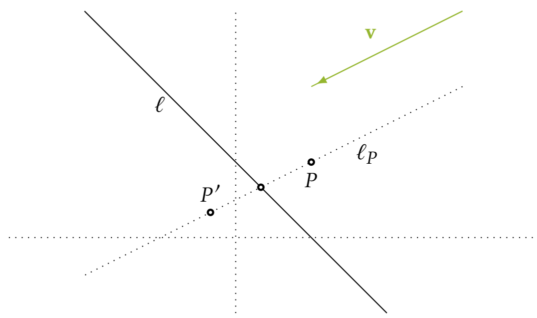



Figure 5.2: Reflection of the point $P$ in the line $\ell$ parallel to the vector $\mathbf{v}$.

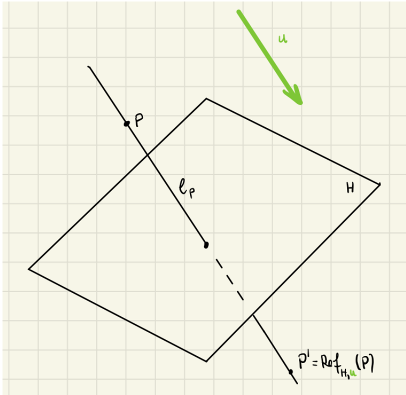
Using the calculation in Example 5.10, the reflection of $P$ is

$$[P']_{\mathcal{K}} = \begin{bmatrix} -\frac{1}{3}x_P - \frac{4}{3}y_p + \frac{4}{3} \\ -\frac{2}{3}x_P + \frac{1}{3}y_p + \frac{2}{3} \end{bmatrix} = \underbrace{\frac{1}{3}\begin{bmatrix} -1 & -4 \\ -2 & 1 \end{bmatrix}}_{A} \begin{bmatrix} x_P \\ y_P \end{bmatrix} + \underbrace{\frac{2}{3}\begin{bmatrix} 2 \\ 1 \end{bmatrix}}_{b}.$$

**Definition 5.14.** Let $H$ be a hyperplane, and let $\mathbf{v}$ be a vector in $\mathbb{V}^n$ that is not parallel to $H$. For any point $P \in \mathbb{A}^n$, there is a unique point $P'$ such that $\mathrm{Pr}_{H,\mathbf{v}}(P)$ is the midpoint of the segment $[PP']$. We denote $P'$ by $\mathrm{Ref}_{H,\mathbf{v}}(P)$ and call it the *reflection of the point $P$ in the hyperplane $H$ parallel to* $\mathbf{v}$. This gives a map

$$\mathrm{Ref}_{H,\mathbf{v}} : \mathbb{A}^n \to \mathbb{A}^n$$

called the *reflection in the hyperplane $H$ parallel to* $\mathbf{v}$.




We keep the notation in the previous section. In particular, the hyperplane $H$ is given by the equation (5.4). The idea here is the same as in Example 5.13: $\mathrm{Pr}_{H,\mathbf{v}}(P)$ is the midpoint of the segment $[PP']$. Thus, identifying points with column matrices of their coordinates relative to $\mathcal{K}$, we have:

$$\mathrm{Pr}_{H,\mathbf{v}}(P) = \frac{P+P'}{2} \quad \Rightarrow \quad \mathrm{Ref}_{H,\mathbf{v}}(P) = P - 2\frac{\mathbf{a}^T \cdot P + a_{n+1}}{\mathbf{v}^T \cdot \mathbf{a}}\mathbf{v}.$$

Again, since $\mathbf{a}^T \cdot P \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{a}^T \cdot P$, in matrix form we have:

$$\mathrm{Ref}_{H,\mathbf{v}}(P) = \left(I_n - 2\frac{\mathbf{v}\cdot\mathbf{a}^T}{\mathbf{v}^T \cdot \mathbf{a}}\right) \cdot P - 2\frac{a_{n+1}}{\mathbf{v}^T \cdot \mathbf{a}}\mathbf{v}.$$

In particular, if $\mathcal{B}$ is orthonormal, the linear part of this map is

$$\mathrm{M}_{\mathcal{B}}(\mathrm{lin}(\mathrm{Ref}_{H,\mathbf{v}})) = \left(I_n - 2\frac{\mathbf{v}\otimes\mathbf{a}}{\langle \mathbf{v},\mathbf{a}\rangle}\right).$$

Parallel reflections in hyperplanes are affine maps. Obviously, they are bijective, so

$$\mathrm{Ref}_{H,\mathbf{v}} \in \mathrm{AGL}(\mathbb{A}^n) \subseteq \mathrm{End}_{\mathrm{aff}}(\mathbb{A}^n).$$

**Definition 5.15.** The *orthogonal reflection* $\mathrm{Ref}_H^{\perp}$ in the hyperplane $H \subseteq \mathbb{E}^n$ is the reflection in $H$ parallel to a vector that is orthogonal to $H$, i.e.,

$$\mathrm{Ref}_H^{\perp} = \mathrm{Ref}_{H,\mathbf{v}}$$

where $\mathbf{v}$ is a normal vector of $H$. With the above notation we see that

$$\mathrm{Ref}_H^{\perp}(P) = \left(I_n - 2\frac{\mathbf{a}\otimes\mathbf{a}}{|\mathbf{a}|^2}\right) \cdot P - 2\frac{a_{n+1}}{|\mathbf{a}|^2}\mathbf{a} \tag{5.7}$$

since we may choose $\mathbf{v} = \mathbf{a}$.

### 5.2.4 Parallel projection on a line

**Example 5.16.** Consider again Example 5.10. We have a line $\ell$ and a point $P$ that we want to project onto $\ell$ in the direction of $\mathbf{v}(-2,-1)$. We know how to do this, but let us exchange the roles of the Cartesian equation and the parametric equations. Parametric equations for $\ell$ are

$$\ell : \begin{cases} x = 1 - t \\ y = t \end{cases} .$$

For an arbitrary point $P(x_P, y_P)$, the line $\ell_P$ passing through $P$ has a direction space given by the equation

$$\mathrm{D}(\ell_P) : x - 2y = 0$$

with respect to the basis $\mathcal{B}$ of the current coordinate system $\mathcal{K}$. Thus, $\ell_P$ is described by

$$\ell_P : (x - x_P) - 2(y - y_P) = 0.$$

The projection of $P$ onto $\ell$ in the direction of $\mathbf{v}$ is the point $P' = \ell \cap \ell_P$, and the corresponding figure is Figure 5.1. The only difference is that we describe $\ell$ and $\ell_P$ with different types of equations. To determine $P'$, we find the intersection by plugging the points of $\ell$ into the equation of $\ell_P$:

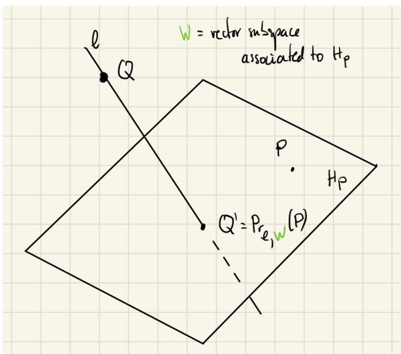
$$(1 - t - x_P) - 2(t - y_P) = 0 \quad \Rightarrow \quad t = -\frac{1}{3}(x_P - 2y_P - 1).$$

As expected, a short calculation shows that $P'$ has the same expression as in Example 5.10.

**Definition 5.17.** Let $\ell$ be a line, and let $\mathbb{W}$ be an $(n-1)$-dimensional vector subspace in $\mathbb{V}^n$ that is not parallel to $\ell$. For any point $P \in \mathbb{A}^n$, there is a unique hyperplane $H_P$ passing through $P$ and having $\mathbb{W}$ as its associated vector subspace. The hyperplane $H_P$ is not parallel to $\ell$; hence, it intersects $\ell$ in a unique point $P'$. We denote $P'$ by $\mathrm{Pr}_{\ell,\mathbb{W}}(P)$ and call it the *projection of the point $P$ onto the line $\ell$ parallel to $\mathbb{W}$*. This gives a map

$$\mathrm{Pr}_{\ell,\mathbb{W}} : \mathbb{A}^n \to \mathbb{A}^n$$

called the *projection onto the line $\ell$ parallel to $\mathbb{W}$*.



With respect to the reference frame $\mathcal{K}$, the vector subspace $\mathbb{W}$ is given by a homogeneous equation

$$\mathbb{W}: a_1x_1 + a_2x_2 + \cdots + a_nx_n = 0 \quad \Leftrightarrow \quad \mathbf{a}^T \cdot \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = 0 \tag{5.8}$$

where $\mathbf{a} = \mathbf{a}(a_1, \ldots, a_n)$. Thus, for a given point $P(p_1, \ldots, p_n) \in \mathbb{A}^n$, the equation of $H_P$ is

$$H_P : a_1(x_1 - p_1) + a_2(x_2 - p_2) + \cdots + a_n(x_n - p_n) = 0 \quad \Leftrightarrow \quad \mathbf{a}^T \cdot \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \mathbf{a}^T \cdot P$$

and $\mathrm{Pr}_{\ell,\mathbb{W}}(P) = \mathrm{Pr}_{H_P,\mathbf{v}}(Q)$ for a fixed but arbitrary point $Q(q_1, \ldots, q_n) \in \ell$. Hence, if we denote by $p_1', \ldots, p_n'$ the coordinates of the projected point $\mathrm{Pr}_{\ell,\mathbb{W}}(P)$, then, by (5.6),

$$\begin{cases} p_1' = q_1 + v_1\mu \\ \quad\vdots \\ p_n' = q_n + v_n\mu \end{cases} \quad \text{where} \quad \mu = -\frac{\mathbf{a}^T \cdot Q - \mathbf{a}^T \cdot P}{\mathbf{a}^T \cdot \mathbf{v}}.$$

In matrix form, we can rearrange this as follows:

$$\mathrm{Pr}_{\ell,\mathbb{W}}(P) = \frac{\mathbf{v} \cdot \mathbf{a}^T}{\mathbf{v}^T \cdot \mathbf{a}} P + \left( I_n - \frac{\mathbf{v} \cdot \mathbf{a}^T}{\mathbf{v}^T \cdot \mathbf{a}} \right) Q.$$

In particular, if $\mathcal{B}$ is orthonormal, the linear part of this map is

$$\mathrm{M}_{\mathcal{B}}(\mathrm{lin}(\mathrm{Pr}_{\ell,\mathbb{W}})) = \frac{\mathbf{v} \otimes \mathbf{a}}{\langle \mathbf{v}, \mathbf{a} \rangle}.$$

Parallel projections on lines are affine maps. Obviously, they are not bijective, so

$$\mathrm{Pr}_{\ell,\mathbb{W}} \in \mathrm{End}_{\mathrm{aff}}(\mathbb{A}^n) \quad \text{but} \quad \mathrm{Pr}_{\ell,\mathbb{W}} \notin \mathrm{AGL}(\mathbb{A}^n).$$

**Definition 5.18.** The *orthogonal projection* $\mathrm{Pr}_\ell^\perp$ onto the line $\ell \subseteq \mathbb{E}^n$ is the projection onto $\ell$ parallel to vectors that are orthogonal to the line $\ell$, i.e.,

$$\mathrm{Pr}_\ell^\perp = \mathrm{Pr}_{\ell,\mathbf{v}^\perp}$$

where $\mathbf{v}$ is a direction vector of $\ell$. With the above notation we see that

$$\mathrm{Pr}_\ell^\perp(P) = \frac{\mathbf{a} \otimes \mathbf{a}}{|\mathbf{a}|^2} P + \left( I_n - \frac{\mathbf{a} \otimes \mathbf{a}}{|\mathbf{a}|^2} \right) Q$$

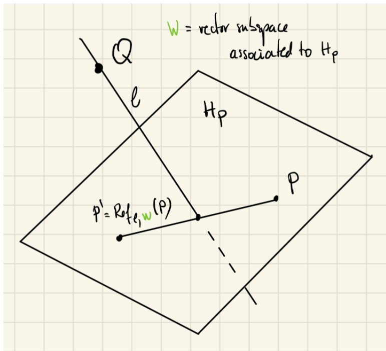
since we may choose $\mathbf{v} = \mathbf{a}$.

### 5.2.5 Parallel reflection in a line

**Definition 5.19.** Let $\ell$ be a line, and let $\mathbb{W}$ be an $(n-1)$-dimensional vector subspace in $\mathbb{V}^n$ that is not parallel to $\ell$. For any point $P \in \mathbb{A}^n$, there is a unique point $P'$ such that $\mathrm{Pr}_{\ell,\mathbb{W}}(P)$ is the midpoint of the segment $[PP']$. We denote $P'$ by $\mathrm{Ref}_{\ell,\mathbb{W}}(P)$ and call it the *reflection of the point $P$ in the line $\ell$ parallel to $\mathbb{W}$*. This gives a map

$$\mathrm{Ref}_{\ell,\mathbb{W}} : \mathbb{A}^n \to \mathbb{A}^n$$

called the *reflection in the line $\ell$ parallel to $\mathbb{W}$*.




As in Section 5.2.4, the vector subspace $\mathbb{W}$ is given by the homogeneous equation (5.8). The idea is similar to the one used in Section 5.2.3: since $\mathrm{Pr}_{\ell,\mathbb{W}}(P)$ is the midpoint of the segment $[PP']$, we have:

$$\mathrm{Ref}_{\ell,\mathbb{W}}(P) = 2\,\mathrm{Pr}_{\ell,\mathbb{W}}(P) - P$$

Here again we use the convention that points and vectors are identified with column matrices of their coordinates and components, respectively. Rearranging this in matrix form, we obtain:

$$\mathrm{Ref}_{\ell,\mathbb{W}}(P) = \left(2\frac{\mathbf{v}\cdot\mathbf{a}^T}{\mathbf{v}^T\cdot\mathbf{a}} - I_n\right)P + 2\left(I_n - \frac{\mathbf{v}\cdot\mathbf{a}^T}{\mathbf{v}^T\cdot\mathbf{a}}\right)Q,$$

where $Q(q_1,\dots,q_n)$ is a point on $\ell$ and $\mathbf{v}(v_1,\dots,v_n)$ is a direction vector for $\ell$. In particular, if $\mathcal{B}$ is orthonormal, the linear part of this map is

$$\mathrm{M}_{\mathcal{B}}(\mathrm{lin}(\mathrm{Ref}_{\ell,\mathbb{W}})) = 2\frac{\mathbf{v}\otimes\mathbf{a}}{\langle\mathbf{v},\mathbf{a}\rangle} - I_n.$$

Parallel reflections in lines are affine maps. Obviously, they are bijective, so

$$\mathrm{Ref}_{\ell,\mathbb{W}} \in \mathrm{AGL}(\mathbb{A}^n) \subseteq \mathrm{End}_{\mathrm{aff}}(\mathbb{A}^n).$$

**Definition 5.20.** The *orthogonal reflection* $\mathrm{Ref}_\ell^\perp$ in the line $\ell \subseteq \mathbb{E}^n$ is the reflection in $\ell$ parallel to vectors that are orthogonal to the line $\ell$, i.e.,

$$\mathrm{Ref}_\ell^\perp = \mathrm{Ref}_{\ell,\mathbf{v}^\perp}$$

where $\mathbf{v}$ is a direction vector of $\ell$. With the above notation we see that for any point $Q \in \ell$

$$\mathrm{Ref}_{\ell}^{\perp}(P) = \left(2\frac{\mathbf{a}\otimes\mathbf{a}}{|\mathbf{a}|^2} - I_n\right)P + 2\left(I_n - \frac{\mathbf{a}\otimes\mathbf{a}}{|\mathbf{a}|^2}\right)Q$$

since we may choose $\mathbf{v} = \mathbf{a}$.

CHAPTER 6

# Isometries

## Contents

## 6.1 Affine form of isometries

**Definition 6.1.** An *isometry* is a map $\phi : \mathbb{E}^n \to \mathbb{E}^n$ which preserves distances, i.e.

$$d(\phi(P), \phi(Q)) = d(P, Q)$$

for any points $P, Q \in \mathbb{E}^n$.

**Proposition 6.2.** Isometries are affine transformations.

*Proof.* Let $\phi : \mathbb{E}^n \to \mathbb{E}^n$ be an isometry. We prove our claim using the characterization of affine transformations in Proposition 5.5.

The map $\phi$ is *injective*: if $\phi(P) = \phi(Q)$ then

$$0 = d(\phi(P), \phi(Q)) = d(P, Q)$$

which implies that $P = Q$.

The map $\phi$ *preserves lines*: consider three collinear points $A, B, C \in \mathbb{E}^n$. One of the points will lie between the other two. Assume that $C \in [AB]$. Then $d(A,B) = d(A,C) + d(C,B)$. Let $A' = \phi(A)$, $B' = \phi(B)$ and $C' = \phi(C)$. Then $d(A', B') = d(A,B)$, $d(A', C') = d(A,C)$ and $d(B', C') = d(B,C)$. Therefore

$$d(A', B') = d(A,B) = d(A,C) + d(C,B) = d(A', C') + d(C', B')$$

which implies $C' \in [A'B']$.

The map $\phi$ *preserves oriented ratios*: notice that

$$\frac{d(A', C')}{d(A', B')} = \frac{d(A,C)}{d(A,B)}$$

hence

$$\frac{\overrightarrow{A'C'}}{\overrightarrow{A'B'}} = \pm\frac{d(A', C')}{d(A', B')} = \pm\frac{d(A,C)}{d(A,B)} = \pm\frac{\overrightarrow{AC}}{\overrightarrow{AB}}.$$

But, the two ratios have the same sign since, as in the previous paragraph, $C$ is between $A$ and $B$ if and only if $C'$ is between $A'$ and $B'$. $\square$

**Remark** (Notation)**.** At this point some simplifications in notation are useful. When it is clear from the context that we are working in a frame $\mathcal{K} = (O, \mathcal{B})$, we will simply write $\mathbf{v}$ instead of $[\mathbf{v}]_{\mathcal{B}}$. We identify points with column matrices. For example, we write $\mathbf{x}$ to mean the column matrix with entries $(x_1, \ldots, x_n)$. This will represent both the coordinates of a point and the components of the position vector.

Moreover, when there is no risk of confusion, we denote the linear map induced by an affine transformation with the same letter, i.e. we write $\psi$ instead of $\mathrm{lin}(\psi)$. The arguments are either points or vectors, and the context will make it clear which one it is.

**Example 6.3.** Translation maps are isometries. Let $\mathbf{v}$ be a vector. Then, the *translation with the vector* $\mathbf{v}$ is the map $T_{\mathbf{v}} : \mathbb{E}^n \to \mathbb{E}^n$ given by $T_{\mathbf{v}}(\mathbf{x}) = \mathbf{x} + \mathbf{v}$. It is just the translation map of the affine structure of $\mathbb{E}^n$ where we fixed the vector argument to be $\mathbf{v}$.

**Proposition 6.4.** Let $\phi \in \mathrm{AGL}(\mathbb{E}^n)$ be an affine transformation given by $\phi(\mathbf{x}) = A\mathbf{x} + b$ with respect to some orthonormal frame. The following are equivalent:

1. $\phi$ is an isometry
2. $\phi$ preserves lengths of vectors
3. $\phi$ preserves the scalar product, i.e. for any vectors $\mathbf{v}, \mathbf{w}$ we have
$$\langle \phi(\mathbf{v}), \phi(\mathbf{w}) \rangle = \langle \mathbf{v}, \mathbf{w} \rangle.$$

*Proof.* The length of a vector $\overrightarrow{AB}$ is by definition the length of the segment $[AB]$ which is the distance between $A$ and $B$, so 1. is equivalent to 2. If $\phi$ is an isometry, it is an affine map, by Proposition 6.2. In particular, it preserves oriented ratios by Proposition 5.5. From this it follows that it preserves the cosine of angles. Then
$$\langle \phi(\mathbf{v}), \phi(\mathbf{w}) \rangle = |\phi(\mathbf{v})| \cdot |\phi(\mathbf{w})| \cdot \cos \measuredangle(\phi(\mathbf{v}), \phi(\mathbf{w})) = |\mathbf{v}| \cdot |\mathbf{w}| \cdot \cos \measuredangle(\mathbf{v}, \mathbf{w}) = \langle \mathbf{v}, \mathbf{w} \rangle.$$
For the converse, let $\mathbf{v} = \overrightarrow{AB}$ and notice that if $\phi$ preserves the scalar product then
$$d(\phi(A), \phi(B)) = |\phi(\overrightarrow{AB})| = \sqrt{\langle \phi(\overrightarrow{AB}), \phi(\overrightarrow{AB}) \rangle} = \sqrt{\langle \overrightarrow{AB}, \overrightarrow{AB} \rangle} = |\overrightarrow{AB}| = d(A, B).$$
This shows that 3. implies 1. and 2. □

**Proposition 6.5.** Let $\phi \in \mathrm{AGL}(\mathbb{E}^n)$ be an affine transformation given by $\phi(\mathbf{x}) = A\mathbf{x} + b$ with respect to some orthonormal frame. The following are equivalent:

1. $\phi$ is an isometry
2. $A^{-1} = A^T$.

*Proof.* Consider $\psi(\mathbf{x}) = \phi(\mathbf{x}) - b = A\mathbf{x}$. Since translations are isometries, $\phi$ is an isometry if and only if $\psi$ is an isometry. Let $\mathcal{K} = (O, \mathcal{B})$ with $\mathcal{B} = (\mathbf{e}_1, \ldots, \mathbf{e}_n)$ be the orthonormal frame with respect to which $\phi$ is given in the form $\phi(\mathbf{x}) = A\mathbf{x} + b$. Let $\mathbf{f}_1 = \psi(\mathbf{e}_1) = A\mathbf{e}_1$, $\mathbf{f}_2 = \psi(\mathbf{e}_2) = A\mathbf{e}_2$, ..., $\mathbf{f}_n = \psi(\mathbf{e}_n) = A\mathbf{e}_n$.

To see that 1. implies 2., notice that if $\psi$ is an isometry, by Proposition 6.4 we have
$$\langle \mathbf{f}_i, \mathbf{f}_j \rangle = \langle \psi(\mathbf{e}_i), \psi(\mathbf{e}_j) \rangle = \langle \mathbf{e}_i, \mathbf{e}_j \rangle = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}.$$
Since the components of $\mathbf{f}_i = A\mathbf{e}_i$ are the entries in the $i$-th column of the matrix $A$, it follows that $A^T A$ is the identity matrix $I_n$.

To see that 2. implies 1. consider two arbitrary points $\mathbf{x}$ and $\mathbf{y}$ in $\mathbb{E}^n$. We have $|\phi(\mathbf{x}) - \phi(\mathbf{y})| = |(A\mathbf{x} + b) - (A\mathbf{y} + b)| = |A(\mathbf{x} - \mathbf{y})|$ and therefore
$$\begin{aligned} |\phi(\mathbf{x}) - \phi(\mathbf{y})|^2 &= |A(\mathbf{x} - \mathbf{y})|^2 \\ &= \langle A(\mathbf{x} - \mathbf{y}), A(\mathbf{x} - \mathbf{y}) \rangle \\ &= (A(\mathbf{x} - \mathbf{y}))^T \cdot A(\mathbf{x} - \mathbf{y}) \\ &= (\mathbf{x} - \mathbf{y})^T A^T A(\mathbf{x} - \mathbf{y}) \\ &= (\mathbf{x} - \mathbf{y})^T (\mathbf{x} - \mathbf{y}) \\ &= |\mathbf{x} - \mathbf{y}|^2 \end{aligned}$$
which is equivalent to $d(\phi(\mathbf{x}), \phi(\mathbf{y})) = d(\mathbf{x}, \mathbf{y})$. □

**Definition 6.6.** A matrix $A \in \mathrm{Mat}_{n\times n}(\mathbb{R})$ such that $A^T A = I_n$ is called *orthogonal* and the set of all such matrices is denoted by $\mathrm{O}(n)$.

**Proposition 6.7.** If $A$ is an orthogonal matrix then $\det(A) = \pm 1$.

*Proof.* Since $A^T A = I_n$ we have $1 = \det(I_n) = \det(A^T A) = \det(A^T)\det(A) = \det(A)^2$. □

**Definition 6.8.** The set of matrices in $\mathrm{O}(n)$ with determinant 1 is denoted by $\mathrm{SO}(n)$. Such matrices are called *special orthogonal*. The set $\mathrm{O}(n)$ is a subgroup of $\mathrm{AGL}(\mathbb{R}^n)$ and $\mathrm{SO}(n)$ is a normal subgroup of $\mathrm{O}(n)$. With group theory notation we write this as follows:

$$\mathrm{SO}(n) \trianglelefteq \mathrm{O}(n) \leq \mathrm{AGL}(\mathbb{R}^n).$$

**Definition 6.9.** Let $\phi$ be an isometry of $\mathbb{E}^n$ given by $\phi(\mathbf{x}) = A\mathbf{x} + b$ with respect to some right-oriented orthonormal frame. Then $\phi$ is called a *displacement*, or a *direct isometry*, if $A \in \mathrm{SO}(n)$. Otherwise, if $\det(A) = -1$, the map $\phi$ is called an *indirect isometry*.

**Remark.** In Section 3.2.1 we have revised $\mathbb{E}^n$ to be $\mathbb{R}^n$ with the standard scalar product. This point of view encapsulates all aspects of Hilbert's axioms [14] but doesn't immediately give a transparent description of the congruence relation. This description can be done now: a segment $[AB]$ is congruent to a segment $[CD]$ if and only if there is an isometry $\phi$ such that $\phi(A) = C$ and $\phi(B) = D$. This shows that the notion of congruence (in dimension $n$) is completely described by translations and by the orthogonal group $\mathrm{O}(n)$.

It is tautological to say that congruence is described by isometries. The point made here is that isometries can be classified and described precisely by focusing the analysis on $\mathrm{O}(n)$. We do this for dimension 2 and 3 in the following two sections.

## 6.2 Isometries in dimension 2

### 6.2.1 Rotations

**Proposition 6.10.** A matrix $A$ is in $\mathrm{SO}(2)$ if and only if $A$ equals

$$R_\theta = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} \tag{6.1}$$

for some $\theta \in \mathbb{R}$.

*Proof.* It is easy to check that $R_\theta$ is special orthogonal. For the converse, consider the matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in \mathrm{SO}(2).$$

We have $\det(A) = ad - bc = 1$ and

$$A^T A = \begin{bmatrix} a^2 + c^2 & ab + cd \\ ab + cd & b^2 + d^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Thus, the entries of the matrix $A$ satisfy the system

$$\begin{cases} a^2 + c^2 = 1 \\ b^2 + d^2 = 1 \\ ab + cd = 0 \\ ad - bc = 1 \end{cases}.$$

From the first two equations it follows that there exist $\theta, \tilde{\theta}$ such that $a = \cos(\theta)$, $c = \sin(\theta)$, $d = \cos(\tilde{\theta})$ and $b = \sin(\tilde{\theta})$. Then, the last two equations become

$$\begin{cases} 0 = ab + cd = \cos(\theta)\sin(\tilde{\theta}) + \sin(\theta)\cos(\tilde{\theta}) = \sin(\theta + \tilde{\theta}) \\ 1 = ad - bc = \cos(\theta)\cos(\tilde{\theta}) - \sin(\theta)\sin(\tilde{\theta}) = \cos(\theta + \tilde{\theta}) \end{cases}$$

and therefore $\tilde{\theta} = -\theta + 2k\pi$ for some integer $k$. □

**Corollary 6.11.** A direct isometry $\phi$ of $\mathbb{E}^2$ that fixes a point is either the identity or a rotation. Moreover, the angle $\theta$ of the rotation is such that

$$\cos(\theta) = \frac{\operatorname{tr}(\operatorname{lin}(\phi))}{2}.$$

*Proof.* Let $\phi(\mathbf{x}) = A\mathbf{x} + \mathbf{b}$ with respect to a right-oriented orthonormal frame. By Proposition 6.10, it is a direct isometry if $A \in \mathrm{SO}(2)$ which is equivalent to $A = R_\theta$ where $R_\theta$ is the rotation matrix (6.1). If $\theta = 0$, then $\phi$ is a (possibly trivial) translation by $b$; thus, it has a fixed point only if $\mathbf{b} = 0$, in which case $\phi$ equals the identity map, i.e., $\phi = \mathrm{Id}$. For the rest of the proof, assume that $\theta \neq 0$.

Let $\mathbf{p}$ be a fixed point for $\phi$. Then

$$\phi(\mathbf{p}) = \mathbf{p} \quad \Leftrightarrow \quad R_\theta \mathbf{p} + \mathbf{b} = \mathbf{p} \quad \Leftrightarrow \quad \mathbf{b} = (I_2 - R_\theta)\mathbf{p}. \tag{6.2}$$

Now notice that

$$\det(I_2 - R_\theta) = \det \begin{bmatrix} 1 - \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & 1 - \cos(\theta) \end{bmatrix} = (1 - \cos(\theta))^2 + \sin(\theta)^2 = 2 - 2\cos(\theta).$$

So, if $\theta \neq 0$, then equation (6.2) has a unique solution $\mathbf{p}$, i.e., the fixed point of $\phi$ is unique. To finish the proof, we show that $\phi$ is a rotation around the point $\mathbf{p}$. We notice that

$$\phi(\mathbf{x}) - \phi(\mathbf{p}) = \phi(\mathbf{x}) - \mathbf{p} = R_\theta \mathbf{x} + \mathbf{b} - \mathbf{p} = R_\theta \mathbf{x} + (I_2 - R_\theta)\mathbf{p} - \mathbf{p} = R_\theta(\mathbf{x} - \mathbf{p})$$

which means that $\phi$ rotates $\overrightarrow{\mathbf{px}}$ with $\theta$ around $\mathbf{p}$. Finally, the trace of the linear map $\operatorname{lin}(\phi)$ is the trace of its matrix with respect to any orthonormal basis. Thus, we may use $A = R_\theta$ to conclude that

$$\operatorname{tr}(\operatorname{lin}(\phi)) = \operatorname{tr}(R_\theta) = 2\cos(\theta).$$

□

**Remark.** Let us notice the effect of rotations on coordinates and on basis vectors in dimension 2. Let $\mathcal{K} = (O, \mathcal{B})$ be a right-oriented orthonormal frame of $\mathbb{E}^2$ with $\mathcal{B} = (\mathbf{i}, \mathbf{j})$. A rotation around the origin with angle $\theta$ is given by the map

$$\begin{bmatrix} x \\ y \end{bmatrix} \mapsto \underbrace{\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}}_{=\mathrm{Rot}_\theta} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x' \\ y' \end{bmatrix}.$$

Thus, $\mathrm{Rot}_\theta$ is a base change matrix $\mathrm{M}_{\mathcal{B}',\mathcal{B}}$ where $\mathcal{B}' = (\mathbf{i}', \mathbf{j}')$ is another orthonormal basis. The components of $\mathbf{i}'$ and $\mathbf{j}'$ with respect to $\mathcal{B}$ are the columns of the matrix $\mathrm{M}^{-1}_{\mathcal{B}',\mathcal{B}} = \mathrm{M}^T_{\mathcal{B}',\mathcal{B}}$:

$$\mathbf{i}' = \begin{bmatrix} \cos\theta \\ -\sin\theta \end{bmatrix}, \quad \mathbf{j}' = \begin{bmatrix} \sin\theta \\ \cos\theta \end{bmatrix}.$$

We notice that the vectors in $\mathcal{B}'$ are obtained by rotating the vectors in $\mathcal{B}$ by $-\theta$. Indeed, rotating points counterclockwise with respect to $\mathcal{K}$ is equivalent to rotating $\mathcal{K}$ clockwise.

### 6.2.2 Classification

**Theorem 6.12** (Chasles)**.** A direct isometry of the plane $\mathbb{E}^2$ is either

a) the identity, or

b) a translation, or

c) a rotation.

*Proof.* Consider a direct isometry $\phi(\mathbf{x}) = A\mathbf{x} + \mathbf{b}$ given with respect to a right-oriented orthonormal frame. By Proposition 6.10, $A = R_\theta$ where $R_\theta$ is the rotation matrix (6.1) for some $\theta$. If $\theta = 0$, then $\phi$ is the identity map if $\mathbf{b} = 0$, and it is a translation if $\mathbf{b} \neq 0$. Suppose, therefore, that $\theta \neq 0$. Then, by the proof of Corollary 6.11, the map $\phi$ has a (unique) fixed point $\mathbf{p}$. Thus, $\phi$ is a direct isometry which fixes a point, and the proof is finished by applying Corollary 6.11. □

**Definition 6.13.** A *glide reflection in the line $\ell$* is the composition of an orthogonal reflection in $\ell$ and a translation in the direction of $\ell$.

**Example 6.14.** A reflection in the $x$-axis followed by a translation with $\lambda\mathbf{i}$ is a glide reflection. In matrix form the map is given by

$$\mathbf{x} \mapsto \mathrm{Ref}^{\perp}_{Ox}(\mathbf{x}) + \lambda\mathbf{i} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} \lambda \\ 0 \end{bmatrix}.$$

**Lemma 6.15.** A rotation with angle $\theta$ around the origin after a reflection in the $x$-axis is a reflection in the line $y = \tan(\theta/2)x$.

*Proof.* The mentioned map, the reflection in the $x$-axis followed by a rotation around the origin, has the form $\phi : \mathbf{x} \mapsto A\mathbf{x}$ where

$$A = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} \cos(\theta) & \sin(\theta) \\ \sin(\theta) & -\cos(\theta) \end{bmatrix}.$$

Notice that $\det(A - I_2) = 0$ thus $\phi$ fixes at least a line. Consider the vector $\mathbf{v}(\cos(\frac{\theta}{2}), \sin(\frac{\theta}{2}))$. A calculation shows that $(A - I_2)\mathbf{v} = 0$ which means that $\mathbf{v}$ is fixed by $A$. Hence the line passing through the origin in the direction of $\mathbf{v}$ is fixed by $\phi$. To see that it is a reflection, check that $\mathbf{J}(\mathbf{v})$ is mapped to $-\mathbf{J}(\mathbf{v})$. □

**Theorem 6.16.** An indirect isometry of the plane $\mathbb{E}^2$ leaves a line $\ell$ invariant and is either

a) a reflection in $\ell$, or

b) a glide-reflection in $\ell$.

*Proof.* Let $\phi(\mathbf{x}) = A\mathbf{x} + \mathbf{b}$ be an indirect isometry given with respect to a right-oriented orthonormal basis. Consider the reflection $\psi(\mathbf{x}) = \mathrm{Ref}^{\perp}_{Ox}(\mathbf{x}) = C\mathbf{x}$, which is also an indirect isometry. Since $(\phi \circ \psi)(\mathbf{x}) = AC\mathbf{x} + \mathbf{b}$ and since $\det(A) = \det(C) = -1$, we see that $AC$ is an orthogonal matrix of determinant 1, i.e., $\phi \circ \psi$ is a direct isometry. Then by Theorem 6.12, we have 3 cases.

If $\phi \circ \psi$ is the identity or a translation, then $AC = I_2$ which implies $A = C^{-1}$. Since $C$ is the matrix of a reflection, we have $C^2 = I_2$, hence $A = C$. Therefore $\phi$ is the composition of $\mathrm{Ref}^{\perp}_{Ox}$ followed by a translation with $\mathbf{b}(b_1, b_2)$. Then

$$\phi(\mathbf{x}) = \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\mathbf{x} + b_2\mathbf{j}}_{\psi'(\mathbf{x})} + b_1\mathbf{i}$$

and one checks (for example with (5.7)) that $\psi'$ is a reflection in the line $y = b_2/2$. Hence $\phi$ is the composition of a reflection and a (possibly trivial) translation along the reflection axis. Thus, if $b_1 = 0$ it is a reflection and if $b_1 \neq 0$ it is a glide reflection.

If $\phi \circ \psi$ is a rotation, then $AC = R_\theta$ for some $\theta$. In this case, we have $A = R_\theta C$. By Lemma 6.15, $\phi$ is a reflection in a line $\ell$ followed by a translation, and the claim follows as in the previous paragraph. □

## 6.3 Isometries in dimension 3

### 6.3.1 Rotations

**Theorem 6.17** (Euler)**.** A direct isometry $\phi$ of $\mathbb{E}^3$ that fixes a point is either the identity or a rotation around an axis that passes through that point. Moreover, the angle $\theta$ of the rotation is such that

$$\cos(\theta) = \frac{\mathrm{tr}(\mathrm{lin}(\phi)) - 1}{2}.$$

*Proof.* By choosing the fixed point of $\phi$ to be the origin, we may assume that $\phi$ has the form $\phi(\mathbf{x}) = A\mathbf{x}$ with respect to a right-oriented orthonormal frame. Since $\phi$ is a direct isometry, we have $A \in \mathrm{SO}(3)$. A rotation around an axis fixes the rotation axis. To see that this is the case for $\phi$, it suffices to show that $A$ has an eigenvector $\mathbf{v}$ for the eigenvalue 1 since then

$$\phi(t\mathbf{v}) = A(t\mathbf{v}) = t(A\mathbf{v}) = t\mathbf{v} \quad \forall t \in \mathbb{R}$$

which means that $\phi$ fixes the line passing through the origin in the direction of $\mathbf{v}$. Notice that

$$\begin{aligned}
\det(A - I_3) &= \det(A^T)\det(A - I_3) && \text{since } \det(A) = 1\\
&= \det(A^T(A - I_3))\\
&= \det(A^T A - A^T)\\
&= \det(I_3 - A^T) && \text{since } A \text{ is orthogonal}\\
&= \det((I_3 - A)^T)\\
&= \det(I_3 - A)
\end{aligned}$$

and since $A - I_3$ is a $3\times 3$ matrix we have $\det(I_3 - A) = -\det(A - I_3)$. Thus

$$\det(A - I_3) = -\det(A - I_3) \quad \Rightarrow \quad \det(A - I_3) = 0.$$

Thus, $A$ admits 1 as an eigenvalue. Let $\mathbf{v}$ be a corresponding eigenvector of length 1.

Next, we show that $\phi$ is a rotation around the axis $\mathbb{R}\mathbf{v}$. Choose a unit vector $\mathbf{u}_1 \perp \mathbf{v}$ and let $\mathbf{u}_2 = \mathbf{v} \times \mathbf{u}_1$. Then $A\mathbf{u}_1$ and $A\mathbf{u}_2$ are also orthogonal to $\mathbf{v}$ since

$$\langle A\mathbf{u}_i, \mathbf{v}\rangle = \langle A\mathbf{u}_i, A\mathbf{v}\rangle = \langle \mathbf{u}_i, \mathbf{v}\rangle = 0.$$

In fact $(\mathbf{u}_1, \mathbf{u}_2)$ is a basis for $\mathbf{v}^\perp$, the orthogonal complement to $\mathbf{v}$. Since $\phi$ is an isometry it maps $\mathbf{v}^\perp$ to itself. In particular, restricting $\phi$ to $\mathbf{v}^\perp$ we have an isometry in dimension 2. Thus, with respect to the basis $\mathcal{B} = (\mathbf{u}_1, \mathbf{u}_2, \mathbf{v})$ we have

$$A = \begin{bmatrix} B & 0 \\ 0 & 1 \end{bmatrix}$$

where $B$ is a $2\times 2$ matrix. Since $\mathcal{B}$ is right-oriented and $\phi$ is a direct isometry, we have $\det(A) = 1$. Therefore $\det(B) = 1$. Hence $\phi$ restricts to a direct isometry on $\mathbf{v}^\perp$. Since it fixes the origin, it cannot be a translation. Therefore, by Theorem 6.12, it must be a (possibly trivial) rotation. Then, the matrix of $\phi$ with respect to $\mathcal{B}$ is

$$[\mathrm{lin}(\phi)]_{\mathcal{B}} = \begin{bmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

for some $\theta \in \mathbb{R}$. This is a rotation around the axis $\mathbb{R}\mathbf{v}$. Finally, the trace of the linear map $\mathrm{lin}(\phi)$ is the trace of the matrix $[\mathrm{lin}(\phi)]_{\mathcal{B}}$. Thus

$$\mathrm{tr}(\mathrm{lin}(\phi)) = \mathrm{tr}([\mathrm{lin}(\phi)]_{\mathcal{B}}) = 2\cos(\theta) + 1.$$

□

**Proposition 6.18** (Euler-Rodrigues)**.** Let $\mathbf{v}$ be a unit vector and $\theta \in \mathbb{R}$. The rotation of angle $\theta$ and axis $\mathbb{R}\mathbf{v}$ is given by

$$\mathrm{Rot}_{\mathbf{v},\theta}(\mathbf{x}) = \cos(\theta)\mathbf{x} + \sin(\theta)(\mathbf{v}\times\mathbf{x}) + (1-\cos(\theta))\langle\mathbf{v},\mathbf{x}\rangle\mathbf{v}. \tag{6.3}$$

*Proof.* Let $\mathbf{x} = \mathbf{x}_{\|} + \mathbf{x}_{\perp}$ where $\mathbf{x}_{\|}$ is parallel to $\mathbf{v}$ and $\mathbf{x}_{\perp}$ is orthogonal to $\mathbf{v}$. Since $\mathbf{v}$ is a unit vector, we know that

$$\mathbf{x}_{\|} = \langle\mathbf{v},\mathbf{x}\rangle\mathbf{v} \quad \text{and} \quad \mathbf{x}_{\perp} = \mathbf{x} - \mathbf{x}_{\|}.$$

Let $\mathbf{x}'_{\perp} = \mathbf{v}\times\mathbf{x}$ and notice that the vector $\mathbf{x}'_{\perp}$ is orthogonal to $\mathbf{x}_{\|}$ and $\mathbf{x}_{\perp}$ and has the same length as $\mathbf{x}_{\perp}$ since

$$\begin{aligned} |\mathbf{x}'_{\perp}|^2 &= \langle\mathbf{v}\times\mathbf{x},\mathbf{v}\times\mathbf{x}\rangle \\ &= \langle\mathbf{v},\mathbf{v}\rangle\langle\mathbf{x},\mathbf{x}\rangle - \langle\mathbf{v},\mathbf{x}\rangle\langle\mathbf{v},\mathbf{x}\rangle \\ &= |\mathbf{x}|^2 - |\mathbf{x}_{\|}|^2 \\ &= |\mathbf{x}_{\perp}|^2 \end{aligned}$$

where for the second equality we used Lagrange's identity. Therefore, since $\mathrm{Rot}_{\mathbf{v},\theta}$ rotates $\mathbf{x}_{\perp}$ by $\theta$ around the axis $\mathbb{R}\mathbf{v}$, we have

$$\mathrm{Rot}_{\mathbf{v},\theta}(\mathbf{x}) = \mathbf{x}_{\|} + \cos(\theta)\mathbf{x}_{\perp} + \sin(\theta)\mathbf{x}'_{\perp}.$$

Replacing $\mathbf{x}_{\|}$, $\mathbf{x}_{\perp}$ and $\mathbf{x}'_{\perp}$ with their expressions in terms of $\mathbf{x}$ we obtain (6.3). □

### 6.3.2 Euler angles

Fix a right oriented orthonormal frame $\mathcal{K} = (O,\mathcal{B})$. Rotations around the coordinate axes are given by the following matrices:

$$\mathrm{Rot}_{x,\theta} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}, \quad \mathrm{Rot}_{y,\theta} = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}, \quad \mathrm{Rot}_{z,\theta} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Consider the composition of the following three rotations $\mathrm{Rot}_{z,\gamma}\,\mathrm{Rot}_{x,\beta}\,\mathrm{Rot}_{z,\alpha} =$

$$\begin{bmatrix} \cos\gamma\cos\alpha - \sin\gamma\cos\beta\sin\alpha & -\cos\gamma\sin\alpha - \sin\gamma\cos\beta\cos\alpha & \sin\gamma\sin\beta \\ \sin\gamma\cos\alpha + \cos\gamma\cos\beta\sin\alpha & -\sin\gamma\sin\alpha + \cos\gamma\cos\beta\cos\alpha & -\cos\gamma\sin\beta \\ \sin\beta\sin\alpha & \sin\beta\cos\alpha & \cos\beta \end{bmatrix}.$$

You may think of the composition of these three rotations as follows: Each of the three rotations is a base change matrix. The first rotation, $\mathrm{Rot}_{z,\alpha} = \mathrm{M}_{\mathcal{K}',\mathcal{K}}$, rotates the unit vector of the $x$-axis and that of the $y$-axis by $-\alpha$ (and therefore rotates points with respect to $\mathcal{K}$ by $\alpha$). The next rotation $\mathrm{Rot}_{x,\beta} = \mathrm{M}_{\mathcal{K}'',\mathcal{K}'}$, rotates the unit vectors of the $y'$-axis and that of the $z'$-axis by $-\beta$ (and therefore rotates points with respect to $\mathcal{K}'$ by $\beta$). Similarly with the last rotation. The observation here is that $\mathrm{Rot}_{x,\beta}$ is a rotation around the *current x-axis*, i.e., a rotation around the first axis of the coordinate system that you are in. If we want to point out that at each step the coordinate system is changing, we may write $\mathrm{Rot}_{z'',\gamma}\,\mathrm{Rot}_{x',\beta}\,\mathrm{Rot}_{z,\alpha}$ for the overall rotation.

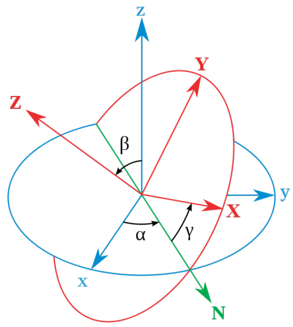



Figure 6.1: Euler angles[1]

**Proposition 6.19.** All rotations in dimension 3 are of this form, i.e., any matrix in SO(3) can be written in this form for some $\alpha,\beta,\gamma \in \mathbb{R}$.

**Definition 6.20.** You may restrict the range of the values $\alpha,\beta,\gamma$ by $\alpha \in [0,2\pi[$, $\beta \in [0,\pi[$ and $\gamma \in [0,2\pi[$. Then, each triple $(\alpha,\beta,\gamma)$ corresponds to a unique rotation $\mathrm{Rot}_{z,\gamma}\,\mathrm{Rot}_{x,\beta}\,\mathrm{Rot}_{z,\alpha} \in \mathrm{SO}(3)$. The angles $\alpha$, $\beta$ and $\gamma$ are called *Euler angles*. Another way of describing rotations in $\mathbb{E}^3$ is via quaternions (see Appendix H).

**Remark.** Euler angles give coordinates on SO(3). Not to be confused with spherical coordinates (see Appendix B).

### 6.3.3 Classification

**Definition 6.21.** A *glide-rotation* (or *helical displacement*) is the composition of a rotation in $\mathbb{E}^3$ with a translation parallel to the rotation axis.

**Theorem 6.22** (Chasles)**.** A direct isometry of the Euclidean space $\mathbb{E}^3$ is either

a) the identity, or

b) a translation, or

c) a rotation, or

d) a glide-rotation.

*Proof.* Let $\phi$ be the direct isometry given by $\phi(\mathbf{x}) = A\mathbf{x} + \mathbf{b}$ with respect to a right-oriented frame. We know that $A \in \mathrm{SO}(3)$. If $A = I_3$, then $\phi$ is a translation or the identity. Suppose that $A \neq I_3$. Applying Theorem 6.17 to $\mathbf{x} \mapsto A\mathbf{x}$, we see that $A$ is a rotation matrix. Let $\mathbf{v}$ be a direction vector of length 1 for the rotation axis of $A$ and decompose the vector $\mathbf{b}$ into its components parallel to $\mathbf{v}$ and orthogonal to $\mathbf{v}$:

$$\mathbf{b} = \mathbf{b}_1 + \mathbf{b}_2 \quad \text{where} \quad \mathbf{b}_1 \| \mathbf{v} \quad \text{and} \quad \mathbf{b}_2 \perp \mathbf{v}.$$

[1]Image source: Wikipedia

We have $\mathbf{b}_1 = \langle \mathbf{v}, \mathbf{b} \rangle \mathbf{v}$ and $\mathbf{b}_2 = (\mathbf{v} \times \mathbf{b}) \times \mathbf{v}$. Now consider the isometries

$$\phi_1(\mathbf{x}) = \mathbf{x} + \mathbf{b}_1 \quad \text{and} \quad \phi_2(\mathbf{x}) = A\mathbf{x} + \mathbf{b}_2.$$

Let $\pi$ be the plane passing through the origin and orthogonal to the rotation axis. Then $\phi_2$ is an isometry which leaves $\pi$ invariant ($\phi_2(\pi) = \pi$). By Chasles' Theorem in dimension 2 (Theorem 6.12), the restriction of $\phi_2$ to $\pi$ is a rotation around a fixed point $\mathbf{p} \in \pi$. Thus, $\phi_2$ is a rotation around the axis $\mathbf{p} + \mathbb{R}\mathbf{v}$. On the other hand, $\phi_1$ is a translation by $\mathbf{b}_1$ parallel to the rotation axis. This finishes the proof since $\phi = \phi_1 \circ \phi_2$. □

**Theorem 6.23.** An indirect isometry of the Euclidean space $\mathbb{E}^3$ leaves a plane $\pi$ invariant and is either

a) a reflection in $\pi$, or

b) the composition of a reflection in $\pi$ with a translation parallel to $\pi$, in which case it is called a *glide-reflection*, or

c) the composition of a reflection in $\pi$ with a rotation of axis orthogonal to $\pi$, in which case it is called a *rotation-reflection*.

*Proof.* Let $\phi$ be the indirect isometry given by $\phi(\mathbf{x}) = A\mathbf{x} + \mathbf{b}$ with respect to a right-oriented frame. One can show that a matrix $A \in \mathrm{O}(3)$ of determinant $-1$ admits $-1$ as an eigenvalue. Let $\mathbf{v}$ be an eigenvector for the eigenvalue $-1$, i.e., $A\mathbf{v} = -\mathbf{v}$. Notice that we also have $A^T\mathbf{v} = A^{-1}\mathbf{v} = -\mathbf{v}$. Calculating, we obtain

$$\begin{aligned} \langle \mathbf{v}, \phi(\mathbf{x}) \rangle &= \langle \mathbf{v}, A\mathbf{x} + \mathbf{b} \rangle \\ &= \langle \mathbf{v}, A\mathbf{x} \rangle + \langle \mathbf{v}, \mathbf{b} \rangle \\ &= \langle A^T\mathbf{v}, \mathbf{x} \rangle + \langle \mathbf{v}, \mathbf{b} \rangle \\ &= \langle -\mathbf{v}, \mathbf{x} \rangle + \langle \mathbf{v}, \mathbf{b} \rangle \end{aligned}$$

Thus, the plane

$$\pi : \langle \mathbf{v}, \mathbf{x} \rangle = \frac{1}{2} \langle \mathbf{v}, \mathbf{b} \rangle$$

is invariant under the isometry $\phi$. Moreover, if we choose the frame with the first basis vectors parallel to $\pi$ then

$$A = \begin{bmatrix} B & 0 \\ 0 & -1 \end{bmatrix}$$

and $\det(A) = \det(B)$. Thus, the restriction of $\phi$ to the plane $\pi$ is a direct isometry. Therefore, by Theorem 6.12, it is either the identity, or a translation or a rotation, which correspond to the three cases stated in the theorem. □

## 6.4 Moving points with isometries

### 6.4.1 Cycloids

Here we restrict to the Euclidean plane $\mathbb{E}^2$. Rotation matrices give an effective way of describing circular motions which can be used to construct curves as trajectories of a particle. Here we look at some cycloids. For this, let us first deduce the homogeneous matrix of a rotation around a point $C(c_1, c_2) \in \mathbb{E}^2$.

$$\begin{bmatrix} 1 & 0 & c_1 \\ 0 & 1 & c_2 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & -c_1 \\ 0 & 1 & -c_2 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \cos(\theta) & -\sin(\theta) & -c_1\cos(\theta) + c_2\sin(\theta) + c_1 \\ \sin(\theta) & \cos(\theta) & -c_1\sin(\theta) - c_2\cos(\theta) + c_2 \\ 0 & 0 & 1 \end{bmatrix}$$

Now, if you choose the center $C$ to be the point $(0,1)$, you have the following homogeneous rotation matrix

$$\begin{bmatrix} \cos(\theta) & -\sin(\theta) & \sin(\theta) \\ \sin(\theta) & \cos(\theta) & -\cos(\theta) + 1 \\ 0 & 0 & 1 \end{bmatrix}$$

If you move the origin with this rotation, you obtain

$$\begin{bmatrix} \cos(\theta) & -\sin(\theta) & \sin(\theta) \\ \sin(\theta) & \cos(\theta) & -\cos(\theta) + 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} \sin(\theta) \\ -\cos(\theta) + 1 \\ 1 \end{bmatrix}$$

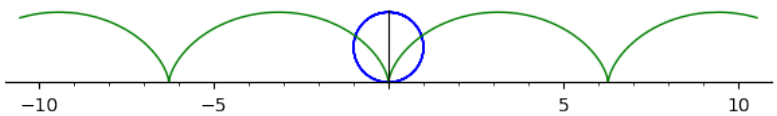
since the homogeneous coordinates of the origin are $(0,0,1)$. If you vary $\theta$ with time $t$, you are rotating the origin around $C$ counterclockwise. If you want to have a clockwise motion, you just change the sign of the angle to obtain

$$\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \mapsto \begin{bmatrix} -\sin(t) \\ -\cos(t) + 1 \\ 1 \end{bmatrix}.$$

Now, if at the same time $t$ you translate the point along the $x$-axis in the direction of $\mathbf{i}$, you get

$$\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \mapsto \begin{bmatrix} 1 & 0 & t \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -\sin(t) \\ -\cos(t) + 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -\sin(t) + t \\ -\cos(t) + 1 \\ 1 \end{bmatrix}.$$

What you obtain is the trajectory of the point $O$ as it rotates on the blue circle while the circle is moving like a wheel on the $x$-axis. The corresponding curve is called a cycloid:




Let's do something else. Instead of rotating the circle on the $x$-axis let us rotate it on a bigger circle centered at the origin. The small circle we can think of as the trajectory of the point $P(3,0)$ rotated around the center $C(4,0)$. The corresponding rotation matrix is

$$\begin{bmatrix} \cos(\theta) & -\sin(\theta) & -x_C\cos(\theta) + y_C\sin(\theta) + x_C \\ \sin(\theta) & \cos(\theta) & -x_C\sin(\theta) - y_C\cos(\theta) + y_C \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \cos(\theta) & -\sin(\theta) & -4\cos(\theta) + 4 \\ \sin(\theta) & \cos(\theta) & -4\sin(\theta) \\ 0 & 0 & 1 \end{bmatrix}$$

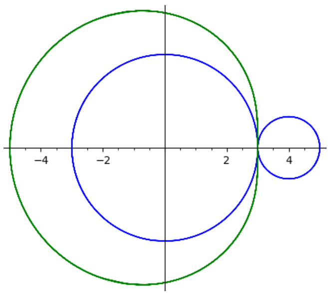
and thus, rotating $P(3,0)$ with time $t$ we obtain

$$\begin{bmatrix} \cos(t) & -\sin(t) & -4\cos(t)+4 \\ \sin(t) & \cos(t) & -4\sin(t) \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 3\cos(t)-4\cos(t)+4 \\ 3\sin(t)-4\sin(t) \\ 1 \end{bmatrix}$$

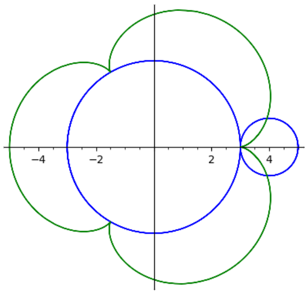
If at the same time we rotate around the origin, $P$ will move on the small circle which rotates around a big circle of radius 3 centered at the origin:

$$\begin{bmatrix} \cos(t') & -\sin(t') & 0 \\ \sin(t') & \cos(t') & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3\cos(t)-4\cos(t)+4 \\ 3\sin(t)-4\sin(t) \\ 1 \end{bmatrix} = \ldots$$

If we do this simultaneously, i.e., if we choose $t' = t$, then we obtain the following trajectory for $P$:




However, if we want the small circle to rotate like a wheel on the big circle, then, after an entire revolution of the small circle we need to have traversed the length of this circle on the big circle, i.e., $2\pi$. But the big circle is 3 times longer, so we need to choose $t = 3t'$, i.e., the rotation on the small circle is 3-times faster:




This is an example of an epicycloid.

### 6.4.2 Surfaces of revolution

Another way of looking at Euler angles $\mathrm{Rot}_{z,\gamma}\,\mathrm{Rot}_{x,\beta}\,\mathrm{Rot}_{z,\alpha}$ (see Section 6.3.2) is by considering the trajectory that you obtain on a given point when you vary $\alpha$, $\beta$ and $\gamma$. For instance

$$\mathrm{Rot}_{z,\gamma}\,\mathrm{Rot}_{x,\beta}\,\mathrm{Rot}_{z,\alpha}\begin{bmatrix}1\\0\\0\end{bmatrix}=\begin{bmatrix}\cos\gamma\cos\alpha-\sin\gamma\cos\beta\sin\alpha\\ \sin\gamma\cos\alpha+\cos\gamma\cos\beta\sin\alpha\\ \sin\beta\sin\alpha\end{bmatrix}.$$

If in this expression you fix $\gamma=0$ and let $\alpha\in[0,2\pi[$, $\beta\in[0,\pi[$ vary, you obtain

$$\begin{bmatrix}1\\0\\0\end{bmatrix}\mapsto\begin{bmatrix}\cos\alpha\\ \cos\beta\sin\alpha\\ \sin\beta\sin\alpha\end{bmatrix},$$

i.e., the trajectory of the point $(1,0,0)$ is a sphere and varying $\alpha$ and $\beta$ corresponds to the map

$$[0,2\pi[\times[0,\pi[\ni(\alpha,\beta)\mapsto\begin{bmatrix}\cos\alpha\\ \cos\beta\sin\alpha\\ \sin\beta\sin\alpha\end{bmatrix},\tag{6.4}$$

which is a parametrization of the sphere. You can get other parametrizations of the sphere if you fix $\alpha$ and let $\beta$ and $\gamma$ vary.

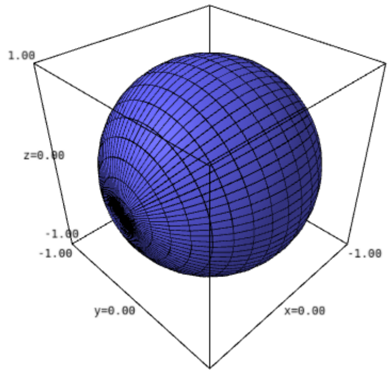
From a different perspective, notice that a plane in $\mathbb{E}^3$ can be described as the set of points which you touch if you translate a line in a given direction. This can be seen with the parametric equations:

$$\pi:\begin{bmatrix}x\\y\\z\end{bmatrix}=\begin{bmatrix}x_Q\\y_Q\\z_Q\end{bmatrix}+s\begin{bmatrix}v_x\\v_y\\v_z\end{bmatrix}+t\begin{bmatrix}w_x\\w_y\\w_z\end{bmatrix}=\underbrace{\left(\begin{bmatrix}x_Q\\y_Q\\z_Q\end{bmatrix}+s\begin{bmatrix}v_x\\v_y\\v_z\end{bmatrix}\right)}_{\text{line }\ell}\quad\underbrace{+t\begin{bmatrix}w_x\\w_y\\w_z\end{bmatrix}}_{\text{translation with }t\mathbf{w}}\tag{6.5}$$

This is a general method of constructing surfaces starting from curves: you start with a curve in $\mathbb{E}^3$ and apply a motion to it. What you obtain, if non-degenerate, is a parametrization of a surface. This can be exemplified with the parametrization of the sphere in (6.4) which you can rewrite as follows:

$$(\alpha,\beta)\mapsto\underbrace{\begin{bmatrix}1&0&0\\0&\cos\beta&-\sin\beta\\0&\sin\beta&\cos\beta\end{bmatrix}}_{\text{rotation }\mathrm{Rot}_{x,\beta}}\underbrace{\begin{bmatrix}\cos\alpha\\ \sin\alpha\\ 0\end{bmatrix}}_{\text{circle in }Oxy\text{-plane}}=\begin{bmatrix}\cos\alpha\\ \cos\beta\sin\alpha\\ \sin\beta\sin\alpha\end{bmatrix}.$$

This describes the unit sphere centered at the origin as the set of points which you touch with the unit circle centered at the origin in the $Oxy$-plane if you rotate the circle around the $x$-axis.



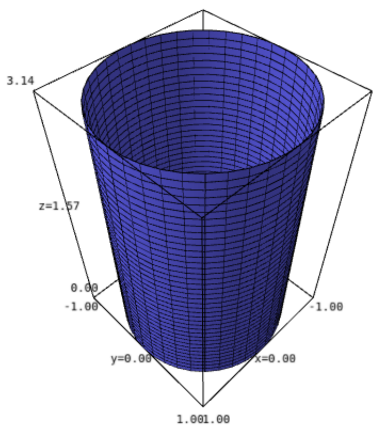

**Definition 6.24.** A *surface of revolution* in $\mathbb{E}^3$ is a surface obtained by rotating a curve around a line $\ell$. The line $\ell$ is called the *axis* of the surface.

**Example 6.25.** In (6.5), instead of translating the line $\ell$ we can rotate it around a line which is parallel to $\ell$. In this way we obtain a cylinder. For example, if

$$\ell : \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + s \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

and if we rotate around the $z$-axis we obtain a parametrization of a cylinder of radius 1 and axis the $z$-axis:

$$(\theta, s) \mapsto \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \left( \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + s \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right) = \begin{bmatrix} \cos\theta \\ \sin\theta \\ s \end{bmatrix}$$



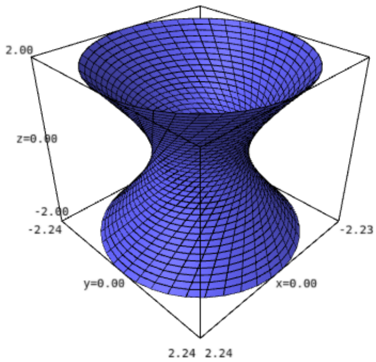

**Example 6.26.** If instead we consider two skew lines and rotate one around the other, we obtain

hyperboloids of revolution. For example, if

$$\ell : \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + s \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

and if we rotate around the $z$-axis we obtain a parametrization of a hyperboloid

$$(\theta, s) \mapsto \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \left( \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + s \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right) = \begin{bmatrix} \cos\theta - s\sin\theta \\ \sin\theta + s\cos\theta \\ s \end{bmatrix}.$$



CHAPTER 7

# Curves and surfaces

## Contents

## 7.1 Smooth curves in $\mathbb{E}^n$

**Definition 7.1.** A *parametrized curve* of the Euclidean space $\mathbb{E}^n$ is a map

$$\gamma : I \to \mathbb{E}^n$$

where $I$ is an interval of $\mathbb{R}$. Then, with respect to an orthonormal coordinate system $\mathcal{K}$ of $\mathbb{E}^n$ we have

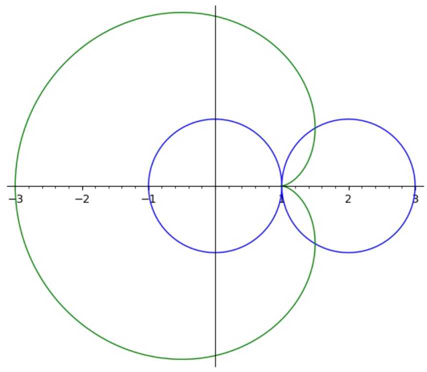
$$[\gamma]_{\mathcal{K}} : \mathbb{R} \supset I \to \mathbb{R}^n \quad [\gamma(t)]_{\mathcal{K}} = \begin{bmatrix} \gamma_1(t) \\ \vdots \\ \gamma_n(t) \end{bmatrix}$$

where $\gamma_1, \ldots, \gamma_n : I \to \mathbb{R}$. When there is no risk of confusion, we write $\gamma(t)$ instead of $[\gamma(t)]_{\mathcal{K}}$. The parametrized curve $\gamma$ is called *smooth* if $\gamma_1, \ldots, \gamma_n$ are smooth, meaning they are infinitely differentiable (i.e., have derivatives of all orders). The parametrized curve $\gamma$ is called *piecewise smooth* if it is smooth except at a finite number of points.

Geometrically, curves are certain sets of points in a Euclidean space $\mathbb{E}^n$. The curve $\mathcal{C}$ corresponding to the parametrization $\gamma$ is the image of the map $\gamma$

$$\mathcal{C} = \left\{ \gamma(t) : t \in I \right\}.$$

It has many different parametrizations. We say that a curve $\mathcal{C}$ is smooth if it admits a smooth parametrization. Similarly, a curve is called piecewise smooth if it admits a piecewise smooth parametrization.




**Example 7.2.** The *cardioid* is a curve obtained as the trajectory of a point on a circle that is rolling around a fixed circle of the same radius. Let $a$ be the radius of the two circles. As in Section 6.4, we may deduce a parametrization of such a curve using rotations. For this, let the fixed circle be centered at the origin, let the rolling circle have initial center $(2a, 0)$ and consider the trajectory of the contact point $(a, 0)$ of the two circles in the initial position. We obtain

$$\gamma(t) = \underbrace{\mathrm{Rot}_t}_{\text{rotation around the fixed circle}} \circ \underbrace{T_{(2a,0)} \circ \mathrm{Rot}_t \circ T_{-(2a,0)}}_{\text{rotation of the moving circle}} \cdot \begin{bmatrix} a \\ 0 \end{bmatrix} = \begin{bmatrix} a + 2a(1-\cos(t))\cos(t) \\ 2a(1-\cos(t))\sin(t) \end{bmatrix}$$

Notice that if we translate the reference frame to the right by $(a, 0)$, the parametrization changes to

$$\gamma(t) = \begin{bmatrix} 2a(1-\cos(t))\cos(t) \\ 2a(1-\cos(t))\sin(t) \end{bmatrix} \tag{7.1}$$

since the coordinates are modified by $(-a, 0)$. We will take this to be the parametrization of our cardioid.

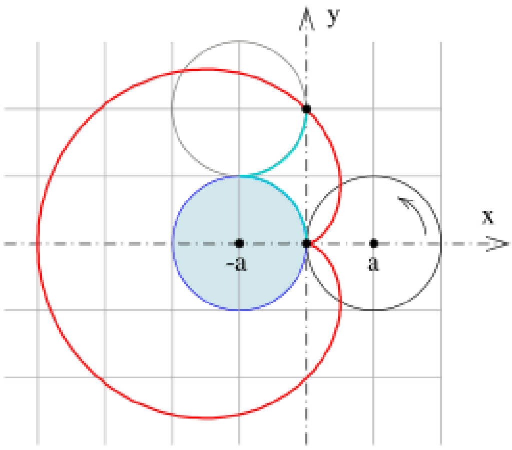
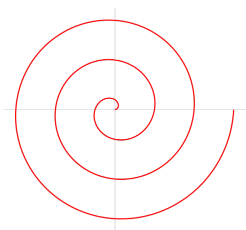



Figure 7.1: Cardioid (Image source: Wikipedia)

**Example 7.3.** The *Archimedean spiral* is defined as the trajectory of a point obtained by a rotation with time $t$ around a point $O$, followed by a homothety with factor $|t|$ centered at the same point $O$. More precisely, if we choose the center $O$ to be the origin, then a parametrization of this curve is given by

$$\gamma(t) = \begin{bmatrix} t\cos(t) \\ t\sin(t) \end{bmatrix} \tag{7.2}$$

The associated *conical Archimedean spiral* is obtained by simultaneously translating with time $t$ in a direction orthogonal to the spiral.



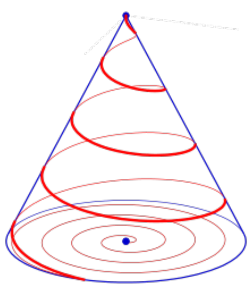


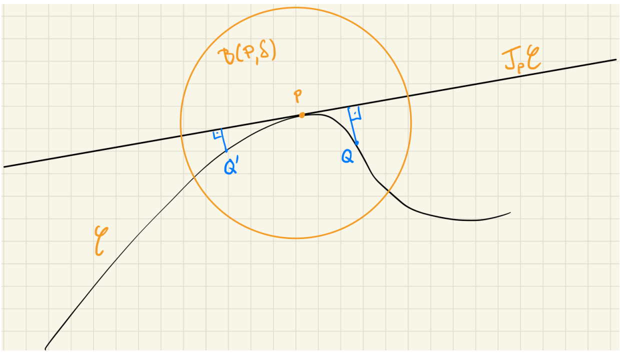
Figure 7.2: Archimedean spiral and associated conical spiral (Image source: Wikipedia)

**Definition 7.4.** Let $\mathcal{C}$ be a piecewise smooth curve and $P$ a point on $\mathcal{C}$. A line $\ell$ is called *tangent at $P$ to $\mathcal{C}$* if the points on the curve are arbitrarily close to $\ell$ in a small enough neighborhood of $P$. More concretely, if for any $\varepsilon > 0$ there exists $\delta > 0$ such that

$$\text{for all } Q \in \mathcal{C} \cap B(P,\delta) \setminus \{P\} \quad \text{we have} \quad \frac{d(Q,\ell)}{d(P,Q)} < \varepsilon$$

where $B(P,\delta)$ is the open ball centered at $P$ and of radius $\delta$. If the line $\ell$ is tangent at the point $P$ to the curve $\mathcal{C}$, then it is called *the tangent line to $\mathcal{C}$ at $P$* and it is denoted by

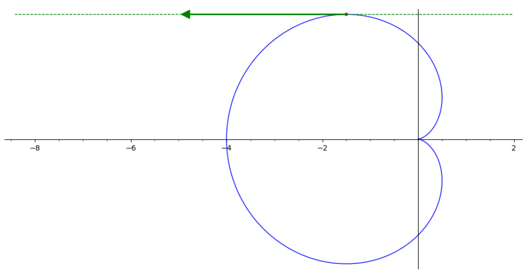
$$\mathcal{T}_P\mathcal{C}.$$




**Proposition 7.5.** Let $\mathcal{C}$ be a smooth curve with parametrization $\gamma : I \to \mathbb{E}^n$. Consider the point $P = \gamma(t_0)$ for some $t_0 \in I$. If $\dot{\gamma}(t_0) \neq \mathbf{0}$, then the *velocity vector at $\gamma(t_0)$*,

$$\dot{\gamma}(t_0) = \begin{bmatrix} \frac{d\gamma_1}{dt}(t_0) \\ \vdots \\ \frac{d\gamma_n}{dt}(t_0) \end{bmatrix}, \quad \text{is a direction vector for} \quad \mathcal{T}_P\mathcal{C}.$$

**Example 7.6.** For the cardioid parametrized with Equation (7.1), one calculates that for $t_0 = 2\pi/3$ we have $\dot{\gamma}(t_0) = (-2a\sqrt{3}, 0)$.




**Definition 7.7.** The *tangent space* to the smooth curve $\mathcal{C}$ is the 1-dimensional vector subspace generated by the velocity vector

$$T_P\mathcal{C} := \langle \dot{\gamma}(t_0) \rangle$$

for any parametrization $\gamma$ of $\mathcal{C}$. It is the direction space of the tangent line at $P$, i.e.,

$$T_P\mathcal{C} = \mathrm{D}(\mathcal{T}_P\mathcal{C}).$$

### 7.1.1 Kinematic versus geometric properties

Let $\mathcal{C}$ be a curve with smooth parametrization $\gamma : I \to \mathbb{E}^n$. Quantities (or objects) defined starting from the curve $\gamma(I) = \mathcal{C}$, that depend only on $\mathcal{C}$ and not on the parametrization $\gamma$, are called *geometric quantities* (or *objects*); whereas those that depend on the parametrization $\gamma$ are called *kinematic quantities* (or *objects*). Note, in particular, that the parametrization $\gamma$ depends on the chosen frame $\mathcal{K}$, whereas geometric properties are independent of $\mathcal{K}$. For example, the velocity vector and the acceleration vector

$$\dot{\gamma}(t_0) := \frac{d\gamma}{dt}(t_0) = \begin{bmatrix} \frac{d\gamma_1}{dt}(t_0) \\ \vdots \\ \frac{d\gamma_n}{dt}(t_0) \end{bmatrix} \quad \text{and} \quad \ddot{\gamma}(t_0) = \frac{d^2\gamma}{dt^2}(t_0) = \begin{bmatrix} \frac{d^2\gamma_1}{dt^2}(t_0) \\ \vdots \\ \frac{d^2\gamma_n}{dt^2}(t_0) \end{bmatrix}$$

are kinematic objects. Therefore, the *velocity* $V_\gamma(t) = |\dot{\gamma}(t)|$ and the *acceleration* $A_\gamma(t) = |\ddot{\gamma}(t)|$ are kinematic quantities.

**Definition 7.8.** The *arc length* of a parametrized curve $\gamma : I \to \mathbb{R}^n$ is the integral of its velocity

$$\ell(\gamma) = \int_I V_\gamma(t)dt.$$

**Example 7.9.** Consider the cardioid in Example 7.2. We have

$$\gamma(t) = 2a\begin{bmatrix}(1-\cos(t))\cos(t) \\ (1-\cos(t))\sin(t)\end{bmatrix} \quad \text{and therefore} \quad \dot{\gamma}(t) = 2a\begin{bmatrix}\sin(2t)-\sin(t) \\ \cos(t)-\cos(2t)\end{bmatrix}.$$

Thus

$$\begin{aligned}
V_\gamma(t)^2 &= 4a^2\Big((\sin(2t)-\sin(t))^2 + (\cos(t)-\cos(2t))^2\Big) \\
&= 4a^2\Big(2 - 2\sin(2t)\sin(t) - 2\cos(t)\cos(2t)\Big) \\
&= 4a^2\Big(2 - 4\cos(t)\sin(t)^2 - 2\cos(t)(\cos(t)^2 - \sin(t)^2)\Big) \\
&= 4a^2\Big(2 - 2\cos(t)\sin(t)^2 - 2\cos(t)^3\Big) \\
&= 4a^2\Big(2 - 2\cos(t)\Big) \\
&= 16a^2\sin(\frac{t}{2})^2 \quad \text{therefore} \quad V_\gamma(t) = 4a\sin(\frac{t}{2})
\end{aligned}$$

for $I = [0, 2\pi]$ since $\frac{t}{2} \in [0, \pi]$. Then

$$\ell(\gamma) = \int_0^{2\pi} 4a\sin(\frac{t}{2})dt = -8a\cos(\frac{t}{2})\Big|_0^{2\pi} = 16a.$$

**Proposition 7.10.** The arc length of a parametrized curve is a geometric quantity.

**Definition 7.11.** The vector field of *normalized velocity vectors* of a parametrized curve $\gamma : I \to \mathbb{R}^n$ is denoted by

$$\mathbf{T}(\gamma, t) := \frac{\dot{\gamma}(t)}{V_\gamma(t)}.$$

**Example 7.12.** Continuing with Example 7.9, for the cardioid $\gamma : [0, 2\pi] \to \mathbb{R}^2$, we have

$$\mathbf{T}(\gamma, t) = \frac{1}{2\sin(\frac{t}{2})} \begin{bmatrix} \sin(2t) - \sin(t) \\ \cos(t) - \cos(2t) \end{bmatrix}.$$

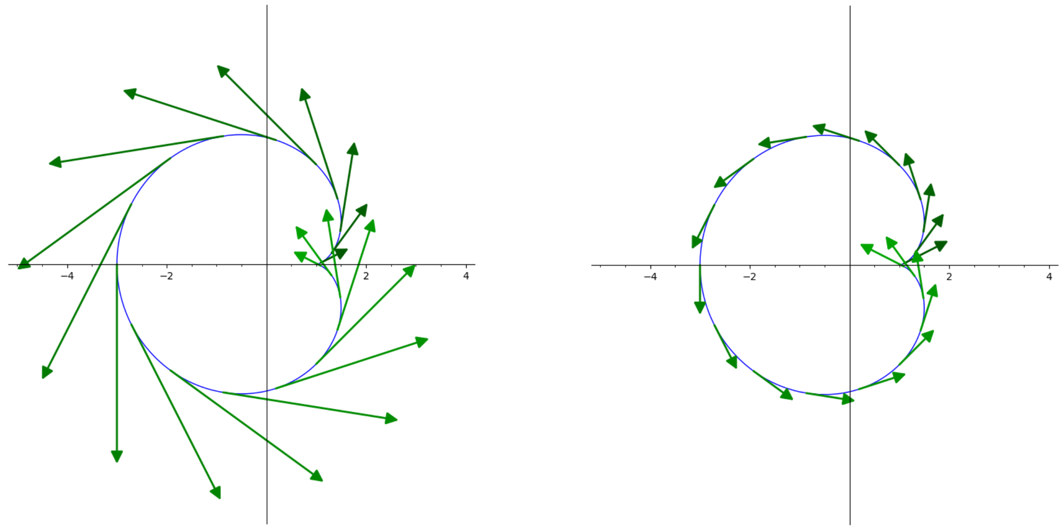



Figure 7.3: Velocity vectors and normalized tangent vectors.

**Proposition 7.13.** Up to sign, the vector field $\mathbf{T}(\gamma, t)$ is a geometric object attached to the curve $\mathcal{C} = \gamma(I)$.

**Definition 7.14.** We say that a parametrization $\gamma : I \to \mathbb{R}^2$ is a *natural parametrization* if $V_\gamma(t) = 1$ for all $t \in I$.

**Proposition 7.15.** Any regular smooth curve (i.e., one with non-zero velocity everywhere) admits a natural parametrization, i.e., it can be reparametrized by arc length.

**Definition 7.16.** The vector field of curvature vectors of a parametrized curve $\gamma : I \to \mathbb{R}^n$ is, by definition,

$$\mathbf{K}(\gamma, t) := \frac{1}{V_\gamma(t)} \frac{d}{dt} \mathbf{T}(\gamma, t).$$

The curvature of the curve $\mathcal{C} = \gamma(I)$ at the point $\gamma(t)$ is

$$\kappa(\gamma, t) = |\mathbf{K}(\gamma, t)|.$$

**Proposition 7.17.** If $\gamma : I \to \mathbb{R}^2$ is a natural parametrization, then

$$\mathbf{K}(\gamma, t) = \ddot{\gamma}(t) \quad \text{and} \quad \kappa(\gamma, t) = |\ddot{\gamma}(t)|.$$

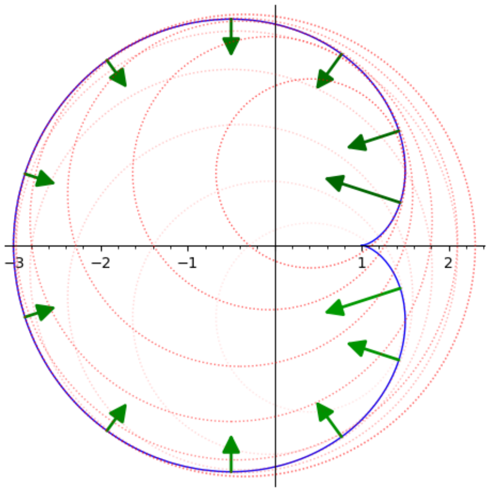


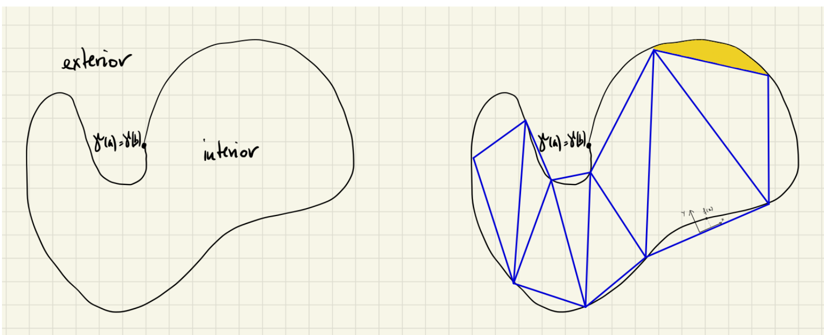

Figure 7.4: Curvature vectors and osculating circles.

**Proposition 7.18.** Up to sign, the vector field $\mathbf{K}(\gamma, t)$ is a geometric object attached to the curve $\mathcal{C} = \gamma(I)$. In particular, the curvature $\kappa(\gamma, t)$ is a geometric quantity.

**Proposition 7.19.** A parametrized curve $\gamma : [a, b] \to \mathbb{R}^n$ has zero curvature if and only if it is a line segment.

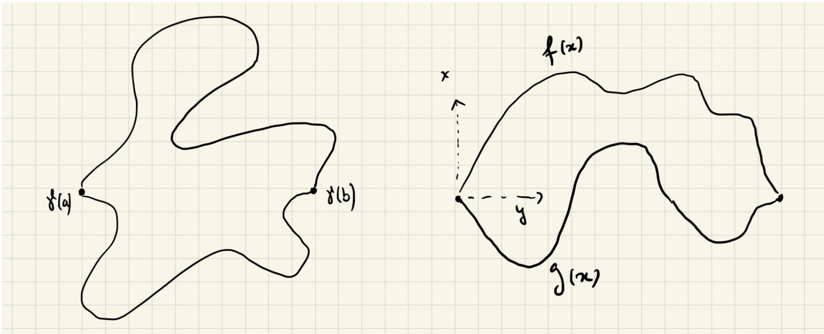
### 7.1.2 Area enclosed by a simple planar loop

**Definition 7.20.** A parametrized curve $\gamma : [a, b] \to \mathbb{E}^2$ is called *simple* if it is injective on $(a, b)$, i.e., if it doesn't have *self-intersection points*. It is a *loop* if $\gamma(a) = \gamma(b)$. A *Jordan curve* is a simple loop.



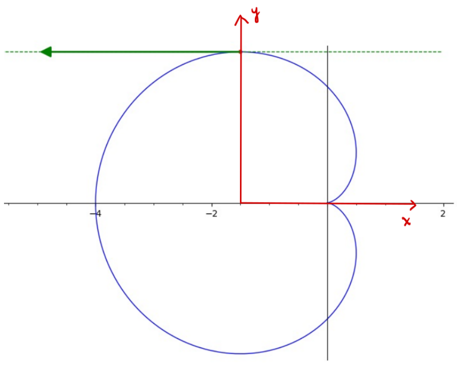

A Jordan curve divides the plane into an interior and an exterior. In some cases, it is possible to compute the area of the interior. One may try to approximate the interior with a polygon (the area of which can be calculated by subdividing into triangles) and the remaining pieces (yellow in the picture) may be calculated by choosing the frame such that the corresponding portion of $\gamma$ is described by a function along the sides of the polygon, which can be integrated using standard calculus techniques. While this approach is theoretically straightforward, it is often impractical in most cases. Instead, one may approximate the interior using polygons with a large number of vertices.

**Example 7.21.** One may think of a parametrization $\gamma : [a,b] \to \mathbb{E}^2$ as a *curved segment*. There are many curved segments between two points. Choosing two such segments that intersect only at their endpoints, we obtain a simple loop.




*If* the two curved segments can be described as graphs of functions – i.e., $\gamma_1(t) = (t, f(t))$ for some function $f : [a,b] \to \mathbb{R}$ and similarly for the second parametrized curved segment $\gamma_2$ – then the area enclosed between them can be computed by ordinary integration.

**Example 7.22.** Consider again the case of the cardioid with parametrization $\gamma : I \to \mathbb{R}^2$ given in (7.1). From the symmetry of the curve, we see that the area of the region enclosed by the cardioid is twice the area above the $x$-axis. This corresponds to $t \in [0, \pi]$.




From Example 7.6, we see that the $x$-values of $\gamma$ are a function of $y$ for $t \in [0, \frac{2\pi}{3}]$ and $[\frac{2\pi}{3}, \pi]$, respectively. Moreover, $\gamma(\frac{2\pi}{3}) = a(-\frac{3}{2}, \frac{3\sqrt{3}}{2})$. We may therefore translate the current frame with the vector $(-\frac{3a}{2}, 0)$ to arrive in the red frame (see above image). In this frame, we have

$$\gamma(t) = \begin{bmatrix} 2a(1-\cos(t))\cos(t) + \frac{3a}{2} \\ 2a(1-\cos(t))\sin(t) \end{bmatrix}.$$

Thus

$$\text{Area}_{\text{int}}(\mathcal{C}) = 2\int_0^{\frac{2\pi}{3}} x(t)\dot{y}(t)dt + 2\int_{\frac{2\pi}{3}}^{\pi} x(t)\dot{y}(t)dt$$

and, ignoring the constant $4a^2$, we calculate:

$$\int_0^{\frac{2\pi}{3}} \frac{x(t)\dot{y}(t)}{4a^2}dt = \int_0^{\frac{2\pi}{3}} \underbrace{\left((1-\cos(t))\cos(t)+\frac{3}{4}\right)}_{x(t)/2a}\underbrace{\left(\sin(t)^2-\cos(t)^2+\cos(t)\right)}_{\dot{y}(t)/2a}dt = \cdots = \frac{\pi}{2} - \frac{9\sqrt{3}}{32}$$

where one opens the parentheses of the integrand and calculates $\int \sin(t)^n\cos(t)^m dt$ based on the different values of $n$ and $m$. Similarly, on the other side of the red $y$-axis, we obtain

$$\int_{\frac{2\pi}{3}}^{\pi} x(t)\dot{y}(t)dt = 4a^2\left(\frac{\pi}{4}+\frac{9\sqrt{3}}{32}\right)$$

and therefore

$$\text{Area}_{\text{int}}(\mathcal{C}) = 6\pi a^2.$$

The above calculation is tedious. In many cases where a curve is defined by rotations, it is much easier to integrate using polar coordinates (Appendix B). Notice that Equation (7.1) easily translates into polar coordinates:

$$\begin{cases} x(t) = 2a(1-\cos(t))\cos(t) \\ y(t) = 2a(1-\cos(t))\sin(t) \end{cases} \quad \Leftrightarrow \quad r(t) = 2a(1-\cos(t)) \quad t\in[0,2\pi[$$

where $r = \sqrt{x^2+y^2}$. Then, with Theorem 7.23 below, we calculate

$$\begin{aligned}\text{Area}_{\text{int}}(\mathcal{C}) &= 2a^2\int_0^{2\pi}(1-\cos(t))^2dt \\ &= 2a^2\int_0^{2\pi}(1-2\cos(t)+\cos(t)^2)dt \\ &= 2a^2\int_0^{2\pi}(\frac{3}{2}-2\cos(t)+\frac{1}{2}\cos(2t))dt \\ &= 2a^2\,(\frac{3}{2}t-2\sin(t)+\frac{1}{4}\sin(2t))\Big|_0^{2\pi} \\ &= 6\pi a^2.\end{aligned}$$

**Theorem 7.23.** Let $\gamma$ be a parametrized curve, given in polar coordinates $(r,\theta)$ by the equation $r = f(\theta)$. Then

$$\text{Area}(\gamma,\theta_1,\theta_2) = \frac{1}{2}\int_{\theta_1}^{\theta_2} f(\theta)^2 d\theta.$$

## 7.2 Smooth hypersurfaces in $\mathbb{E}^n$

The term *hypersurfaces* refers to objects that generalize curves in $\mathbb{E}^2$ and surfaces in $\mathbb{E}^3$ to the $n$-dimensional setting – in the same way that hyperplanes generalize lines and planes. Loosely speaking, hypersurfaces are bent hyperplanes or portions thereof. There are two ways of describing such objects relative to a Cartesian frame: through equations (as zero sets of certain functions) or through parametrizations.

### 7.2.1 Level sets

Given a function $f:\mathbb{E}^n\to\mathbb{R}$, the *zero set of $f$* is

$$Z(f)=\{P\in\mathbb{E}^n : f(P)=0\}.$$

It is a particular case of the *level set of $f$*

$$L(f,c)=\{P\in\mathbb{E}^n : f(P)=c\}$$

*for the value $c\in\mathbb{R}$.*

**Example 7.24.** If we are in dimension 3 and $f(x,y,z)=x^2+y^2+z^2$ relative to some frame, then $Z(f)$ equals the origin, while for $c>0$ we get spheres of radius $\sqrt{c}$:

$$L(f,c)=Z(f-c)=\{(x,y,z): x^2+y^2+z^2=c\}$$

and for $c<0$ we have the empty set.

**Example 7.25.** In $\mathbb{E}^n$, the function $f(x_1,\dots,x_n)=a_1x_1+\dots+a_nx_n$ (where not all coefficients $a_i$ are zero) describes a family of parallel hyperplanes. Indeed,

$$L(f,c)=Z(f-c)=\{(x_1,\dots,x_n): a_1x_1+\dots+a_nx_n=c\}$$

are hyperplanes with normal vector $\mathbf{n}(a_1,\dots,a_n)$; thus, all are parallel to $Z(f)$, which passes through the origin.

This example generalizes to affine maps. If $\phi:\mathbb{A}^n\to\mathbb{A}^m$ is an affine map given by $\phi(\mathbf{x})=A\mathbf{x}+b$, then $Z(\phi)$ is an affine subspace of $\mathbb{A}^n$ defined by the system $A\mathbf{x}+b=0$. Similarly, for any $c\in\mathbb{A}^m$, the level set $L(\phi,c)$ is the affine subspace of $\mathbb{A}^n$ defined by the system $A\mathbf{x}=c-b$ (again a family of parallel subspaces, since the corresponding homogeneous system is the same).

**Definition 7.26.** While in the linear case (Example 7.25) level sets always define hypersurfaces (hyperplanes), this is no longer true in general (as can be seen in Example 7.24). In Example 7.24, only the level sets $L(f,c)$ with $c>0$ are considered surfaces. To avoid empty sets, e.g., $L(f,-1)$, we may simply ask that the level set is non-empty. (However, notice that if we work over the complex numbers $\mathbb{C}$, then $L(f,-1)$ is non-empty; it is a so-called complex sphere.) The second degenerate case, where the level set is a point, i.e., $L(f,0)$, can be avoided by asking that the Jacobian $J(f)=(\partial_{x_1}f,\dots,\partial_{x_n}f)$ is non-zero.

We say that a *smooth hypersurface* is the set of solutions to an equation of the form

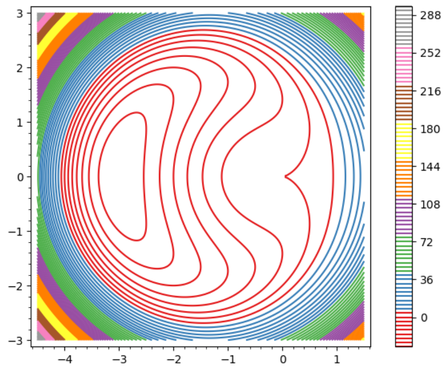
$$L(f,c): f(\mathbf{x}) = c$$

where $f:\mathbb{R}^n \to \mathbb{R}$ is a smooth function such that there exists $\mathbf{p} \in \mathbb{R}^n$ with $f(\mathbf{p}) = c$ and such that $J(f)(\mathbf{q})$ is non-zero for all $\mathbf{q} \in L(f,c)$.

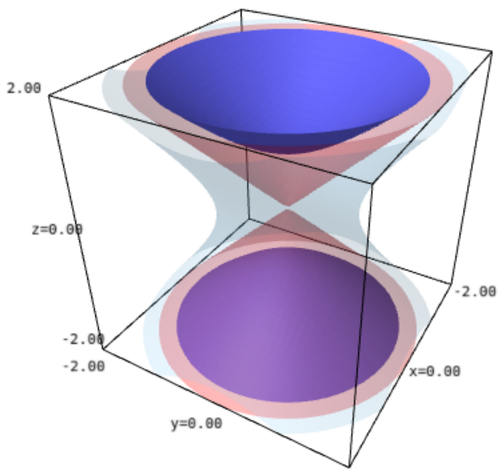
**Example 7.27.** Consider again the case of the cardioid $\mathcal{C}$ given via the parametrization (7.1). One can show that it satisfies the equation

$$\mathcal{C}: f(x,y) = 0 \quad \text{where} \quad f(x,y) = (x^2+y^2)^2 + 4ax(x^2+y^2) - 4a^2y^2.$$

Various level sets of the function $f$ are visible in the following image.




**Example 7.28.** In 3-dimensional space, consider the function $f(x,y,z) = x^2+y^2-z^2$. Then the level set $L(f,0)$ is a cone, $L(f,c)$ is a hyperboloid of two sheets if $c<0$, and a hyperboloid of one sheet if $c>0$.



### 7.2.2 Local parametrization

The second way of describing a hypersurface is via parameters. We have already seen how to construct surfaces via motion (e.g., the cone and the hyperboloid discussed in Section 6.4), which produced parametrizations $\gamma : V \to \mathbb{R}^3$ with $V$ a product of intervals.

In view of the definition in terms of equations (Definition 7.26), it is natural to ask how the two descriptions are related. In particular, if a hypersurface is given by an equation, can we describe it via a parametrization? The following proposition gives an affirmative answer locally. It is a particular case of the Implicit Function Theorem.

**Proposition 7.29.** Let $f : \mathbb{R}^n \to \mathbb{R}$ be a smooth function, and consider the zero set

$$S : f(\mathbf{x}) = 0 \quad \text{and a point} \quad \mathbf{p}(p_1, \dots, p_n) \in S.$$

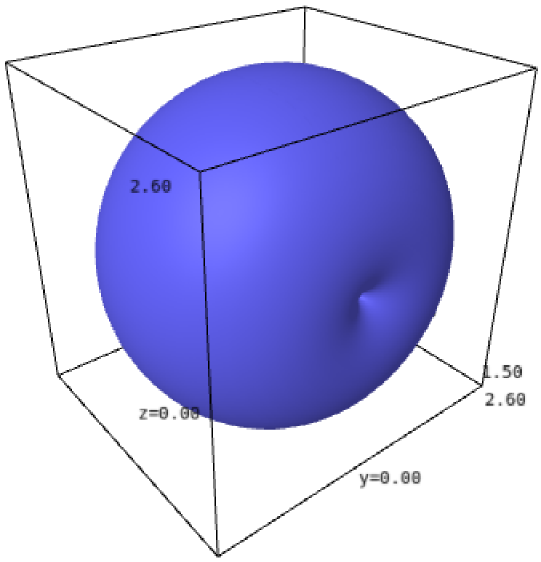
If the Jacobian $J(f)(\mathbf{p})$ is non-zero, then there exists a neighborhood $U$ of $\mathbf{p}$, an open subset $V \subseteq \mathbb{R}^{n-1}$, and a smooth injective map $\gamma : V \to \mathbb{R}^n$ such that

$$\forall \mathbf{q} \in U : \quad f(\mathbf{q}) = 0 \quad \Leftrightarrow \quad \exists (t_1, \dots, t_{n-1}) \in V : \quad \mathbf{q} = \gamma(t_1, \dots, t_{n-1})$$

i.e., locally, in the neighborhood $U$ of the point $\mathbf{p}$, the map $\gamma$ is a parametrization of $S$.

### 7.2.3 Volume enclosed by a simple surface of revolution

**Example 7.30.** Rotating the cardioid around the $x$-axis yields the following apple-shaped surface. Its volume is $\frac{64\pi a^3}{3}$.




**Example 7.31.** Consider a circle of radius $r$ in the $Oyz$-plane centered at $(0, R, 0)$. A torus $\mathcal{T}$ is obtained by rotating (revolving) this circle around the $z$-axis. Then $r$ is the *minor radius* and $R$ is the *major*

*radius* of the torus. From this description, it is not difficult to obtain a parametrization. Such a torus has volume $2\pi^2 R r^2$. Moreover, one may show that

$$\mathcal{T} : f(x,y,z) = 0 \quad \text{where} \quad f(x,y,z) = (x^2+y^2+z^2+R^2-r^2)^2 - 4R^2(x^2+y^2).$$

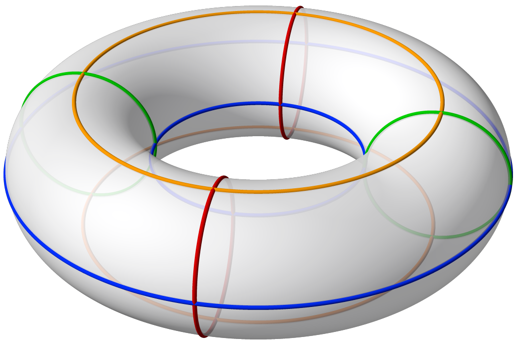


Figure 7.5: A torus[1]

[1] Image source: Wikipedia

# CHAPTER 8

## Quadratic curves (conics)

### Contents

**Definition 8.1.** A *quadratic curve* (or *conic*) in $\mathbb{E}^2$ is a curve defined by a quadratic equation

$$ax^2 + bxy + cy^2 + dx + ey + f = 0$$

where $a, b, c, d, e, f \in \mathbb{R}$.

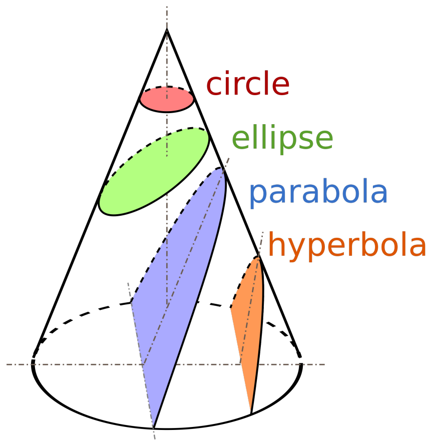
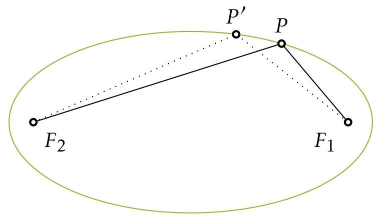



Figure 8.1: Conic sections[1]

## 8.1 Ellipse

### 8.1.1 Geometric description




**Definition 8.2.** An *ellipse* is the locus of points in $\mathbb{E}^2$ for which the sum of the distances from two fixed points, called *focal points* (or *foci*), is constant.

### 8.1.2 Canonical equation

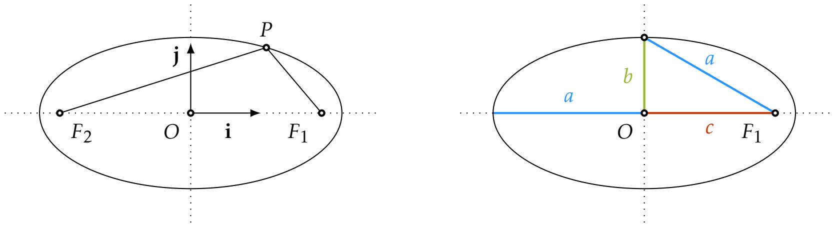
In general, we can describe a conic with a *canonical equation*, obtained with respect to a suitably chosen coordinate frame.

[1]Image source: Wikipedia

**Proposition 8.3.** Let $F_1$ and $F_2$ be two points in $\mathbb{E}^2$ and let $a > 0$ be a real number. Choose the coordinate system $Oxy = (O, \mathbf{i}, \mathbf{j})$ such that $F_1$ and $F_2$ are on the $x$-axis, such that $\overrightarrow{F_2F_1}$ has the same direction as $\mathbf{i}$, and such that $O$ is the midpoint of $[F_1F_2]$. With these choices, the ellipse with focal points $F_1$ and $F_2$ for which the sum of distances from the focal points is $2a$ has the equation

$$\mathcal{E}_{a,b} : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \tag{8.1}$$

for some number $b > 0$. We denote this ellipse by $\mathcal{E}_{a,b}$ and call $a$ the *semi-major axis* and $b$ the *semi-minor axis*.




- Equation (8.1) is called the *canonical equation of the ellipse* $\mathcal{E}_{a,b}$. Clearly, with respect to some other coordinate system, the same ellipse will have a different equation.
- If $2c$ denotes the distance between $F_1$ and $F_2$, then $b^2 = a^2 - c^2$.
- The intersections of $\mathcal{E}_{a,b}$ with the coordinate axes are the points $(\pm a, 0)$ and $(0, \pm b)$.
- The numerical quantity
$$\varepsilon = \frac{c}{a} = \sqrt{1 - \frac{b^2}{a^2}} \in [0,1)$$
is called the *eccentricity* of the ellipse $\mathcal{E}_{a,b}$. It measures the flatness or roundness of the ellipse.
- The canonical equation shows that $M(x_M, y_M) \in \mathcal{E}_{a,b}$ if and only if $(\pm x_M, \pm y_M) \in \mathcal{E}_{a,b}$.

### 8.1.3 Parametric equations

Parametric equations are never unique. Depending on the intended application, one parametrization may be preferred over another. Equation (8.1) allows us to express $y$ in terms of $x$:

$$y(x) = \pm \frac{b}{a}\sqrt{a^2 - x^2}.$$

This gives a partial parametrization of $\mathcal{E}_{a,b}$. For the "northern part" we have the parametrization

$$\phi : [-a, a] \to \mathbb{E}^2 \quad \text{given by} \quad \phi(x) = \big(x, y(x)\big) = \Big(x, \frac{b}{a}\sqrt{a^2 - x^2}\Big).$$

This is the graph of the function

$$y(x) = \frac{b}{a}\sqrt{a^2 - x^2} \quad \text{for which} \quad y'(x) = \frac{-bx}{a\sqrt{a^2 - x^2}} \quad \text{and} \quad y''(x) = -\frac{ab}{(a^2 - x^2)\sqrt{a^2 - x^2}}.$$

Thus, we can use the known methods to verify the monotonicity and the convexity of $y(x)$, which describes this part of the ellipse.

A second way of parametrizing the ellipse $\mathcal{E}_{a,b}$ is with

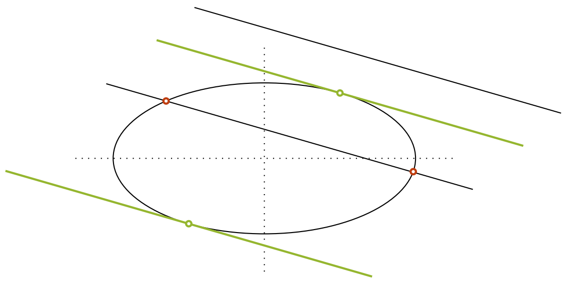
$$\phi : \mathbb{R} \to \mathbb{E}^2 \quad \text{defined by} \quad \phi(t) = (a\cos(t), b\sin(t)).$$

It is easy to check using equation (8.1) that this is a parametrization. Moreover, with this parametrization, we may view the ellipse as the orbit of a rotation followed by a dilation along the coordinate axes:

$$t \mapsto \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \begin{bmatrix} \cos(t) & -\sin(t) \\ \sin(t) & \cos(t) \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

We call such a transformation *elliptical motion*.

### 8.1.4 Relative position of a line



Consider the canonical equation (8.1) of the ellipse $\mathcal{E}_{a,b}$. Let $\ell$ be a line with equation $y = kx + m$. The intersection of the two objects is the set of points whose coordinates are solutions to the system

$$\begin{cases} \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \\ y = kx + m \end{cases} \Leftrightarrow \begin{cases} \frac{x^2}{a^2} + \frac{(kx+m)^2}{b^2} = 1 \\ y = kx + m \end{cases}.$$

The solutions to this system are $(x, y) = (x, kx + m)$ where $x$ is a solution to the first equation. Let us now discuss that equation after substituting $y = kx + m$.

$$(b^2 + a^2k^2)x^2 + 2kma^2x + a^2(m^2 - b^2) = 0.$$

This is a quadratic equation in $x$ since $a, b, k, m$ are fixed. The discriminant of this equation is

$$\Delta = 4k^2m^2a^4 - 4a^2(m^2 - b^2)(b^2 + a^2k^2) = 4a^2b^2(a^2k^2 + b^2 - m^2).$$

So, the number of real solutions is controlled by $a^2k^2 + b^2 - m^2$:

- $-\sqrt{a^2k^2+b^2} < m < \sqrt{a^2k^2+b^2}$, in which case $\ell$ intersects $\mathcal{E}_{a,b}$ in two distinct points.
- $m = \pm\sqrt{a^2k^2+b^2}$, in which case $\ell$ intersects $\mathcal{E}_{a,b}$ in a unique point. Such a point is a *double intersection point* because it is obtained as a double solution to the algebraic equation. For these two values of $m$, the line $\ell : y = kx + m$ is tangent to the ellipse. Therefore, if a slope $k$ is given, there are two tangent lines to the ellipse with this slope:
$$y = kx \pm \sqrt{a^2k^2+b^2}.$$
- $m < -\sqrt{a^2k^2+b^2}$ or $m > \sqrt{a^2k^2+b^2}$, in which case there is no intersection point between $\ell$ and $\mathcal{E}_{a,b}$.

### 8.1.5 Tangent line at a given point - algebraic

Consider an ellipse and a line:
$$\mathcal{E}_{a,b} : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad \text{and} \quad \ell : \begin{cases} x = x_0 + tv_x \\ y = y_0 + tv_y \end{cases}$$
which have the point $(x_0, y_0)$ in common. When is $\ell$ tangent to the ellipse? This occurs if the intersection $\mathcal{E}_{a,b} \cap \ell$ consists of a unique point. In order to determine when this is the case, we check which points on $\ell$ satisfy the equation of the ellipse:
$$\frac{(x_0+v_xt)^2}{a^2} + \frac{(y_0+v_yt)^2}{b^2} = 1.$$
The parameter values $t$ satisfying the above equation correspond to points on $\ell$ that lie on $\mathcal{E}_{a,b}$. The equation is equivalent to
$$\left(\frac{v_x^2}{a^2} + \frac{v_y^2}{b^2}\right)t^2 + 2\left(\frac{x_0v_x}{a^2} + \frac{y_0v_y}{b^2}\right)t + \underbrace{\frac{x_0^2}{a^2} + \frac{y_0^2}{b^2} - 1}_{=0} = 0.$$
In order for $\ell$ to be tangent to $\mathcal{E}_{a,b}$, there needs to be a unique solution $t$ to the above equation. Since $t = 0$ is obviously a solution, this needs to be the *only* solution. In other words, $t = 0$ should be a double solution. For this to happen we must have
$$\frac{x_0v_x}{a^2} + \frac{y_0v_y}{b^2} = 0 \quad \Leftrightarrow \quad \langle \mathbf{n}, \mathbf{v} \rangle = 0$$
where $\mathbf{n} = (\frac{x_0}{a^2}, \frac{y_0}{b^2})$. Thus, $\ell$ is tangent to the ellipse if and only if the vector $\mathbf{n}$ is orthogonal to $\ell$, i.e., if and only if $\mathbf{n}$ is a normal vector for $\ell$. It follows that $\ell$ is tangent to $\mathcal{E}_{a,b}$ at the point $(x_0, y_0) \in \mathcal{E}_{a,b}$ if and only if it satisfies the Cartesian equation:
$$\ell : \frac{x_0}{a^2}(x - x_0) + \frac{y_0}{b^2}(y - y_0) = 0.$$
We call this line the *tangent line to $\mathcal{E}_{a,b}$ at the point $(x_0, y_0) \in \mathcal{E}_{a,b}$* and denote it by $\mathcal{T}_{(x_0,y_0)}\mathcal{E}_{a,b}$. Rearranging the above equation, we see that:
$$\mathcal{T}_{(x_0,y_0)}\mathcal{E}_{a,b} : \frac{x_0x}{a^2} + \frac{y_0y}{b^2} = 1. \tag{8.2}$$

### 8.1.6 Tangent line at a given point - via gradients

It is possible to describe the tangent line $\mathcal{T}_{(x_0,y_0)}\mathcal{E}_{a,b}$ to $\mathcal{E}_{a,b}$ at the point $(x_0,y_0)\in\mathcal{E}_{a,b}$ using gradients. For this consider the map

$$\psi:\mathbb{E}^2\to\mathbb{R}\quad\text{defined by}\quad \psi(x,y)=\frac{x^2}{a^2}+\frac{y^2}{b^2}$$

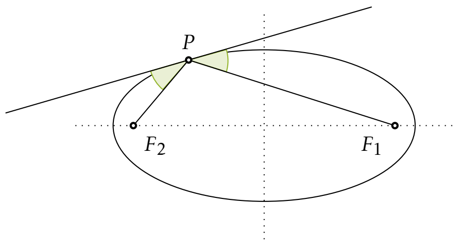
and notice that $\mathcal{E}_{a,b}=\psi^{-1}(1)$. The gradient at a point $(x_0,y_0)\in\mathcal{E}_{a,b}$ is

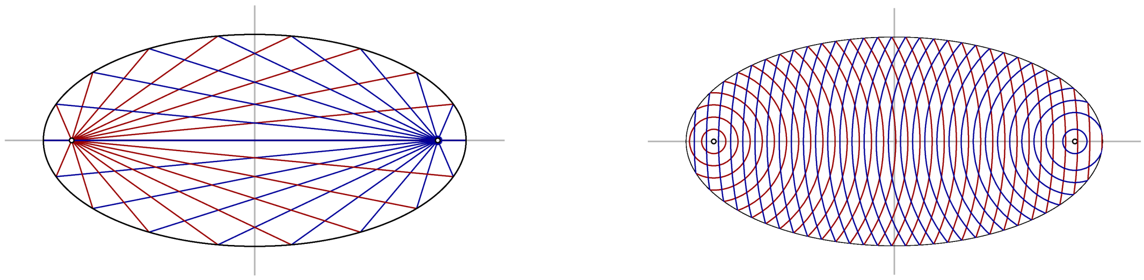
$$\nabla_{(x_0,y_0)}(\psi)=\left(2\frac{x}{a^2},2\frac{y}{b^2}\right)_{(x_0,y_0)}=2\left(\frac{x_0}{a^2},\frac{y_0}{b^2}\right).$$

By using a parametrization $\phi:I\to\mathbb{E}^2$ of $\mathcal{E}_{a,b}$ and applying the chain rule to $\partial_t\psi(\phi(t))$, one shows that $\nabla_{(x_0,y_0)}(\psi)$ is orthogonal to the tangent vectors at the point $(x_0,y_0)$. In other words, $\nabla_{(x_0,y_0)}(\psi)$ is a normal vector at the point $(x_0,y_0)\in\mathcal{E}_{a,b}$. This gives a different way of obtaining the equation (8.2).

### 8.1.7 Reflective properties

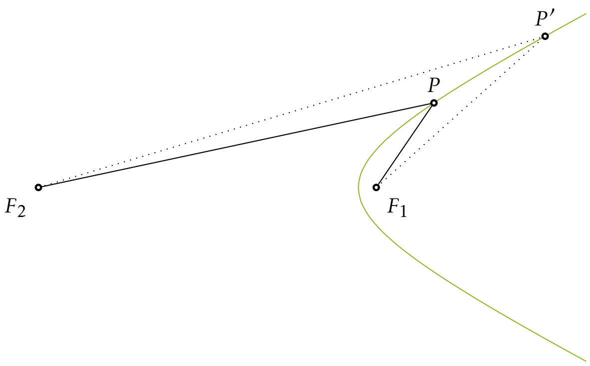
An ellipse has the following reflective properties: a ray starting at one focal point is reflected by the ellipse to the other focal point; i.e., if $P$ is a point on the ellipse, then the tangent line at $P$ is the exterior angle bisector of the angle $\measuredangle F_1PF_2$.



## 8.2 Hyperbola

### 8.2.1 Geometric description



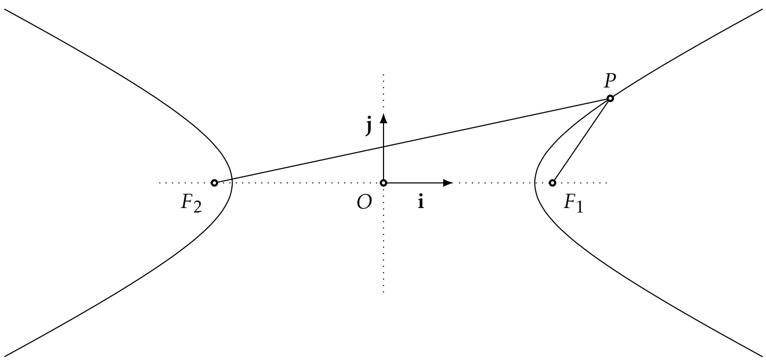

**Definition 8.4.** A *hyperbola* is the locus of points in $\mathbb{E}^2$ for which the difference of the distances from two given points, the *focal points*, is constant.

### 8.2.2 Canonical equation

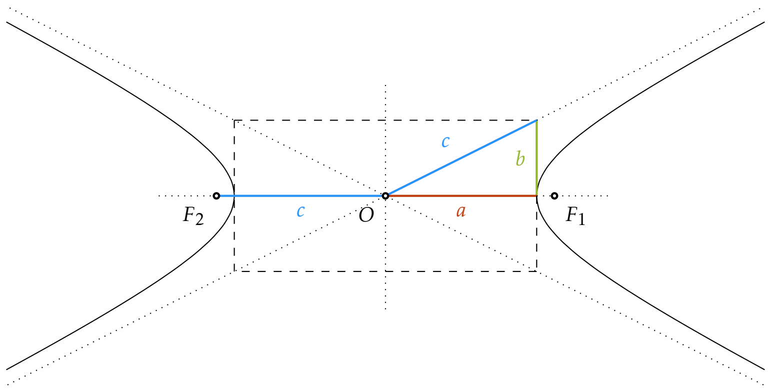
**Proposition 8.5.** Let $F_1$ and $F_2$ be two points in $\mathbb{E}^2$ and let $a > 0$ be a real number. Choose the coordinate system $Oxy = (O, \mathbf{i}, \mathbf{j})$ such that $F_1$ and $F_2$ are on the $x$-axis, such that $\overrightarrow{F_2F_1}$ has the same direction as $\mathbf{i}$, and such that $O$ is the midpoint of $[F_1F_2]$. With these choices, the hyperbola with focal points $F_1$ and $F_2$ for which the absolute value of the difference of distances from the focal points equals $2a$ has the equation

$$\mathcal{H}_{a,b} : \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \tag{8.3}$$

for some positive scalar $b \in \mathbb{R}$. We denote this hyperbola by $\mathcal{H}_{a,b}$ and call $a$ the *semi-major axis* and $b$ the *semi-minor axis*.



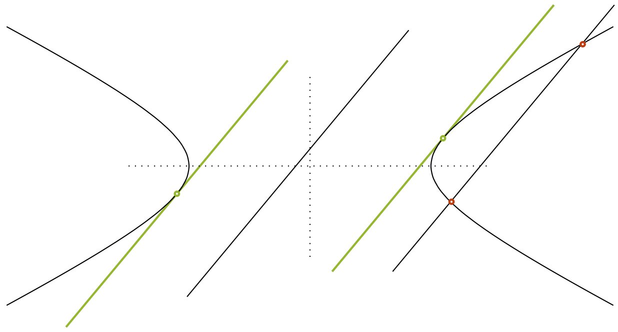
- Equation (8.3) is called the *canonical equation of the hyperbola* $\mathcal{H}_{a,b}$. Clearly, the same hyperbola may be represented by a different equation in another frame.
- If $2c$ denotes the distance between $F_1$ and $F_2$, then $b^2 = c^2 - a^2$.
- The intersections of $\mathcal{H}_{a,b}$ with the coordinate axes are the points $(\pm a, 0)$.
- The numerical quantity
$$\varepsilon = \frac{c}{a} = \sqrt{1 + \frac{b^2}{a^2}} \in (1, \infty)$$
is called the *eccentricity* of the hyperbola $\mathcal{H}_{a,b}$. It measures how widely the branches of the hyperbola open.
- The canonical equation shows that $M(x_M, y_M) \in \mathcal{H}_{a,b}$ if and only if $(\pm x_M, \pm y_M) \in \mathcal{H}_{a,b}$.




### 8.2.3 Parametric equations

Parametric equations are never unique. Depending on the intended application, one parametrization may be preferred over another. Equation (8.3) allows us to express $y$ in terms of $x$:
$$y(x) = \pm \frac{b}{a}\sqrt{x^2 - a^2}.$$
This gives a partial parametrization of $\mathcal{H}_{a,b}$. For the "northern part" we have the parametrization
$$\phi : (-\infty, -a] \cup [a, \infty) \to \mathbb{E}^2 \quad \text{given by} \quad \phi(x) = \big(x, y(x)\big) = \Big(x, \frac{b}{a}\sqrt{x^2 - a^2}\Big).$$
This is the graph of the function
$$y(x) = \frac{b}{a}\sqrt{x^2 - a^2} \quad \text{for which} \quad y'(x) = \frac{bx}{a\sqrt{x^2 - a^2}} \quad \text{and} \quad y''(x) = -\frac{ab}{(x^2 - a^2)\sqrt{x^2 - a^2}}$$
Thus, we can use the known methods to verify the monotonicity and the convexity of $y(x)$, which describes this part of the hyperbola.

### 8.2.4 Relative position of a line



Consider the canonical equation (8.3) of the hyperbola $\mathcal{H}_{a,b}$. Let $\ell$ be a line with equation $y = kx+m$. The intersection of the two objects is the set of points whose coordinates are solutions to the system

$$\begin{cases} \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \\ y = kx + m \end{cases} \Leftrightarrow \begin{cases} \frac{x^2}{a^2} - \frac{(kx+m)^2}{b^2} = 1 \\ y = kx + m \end{cases}.$$

The solutions to this system are $(x,y) = (x, kx+m)$ where $x$ is a solution to the first equation. Let us discuss that equation:

$$(b^2 - a^2k^2)x^2 - 2kma^2x - a^2(m^2 + b^2) = 0 \tag{8.4}$$

This is a quadratic equation in $x$ since $a,b,k,m$ are fixed. The discriminant of this equation is

$$\Delta = 4k^2m^2a^4 + 4a^2(m^2 + b^2)(b^2 - a^2k^2) = 4a^2b^2(m^2 + b^2 - a^2k^2).$$

So, the number of real solutions is controlled by $m^2 + b^2 - a^2k^2$ ... *if* the equation is quadratic. Suppose equation (8.4) is quadratic, i.e., $b^2 - a^2k^2 \neq 0$. If $a^2k^2 - b^2 > 0$ (i.e., $|k| > \frac{b}{a}$), the number of solutions depends on $m$:

- $m < -\sqrt{a^2k^2 - b^2}$ or $m > \sqrt{a^2k^2 - b^2}$, in which case $\ell$ intersects $\mathcal{H}_{a,b}$ in two distinct points.
- $m = \pm\sqrt{a^2k^2 - b^2}$, in which case $\ell$ intersects $\mathcal{H}_{a,b}$ in a unique point. Such a point is a *double intersection point* because it is obtained as a double solution to the algebraic equation. For these two values of $m$, the line $\ell : y = kx + m$ is tangent to the hyperbola. Therefore, if a slope $k$ is given such that $|k| > \frac{b}{a}$, there are two tangent lines to the hyperbola with slope $k$:

$$y = kx \pm \sqrt{a^2k^2 - b^2}.$$

- $-\sqrt{a^2k^2 - b^2} < m < \sqrt{a^2k^2 - b^2}$, in which case there is no intersection point between $\ell$ and $\mathcal{H}_{a,b}$.

If $a^2k^2 - b^2 < 0$ (i.e., $|k| < \frac{b}{a}$), then $m^2 + b^2 - a^2k^2 > 0$ for all $m$. In this case, $\ell$ always intersects $\mathcal{H}_{a,b}$ in two distinct points, and there are no tangent lines with such a slope.

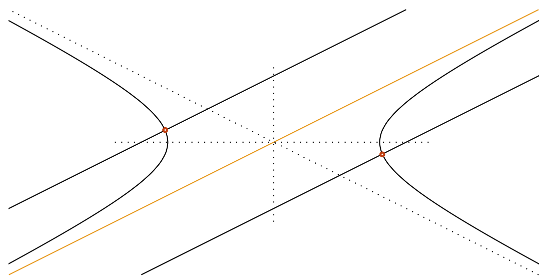
Suppose equation (8.4) is not quadratic, i.e., $b^2 - a^2k^2 = 0$, and the equation is

$$-2kma^2x - a^2(m^2 + b^2) = 0$$

Notice that $k \neq 0$ and $a \neq 0$ and that $k = \pm\frac{b}{a}$. We have two cases:

- $m \neq 0$, hence the unique solution $x = -\frac{m^2+b^2}{2km}$, which corresponds to a unique intersection point. In this case, it is a *simple intersection point*, corresponding to a simple solution of the algebraic equation (not a double solution).

- $m = 0$, in which case there is no intersection point and $\ell$ is either $y = \frac{b}{a}x$ or $y = -\frac{b}{a}x$. These are the two asymptotes of the hyperbola $\mathcal{H}_{a,b}$. One can check with the parametrization in the previous section that these two lines really are asymptotes.



### 8.2.5 Tangent line at a given point

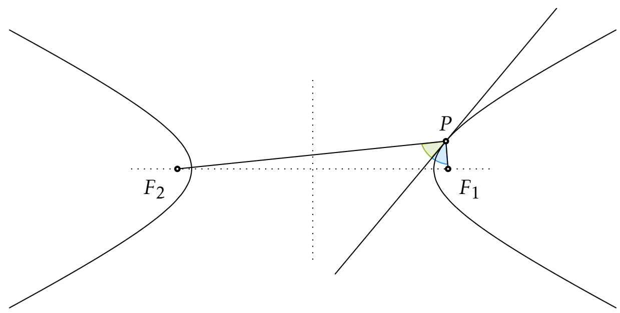
The *tangent line to $\mathcal{H}_{a,b}$ at the point* $(x_0, y_0) \in \mathcal{H}_{a,b}$ has the equation

$$\mathcal{T}_{(x_0,y_0)}\mathcal{H}_{a,b} : \frac{x_0 x}{a^2} - \frac{y_0 y}{b^2} = 1. \tag{8.5}$$

This can be deduced either with the algebraic method or via the gradient as in the case of the ellipse.

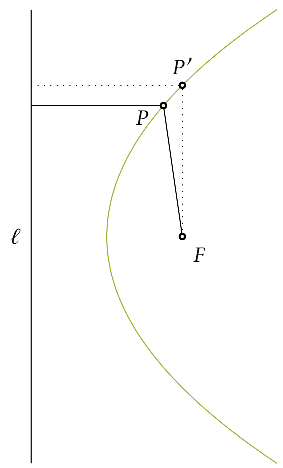
### 8.2.6 Reflective properties

A hyperbola has the following reflective properties: for a point $P$ on the hyperbola, the tangent line at $P$ is the angle bisector of the angle $\measuredangle F_1 P F_2$.



## 8.3 Parabola

### 8.3.1 Geometric description




**Definition 8.6.** A *parabola* is the locus of points in $\mathbb{E}^2$ for which the distance from a given point, the *focal point*, equals the distance to a given line, the *directrix*.

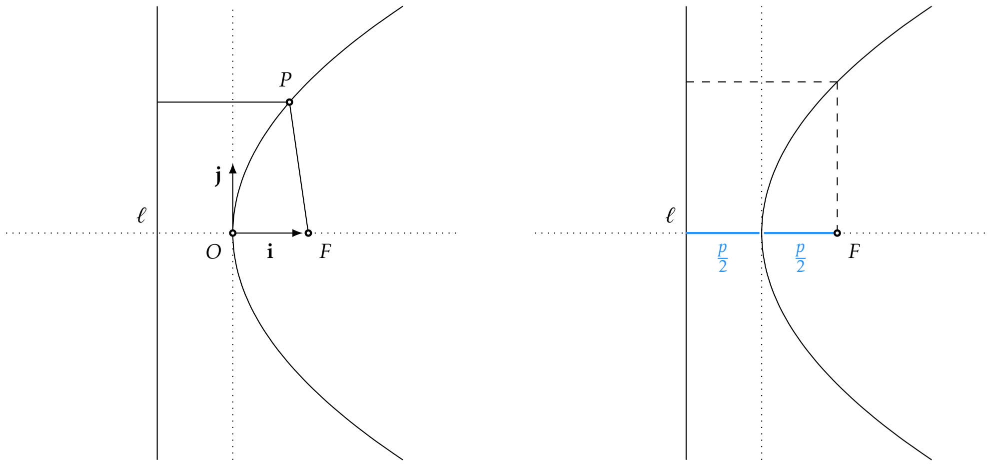
### 8.3.2 Canonical equation

**Proposition 8.7.** Let $F$ be a point, $d$ be a line in $\mathbb{E}^2$ and $p > 0$ be a real number. Choose the coordinate system $Oxy = (O, \mathbf{i}, \mathbf{j})$ such that $F$ lies on the $x$-axis, the $x$-axis is orthogonal to $d$, the origin is equidistant from $d$ and $F$, and the vector $\mathbf{i}$ has the same direction as $\overrightarrow{OF}$. With these choices, the parabola with focal point $F$ and directrix $d$ for which $d(F, d) = p$ has the equation

$$\mathcal{P}_p : y^2 = 2px \tag{8.6}$$

We denote this parabola by $\mathcal{P}_p$ and call $p$ the parameter of the parabola.

- Equation (8.6) is called the *canonical equation of the parabola* $\mathcal{P}_p$. Clearly, with respect to some other coordinate system, the same parabola will have a different equation.
- The focal point is $F(\frac{p}{2}, 0)$ and the directrix has equation $d : x = -\frac{p}{2}$.
- The intersection of $\mathcal{P}_p$ with the coordinate axes is the point $(0, 0)$.
- The canonical equation shows that $M(x_M, y_M) \in \mathcal{P}_p$ if and only if $(x_M, \pm y_M) \in \mathcal{P}_p$.




### 8.3.3 Parametric equations

Parametric equations are never unique. Depending on the intended application, one parametrization may be preferred over another. Equation (8.6) allows us to express $y$ in terms of $x$:

$$y(x) = \pm\sqrt{2px}.$$

This gives a partial parametrization of $\mathcal{P}_p$. For the "northern part" we have the parametrization

$$\phi : [0,\infty) \to \mathbb{E}^2 \quad \text{given by} \quad \phi(x) = \big(x, y(x)\big) = \big(x, \sqrt{2px}\big).$$

This is the graph of the function

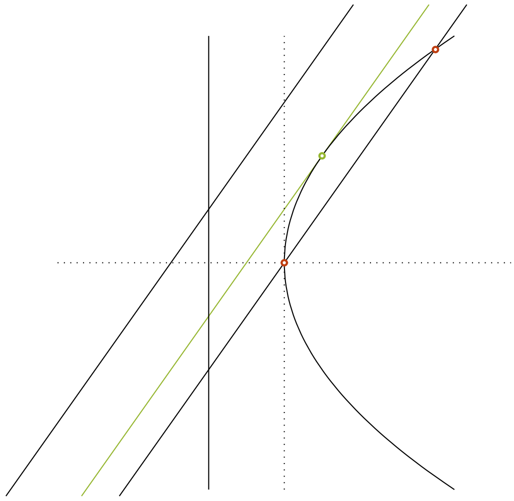
$$y(x) = \sqrt{2px} \quad \text{for which} \quad y'(x) = \frac{\sqrt{2p}}{2\sqrt{x}} \quad \text{and} \quad y''(x) = -\frac{\sqrt{2p}}{4x^{3/2}}$$

Thus, we can use the known methods to verify the monotonicity and the convexity of $y(x)$, which describes this part of the parabola.

We can, in fact, parametrize the whole parabola if we express $x$ in terms of $y$, which is another way of reading equation (8.6). We then have the parametrization

$$\phi : \mathbb{R} \to \mathbb{E}^2 \quad \text{given by} \quad \phi(y) = \big(x(y), y\big) = \Big(\frac{y^2}{2p}, y\Big).$$

### 8.3.4 Relative position of a line



Consider the canonical equation (8.6) of the parabola $\mathcal{P}_p$. Let $\ell$ be a line with equation $y = kx + m$. The intersection of the two objects is the set of points whose coordinates are solutions to the system

$$\begin{cases} y^2 = 2px \\ y = kx + m \end{cases} \Leftrightarrow \begin{cases} (kx+m)^2 = 2px \\ y = kx + m \end{cases}.$$

The solutions to this system are $(x, y) = (x, kx + m)$ where $x$ is a solution to the first equation. Let us discuss that equation:

$$k^2x^2 + 2(km - p)x + m^2 = 0 \tag{8.7}$$

Assuming $k \neq 0$, this is a quadratic equation in $x$ since $p, k, m$ are fixed. The discriminant of this equation is

$$\Delta = 4p(p - 2km).$$

So, the number of real solutions is controlled by $p - 2km$:

- $km < p/2$, in which case $\ell$ intersects $\mathcal{P}_p$ in two distinct points.
- $km = p/2$, in which case $\ell$ intersects $\mathcal{P}_p$ in a unique point. Such a point is a *double intersection point* because it is obtained as a double solution to the algebraic equation. For this value of $m$, the line $\ell : y = kx + m$ is tangent to the parabola. Therefore, if a slope $k \neq 0$ is given, there is one tangent line to the parabola having the given slope:

$$y = kx + \frac{p}{2k}.$$

- $km > p/2$, in which case there is no intersection point between $\ell$ and $\mathcal{P}_p$.

If $k = 0$, the line is $y = m$, which is horizontal and intersects the parabola in exactly one point, $(\frac{m^2}{2p}, m)$. This is a simple intersection point, not a tangency point.

### 8.3.5 Tangent line at a given point

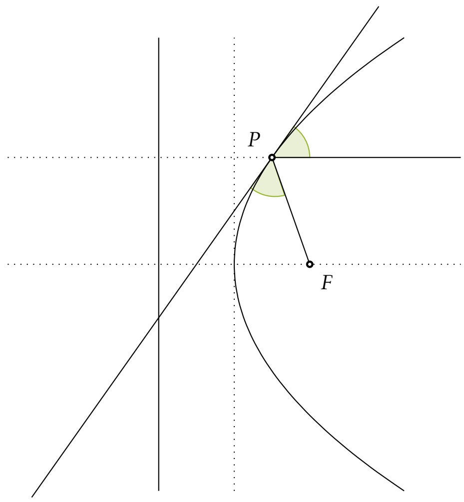
The *tangent line to* $\mathcal{P}_p$ *at the point* $(x_0, y_0) \in \mathcal{P}_p$ has the equation

$$\mathcal{T}_{(x_0,y_0)}\mathcal{P}_p : yy_0 = p(x + x_0) \tag{8.8}$$

This can be deduced either with the algebraic method or via the gradient as in the case of the ellipse.

### 8.3.6 Reflective properties

A parabola has the following reflective properties: rays starting at the focal point are reflected by the parabola into rays parallel to the axis of the parabola. Equivalently, rays inside the parabola which are parallel to the axis are reflected toward the focal point.




These properties are used for lenses, parabolic reflectors, satellite dishes, etc.

# CHAPTER 9

# Hyperquadrics

**Contents**

## 9.1 Hyperquadrics

**Definition 9.1.** A *hyperquadric* $\mathcal{Q}$ in $\mathbb{E}^n$ is the set of points whose coordinates satisfy a quadratic equation, i.e.,

$$\mathcal{Q}: \sum_{i,j=1}^{n} a_{ij}x_ix_j + \sum_{i=1}^{n} b_ix_i + c = 0 \tag{9.1}$$

with respect to some Cartesian frame.

- Hyperquadrics in $\mathbb{E}^2$ are called *conic sections* (see the previous chapter).
- In Chapter 10, we consider hyperquadrics in $\mathbb{E}^3$. In dimension 3 they are called *quadrics*.
- Notice that Equation (9.1) is equivalent to

$$\mathcal{Q}: \sum_{i,j=1}^{n} q_{ij}x_ix_j + \sum_{i=1}^{n} b_ix_i + c = 0 \tag{9.2}$$

  where $q_{ii} = a_{ii}$ and $q_{ij} = q_{ji} = \frac{a_{ij}+a_{ji}}{2}$. The matrix $Q = (q_{ij})$ is symmetric, and we call it the *symmetric matrix associated with Equation* (9.2) *of the quadric* $\mathcal{Q}$.
- The matrix $Q$ defines a homogeneous polynomial of degree 2 in the above equation.
- Notice that Equation (9.2) can be rearranged in matrix form as follows:

$$\mathcal{Q}: \mathbf{x}^T \cdot Q \cdot \mathbf{x} + \mathbf{b}^T \cdot \mathbf{x} + c = 0 \tag{9.3}$$

  where $\mathbf{x} = (x_1, \ldots, x_n)$ and $\mathbf{b} = (b_1, \ldots, b_n)$.

## 9.2 Reducing to canonical form

Let $\mathcal{Q}$ be the hyperquadric described by Equation (9.2) with respect to the right-oriented orthonormal frame $\mathcal{K} = (O, \mathcal{B})$, where $\mathcal{B} = (\mathbf{e}_1, \ldots, \mathbf{e}_n)$. Let $Q$ be the matrix associated with Equation (9.2) of $\mathcal{Q}$.

**[Step 1 - Rotation]** By the Spectral Theorem (see Theorem F.15), there is an orthonormal basis $\mathcal{B}'$ of eigenvectors for $Q$ that diagonalizes $Q$. Changing the frame from $\mathcal{K} = (O, \mathcal{B})$ to $\mathcal{K}' = (O, \mathcal{B}')$, the equation of the hyperquadric becomes

$$\mathcal{Q}: \mathbf{y}^T \cdot D \cdot \mathbf{y} + \mathbf{v}^T \cdot \mathbf{y} + c = 0 \quad \text{where} \quad D = \begin{bmatrix} \lambda_1 & 0 & \ldots & 0 \\ 0 & \lambda_2 & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & \lambda_n \end{bmatrix} \tag{9.4}$$

and where $\mathbf{y} = (y_1, \ldots, y_n)$ and $\mathbf{v} = (v_1, \ldots, v_n)$. This change of coordinates consists of replacing $\mathbf{x}$ with $\mathrm{M}_{\mathcal{B},\mathcal{B}'}\mathbf{y}$, since $\mathbf{x} = \mathrm{M}_{\mathcal{B},\mathcal{B}'}\mathbf{y}$. Notice that, since the two bases $\mathcal{B}$ and $\mathcal{B}'$ are orthonormal, the matrix $\mathrm{M}_{\mathcal{B},\mathcal{B}'}$ is an orthogonal matrix, i.e., $\mathrm{M}_{\mathcal{B},\mathcal{B}'} \in \mathrm{O}(n)$.

For the proof of the Spectral Theorem, one uses the fact that, since $Q$ is symmetric, the eigenvalues $\lambda_1,\dots,\lambda_n$ are all real. Hence, possibly after permuting the basis vectors in $\mathcal{B}'$, we may assume that $\lambda_1,\dots,\lambda_p>0$, $\lambda_{p+1},\dots,\lambda_r<0$, and $\lambda_{r+1},\dots,\lambda_n=0$, where $r$ is the rank of $Q$. This permutation corresponds to changing $\mathcal{B}'=(\mathbf{e}'_1,\dots,\mathbf{e}'_n)$ to $\mathcal{B}''=(\mathbf{e}'_{\sigma(1)},\dots,\mathbf{e}'_{\sigma(n)})$ for some permutation $\sigma$ of $\{1,\dots,n\}$. The change of basis matrix $\mathrm{M}_{\mathcal{B}',\mathcal{B}''}$ is again orthogonal. Thus, so far, we have a change of coordinates from $\mathcal{K}=(O,\mathcal{B})$ to $\mathcal{K}''=(O,\mathcal{B}'')$ given by the change of basis matrix $\mathrm{M}_{\mathcal{B},\mathcal{B}''}=\mathrm{M}_{\mathcal{B},\mathcal{B}'}\mathrm{M}_{\mathcal{B}',\mathcal{B}''}\in\mathrm{O}(n)$.

Recall that $\det(\mathrm{M}_{\mathcal{B},\mathcal{B}''})=1$ if $\mathrm{M}_{\mathcal{B},\mathcal{B}''}\in\mathrm{SO}(n)$ and $\det(\mathrm{M}_{\mathcal{B},\mathcal{B}''})=-1$ if $\mathrm{M}_{\mathcal{B},\mathcal{B}''}$ is not special orthogonal. In the latter case, the matrix $\mathrm{M}_{\mathcal{B},\mathcal{B}''}$ changes the orientation of the basis $\mathcal{B}''$, i.e., the change of coordinates is an indirect isometry. So, if we replace one vector in $\mathcal{B}''$ by minus that vector, for example $\mathbf{e}''_1\leftrightarrow-\mathbf{e}''_1$, then $\mathcal{B}''$ has the same orientation as $\mathcal{B}$ and $\mathrm{M}_{\mathcal{B},\mathcal{B}''}\in\mathrm{SO}(n)$, i.e., the change of coordinates is a displacement. In conclusion, we may assume that $\mathcal{B}''$ is such that $\mathrm{M}_{\mathcal{B},\mathcal{B}''}\in\mathrm{SO}(n)$.

Writing out the equation of $\mathcal{Q}$ in $\mathcal{K}''$, we have:

$$\mathcal{Q}:\lambda_1y_1^2+\lambda_2y_2^2+\cdots+\lambda_ry_r^2+v_1y_1+v_2y_2+\cdots+v_ny_n+c=0. \tag{9.5}$$

**[Step 2 - Translation]** Notice that $r>0$, i.e., there is at least one eigenvalue distinct from 0; otherwise, $Q=0_n$ (the $n\times n$ zero matrix) and Equation (9.2) is not a quadratic polynomial. Now, if $v_1\neq0$, then

$$\lambda_1y_1^2+v_1y_1=\lambda_1\Big(y_1^2+2\frac{v_1}{2\lambda_1}y_1\Big)=\lambda_1\Big(y_1+\frac{v_1}{2\lambda_1}\Big)^2-\frac{v_1^2}{4\lambda_1}.$$

Thus, if for $i\in\{1,\dots,r\}$, we let $z_i=y_i+\frac{v_i}{2\lambda_i}$, and $z_i=y_i$ for $i>r$, then Equation (9.5) becomes

$$\mathcal{Q}:\lambda_1z_1^2+\lambda_2z_2^2+\cdots+\lambda_rz_r^2+v_{r+1}z_{r+1}+\cdots+v_nz_n=k \tag{9.6}$$

for some $k\in\mathbb{R}$. This change of variables corresponds to a change of coordinates from $\mathcal{K}''=(O,\mathcal{B}'')$ to $\mathcal{K}'''=(O',\mathcal{B}'')$ given by the translation

$$\begin{bmatrix} z_1\\ \vdots\\ z_r\\ z_{r+1}\\ \vdots\\ z_n\end{bmatrix}=\begin{bmatrix} y_1\\ \vdots\\ y_r\\ y_{r+1}\\ \vdots\\ y_n\end{bmatrix}+\begin{bmatrix} \frac{v_1}{2\lambda_1}\\ \vdots\\ \frac{v_r}{2\lambda_r}\\ 0\\ \vdots\\ 0\end{bmatrix}.$$

Thus, so far, we changed the coordinates from $\mathcal{K}$ to $\mathcal{K}''$ using a direct isometry that fixes the origin (corresponding to a matrix in $\mathrm{SO}(n)$) and from $\mathcal{K}''$ to $\mathcal{K}'''$ using a translation. These are isometries, so the composition of these two transformations is an isometry.

**[Step 3 - non-isometric deformation]** It is possible to further simplify Equation (9.6) of $\mathcal{Q}$ by making other changes of coordinates that scale the axes such that all the coefficients are $\pm1$. However, in general, these do not correspond to isometries (see the discussion in Section 9.3.3).

We *reduced* Equation (9.2) to Equation (9.6). Equation (9.6) is not yet the canonical form, but it is an important step toward the canonical form. We may refer to Equation (9.6) as an *intermediate canonical form*. In what follows, we explore what more can be done if we restrict to conics and quadrics, i.e., if we look at hyperquadrics in $\mathbb{E}^2$ and $\mathbb{E}^3$.

## 9.3 Classification of conics

### 9.3.1 Isometric classification

A hyperquadric in $\mathbb{E}^2$ is a curve given by an equation of the form

$$\mathcal{C} : q_{11}x^2 + 2q_{12}xy + q_{22}y^2 + b_1x + b_2y + c = 0. \tag{9.7}$$

From the discussion in Section 9.2, we may apply a rotation and a translation to change the frame so that Equation (9.7) becomes

$$\mathcal{C} : \lambda_1 x^2 + \lambda_2 y^2 = k \quad \text{or} \quad \mathcal{C} : \lambda_1 x^2 + v_2 y = k. \tag{9.8}$$

We may also assume that $\lambda_1 > 0$, since if $\lambda_1, \lambda_2 < 0$, we can multiply the entire equation by $-1$.

**Case 1:** If $\lambda_2 > 0$ and $k = 0$, then the equation has only $(0,0)$ as a solution; in this case, the curve degenerates to a point: the origin.
**Case 2:** If $\lambda_2 > 0$ and $k < 0$, then there are no real solutions to the equation.
**Case 3:** If $\lambda_2 > 0$ and $k > 0$, after dividing by $k$, the equation becomes

$$\frac{x^2}{\frac{k}{\lambda_1}} + \frac{y^2}{\frac{k}{\lambda_2}} = 1$$

which is the equation of an ellipse. The ellipse is in canonical form if $\frac{k}{\lambda_1} > \frac{k}{\lambda_2}$. If this is not the case, we need to make one additional change of coordinates by rotating the reference frame by $90^\circ$, which is a direct isometry.
**Case 4:** If $\lambda_2 < 0$ and $k = 0$, then the equation becomes

$$(\sqrt{\lambda_1}x - \sqrt{-\lambda_2}y)(\sqrt{\lambda_1}x + \sqrt{-\lambda_2}y) = 0.$$

This is the union of two lines.
**Case 5:** If $\lambda_2 < 0$ and $k < 0$, we may multiply the whole equation by $-1$ and interchange the roles of $\lambda_1$ and $\lambda_2$. Doing so requires one additional coordinate change obtained by rotating the reference frame by $90^\circ$, which is a direct isometry. Then we end up in the following case.
**Case 6:** If $\lambda_2 < 0$ and $k > 0$, after dividing by $k$, the equation becomes

$$\frac{x^2}{\frac{k}{\lambda_1}} - \frac{y^2}{\frac{k}{-\lambda_2}} = 1$$

which is the equation of a hyperbola.
**Case 7:** If $\lambda_2 = 0$, we have the equation

$$\mathcal{C} : \lambda_1 x^2 + v_2 y = k.$$

If $v_2 = 0$ and $k \geq 0$, then we have two lines described by the equation $(\sqrt{\lambda_1}x - \sqrt{k})(\sqrt{\lambda_1}x + \sqrt{k}) = 0$. If $k = 0$, this is a double line, $x^2 = 0$.

**Case 8:** If $v_2 = 0$ and $k < 0$, then we have no real solutions.
**Case 9:** If $v_2 \neq 0$, then after dividing by $|v_2|$, the equation becomes

$$\frac{\lambda_1}{|v_2|}x^2 = \frac{k}{|v_2|} - \frac{v_2}{|v_2|}y$$

and we may change the coordinates using the translation/glide reflection $(x, \frac{k}{|v_2|} - \frac{v_2}{|v_2|}y) \to (x, y)$ so that the equation simplifies to

$$x^2 = \frac{|v_2|}{\lambda_1}y.$$

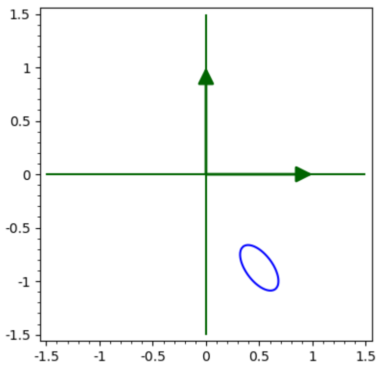
This change of coordinates corresponds to a reflection and a translation (again, an isometric change of coordinates), and we recognize the equation of a parabola with parameter $p = \frac{|v_2|}{2\lambda_1}$. However, here again, we do not yet have the canonical form of a parabola. To obtain $\mathcal{P}_p$, we need to interchange the $x$-axis with the $y$-axis.

**[Conclusion]** Starting with an equation of the form (9.7), we may use rotations, translations, and reflections to transform the equation into one of the following:

- degenerate cases: two lines, double lines, points, or
- non-degenerate cases:

$$\mathcal{E}_{a,b} : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad \text{or} \quad \mathcal{H}_{a,b} : \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \quad \text{or} \quad \mathcal{P}_p : y^2 = 2px.$$

Moreover, we can either carefully select direct isometries at each step or ensure at the end that the composition of all isometries used is direct. So, the curves described by an equation of the form (9.7) are conic sections and can be reduced to such curves via direct isometries (displacements).




**Example 9.2** (Ellipse)**.** Consider the curve with equation

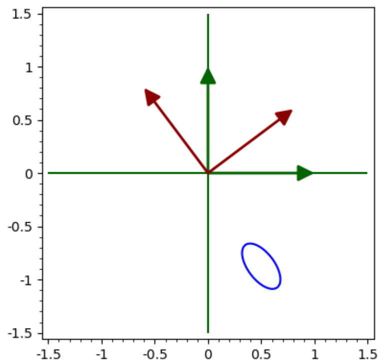
$$\mathcal{C} : 73x^2 + 72xy + 52y^2 - 10x + 55y + 25 = 0. \tag{9.9}$$

It is the ellipse in the image above. However, a priori, it is not at all clear that $\mathcal{C}$ is an ellipse. The symmetric matrix associated with this equation is

$$Q = \begin{bmatrix} 73 & 36 \\ 36 & 52 \end{bmatrix}$$

and, in matrix form, Equation (9.9) becomes

$$\mathcal{C} : \begin{bmatrix} x & y \end{bmatrix} \underbrace{\begin{bmatrix} 73 & 36 \\ 36 & 52 \end{bmatrix}}_{Q} \begin{bmatrix} x \\ y \end{bmatrix} + \underbrace{\begin{bmatrix} -10 & 55 \end{bmatrix}}_{b} \begin{bmatrix} x \\ y \end{bmatrix} + 25 = 0. \tag{9.10}$$




The eigenvalues of $Q$ are $\lambda_1 = 100$ and $\lambda_2 = 25$. The corresponding eigenvectors are $(4, 3)$ for $\lambda_1$ and $(3, -4)$ for $\lambda_2$. These two vectors form an orthogonal basis. Thus, an orthonormal basis is $\mathcal{B}' = (\mathbf{e}_1'(4/5, 3/5), \mathbf{e}_2'(3/5, -4/5))$. The change of basis matrix from $\mathcal{B}'$ to $\mathcal{B}$ is

$$\mathrm{M}_{\mathcal{B},\mathcal{B}'} = \frac{1}{5} \begin{bmatrix} 4 & 3 \\ 3 & -4 \end{bmatrix}.$$

We know that this matrix is orthogonal, which in this example is easy to check directly. In particular, $\mathrm{M}_{\mathcal{B},\mathcal{B}'}^{-1} = \mathrm{M}_{\mathcal{B},\mathcal{B}'}^{T}$. Moreover, the determinant is $-1$, which means that $\mathrm{M}_{\mathcal{B},\mathcal{B}'}$ is not a direct isometry, i.e., $\mathrm{M}_{\mathcal{B},\mathcal{B}'} \in \mathrm{O}(2) \setminus \mathrm{SO}(2)$. If we change the direction of the second eigenvector, we still have an eigenvector for the eigenvalue $\lambda_2$, but now, with respect to the basis $\mathcal{B}' = (\mathbf{e}_1'(4/5, 3/5), \mathbf{e}_2'(-3/5, 4/5))$, we have

$$\mathrm{M}_{\mathcal{B},\mathcal{B}'} = \frac{1}{5} \begin{bmatrix} 4 & -3 \\ 3 & 4 \end{bmatrix} \in \mathrm{SO}(2).$$

Changing the frame from $\mathcal{K}$ to $\mathcal{K}' = (O, \mathcal{B}')$, Equation (9.9) becomes

$$\mathcal{C} : \begin{bmatrix} x' & y' \end{bmatrix} \mathrm{M}_{\mathcal{B},\mathcal{B}'}^{T} \, Q \, \mathrm{M}_{\mathcal{B},\mathcal{B}'} \begin{bmatrix} x' \\ y' \end{bmatrix} + b \, \mathrm{M}_{\mathcal{B},\mathcal{B}'} \begin{bmatrix} x' \\ y' \end{bmatrix} + 25 = 0$$

which one calculates to be

$$\mathcal{C}: \begin{bmatrix} x' & y' \end{bmatrix} \underbrace{\begin{bmatrix} 100 & 0 \\ 0 & 25 \end{bmatrix}}_{Q'} \begin{bmatrix} x' \\ y' \end{bmatrix} + \underbrace{\begin{bmatrix} 25 & 50 \end{bmatrix}}_{b'} \begin{bmatrix} x' \\ y' \end{bmatrix} + 25 = 0$$

and we have

$$\mathcal{C}: 100x'^2 + 25y'^2 + 25x' + 50y' + 25 = 0$$

equivalently

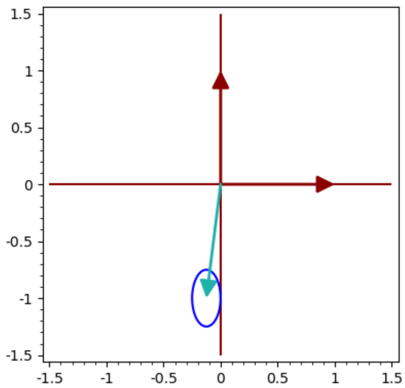
$$\mathcal{C}: 4x'^2 + y'^2 + x' + 2y' + 1 = 0$$

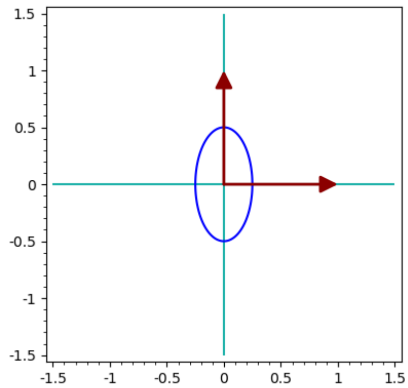
equivalently

$$\mathcal{C}: 4\left(x'^2 + \frac{x'}{4} + \frac{1}{64}\right) - \frac{1}{16} + \left(y'^2 + 2y' + 1\right) - 1 + 1 = 0$$

equivalently

$$\mathcal{C}: 4\left(x' + \frac{1}{8}\right)^2 + \left(y' + 1\right)^2 - \frac{1}{16} = 0.$$





Now, let us change the frame again, using a translation by the vector $(-\frac{1}{8}, -1)$. The new frame is $\mathcal{K}'' = (O'', \mathcal{B}'')$, where $O'' = (-\frac{1}{8}, -1)$, and the basis $\mathcal{B}'' = \mathcal{B}'$ does not change. In $\mathcal{K}''$, the equation of $\mathcal{C}$ becomes

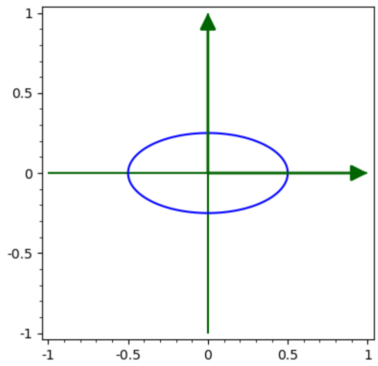
$$\mathcal{C}: 4x''^2 + y''^2 - \frac{1}{16} = 0 \quad \Leftrightarrow \quad \frac{x''^2}{\frac{1}{64}} + \frac{y''^2}{\frac{1}{16}} - 1 = 0.$$

Clearly, this is the equation of an ellipse. However, it is not yet in canonical form because the focal points are on the $y$-axis. To obtain the canonical form, we perform a final coordinate change and permute the coordinate axes, for instance, using

$$\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \in \mathrm{SO}(2) \quad \text{or with} \quad \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \in \mathrm{O}(2).$$

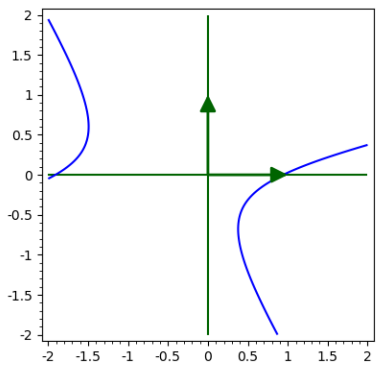
With either of the two transformations, we obtain

$$\mathcal{C} = \mathcal{E}_{\frac{1}{4},\frac{1}{8}} : \frac{x'''^2}{\frac{1}{16}} + \frac{y'''^2}{\frac{1}{64}} - 1 = 0.$$




To recap:

- We changed the coordinates from $\mathcal{K}$ to $\mathcal{K}'$ using the rotation $M_{\mathcal{B},\mathcal{B}'}$ of angle $\theta$, where $\cos(\theta) = \frac{4}{5}$.
- We changed the coordinates from $\mathcal{K}'$ to $\mathcal{K}''$ using a translation by the vector $(-\frac{1}{8}, -1)$.
- We changed the coordinates from $\mathcal{K}''$ to $\mathcal{K}'''$ in order to interchange the variables.
- We obtained $\mathcal{E}_{\frac{1}{4},\frac{1}{8}}$.




**Example 9.3** (Hyperbola)**.** Consider the curve with equation

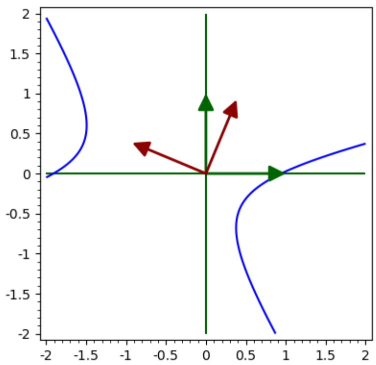
$$\mathcal{C} : -94x^2 + 360xy + 263y^2 - 91x + 221y + 169 = 0. \tag{9.11}$$

It is the hyperbola in the image above. However, a priori, it is not at all clear that $\mathcal{C}$ is a hyperbola. The symmetric matrix associated with this equation is

$$Q = \begin{bmatrix} -94 & 180 \\ 180 & 263 \end{bmatrix}$$

and, in matrix form, Equation (9.11) becomes

$$\mathcal{C}: \begin{bmatrix} x & y \end{bmatrix} \underbrace{\begin{bmatrix} -94 & 180 \\ 180 & 263 \end{bmatrix}}_{Q} \begin{bmatrix} x \\ y \end{bmatrix} + \underbrace{\begin{bmatrix} -91 & 221 \end{bmatrix}}_{b} \begin{bmatrix} x \\ y \end{bmatrix} + 169 = 0. \tag{9.12}$$




The eigenvalues of $Q$ are $\lambda_1 = 338$ and $\lambda_2 = -169$. The corresponding eigenvectors are $(5,12)$ for $\lambda_1$ and $(12,-5)$ for $\lambda_2$. These two vectors form an orthogonal basis. Thus, an orthonormal basis is $\mathcal{B}' = (\mathbf{e}_1'(5/13, 12/13), \mathbf{e}_2'(12/13, -5/13))$. The change of basis matrix from $\mathcal{B}'$ to $\mathcal{B}$ is

$$\mathrm{M}_{\mathcal{B},\mathcal{B}'} = \frac{1}{13}\begin{bmatrix} 5 & 12 \\ 12 & -5 \end{bmatrix}.$$

We know that this matrix is orthogonal, which in this example is easy to check directly. In particular, $\mathrm{M}_{\mathcal{B},\mathcal{B}'}^{-1} = \mathrm{M}_{\mathcal{B},\mathcal{B}'}^{T}$. Moreover, the determinant is $-1$, which means that $\mathrm{M}_{\mathcal{B},\mathcal{B}'}$ is not a direct isometry, i.e., $\mathrm{M}_{\mathcal{B},\mathcal{B}'} \in \mathrm{O}(2) \setminus \mathrm{SO}(2)$. If we change the direction of the second eigenvector, we still have an eigenvector for the eigenvalue $\lambda_2$, but now, with respect to the basis $\mathcal{B}' = (\mathbf{e}_1'(5/13, 12/13), \mathbf{e}_2'(-12/13, 5/13))$, we have

$$\mathrm{M}_{\mathcal{B},\mathcal{B}'} = \frac{1}{13}\begin{bmatrix} 5 & -12 \\ 12 & 5 \end{bmatrix} \in \mathrm{SO}(2).$$

Changing the frame from $\mathcal{K}$ to $\mathcal{K}' = (O, \mathcal{B}')$, Equation (9.11) becomes

$$\mathcal{C}: \begin{bmatrix} x' & y' \end{bmatrix} \mathrm{M}_{\mathcal{B},\mathcal{B}'}^{T} Q \mathrm{M}_{\mathcal{B},\mathcal{B}'} \begin{bmatrix} x' \\ y' \end{bmatrix} + b\,\mathrm{M}_{\mathcal{B},\mathcal{B}'} \begin{bmatrix} x' \\ y' \end{bmatrix} + 169 = 0$$

which one calculates to be

$$\mathcal{C}: \begin{bmatrix} x' & y' \end{bmatrix} \underbrace{\begin{bmatrix} 338 & 0 \\ 0 & -169 \end{bmatrix}}_{Q'} \begin{bmatrix} x' \\ y' \end{bmatrix} + \underbrace{\begin{bmatrix} 169 & 169 \end{bmatrix}}_{b'} \begin{bmatrix} x' \\ y' \end{bmatrix} + 169 = 0$$

and we have

$$\mathcal{C}: 338x'^2 - 169y'^2 + 169x' + 169y' + 169 = 0$$

equivalently

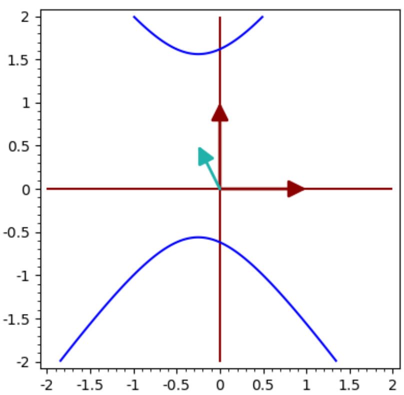
$$\mathcal{C}: 2x'^2 - y'^2 + x' + y' + 1 = 0$$

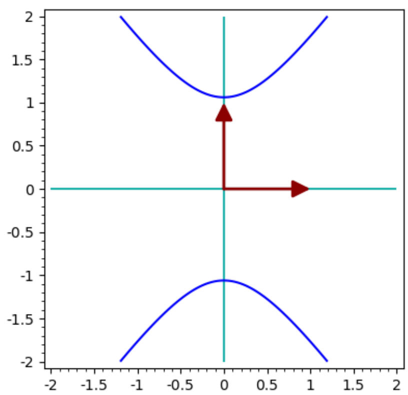
equivalently

$$\mathcal{C}: 2\Big(x'^2 + \frac{x'}{2} + \frac{1}{16}\Big) - \frac{1}{8} - \Big(y'^2 - y' + \frac{1}{4}\Big) + \frac{1}{4} + 1 = 0$$

equivalently

$$\mathcal{C}: 2\Big(x' + \frac{1}{4}\Big)^2 - \Big(y' - \frac{1}{2}\Big)^2 + \frac{9}{8} = 0.$$





Now, let us change the frame again, using a translation by the vector $(-\frac{1}{4}, \frac{1}{2})$. The new frame is $\mathcal{K}'' = (O'', \mathcal{B}'')$, where $O'' = (-\frac{1}{4}, \frac{1}{2})$, and the basis $\mathcal{B}'' = \mathcal{B}'$ does not change. In $\mathcal{K}''$, the equation of $\mathcal{C}$ becomes

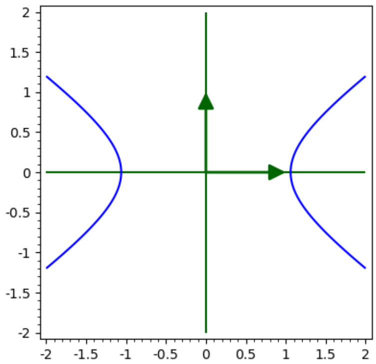
$$\mathcal{C}: 2x''^2 - y''^2 + \frac{9}{8} = 0 \quad \Leftrightarrow \quad -\frac{x''^2}{\frac{9}{16}} + \frac{y''^2}{\frac{9}{8}} - 1 = 0.$$

Clearly, this is the equation of a hyperbola. However, it is not yet in canonical form because the focal points are on the $y$-axis. To obtain the canonical form, we perform a final coordinate change and permute the coordinate axes, for instance, using

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \in \mathrm{SO}(2) \quad \text{or with} \quad \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \in \mathrm{O}(2).$$

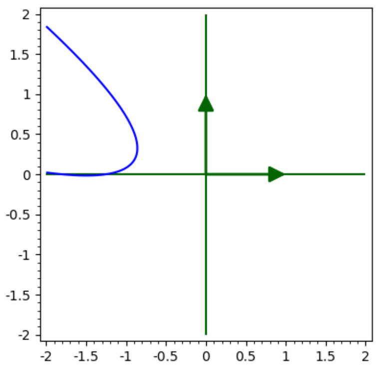
With either of the two transformations, we obtain

$$\mathcal{C} = \mathcal{H}_{\frac{3}{2\sqrt{2}}, \frac{3}{4}} : \frac{x'''^2}{\frac{9}{8}} - \frac{y'''^2}{\frac{9}{16}} - 1 = 0.$$




To recap:

- We changed the coordinates from $\mathcal{K}$ to $\mathcal{K}'$ using the rotation $\mathrm{M}_{\mathcal{B},\mathcal{B}'}$ of angle $\theta$, where $\cos(\theta) = \frac{5}{13}$.
- We changed the coordinates from $\mathcal{K}'$ to $\mathcal{K}''$ using a translation by the vector $(-\frac{1}{4}, \frac{1}{2})$.
- We changed the coordinates from $\mathcal{K}''$ to $\mathcal{K}'''$ in order to interchange the variables.
- We obtained $\mathcal{H}_{\frac{3}{2\sqrt{2}},\frac{3}{4}}$.




**Example 9.4** (Parabola)**.** Consider the curve with equation

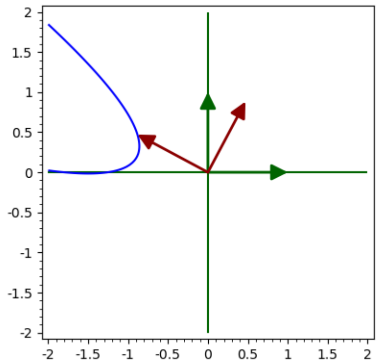
$$\mathcal{C} : 128x^2 + 480xy + 450y^2 + 391x + 119y + 289 = 0. \tag{9.13}$$

It is the parabola in the image above. However, a priori, it is not at all clear that $\mathcal{C}$ is a parabola. The symmetric matrix associated with this equation is

$$Q = \begin{bmatrix} 128 & 240 \\ 240 & 450 \end{bmatrix}$$

and, in matrix form, Equation (9.13) becomes

$$\mathcal{C} : \begin{bmatrix} x & y \end{bmatrix} \underbrace{\begin{bmatrix} 128 & 240 \\ 240 & 450 \end{bmatrix}}_{Q} \begin{bmatrix} x \\ y \end{bmatrix} + \underbrace{\begin{bmatrix} 391 & 119 \end{bmatrix}}_{b} \begin{bmatrix} x \\ y \end{bmatrix} + 289 = 0. \tag{9.14}$$




The eigenvalues of $Q$ are $\lambda_1 = 578$ and $\lambda_2 = 0$. The corresponding eigenvectors are $(8,15)$ for $\lambda_1$ and $(15,-8)$ for $\lambda_2$. These two vectors form an orthogonal basis. Thus, an orthonormal basis is $\mathcal{B}' = (\mathbf{e}_1'(8/17,15/17), \mathbf{e}_2'(15/17,-8/17))$. The change of basis matrix from $\mathcal{B}'$ to $\mathcal{B}$ is

$$\mathrm{M}_{\mathcal{B},\mathcal{B}'} = \frac{1}{17}\begin{bmatrix} 8 & 15 \\ 15 & -8 \end{bmatrix}.$$

We know that this matrix is orthogonal, which in this example is easy to check directly. In particular, $\mathrm{M}_{\mathcal{B},\mathcal{B}'}^{-1} = \mathrm{M}_{\mathcal{B},\mathcal{B}'}^{T}$. Moreover, the determinant is $-1$, which means that $\mathrm{M}_{\mathcal{B},\mathcal{B}'}$ is not a direct isometry, i.e., $\mathrm{M}_{\mathcal{B},\mathcal{B}'} \in \mathrm{O}(2) \setminus \mathrm{SO}(2)$. If we change the direction of the second eigenvector, we still have an eigenvector for the eigenvalue $\lambda_2$, but now, with respect to the basis $\mathcal{B}' = (\mathbf{e}_1'(8/17,15/17), \mathbf{e}_2'(-15/17,8/17))$, we have

$$\mathrm{M}_{\mathcal{B},\mathcal{B}'} = \frac{1}{17}\begin{bmatrix} 8 & -15 \\ 15 & 8 \end{bmatrix} \in \mathrm{SO}(2).$$

Changing the frame from $\mathcal{K}$ to $\mathcal{K}' = (O, \mathcal{B}')$, Equation (9.13) becomes

$$\mathcal{C} : \begin{bmatrix} x' & y' \end{bmatrix} \mathrm{M}_{\mathcal{B},\mathcal{B}'}^{T} Q \mathrm{M}_{\mathcal{B},\mathcal{B}'} \begin{bmatrix} x' \\ y' \end{bmatrix} + b\, \mathrm{M}_{\mathcal{B},\mathcal{B}'} \begin{bmatrix} x' \\ y' \end{bmatrix} + 289 = 0$$

which one calculates to be

$$\mathcal{C} : \begin{bmatrix} x' & y' \end{bmatrix} \underbrace{\begin{bmatrix} 578 & 0 \\ 0 & 0 \end{bmatrix}}_{Q'} \begin{bmatrix} x' \\ y' \end{bmatrix} + \underbrace{\begin{bmatrix} 289 & -289 \end{bmatrix}}_{b'} \begin{bmatrix} x' \\ y' \end{bmatrix} + 289 = 0$$

and we have

$$\mathcal{C} : 578x'^2 + 289x' - 289y' + 289 = 0$$

equivalently

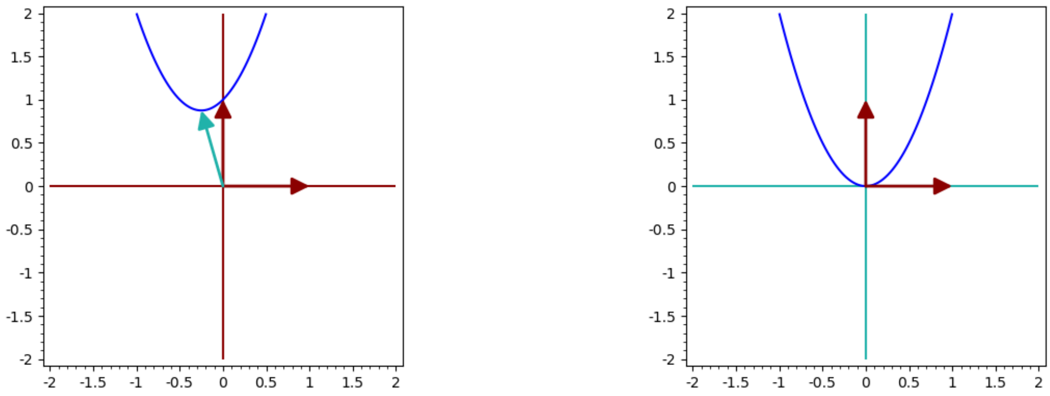
$$\mathcal{C} : 2x'^2 + x' - y' + 1 = 0$$

equivalently

$$\mathcal{C} : 2\Big(x'^2 + \frac{x'}{2} + \frac{1}{16}\Big) - \frac{1}{8} - y' + 1 = 0$$

equivalently

$$\mathcal{C} : 2\Big(x' + \frac{1}{4}\Big)^2 - \Big(y' - \frac{7}{8}\Big) = 0.$$



Now, let us change the frame again, using a translation by the vector $(-\frac{1}{4}, \frac{7}{8})$. The new frame is $\mathcal{K}'' = (O'', \mathcal{B}'')$, where $O'' = (-\frac{1}{4}, \frac{7}{8})$, and the basis $\mathcal{B}'' = \mathcal{B}'$ does not change. In $\mathcal{K}''$, the equation of $\mathcal{C}$ becomes

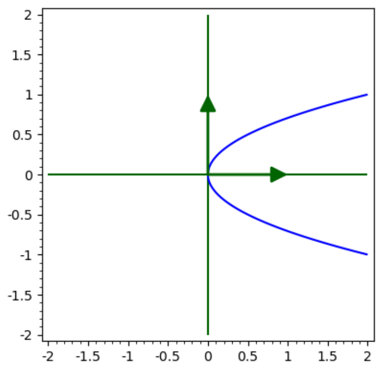
$$\mathcal{C} : 2x''^2 - y'' = 0 \quad \Leftrightarrow \quad x''^2 = \frac{1}{2}y''.$$

Clearly, this is the equation of a parabola. However, it is not yet in canonical form because the focal point is on the $y$-axis. To obtain the canonical form, we perform a final coordinate change and permute the coordinate axes, for instance, using

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \in \mathrm{SO}(2) \quad \text{or with} \quad \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \in \mathrm{O}(2).$$

With either of the two transformations, we obtain

$$\mathcal{C} = \mathcal{P}_{\frac{1}{4}} : y'''^2 = 2\frac{1}{4}x'''.$$



To recap:

- We changed the coordinates from $\mathcal{K}$ to $\mathcal{K}'$ using the rotation $\mathrm{M}_{\mathcal{B},\mathcal{B}'}$ of angle $\theta$, where $\cos(\theta) = \frac{8}{17}$.

- We changed the coordinates from $\mathcal{K}'$ to $\mathcal{K}''$ using a translation by the vector $(-\frac{1}{4}, \frac{7}{8})$.
- We changed the coordinates from $\mathcal{K}''$ to $\mathcal{K}'''$ in order to interchange the variables.
- We obtained $\mathcal{P}_{\frac{1}{4}}$.

### 9.3.2 Algorithm 1: Isometric Invariants

The discussion in Subsection 9.3.1 is an algebraic case-by-case analysis which may appear lengthy. It is the honest way of doing mathematics. However, it may be difficult to carry out such steps.

If we only want to know what type of conic we are dealing with, it is not difficult to extract a recipe that allows us to determine what type of curve a given quadratic equation describes. For this, notice that we may write Equation (9.1) of $\mathcal{Q}$ in a slightly different matrix form. We only do this in dimension 2, so consider the quadratic curve:

$$\mathcal{C} : q_{11}x^2 + 2q_{12}xy + q_{22}y^2 + 2b_1x + 2b_2y + c = 0. \tag{9.15}$$

The equation can be rewritten as

$$\begin{bmatrix} x & y & 1 \end{bmatrix} \underbrace{\begin{bmatrix} q_{11} & q_{12} & b_1 \\ q_{21} & q_{22} & b_2 \\ b_1 & b_2 & c \end{bmatrix}}_{=\widehat{Q}} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = 0$$

The matrix $\widehat{Q}$ is symmetric, and we call it the *extended symmetric matrix associated with Equation* (9.15) *of the conic* $\mathcal{C}$. Notice that changing coordinates via a rotation $R$ [Step 1], followed by a translation by a vector $\mathbf{v}$ [Step 2], amounts to a single matrix multiplication:

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} R & \mathbf{v} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \quad \Leftrightarrow \quad \underbrace{\begin{bmatrix} R^{-1} & -R^{-1}\mathbf{v} \\ 0 & 1 \end{bmatrix}}_{=M} \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

after which the equation becomes

$$\begin{bmatrix} x' & y' & 1 \end{bmatrix} M^T \widehat{Q} M \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = 0.$$

Observe that $\det(\widehat{Q})$ *does not change* when reducing to the canonical form using displacements. We say that $\det(\widehat{Q})$ is *invariant*. Indeed, $\det(M) = \det(R^{-1}) = 1$, so it does not affect the determinant of $\widehat{Q}$:

$$\det(M^T \widehat{Q} M) = \det(M^T)\det(\widehat{Q})\det(M) = \det(\widehat{Q}).$$

To distinguish between the different conics, we use three invariants. As before, $Q$ denotes the symmetric matrix associated with Equation (9.15) of the conic $\mathcal{C}$. The invariants are:

$$\widehat{D} = \det(\widehat{Q}), \quad D = \det(Q) \quad \text{and} \quad T = \mathrm{tr}(Q).$$

By checking Cases 1-9 in Section 9.3.1, and since $\widehat{D}$, $D$ and $T$ do not change under displacements, we have the following result:

**Proposition 9.5.** The type of curve described by Equation (9.15) is determined by the following table.

| $\widehat{D}$ | $D$ | $T$ | curve $C$ |
|---|---|---|---|
| $\widehat{D}=0$ | $D>0$ | - | A point |
| | $D=0$ | - | Two lines or the empty set |
| | $D<0$ | - | Two lines |
| $\widehat{D}\neq 0$ | $D>0$ | $\widehat{D}T<0$ | An ellipse |
| | $D>0$ | $\widehat{D}T>0$ | The empty set |
| | $D=0$ | - | A parabola |
| | $D<0$ | - | A hyperbola |

Table 9.1: Classification in dimension 2 via invariants.

### 9.3.3 Affine classification

In the isometric classification, we allowed only coordinate changes that are isometries. In fact, we noticed that it suffices to use direct isometries to reduce to the canonical form. The non-degenerate curves we obtained are:

$$\mathcal{E}_{a,b}:\frac{x^2}{a^2}+\frac{y^2}{b^2}=1\quad\text{or}\quad\mathcal{H}_{a,b}:\frac{x^2}{a^2}-\frac{y^2}{b^2}=1\quad\text{or}\quad\mathcal{P}_p:y^2=2px.$$

It is possible to further change the coordinates and rescale the basis vectors of the coordinate axes. Rescalings are clearly not isometries.

**Example 9.6.** Let us consider the case of $\mathcal{E}_{a,b}$. If we replace $x$ with $ax$ and $y$ with $by$, we are performing the following coordinate change:

$$\begin{bmatrix}x'\\y'\end{bmatrix}=\begin{bmatrix}a&0\\0&b\end{bmatrix}\begin{bmatrix}x\\y\end{bmatrix}$$

Then, the equation of $\mathcal{E}_{a,b}$ becomes:

$$\mathcal{E}_{a,b}:\begin{bmatrix}x&y\end{bmatrix}\begin{bmatrix}\frac{1}{a^2}&0\\0&\frac{1}{b^2}\end{bmatrix}\begin{bmatrix}x\\y\end{bmatrix}=1\quad\Leftrightarrow\quad\begin{bmatrix}x'&y'\end{bmatrix}\begin{bmatrix}a&0\\0&b\end{bmatrix}\begin{bmatrix}\frac{1}{a^2}&0\\0&\frac{1}{b^2}\end{bmatrix}\begin{bmatrix}a&0\\0&b\end{bmatrix}\begin{bmatrix}x'\\y'\end{bmatrix}=1.$$

Thus, after changing coordinates, the equation of $\mathcal{E}_{a,b}$ becomes

$$x'^2+y'^2=1.$$

Such affine transformations can, in general, be applied to Equation (9.6) for hypersurfaces. However, if we restrict our attention to $\mathbb{E}^2$, a case-by-case analysis shows that curves defined by a quadratic equation correspond to the possibilities listed in the following table. In this table, the second column indicates the signature of $Q$.

| $r = \operatorname{rank} Q$ | $(p, r-p)$ | equation | name |
|---|---|---|---|
| 2 | $(0,2)$ or $(2,0)$ | $x^2+y^2+1=0$ | imaginary ellipse |
| 2 | $(1,1)$ | $x^2-y^2-1=0$ | **hyperbola** |
| 2 | $(0,2)$ or $(2,0)$ | $x^2+y^2-1=0$ | **ellipse** |
| 2 | $(0,2)$ or $(2,0)$ | $x^2+y^2=0$ | two intersecting complex lines |
| 2 | $(1,1)$ | $x^2-y^2=0$ | **two intersecting real lines** |
| 1 | $(0,1)$ or $(1,0)$ | $x^2+1=0$ | two parallel complex lines |
| 1 | $(0,1)$ or $(1,0)$ | $x^2-1=0$ | **two parallel real lines** |
| 1 | $(0,1)$ or $(1,0)$ | $x^2=0$ | **a real double line** |
| 1 | $(0,1)$ or $(1,0)$ | $x^2-y=0$ | **parabola** |

Table 9.2: Affine classification in dimension 2.

### 9.3.4 Algorithm 2: Lagrange's Method

One reason the discussion in Subsection 9.3.1 is lengthy is due to the interpretation we added to each step (such as rotations, translations, and the fact that eigenvectors determine the directions of the new axes). We did this because we aimed to change the reference frame via displacements, without metrically altering the objects.

If we are only interested in the shape of the object, we are allowed to use affine transformations, which are changes of coordinates. This allows us to focus on the algebra behind the calculations, and the method of bringing it to canonical form becomes particularly simple. This point of view is sometimes attributed to Lagrange and works in any dimension.

Fix a quadratic curve, i.e., a hyperquadric in dimension 2:

$$\mathcal{C} : q_{11}x^2 + 2q_{12}xy + q_{22}y^2 + b_1x + b_2y + c = 0. \tag{9.16}$$

(Step 1) Eliminate the mixed terms by completing the squares.

(Step 2) Eliminate the linear terms by completing the squares.

(Step 3) Then, with respect to some frame, the curve $\mathcal{C}$ has the equation

$$ax^2 + by^2 + c = 0 \quad \text{or} \quad ax^2 + by + c = 0$$

where $a, b$, and $c$ are obtained in Steps 1 and 2. It is now easy to identify the type of curve (see Table 9.2).

This works because Step 1 and Step 2 correspond to affine changes of coordinates.

## 9.4 Classification of quadrics

A hyperquadric in $\mathbb{E}^3$ is a surface given by an equation of the form:

$$\mathcal{C} : q_{11}x^2 + q_{22}y^2 + q_{33}z^2 + 2q_{12}xy + 2q_{23}yz + 2q_{13}xz + b_1x + b_2y + b_3z + c = 0. \tag{9.17}$$

From the discussion in Section 9.2, we may apply an orthogonal change of coordinates and a translation to change the frame so that Equation (9.17) becomes

$$\mathcal{Q} : \lambda_1 x_1^2 + \lambda_2 x_2^2 + \cdots + \lambda_r x_r^2 + v_{r+1} x_{r+1} + \cdots + v_n x_n = k. \tag{9.18}$$

The isometric classification in this case is similar: we work out all possible cases to see what we obtain. One important remark is that the change of basis matrix in Step 1 – the matrix $M_{\mathcal{B},\mathcal{B}''}$ used to obtain Equation (9.18) – is an element of the group SO(3). Thus, by Euler's theorem (Theorem 6.17), this transformation is a rotation around an axis.

However, as in Section 9.3.3 and Section 9.3.4, one can "stretch" the coordinate axes using affine transformations, which are not isometries, to show that one may change the frame so that Equation (9.17) becomes one of the possibilities listed in the following table.

| $r = \operatorname{rank} Q$ | $(p, r-p)$ | equation | name |
|---:|---:|:---:|---:|
| 3 | $(3,0)$ or $(0,3)$ | $x^2+y^2+z^2-1=0$ | **ellipsoid** |
| 3 | $(2,1)$ or $(1,2)$ | $x^2+y^2-z^2-1=0$ | **hyperboloid of one sheet** |
| 3 | $(2,1)$ or $(1,2)$ | $x^2-y^2-z^2-1=0$ | **hyperboloid of two sheets** |
| 3 | $(3,0)$ or $(0,3)$ | $x^2+y^2+z^2+1=0$ | imaginary ellipsoid |
| 3 | $(3,0)$ or $(0,3)$ | $x^2+y^2+z^2=0$ | imaginary cone |
| 3 | $(2,1)$ or $(1,2)$ | $x^2+y^2-z^2=0$ | (real, elliptic) **cone** |
| 2 | $(2,0)$ or $(0,2)$ | $x^2+y^2+1=0$ | cylinder on imaginary ellipse |
| 2 | $(1,1)$ | $x^2-y^2-1=0$ | **cylinder** on hyperbola |
| 2 | $(2,0)$ or $(0,2)$ | $x^2+y^2-1=0$ | **cylinder** on ellipse |
| 2 | $(2,0)$ or $(0,2)$ | $x^2+y^2=0$ | cylinder on two complex lines |
| 2 | $(1,1)$ | $x^2-y^2=0$ | **cylinder** on two real lines |
| 1 | $(1,0)$ or $(0,1)$ | $x^2+1=0$ | two parallel complex planes |
| 1 | $(1,0)$ or $(0,1)$ | $x^2-1=0$ | **two parallel real planes** |
| 1 | $(1,0)$ or $(0,1)$ | $x^2=0$ | **a double plane** |
| 2 | $(2,0)$ or $(0,2)$ | $x^2+y^2-z=0$ | **elliptic paraboloid** (EP) |
| 2 | $(1,1)$ | $x^2-y^2-z=0$ | **hyperbolic paraboloid** (HP) |
| 1 | $(1,0)$ or $(0,1)$ | $x^2+y=0$ | **cylinder** on parabola |

CHAPTER 10

# Canonical equations of real quadrics

## Contents

Here we look at the main examples of *quadrics* in $\mathbb{E}^3$, sometimes called *quadratic surfaces*. These are surfaces which satisfy a quadratic equation of the form

$$\mathcal{S} : q_{11}x^2 + q_{22}y^2 + q_{33}z^2 + 2q_{12}xy + 2q_{13}xz + 2q_{23}yz + a_1x + a_2y + a_3z + c = 0. \tag{10.1}$$

Notice that an equation as above may not define a surface. It could happen that there are no solutions or that the coordinates of only one point satisfy a given quadratic equation.

## 10.1 Ellipsoid

### 10.1.1 Canonical equation - global description

An *ellipsoid* is a surface which (in some coordinate system) satisfies an equation of the form

$$\mathcal{E}_{a,b,c} : \underbrace{\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2}}_{\varphi(x,y,z)} = 1 \quad \text{so} \quad \mathcal{E}_{a,b,c} = \varphi^{-1}(1)$$

for some positive constants $a, b, c \in \mathbb{R}$.

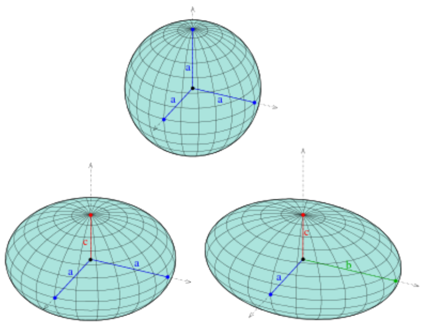


Figure 10.1: Ellipsoid[1]

If we fix one variable, we obtain intersections with planes parallel to the coordinate axes. For this surface the intersections are either ellipses, points or the empty set. Check this for $z = h$ and deduce the axes of the ellipses that you obtain.

### 10.1.2 Tangent planes

Using the gradient or the algebraic method we obtain the tangent plane at a point $p = (x_p, y_p, z_p) \in \mathcal{E}_{a,b,c}$ to be

$$\mathcal{T}_p\mathcal{E}_{a,b,c} : \frac{x_p x}{a^2} + \frac{y_p y}{b^2} + \frac{z_p z}{c^2} = 1.$$

[1] Image source: Wikipedia

We will check this with the algebraic method. Let us consider a line passing through $p$ in parametric form

$$l:\begin{bmatrix}x\\y\\z\end{bmatrix}=\begin{bmatrix}x_p\\y_p\\z_p\end{bmatrix}+t\underbrace{\begin{bmatrix}v_x\\v_y\\v_z\end{bmatrix}}_{=v}\quad\Leftrightarrow\quad l=p+\langle v\rangle.$$

Recall that the tangent plane $\mathcal{T}_p\mathcal{E}_{a,b,c}$ is the union of all lines intersecting the quadric $\mathcal{E}_{a,b,c}$ at $p$ in a special way, i.e., in a double point. This is why we investigate the intersection of our quadric with $l$. How do we obtain the intersection $\mathcal{E}_{a,b,c}\cap l$? We look at those points of $l$ which satisfy the equation of the ellipsoid, i.e., we look for solutions $t$ for the equation

$$\varphi(\begin{bmatrix}x_p\\y_p\\z_p\end{bmatrix}+t\begin{bmatrix}v_x\\v_y\\v_z\end{bmatrix})=1\quad\Leftrightarrow\quad\frac{(x_p+tv_x)^2}{a^2}+\frac{(y_p+tv_y)^2}{b^2}+\frac{(z_p+tv_z)^2}{c^2}-1=0.$$

If we rearrange the left-hand side as a polynomial in $t$ we get

$$\Big(\frac{v_x^2}{a^2}+\frac{v_y^2}{b^2}+\frac{v_z^2}{c^2}\Big)t^2+2\Big(\frac{x_pv_x}{a^2}+\frac{y_pv_y}{b^2}+\frac{z_pv_z}{c^2}\Big)t+\underbrace{\frac{x_p^2}{a^2}+\frac{y_p^2}{b^2}+\frac{z_p^2}{c^2}}_{=1}-1=0$$

$$\Leftrightarrow\quad\underbrace{\Big(\frac{v_x^2}{a^2}+\frac{v_y^2}{b^2}+\frac{v_z^2}{c^2}\Big)}_{\neq 0}t^2+2\Big(\frac{x_pv_x}{a^2}+\frac{y_pv_y}{b^2}+\frac{z_pv_z}{c^2}\Big)t=0.$$

This equation in $t$ admits the solution $t=0$. That is clear, the point on $l$ corresponding to $t=0$ is the point $p$ which, by assumption, lies on $\mathcal{E}_{a,b,c}$. Furthermore, the second solution will correspond to a second point of intersection. However, the line $l$ is tangent to our quadric if $p$ is a *double point of intersection*. This happens if and only if the above equation has $t=0$ as double solution, i.e., if and only if

$$\frac{x_pv_x}{a^2}+\frac{y_pv_y}{b^2}+\frac{z_pv_z}{c^2}=0.$$

How do we interpret this? In the Euclidean setting this can be interpreted as saying that $(\frac{x_p}{a^2},\frac{y_p}{b^2},\frac{z_p}{c^2})$ is perpendicular to the direction vector $v$ of the line. All lines which are tangent to the surface and contain $p$ need to satisfy this condition, so

$$\mathcal{T}_p\mathcal{E}_{a,b,c}:\begin{bmatrix}\frac{x_p}{a^2}\\\frac{y_p}{b^2}\\\frac{z_p}{c^2}\end{bmatrix}\cdot(\begin{bmatrix}x\\y\\z\end{bmatrix}-\begin{bmatrix}x_p\\y_p\\z_p\end{bmatrix})=0\quad\Leftrightarrow\quad\frac{x_px}{a^2}+\frac{y_py}{b^2}+\frac{z_pz}{c^2}-1=0.$$

Deduce this equation also with the gradient.

### 10.1.3 Parametrizations - local description

A parametrization of this surface is

$$\begin{cases} x(\theta_1,\theta_2) = a\cos(\theta_1)\cos(\theta_2) \\ y(\theta_1,\theta_2) = b\sin(\theta_1)\cos(\theta_2) \\ z(\theta_1,\theta_2) = c\sin(\theta_2) \end{cases} \quad \theta_1 \in [0,2\pi[ \quad \theta_2 \in [-\frac{\pi}{2},\frac{\pi}{2}[$$

Why? Check this and deduce a parametrization of the tangent plane $\mathcal{T}_p\mathcal{E}_{a,b,c}$ at the point

$$p = \begin{bmatrix} x(\theta_{1,p},\theta_{2,p}) \\ y(\theta_{1,p},\theta_{2,p}) \\ z(\theta_{1,p},\theta_{2,p}) \end{bmatrix} \in \mathcal{E}_{a,b,c}.$$

## 10.2 Elliptic Cone

### 10.2.1 Canonical equation - global description

An *elliptic cone* is a surface which (in some coordinate system) satisfies an equation of the form

$$\mathcal{C}_{a,b,c} : \underbrace{\frac{x^2}{a^2}+\frac{y^2}{b^2}-\frac{z^2}{c^2}}_{\varphi(x,y,z)} = 0 \quad \text{so} \quad \mathcal{C}_{a,b,c} = \varphi^{-1}(0)$$

for some positive constants $a,b,c \in \mathbb{R}$.

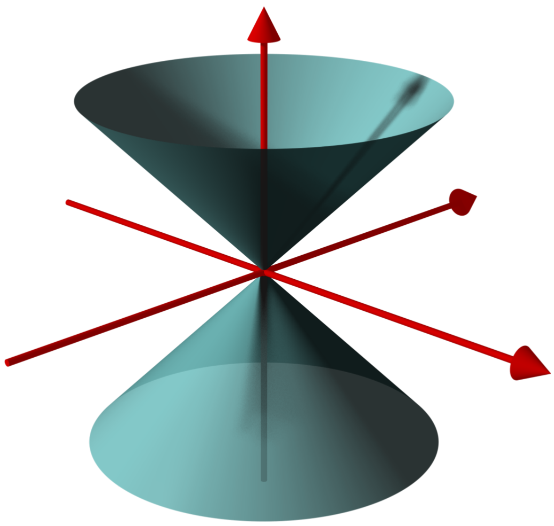


Figure 10.2: Elliptic cone[2]

If we fix one variable, we obtain intersections with planes parallel to the coordinate axes. For this surface the intersections are either ellipses, hyperbolas or a point. Check this for $z = h$ and deduce the axes of the ellipses that you obtain.

[2]Image source: Wikipedia

### 10.2.2 Conic sections

Above we noticed that the intersection of an elliptic cone with planes parallel to the coordinate axes are quadratic surfaces (possibly degenerate). In fact we have the following result.

**Proposition 10.1.** The intersection of an elliptic cone with an arbitrary plane is a (possibly degenerate) quadratic curve.

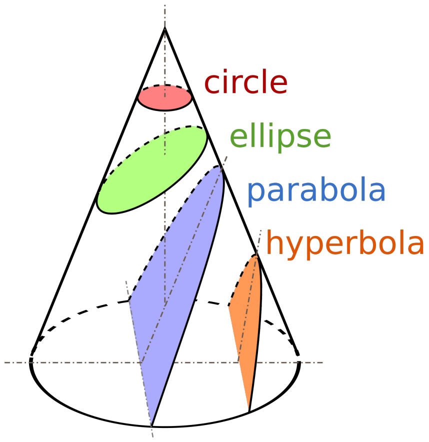



Figure 10.3: Conic sections[3]

### 10.2.3 Tangent planes

As in the case of the ellipsoid, using the gradient or the algebraic method we obtain the tangent plane at a point $p = (x_p, y_p, z_p) \in \mathcal{C}_{a,b,c}$ to be

$$\mathcal{T}_p\mathcal{C}_{a,b,c} : \frac{x_p x}{a^2} + \frac{y_p y}{b^2} - \frac{z_p z}{c^2} = 0.$$

### 10.2.4 $\mathcal{C}_{a,b,c}$ as ruled surface

A *ruled surface* is a surface $\mathcal{S}$ such that for any point $p$ on the surface $S$ there is a line $l$ which contains $p$ and which is contained in $S$:

$$\forall p \in \mathcal{S}, \exists \text{ a line } l \text{ such that } p \in l \text{ and } l \subseteq S.$$

If we denote by $\mathcal{L}$ the family of all these lines, it is easy to see that the surface is the union of them:

$$\mathcal{S} = \bigcup_{l \in \mathcal{L}} l.$$

[3]Image source: Wikipedia

The lines in $\mathcal{L}$ are called *rectilinear generators of the surface* $\mathcal{S}$. We refer to them as *generators* since we don't consider here non-rectilinear generators.

So, how is the cone a ruled surface? Fix a point $(x_0, y_0, z_0) \in \mathcal{C}_{a,b,c}$ and notice that for any $t \in \mathbb{R}$ we have

$$\frac{(tx_0)^2}{a^2} + \frac{(ty_0)^2}{b^2} - \frac{(tz_0)^2}{c^2} = t^2\left(\frac{x_0^2}{a^2} + \frac{y_0^2}{b^2} - \frac{z_0^2}{c^2}\right) = 0.$$

Thus, the line $\{(tx_0, ty_0, tz_0) : t \in \mathbb{R}\}$, which passes through the given point and the origin is contained in $\mathcal{C}_{a,b,c}$. The set of all lines obtained in this way forms the generators $\mathcal{L}$ of the cone.

### 10.2.5 Parametrizations - local description

Describing a parametrization of an elliptic cone can be generalized to any planar curve. Suppose you have a parametrization of a curve in the plane $Oxy$. In our case, for the ellipse

$$\begin{cases} x(\theta) = a\cos(\theta) \\ y(\theta) = b\sin(\theta) \end{cases} \quad \theta \in [0, 2\pi[.$$

You want to rescale this curve with the height such that when the height $z = 0$ you have a point, and for all other values of $z$ you have a rescaled version of your curve:

$$\begin{cases} x(\theta, h) = ha\cos(\theta) \\ y(\theta, h) = hb\sin(\theta) \\ z(\theta, h) = hc \end{cases} \quad \theta \in [0, 2\pi[ \quad h \in \mathbb{R}.$$

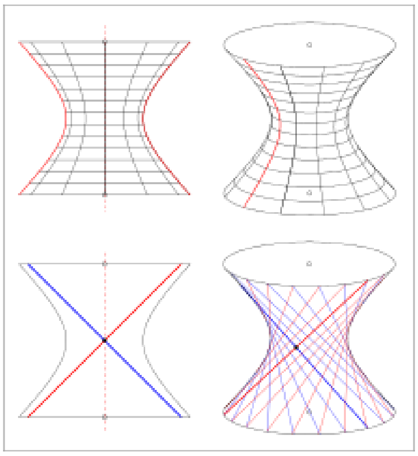
## 10.3 Hyperboloid of one sheet

### 10.3.1 Canonical equation - global description

A *hyperboloid of one sheet* is a surface which (in some coordinate system) satisfies an equation of the form

$$\mathcal{H}^1_{a,b,c} : \underbrace{\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2}}_{\varphi(x,y,z)} = 1 \quad \text{so} \quad \mathcal{H}^1_{a,b,c} = \varphi^{-1}(1) \tag{10.2}$$

for some positive constants $a, b, c \in \mathbb{R}$.



(a) Hyperboloid of one sheet[4]

(b) Kobe Port Tower

If we fix one variable, we obtain intersections with planes parallel to the coordinate axes. For this surface the intersections are either ellipses, hyperbolas or two lines. Check this for $y = h$ and deduce the axes of the hyperbolas that you obtain.

### 10.3.2 Tangent planes

Using the gradient or the algebraic method we obtain the tangent plane at a point $p = (x_p, y_p, z_p) \in \mathcal{H}^1_{a,b,c}$ to be

$$\mathcal{T}_p\mathcal{H}^1_{a,b,c} : \frac{x_p x}{a^2} + \frac{y_p y}{b^2} - \frac{z_p z}{c^2} = 1.$$

We will check this with the algebraic method. Let us consider a line passing through $p$ in parametric form

$$l : \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x_p \\ y_p \\ z_p \end{bmatrix} + t \underbrace{\begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix}}_{=v} \quad \Leftrightarrow \quad l = p + \langle v \rangle.$$

Recall that the tangent plane $\mathcal{T}_p\mathcal{H}^1_{a,b,c}$ is the union of all lines intersecting the quadric $\mathcal{H}^1_{a,b,c}$ at $p$ in a special way, i.e., in a double point. This is why we investigate the intersection of our quadric with $l$. How do we obtain the intersection $\mathcal{H}^1_{a,b,c} \cap l$? We look at those points of $l$ which satisfy the equation of our hyperboloid, i.e., we look for solutions $t$ for the equation

$$\varphi(\begin{bmatrix} x_p \\ y_p \\ z_p \end{bmatrix} + t \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix}) = 1 \quad \Leftrightarrow \quad \frac{(x_p + tv_x)^2}{a^2} + \frac{(y_p + tv_y)^2}{b^2} - \frac{(z_p + tv_z)^2}{c^2} - 1 = 0. \tag{10.3}$$

[4]Image source: Wikipedia

If we rearrange the left-hand side as a polynomial in $t$ we get

$$\Big(\frac{v_x^2}{a^2}+\frac{v_y^2}{b^2}-\frac{v_z^2}{c^2}\Big)t^2+2\Big(\frac{x_pv_x}{a^2}+\frac{y_pv_y}{b^2}-\frac{z_pv_z}{c^2}\Big)t+\underbrace{\frac{x_p^2}{a^2}+\frac{y_p^2}{b^2}-\frac{z_p^2}{c^2}}_{=1}-1=0$$

$$\Leftrightarrow\quad \underbrace{\Big(\frac{v_x^2}{a^2}+\frac{v_y^2}{b^2}-\frac{v_z^2}{c^2}\Big)}_{=0?}t^2+2\Big(\frac{x_pv_x}{a^2}+\frac{y_pv_y}{b^2}-\frac{z_pv_z}{c^2}\Big)t=0. \tag{10.4}$$

This equation in $t$ admits the solution $t=0$. That is clear, the point on $l$ corresponding to $t=0$ is the point $p$ which, by assumption, lies on $\mathcal{H}^1_{a,b,c}$. Furthermore, a second solution will correspond to a second point of intersection. However, the line $l$ is tangent to our quadric if $p$ is a *double point of intersection*. This happens if and only if the above equation has $t=0$ as double solution, i.e., if and only if

$$\frac{v_x^2}{a^2}+\frac{v_y^2}{b^2}-\frac{v_z^2}{c^2}\neq 0 \quad\text{and}\quad \frac{x_pv_x}{a^2}+\frac{y_pv_y}{b^2}-\frac{z_pv_z}{c^2}=0.$$

How do we interpret the second condition? In the Euclidean setting this can also be interpreted as saying that $(\frac{x_p}{a^2},\frac{y_p}{b^2},-\frac{z_p}{c^2})$ is perpendicular to the direction vector $v$ of the line. In both cases, all lines which are tangent to the surface and contain $p$ need to satisfy this equation, so the tangent plane is

$$\mathcal{T}_p\mathcal{H}^1_{a,b,c}:\begin{bmatrix}\frac{x_p}{a^2}\\ \frac{y_p}{b^2}\\ -\frac{z_p}{c^2}\end{bmatrix}\cdot\Big(\begin{bmatrix}x\\ y\\ z\end{bmatrix}-\begin{bmatrix}x_p\\ y_p\\ z_p\end{bmatrix}\Big)=0 \quad\Leftrightarrow\quad \frac{x_px}{a^2}+\frac{y_py}{b^2}-\frac{z_pz}{c^2}-1=0.$$

Deduce this equation also with the gradient.

How do we interpret the first condition? If

$$\frac{v_x^2}{a^2}+\frac{v_y^2}{b^2}-\frac{v_z^2}{c^2}=0 \tag{10.5}$$

then the $t^2$ term in equation (10.4) vanishes. If the coefficient of $t$ is non-zero, then equation (10.4) is linear, and has one simple solution $t=0$. This means that $l$ intersects $\mathcal{H}^1_{a,b,c}$ only once (it punctures the surface in one point). Such lines are not tangent to the surface. How can we visualize this? Let us start with what we know: the vector $v=(v_x,v_y,v_z)$ satisfies the equation (10.5), so we can think of it as the position vector of some point on the cone $\mathcal{C}_{a,b,c}$. How does this cone relate to our hyperboloid? Our surface $\mathcal{H}^1_{a,b,c}$ is the union of hyperbolas (revolving on ellipses around the $z$-axis) and if we take the union of all the asymptotes to these hyperbolas we get $\mathcal{C}_{a,b,c}$ (see Figure 10.5).

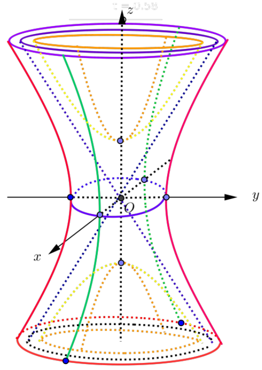



Figure 10.5: Hyperboloid and asymptotic cone[5]

This should help to see that, when the vector $v = (v_x, v_y, v_z)$ satisfies equation (10.5), the line $l$ is parallel to a line contained in the cone $\mathcal{C}_{a,b,c}$. It will therefore intersect $\mathcal{H}^1_{a,b,c}$ in at most one point.

Notice also that if $l \subseteq \mathcal{C}_{a,b,c}$, it will not intersect $\mathcal{H}^1_{a,b,c}$ at all, but this cannot happen in our setting because we chose the point $p$ such that it lies both on $l$ and on our quadric. In fact, the related question 'does a given line $l$ intersect $\mathcal{H}^1_{a,b,c}$?' can be answered by investigating equation (10.3) without the assumption that $p$ lies on the surface $\mathcal{H}^1_{a,b,c}$. How would you do this?

### 10.3.3 $\mathcal{H}^1_{a,b,c}$ as ruled surface

Hmm... I know what a ruled surface is, because cones and cylinders are ruled surfaces, but, 'doubly ruled'? *Doubly ruled* just means that it is a ruled surface in two ways. So, there are two distinct families of lines, $\mathcal{L}_1$ and $\mathcal{L}_2$ respectively, such that

$$\mathcal{H}^1_{a,b,c} = \bigcup_{l \in \mathcal{L}_1} l \quad \text{and} \quad \mathcal{H}^1_{a,b,c} = \bigcup_{l \in \mathcal{L}_2} l.$$

One way to see where the two families of lines come from is to rearrange Equation (10.2):

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1 \quad \Leftrightarrow \quad \frac{x^2}{a^2} - \frac{z^2}{c^2} = 1 - \frac{y^2}{b^2} \quad \Leftrightarrow \quad \left(\frac{x}{a} - \frac{z}{c}\right)\left(\frac{x}{a} + \frac{z}{c}\right) = \left(1 - \frac{y}{b}\right)\left(1 + \frac{y}{b}\right) \tag{10.6}$$

Now, assume that the factors in the last equation are not 0, then we can divide to obtain

$$\Leftrightarrow \quad \frac{\frac{x}{a} - \frac{z}{c}}{1 - \frac{y}{b}} = \frac{1 + \frac{y}{b}}{\frac{x}{a} + \frac{z}{c}} = \frac{\mu}{\lambda}$$

for some parameters $\lambda$ and $\mu$. We introduced these parameters in order to separate the above equation:

$$\Leftrightarrow \quad l_{\lambda,\mu} : \begin{cases} \lambda\left(\frac{x}{a} - \frac{z}{c}\right) = \mu\left(1 - \frac{y}{b}\right) \\ \mu\left(\frac{x}{a} + \frac{z}{c}\right) = \lambda\left(1 + \frac{y}{b}\right) \end{cases}.$$

[5]Prof. C. Pintea - lecture notes

What we end up with is a system of two equations, which are linear in $x, y, z$ and which depend on the parameters $\lambda$ and $\mu$. For each fixed pair of parameters, $\lambda$ and $\mu$, we get a line which we denote with $l_{\lambda,\mu}$. Reading the above deduction backwards it is easy to see that all points on such a line satisfy the equation of $\mathcal{H}^1_{a,b,c}$. So, we have a family of lines contained in your hyperboloid.

We assumed that the factors in (10.6) are not zero. In fact, you only divide by two of them, so .. if one of those two is zero, you can flip the above fraction and divide by the other two. That will lead to the same family of lines $\mathcal{L}_1 = \{l_{\lambda,\mu} : \lambda, \mu \in \mathbb{R}, \lambda^2 + \mu^2 \neq 0\}$.

OK, what about $\mathcal{L}_2$? The second family of generators (these lines are called generators), is obtained if you group the terms differently:

$$\frac{\frac{x}{a} - \frac{z}{c}}{1 + \frac{y}{b}} = \frac{1 - \frac{y}{b}}{\frac{x}{a} + \frac{z}{c}} = \frac{\mu}{\lambda}.$$

Then, you obtain:

$$\tilde{l}_{\lambda,\mu} : \begin{cases} \lambda\left(\frac{x}{a} - \frac{z}{c}\right) = \mu\left(1 + \frac{y}{b}\right) \\ \mu\left(\frac{x}{a} + \frac{z}{c}\right) = \lambda\left(1 - \frac{y}{b}\right) \end{cases}.$$

As above, one can check that points on these lines satisfy the equation of our hyperboloid.

One important thing to notice is that, although we write down two parameters, $\lambda$ and $\mu$, we don't necessarily get distinct lines for distinct parameters: $l_{\lambda,\mu} = l_{t\lambda,t\mu}$ for any nonzero scalar $t$. So, in fact, $\mathcal{L}_1$ depends on one parameter. More concretely

$$\mathcal{L}_1 = \left\{ l_\alpha : \begin{cases} \left(\frac{x}{a} - \frac{z}{c}\right) = \alpha\left(1 - \frac{y}{b}\right) \\ \alpha\left(\frac{x}{a} + \frac{z}{c}\right) = \left(1 + \frac{y}{b}\right) \end{cases} \right\} \bigcup \left\{ l_\infty : \begin{cases} 0 = \left(1 - \frac{y}{b}\right) \\ \left(\frac{x}{a} + \frac{z}{c}\right) = 0 \end{cases} \right\}$$

and similarly for $\mathcal{L}_2$.

### 10.3.4 Parametrizations - local description

Two parametrizations of this surface are

$$\sigma_1(\theta_1, \theta_2) = \begin{bmatrix} a\sqrt{1+\theta_2^2}\cos(\theta_1) \\ b\sqrt{1+\theta_2^2}\sin(\theta_1) \\ c\theta_2 \end{bmatrix} \quad \text{and} \quad \sigma_2(\theta_1, \theta_2) = \begin{bmatrix} a\cosh(\theta_2)\cos(\theta_1) \\ b\cosh(\theta_2)\sin(\theta_1) \\ c\sinh(\theta_2) \end{bmatrix}$$

for $\theta_1 \in [0, 2\pi[$ and $\theta_2 \in \mathbb{R}$. Why? The parameter $\theta_1$ is used to rotate on ellipses the curve obtained for $\theta_1 = 0$. What is this curve that we 'rotate'? Check this and deduce a parametrization of the tangent plane $T_p\mathcal{H}^1_{a,b,c}$ at the point

$$p = \begin{bmatrix} x(\theta_{1,p}, \theta_{2,p}) \\ y(\theta_{1,p}, \theta_{2,p}) \\ z(\theta_{1,p}, \theta_{2,p}) \end{bmatrix} \in \mathcal{H}^1_{a,b,c}$$

where $\theta_{1,p}$ and $\theta_{2,p}$ are the parameters of the point $p$, i.e., $p = p(x(\theta_{1,p}, \theta_{2,p}), y(\theta_{1,p}, \theta_{2,p}), z(\theta_{1,p}, \theta_{2,p}))$.

## 10.4 Hyperboloid of two sheets

### 10.4.1 Canonical equation - global description

A *hyperboloid of two sheets* is a surface which (in some coordinate system) satisfies an equation of the form

$$\mathcal{H}^2_{a,b,c} : \underbrace{\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2}}_{\varphi(x,y,z)} = -1 \quad \text{so} \quad \mathcal{H}^2_{a,b,c} = \varphi^{-1}(-1)$$

for some positive constants $a, b, c \in \mathbb{R}$.

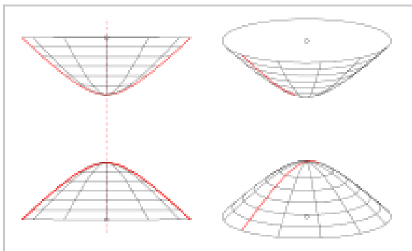


Figure 10.6: Hyperboloid of two sheets[6]

If we fix one variable, we obtain intersections with planes parallel to the coordinate axes. For this surface the intersections are either ellipses, hyperbolas or the empty set. Check this for $y = h$ and deduce the axes of the hyperbolas that you obtain.

### 10.4.2 Tangent planes

Using the gradient or the algebraic method we obtain the tangent plane at a point $p = (x_p, y_p, z_p) \in \mathcal{H}^2_{a,b,c}$ to be

$$\mathcal{T}_p\mathcal{H}^2_{a,b,c} : \frac{x_p x}{a^2} + \frac{y_p y}{b^2} - \frac{z_p z}{c^2} = -1.$$

We will check this with the algebraic method. Let us consider a line passing through $p$ in parametric form

$$l : \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x_p \\ y_p \\ z_p \end{bmatrix} + t \underbrace{\begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix}}_{=v} \quad \Leftrightarrow \quad l = p + \langle v \rangle.$$

Recall that the tangent plane $\mathcal{T}_p\mathcal{H}^2_{a,b,c}$ is the union of all lines intersecting the quadric $\mathcal{H}^2_{a,b,c}$ at $p$ in a special way, i.e., in a double point. This is why we investigate the intersection of our quadric with $l$. How do we obtain the intersection $\mathcal{H}^2_{a,b,c} \cap l$? We look at those points of $l$ which satisfy the equation of

[6]Image source: Wikipedia

our hyperboloid, i.e., we look for solutions $t$ for the equation

$$\varphi(\begin{bmatrix}x_p\\y_p\\z_p\end{bmatrix}+t\begin{bmatrix}v_x\\v_y\\v_z\end{bmatrix})=-1 \quad\Leftrightarrow\quad \frac{(x_p+tv_x)^2}{a^2}+\frac{(y_p+tv_y)^2}{b^2}-\frac{(z_p+tv_z)^2}{c^2}+1=0.$$

If we rearrange the left-hand side as a polynomial in $t$ we get

$$\Big(\frac{v_x^2}{a^2}+\frac{v_y^2}{b^2}-\frac{v_z^2}{c^2}\Big)t^2+2\Big(\frac{x_pv_x}{a^2}+\frac{y_pv_y}{b^2}-\frac{z_pv_z}{c^2}\Big)t+\underbrace{\frac{x_p^2}{a^2}+\frac{y_p^2}{b^2}-\frac{z_p^2}{c^2}}_{=-1}+1=0$$

$$\Leftrightarrow\quad \underbrace{\Big(\frac{v_x^2}{a^2}+\frac{v_y^2}{b^2}-\frac{v_z^2}{c^2}\Big)}_{=0?}t^2+2\Big(\frac{x_pv_x}{a^2}+\frac{y_pv_y}{b^2}-\frac{z_pv_z}{c^2}\Big)t=0. \tag{10.7}$$

This equation in $t$ admits the solution $t=0$. That is clear, the point on $l$ corresponding to $t=0$ is the point $p$ which, by assumption, lies on $\mathcal{H}^2_{a,b,c}$. Furthermore, a second solution will correspond to a second point of intersection. However, the line $l$ is tangent to our quadric if $p$ is a *double point of intersection*. This happens if and only if the above equation has $t=0$ as double solution, i.e., if and only if

$$\frac{v_x^2}{a^2}+\frac{v_y^2}{b^2}-\frac{v_z^2}{c^2}\neq 0 \quad\text{and}\quad \frac{x_pv_x}{a^2}+\frac{y_pv_y}{b^2}-\frac{z_pv_z}{c^2}=0.$$

How do we interpret the second condition? Similar to the case of the hyperboloid of one sheet:

$$T_p\mathcal{H}^2_{a,b,c}:\begin{bmatrix}\frac{x_p}{a^2}\\\frac{y_p}{b^2}\\-\frac{z_p}{c^2}\end{bmatrix}\cdot\Big(\begin{bmatrix}x\\y\\z\end{bmatrix}-\begin{bmatrix}x_p\\y_p\\z_p\end{bmatrix}\Big)=0 \quad\Leftrightarrow\quad \frac{x_px}{a^2}+\frac{y_py}{b^2}-\frac{z_pz}{c^2}+1=0.$$

Deduce this equation also with the gradient.

How do we interpret the first condition? If

$$\frac{v_x^2}{a^2}+\frac{v_y^2}{b^2}-\frac{v_z^2}{c^2}=0 \tag{10.8}$$

then the $t^2$ term in equation (10.7) vanishes. If the coefficient of $t$ is non-zero, then equation (10.7) is linear, and has one simple solution $t=0$. This means that $l$ intersects $\mathcal{H}^2_{a,b,c}$ only once (it punctures the surface in one point). Such lines are not tangent to the surface. How can we visualize this? Let us start with what we know: the vector $v=(v_x,v_y,v_z)$ satisfies the equation (10.8), so we can think of it as the position vector of some point on the cone $\mathcal{C}_{a,b,c}$. How does this cone relate to our hyperboloid? Our surface $\mathcal{H}^2_{a,b,c}$ is the union of hyperbolas and if we take the union of all the asymptotes to these hyperbolas we get $\mathcal{C}_{a,b,c}$ (see Figure 10.5). So, when $l$ is parallel to a line contained in $\mathcal{C}_{a,b,c}$, it will intersect $\mathcal{H}^2_{a,b,c}$ in at most one point. Notice also that if $l\subseteq\mathcal{C}_{a,b,c}$, it will not intersect $\mathcal{H}^2_{a,b,c}$ at all, but this cannot happen because we chose the point $p$ such that it lies both on $l$ and on our quadric.

### 10.4.3 Parametrizations - local description

Two parametrizations of this surface are

$$\sigma_1(\theta_1,\theta_2)=\begin{bmatrix} a\sqrt{\theta_2^2-1}\cos(\theta_1)\\ b\sqrt{\theta_2^2-1}\sin(\theta_1)\\ c\theta_2\end{bmatrix}\quad\text{and}\quad \sigma_2(\theta_1,\theta_2)=\begin{bmatrix} a\sinh(\theta_2)\cos(\theta_1)\\ b\sinh(\theta_2)\sin(\theta_1)\\ \varepsilon c\cosh(\theta_2)\end{bmatrix}$$

for $\theta_1\in[0,2\pi[$, $\theta_2\in\mathbb{R}$ and $\varepsilon\in\{\pm1\}$. Why? The parameter $\theta_1$ is used to 'rotate' on ellipses the curve obtained for $\theta_1=0$. What is this curve? Check this and deduce a parametrization of the tangent plane $\mathcal{T}_p\mathcal{H}^2_{a,b,c}$ at the point

$$p=\begin{bmatrix} x(\theta_{1,p},\theta_{2,p})\\ y(\theta_{1,p},\theta_{2,p})\\ z(\theta_{1,p},\theta_{2,p})\end{bmatrix}\in\mathcal{H}^2_{a,b,c}$$

where $\theta_{1,p}$ and $\theta_{2,p}$ are the parameters of the point $p$, i.e., $p=p(x(\theta_{1,p},\theta_{2,p}),y(\theta_{1,p},\theta_{2,p}),z(\theta_{1,p},\theta_{2,p}))$. Notice also that with $\sigma_2$ we have a parametrization for each sheet of this hyperboloid, with $\varepsilon=1$ we get one sheet and with $\varepsilon=-1$ we get the other sheet. One should also be careful with where the parameters live: for $\sigma_1$ you want to choose $\theta_2$ in $]-\infty,-1]\cup[1,\infty[$ so that the square root is defined.

## 10.5 Elliptic paraboloid

### 10.5.1 Canonical equation - global description

An *elliptic paraboloid* is a surface which (in some coordinate system) satisfies an equation of the form

$$\mathcal{P}^e_{a,b}:\underbrace{\frac{x^2}{a}+\frac{y^2}{b}-2z}_{\varphi(x,y,z)}=0\quad\text{so}\quad \mathcal{P}^e_{a,b}=\varphi^{-1}(0)$$

for some positive constants $a,b\in\mathbb{R}$.

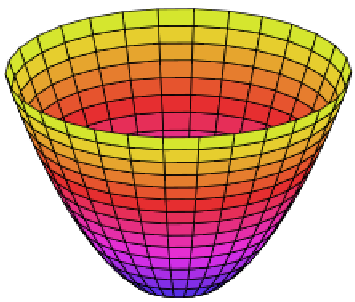


Figure 10.7: Elliptic paraboloid[7]

[7] Image source: Wikipedia

If we fix one variable, we obtain intersections with planes parallel to the coordinate axes. For this surface the intersections are either ellipses, parabolas or the empty set. Check this for $y = h$ and see what parabolas you obtain.

### 10.5.2 Tangent planes

Using the gradient or the algebraic method we obtain the tangent plane at a point $p = (x_p, y_p, z_p) \in \mathcal{P}^e_{a,b}$ to be

$$\mathcal{T}_p\mathcal{P}^e_{a,b} : \frac{x_p x}{a} + \frac{y_p y}{b} - z_p - z = 0.$$

We will check this with the algebraic method. Let us consider a line passing through $p$ in parametric form

$$l : \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x_p \\ y_p \\ z_p \end{bmatrix} + t \underbrace{\begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix}}_{=v} \quad \Leftrightarrow \quad l = p + \langle v \rangle.$$

Recall that the tangent plane $\mathcal{T}_p\mathcal{P}^e_{a,b}$ is the union of all lines intersecting the quadric $\mathcal{P}^e_{a,b}$ at $p$ in a special way, i.e., in a double point. This is why we investigate the intersection of our quadric with $l$. How do we obtain the intersection $\mathcal{P}^e_{a,b} \cap l$? We look at those points of $l$ which satisfy the equation of our paraboloid, i.e., we look for solutions $t$ for the equation

$$\varphi\left(\begin{bmatrix} x_p \\ y_p \\ z_p \end{bmatrix} + t \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix}\right) = 0 \quad \Leftrightarrow \quad \frac{(x_p + tv_x)^2}{a} + \frac{(y_p + tv_y)^2}{b} - 2(z_p + tv_z) = 0.$$

If we rearrange the left-hand side as a polynomial in $t$ we get

$$\Big(\frac{v_x^2}{a} + \frac{v_y^2}{b}\Big)t^2 + 2\Big(\frac{x_p v_x}{a} + \frac{y_p v_y}{b} - v_z\Big)t + \underbrace{\frac{x_p^2}{a} + \frac{y_p^2}{b} - 2z_p}_{=0} = 0$$

$$\Leftrightarrow \quad \underbrace{\Big(\frac{v_x^2}{a} + \frac{v_y^2}{b}\Big)}_{\neq 0}t^2 + 2\Big(\frac{x_p v_x}{a} + \frac{y_p v_y}{b} - v_z\Big)t = 0. \tag{10.9}$$

This equation in $t$ admits the solution $t = 0$. That is clear, the point on $l$ corresponding to $t = 0$ is the point $p$ which, by assumption, lies on $\mathcal{P}^e_{a,b}$. Furthermore, a second solution will correspond to a second point of intersection. However, the line $l$ is tangent to our quadric if $p$ is a *double point of intersection*. This happens if and only if the above equation has $t = 0$ as double solution, i.e., if and only if

$$\frac{x_p v_x}{a} + \frac{y_p v_y}{b} - v_z = 0.$$

How do we interpret this condition? Similar to the quadrics treated in the previous sections, so

$$\mathcal{T}_p\mathcal{P}^e_{a,b} : \begin{bmatrix} \frac{x_p}{a} \\ \frac{y_p}{b} \\ -1 \end{bmatrix} \cdot \left(\begin{bmatrix} x \\ y \\ z \end{bmatrix} - \begin{bmatrix} x_p \\ y_p \\ z_p \end{bmatrix}\right) = 0 \quad \Leftrightarrow \quad \frac{x_p x}{a} + \frac{y_p y}{b} - z_p - z = 0.$$

Deduce this equation also with the gradient.

### 10.5.3 Parametrizations - local description

A parametrization of this surface is

$$\sigma(\theta_1, \theta_2) = \begin{bmatrix} \sqrt{a\theta_2}\cos(\theta_1) \\ \sqrt{b\theta_2}\sin(\theta_1) \\ \theta_2/2 \end{bmatrix} \quad \theta_1 \in [0, 2\pi[ \quad \theta_2 \in [0, \infty[$$

Why? Check this and deduce a parametrization of the tangent plane $\mathcal{T}_p\mathcal{P}^e_{a,b}$ at the point

$$p = \begin{bmatrix} x(\theta_{1,p}, \theta_{2,p}) \\ y(\theta_{1,p}, \theta_{2,p}) \\ z(\theta_{1,p}, \theta_{2,p}) \end{bmatrix} \in \mathcal{P}^e_{a,b}$$

where $\theta_{1,p}$ and $\theta_{2,p}$ are the parameters of the point $p$, i.e., $p = p(x(\theta_{1,p}, \theta_{2,p}), y(\theta_{1,p}, \theta_{2,p}), z(\theta_{1,p}, \theta_{2,p}))$.

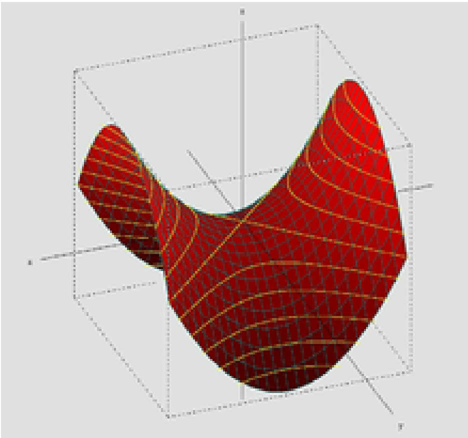
## 10.6 Hyperbolic paraboloid

### 10.6.1 Canonical equation - global description

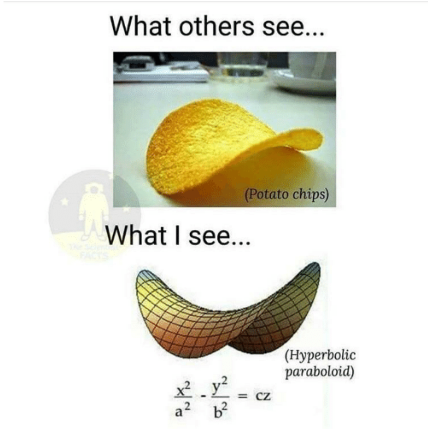
A *hyperbolic paraboloid* is a surface which (in some coordinate system) satisfies an equation of the form

$$\mathcal{P}^h_{a,b} : \underbrace{\frac{x^2}{a} - \frac{y^2}{b} - 2z}_{\varphi(x,y,z)} = 0 \quad \text{so} \quad \mathcal{P}^h_{a,b} = \varphi^{-1}(0) \tag{10.10}$$

for some positive constants $a, b \in \mathbb{R}$.



(a) Hyperbolic paraboloid[8]




(b) Potato chips[9]

If we fix one variable, we obtain intersections with planes parallel to the coordinate axes. For this surface the intersections are either parabolas, hyperbolas or two lines. Check this for $y = h$ and deduce the parabolas that you obtain.

### 10.6.2 Tangent planes

Using the gradient or the algebraic method we obtain the tangent plane at a point $p = (x_p, y_p, z_p) \in \mathcal{P}^h_{a,b}$ to be

$$\mathcal{T}_p\mathcal{P}^h_{a,b} : \frac{x_p x}{a} - \frac{y_p y}{b} - z_p - z = 0.$$

We will check this with the algebraic method. Let us consider a line passing through $p$ in parametric form

$$l : \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x_p \\ y_p \\ z_p \end{bmatrix} + t \underbrace{\begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix}}_{=v} \quad \Leftrightarrow \quad l = p + \langle v \rangle.$$

Recall that the tangent plane $\mathcal{T}_p\mathcal{P}^h_{a,b}$ is the union of all lines intersecting the quadric $\mathcal{P}^h_{a,b}$ at $p$ in a special way, i.e., in a double point. This is why we investigate the intersection of our quadric with $l$. How do we obtain the intersection $\mathcal{P}^h_{a,b} \cap l$? We look at those points of $l$ which satisfy the equation of our paraboloid, i.e., we look for solutions $t$ for the equation

$$\varphi(\begin{bmatrix} x_p \\ y_p \\ z_p \end{bmatrix} + t \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix}) = 0 \quad \Leftrightarrow \quad \frac{(x_p + tv_x)^2}{a} - \frac{(y_p + tv_y)^2}{b} - 2(z_p + tv_z) = 0.$$

[6]Image source: Wikipedia
[9]Image source: the internet

If we rearrange the left-hand side as a polynomial in $t$ we get

$$\Big(\frac{v_x^2}{a}-\frac{v_y^2}{b}\Big)t^2+2\Big(\frac{x_pv_x}{a}-\frac{y_pv_y}{b}-v_z\Big)t+\underbrace{\frac{x_p^2}{a}-\frac{y_p^2}{b}-2z_p}_{=0}=0$$

$$\Leftrightarrow\quad \underbrace{\Big(\frac{v_x^2}{a}-\frac{v_y^2}{b}\Big)}_{=0?}t^2+2\Big(\frac{x_pv_x}{a}-\frac{y_pv_y}{b}-v_z\Big)t=0. \tag{10.11}$$

This equation in $t$ admits the solution $t=0$. That is clear, the point on $l$ corresponding to $t=0$ is the point $p$ which, by assumption, lies on $\mathcal{P}^h_{a,b}$. Furthermore, a second solution will correspond to a second point of intersection. However, the line $l$ is tangent to our quadric if $p$ is a *double point of intersection*. This happens if and only if the above equation has $t=0$ as double solution, i.e., if and only if

$$\frac{v_x^2}{a}-\frac{v_y^2}{b}\neq 0\quad\text{and}\quad \frac{x_pv_x}{a}-\frac{y_pv_y}{b}-v_z=0.$$

How do we interpret the second condition? Similar to the quadrics treated in the previous sections, so

$$T_p\mathcal{P}^h_{a,b}:\begin{bmatrix}\frac{x_p}{a}\\ -\frac{y_p}{b}\\ -1\end{bmatrix}\cdot\left(\begin{bmatrix}x\\ y\\ z\end{bmatrix}-\begin{bmatrix}x_p\\ y_p\\ z_p\end{bmatrix}\right)=0\quad\Leftrightarrow\quad \frac{x_px}{a}-\frac{y_py}{b}-z_p-z=0.$$

Deduce this equation also with the gradient.

How do we interpret the first condition? If

$$\frac{v_x^2}{a}-\frac{v_y^2}{b}=0\quad\Leftrightarrow\quad\Big(\frac{v_x}{\sqrt{a}}-\frac{v_y}{\sqrt{b}}\Big)\Big(\frac{v_x}{\sqrt{a}}+\frac{v_y}{\sqrt{b}}\Big)=0 \tag{10.12}$$

then the $t^2$ term in equation (10.11) vanishes. If the coefficient of $t$ is non-zero, equation (10.11) is linear, and has one simple solution $t=0$. This means that $l$ intersects $\mathcal{P}^h_{a,b}$ only once (it punctures the surface in one point). Such lines are not tangent to the surface. How can we visualize this? Let us start with what we know: the vector $v=(v_x,v_y,v_z)$ satisfies the equation (10.12), so we can think of it as the position vector of some point on the cylinder $\mathrm{Cyl}(\mathcal{A},\mathbf{k})$ where $\mathcal{A}$ is the union of two lines given by the equation (10.12) and $z=0$. These two lines are the intersection of our quadric $\mathcal{P}^h_{a,b}$ with the coordinate plane $z=0$ (they are visible in Figure 10.8a). If $v_z=0$, $v$ is parallel to one of the lines in $\mathcal{A}$. Three things can happen here:

1. $l$ is one of the lines in $\mathcal{A}$, then all points of $l$ lie in $\mathcal{P}^h_{a,b}$, so we have infinitely many solutions $t$ for equation (10.11), or
2. $l$ is one of the lines in $\mathcal{A}$ translated in the positive direction of the $z$-axis, in which case $l$ will not intersect the surface $\mathcal{P}^h_{a,b}$ (this cannot happen for our choice of $l$ because we assume that $p\in l\cap\mathcal{P}^h_{a,b}$), or
3. $l$ is parallel to one of the lines in $\mathcal{A}$ (excluding the previous two cases), in which case $l$ will puncture the surface $\mathcal{P}^h_{a,b}$ in a simple point (it will not be tangent to the surface).

### 10.6.3 $\mathcal{P}^h_{a,b}$ as ruled surface

Here is another fact: the hyperbolic paraboloid is a *doubly ruled surface* (like the hyperboloid of one sheet). In other words, there are two families of lines, $\mathcal{L}_1$ and $\mathcal{L}_2$ respectively, such that

$$\mathcal{P}^h_{a,b} = \bigcup_{l\in\mathcal{L}_1} l \quad \text{and} \quad \mathcal{P}^h_{a,b} = \bigcup_{l\in\mathcal{L}_2} l.$$

Every point on this surface lies on one line in $\mathcal{L}_1$ and on one line in $\mathcal{L}_2$. The generators containing the *saddle point* (with our equation, this point is the origin of the coordinate system) are visible in Figure 10.8a.

Again, one way to see where the two families of lines come from is to rearrange (10.10)

$$\frac{x^2}{a} - \frac{y^2}{b} - 2z = 0 \quad \Leftrightarrow \quad \frac{x^2}{a} - \frac{y^2}{b} = 2z \quad \Leftrightarrow \quad \left(\frac{x}{\sqrt{a}} - \frac{y}{\sqrt{b}}\right)\left(\frac{x}{\sqrt{a}} + \frac{y}{\sqrt{b}}\right) = 2z \tag{10.13}$$

Similar to the case of the hyperboloid of one sheet, we can introduce two parameters $\lambda$ and $\mu$, in order to separate the above equation:

$$\Leftrightarrow \quad l_{\lambda,\mu} : \begin{cases} \lambda\left(\frac{x}{\sqrt{a}} - \frac{y}{\sqrt{b}}\right) = 2\mu z \\ \mu\left(\frac{x}{\sqrt{a}} + \frac{y}{\sqrt{b}}\right) = \lambda \end{cases}.$$

What we end up with is a system of two equations, which are linear in $x, y, z$ and which depend on the parameters $\lambda$ and $\mu$. For each fixed pair of parameters, $\lambda$ and $\mu$, we get a line which we denote with $l_{\lambda,\mu}$. It is easy to check that all points on such a line satisfy the equation of $\mathcal{P}^h_{a,b}$. This is the first family of lines $\mathcal{L}_1 = \{l_{\lambda,\mu} : \lambda, \mu \text{ not both zero}\}$.

The second family of generators, $\mathcal{L}_2$, is obtained if you group the terms differently:

$$\tilde{l}_{\lambda,\mu} : \begin{cases} \lambda\left(\frac{x}{\sqrt{a}} + \frac{y}{\sqrt{b}}\right) = 2\mu z \\ \mu\left(\frac{x}{\sqrt{a}} - \frac{y}{\sqrt{b}}\right) = \lambda \end{cases}.$$

As above, one can check that points on these lines satisfy the equation of our paraboloid.

Again, one important thing to notice is that, although we write down two parameters, $\lambda$ and $\mu$, we don't necessarily get distinct lines for distinct parameters: $l_{\lambda,\mu} = l_{t\lambda,t\mu}$ for any nonzero scalar $t$. So, in fact, $\mathcal{L}_1$ depends on one parameter. More concretely

$$\mathcal{L}_1 := \left\{ l_\alpha : \begin{cases} \left(\frac{x}{\sqrt{a}} - \frac{y}{\sqrt{b}}\right) = 2\alpha z \\ \alpha\left(\frac{x}{\sqrt{a}} + \frac{y}{\sqrt{b}}\right) = 1 \end{cases} \right\} \bigcup \left\{ l_\infty : \begin{cases} 0 = 2z \\ \left(\frac{x}{\sqrt{a}} + \frac{y}{\sqrt{b}}\right) = 0 \end{cases} \right\}$$

and similarly for $\mathcal{L}_2$. You might have noticed that $l_\infty$ is one of the lines visible in Figure 10.8a, since it lies in the plane $z = 0$. The other one, $\tilde{l}_\infty$, belongs to the family $\mathcal{L}_2$.

### 10.6.4 Parametrizations - local description

A parametrization of this surface is

$$\sigma(\theta_1,\theta_2)=\begin{bmatrix}\sqrt{a}\theta_1\\ \sqrt{b}\theta_2\\ \frac{1}{2}(\theta_1^2-\theta_2^2)\end{bmatrix}\quad \theta_1,\theta_2\in\mathbb{R}$$

Check this and deduce a parametrization of the tangent plane $\mathcal{T}_p\mathcal{P}^h_{a,b}$ at the point

$$p=\begin{bmatrix}x(\theta_{1,p},\theta_{2,p})\\ y(\theta_{1,p},\theta_{2,p})\\ z(\theta_{1,p},\theta_{2,p})\end{bmatrix}\in\mathcal{P}^h_{a,b}$$

where $\theta_{1,p}$ and $\theta_{2,p}$ are the parameters of the point $p$, i.e., $p=p(x(\theta_{1,p},\theta_{2,p}),y(\theta_{1,p},\theta_{2,p}),z(\theta_{1,p},\theta_{2,p}))$.

APPENDIX A

# Changing the basis in a vector space

Let $\phi : V \to W$ be a linear map between the vector spaces $V$ and $W$. Let $\mathcal{E} = (\mathbf{e}_1, \ldots, \mathbf{e}_n)$ be a basis for $V$ and let $\mathcal{F}$ be a basis for $W$. In your algebra course, you used the notation $[\phi]_{\mathcal{E},\mathcal{F}}$ for the matrix of the linear map $\phi$ with respect to the bases $\mathcal{E}$ and $\mathcal{F}$ [9, Definition 3.4.1]. We will use the notation

$$\mathrm{M}_{\mathcal{F},\mathcal{E}}(\phi) = [\phi]_{\mathcal{E},\mathcal{F}}.$$

Notice that the indices $\mathcal{E},\mathcal{F}$ are *reversed*. Recall that this is the matrix whose columns are the components of the $\phi(\mathbf{e}_i)$'s with respect to the basis $\mathcal{F}$:

$$\mathrm{M}_{\mathcal{F},\mathcal{E}}(\phi) = \begin{bmatrix} \uparrow & \cdots & \uparrow \\ [\phi(\mathbf{e}_1)]_{\mathcal{F}} & \cdots & [\phi(\mathbf{e}_n)]_{\mathcal{F}} \\ \downarrow & \cdots & \downarrow \end{bmatrix}.$$

You have also learned [9, Theorem 3.4.8] that if $\psi : W \to U$ is another linear map to some vector space $U$ with basis $\mathcal{G}$, then

$$\mathrm{M}_{\mathcal{G},\mathcal{E}}(\psi \circ \phi) = \mathrm{M}_{\mathcal{G},\mathcal{F}}(\psi) \cdot \mathrm{M}_{\mathcal{F},\mathcal{E}}(\phi).$$

In particular, if $V = W = U$, $\mathcal{G} = \mathcal{F}$, and $\phi = \psi = \mathrm{Id}_V$, then

$$I_n = \mathrm{M}_{\mathcal{F},\mathcal{F}}(\mathrm{Id}_V) = \mathrm{M}_{\mathcal{F},\mathcal{F}}(\mathrm{Id}_V \circ \mathrm{Id}_V) = \mathrm{M}_{\mathcal{F},\mathcal{E}}(\mathrm{Id}_V) \cdot \mathrm{M}_{\mathcal{E},\mathcal{F}}(\mathrm{Id}_V)$$

hence, $\mathrm{M}_{\mathcal{F},\mathcal{E}}(\mathrm{Id}_V) = \mathrm{M}_{\mathcal{E},\mathcal{F}}(\mathrm{Id}_V)^{-1}$, and thus

$$\mathrm{M}_{\mathcal{F},\mathcal{F}}(\phi) = \mathrm{M}_{\mathcal{F},\mathcal{E}}(\mathrm{Id}_V) \cdot \mathrm{M}_{\mathcal{E},\mathcal{E}}(\phi) \cdot \mathrm{M}_{\mathcal{E},\mathcal{F}}(\mathrm{Id}_V) = \mathrm{M}_{\mathcal{E},\mathcal{F}}(\mathrm{Id}_V)^{-1} \cdot \mathrm{M}_{\mathcal{E},\mathcal{E}}(\phi) \cdot \mathrm{M}_{\mathcal{E},\mathcal{F}}(\mathrm{Id}_V).$$

So, the matrix of $\phi$ with respect to the basis $\mathcal{F}$ is obtained from the matrix of $\phi$ with respect to the basis $\mathcal{E}$ by conjugating with the matrix $\mathrm{M}_{\mathcal{E},\mathcal{F}}(\mathrm{Id}_V)$.

**Definition A.1.** The matrix $\mathrm{M}_{\mathcal{E},\mathcal{F}} := \mathrm{M}_{\mathcal{E},\mathcal{F}}(\mathrm{Id}_V)$ is called the *change of basis matrix from the basis $\mathcal{F}$ to the basis $\mathcal{E}$*. It is the matrix whose columns are the components of the vectors in $\mathcal{F}$ with respect to $\mathcal{E}$. If $\mathcal{F} = (\mathbf{f}_1, \dots, \mathbf{f}_n)$, then

$$\mathrm{M}_{\mathcal{E},\mathcal{F}} = \begin{bmatrix} \uparrow & \dots & \uparrow \\ [\mathbf{f}_1]_{\mathcal{E}} & \dots & [\mathbf{f}_n]_{\mathcal{E}} \\ \downarrow & \dots & \downarrow \end{bmatrix}.$$

Moreover, we have $\mathrm{M}_{\mathcal{F},\mathcal{E}} = (\mathrm{M}_{\mathcal{E},\mathcal{F}})^{-1}$.

**Proposition A.2.** Let $\mathbf{v} \in V$. The coordinates of $\mathbf{v}$ with respect to the basis $\mathcal{E}$ are obtained from the coordinates of $\mathbf{v}$ with respect to the basis $\mathcal{F}$ by multiplication with the change of basis matrix:

$$[\mathbf{v}]_{\mathcal{E}} = \mathrm{M}_{\mathcal{E},\mathcal{F}} \cdot [\mathbf{v}]_{\mathcal{F}}.$$

APPENDIX B

# Coordinate systems

## B.1 Polar coordinates

Polar coordinates use oriented angles with a line. They establish a bijection between points distinct from the origin and pairs of numbers $(r, \varphi)$ in $\mathbb{R}_{>0} \times [0, 2\pi)$.

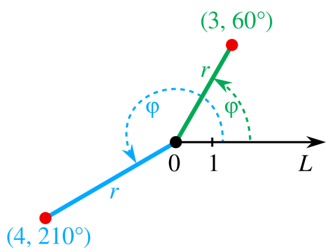



Figure B.1: Polar coordinates[1].

The correspondence between Cartesian coordinates $(x, y)$ and polar coordinates $(r, \varphi)$ is as follows:

$$\begin{cases} x = r\cos\varphi \\ y = r\sin\varphi \end{cases}, \quad \begin{cases} r = r(x,y) = \sqrt{x^2+y^2} \\ \varphi = f(x,y) \end{cases}$$

where, with $r \neq 0$,

$$f(x,y) = \begin{cases} \arccos(\frac{x}{r}) & \text{if } y \geq 0 \\ -\arccos(\frac{x}{r}) & \text{if } y < 0 \end{cases}.$$

[1] Image source: Wikipedia

## B.2 Cylindrical coordinates

Cylindrical coordinates extend polar coordinates to dimension 3. They establish a bijection between points distinct from the origin and triples of numbers $(r,\varphi,z)$ in $\mathbb{R}_{>0}\times[0,2\pi)\times\mathbb{R}$.

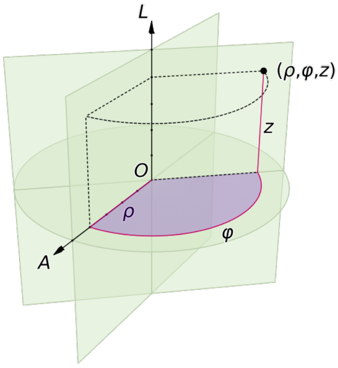



Figure B.2: Cylindrical coordinates[2].

The correspondence between Cartesian coordinates $(x,y,z)$ and cylindrical coordinates $(r,\varphi,z)$ is as follows:

$$\begin{cases} x = r\cos\varphi \\ y = r\sin\varphi \\ z = z \end{cases}, \quad \begin{cases} r = r(x,y) = \sqrt{x^2+y^2} \\ \varphi = f(x,y) \\ z = z \end{cases}$$

where $f$ is as in Section B.1.

## B.3 Spherical coordinates

Spherical coordinates also extend polar coordinates to dimension 3 but use angles for the third coordinate. They establish a bijection between points distinct from the origin and triples of numbers $(r,\varphi,\theta)$ in $\mathbb{R}_{>0}\times[0,2\pi)\times[0,\pi)$.

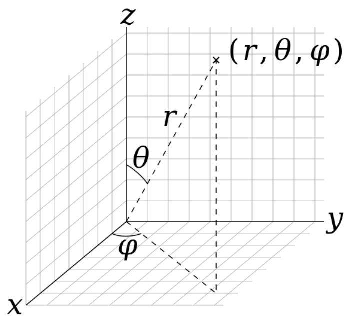



Figure B.3: Spherical coordinates[3].

[2]Image source: Wikipedia

The correspondence between Cartesian coordinates $(x,y,z)$ and spherical coordinates $(r,\varphi,\theta)$ is as follows:

$$\begin{cases} x = r\cos\varphi\sin\theta \\ y = r\sin\varphi\sin\theta \\ z = r\cos\theta \end{cases}, \quad \begin{cases} r = r(x,y,z) = \sqrt{x^2+y^2+z^2} \\ \varphi = f(x,y) \\ \theta = \arccos(\frac{z}{r}) \end{cases}$$

where $f$ is as in Section B.1.

## B.4 Barycentric coordinates

In the $n$-dimensional affine space $\mathbb{A}^n$, let $P_0,P_1,\dots,P_n$ be points such that $\mathcal{B} = (\overrightarrow{P_0P_1},\dots,\overrightarrow{P_0P_n})$ is a basis of the direction space $\mathrm{D}(\mathbb{A}^n)$. This is the case if and only if the given points are the vertices of an $n$-simplex. Then $\mathcal{K} = (P_0,\mathcal{B})$ is a Cartesian frame. Every point $A$ has a unique expression of the form

$$\begin{aligned} A &= P_0 + a_1\overrightarrow{P_0P_1} + \cdots + a_n\overrightarrow{P_0P_n} \\ &= (1-a_1-\cdots-a_n)P_0 + a_1P_1 + \cdots + a_nP_n \\ &= \lambda_0P_0 + \lambda_1P_1 + \cdots + \lambda_nP_n. \end{aligned}$$

The $n+1$ values $(\lambda_0,\lambda_1,\dots,\lambda_n)$ are the barycentric coordinates of the point $A$. Notice that the sum of these coordinates equals 1. The above equation also shows how to translate from Cartesian coordinates to barycentric coordinates. This is the algebraic point of view.

The geometric interpretation in dimension 2 is the following. Let $ABC$ be a triangle of area 1. A point $P$ in the plane of the triangle has barycentric coordinates

$$\lambda_1 = \mathrm{Area}_{\mathrm{or}}(\triangle PAB), \quad \lambda_2 = \mathrm{Area}_{\mathrm{or}}(\triangle PBC), \quad \lambda_3 = \mathrm{Area}_{\mathrm{or}}(\triangle PCA)$$

where $\mathrm{Area}_{\mathrm{or}}$ stands for the oriented area of a triangle. Similarly, in higher dimensions, the barycentric coordinates are given by oriented volume relative to an $n$-simplex.

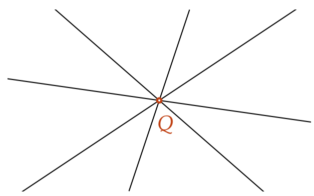
[3] Image source: Wikipedia

APPENDIX C

# Pencils of lines and planes

## C.1 Pencil of lines in $\mathbb{A}^2$

**Definition C.1.** Fix a point $Q \in \mathbb{A}^2$. The set $\mathcal{L}_Q$ of all lines in $\mathbb{A}^2$ passing through $Q$ is called a *pencil of lines*, and $Q$ is called the *center* of the pencil $\mathcal{L}_Q$.




**Proposition C.2.** If $\ell_1 : a_1x + b_1y + c_1 = 0$ and $\ell_2 : a_2x + b_2y + c_2 = 0$ are two distinct lines in the pencil $\mathcal{L}_Q$, then $\mathcal{L}_Q$ consists of lines having equations of the form

$$\ell_{\lambda,\mu} : \lambda(a_1x + b_1y + c_1) + \mu(a_2x + b_2y + c_2) = 0,$$

where $\lambda, \mu \in \mathbb{R}$ are not both zero. In particular, if $Q = Q(x_0, y_0)$, $\ell_1 : x = x_0$, and $\ell_2 : y = y_0$, then

$$\mathcal{L}_Q = \Big\{\ell_{\lambda,\mu} : \lambda(x - x_0) + \mu(y - y_0) = 0 : \lambda, \mu \in \mathbb{R} \text{ not both zero}\Big\}.$$

Pencils of lines are useful when a point $Q$ is given as the intersection of two lines, but its coordinates are not known explicitly, and one wants to find the equation of a line passing through $Q$ and satisfying some other conditions. For example, one might require that it contains some point $P$ distinct from $Q$ or that it is parallel to a given line.

Notice that there is redundancy in the two parameters $\lambda$ and $\mu$, meaning that we don't have two independent parameters here. If $\lambda \neq 0$, then one can divide the equation of $\ell_{\lambda,\mu}$ by $\lambda$ to obtain

$$\ell_{1,t} : (a_1 x + b_1 y + c_1) + t(a_2 x + b_2 y + c_2) = 0,$$

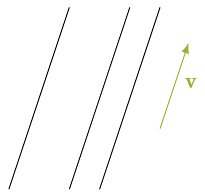
where $t = \frac{\mu}{\lambda} \in \mathbb{R}$. So $\ell_{1,\frac{\mu}{\lambda}}$ and $\ell_{\lambda,\mu}$ are in fact the same line.

**Definition C.3.** A *reduced pencil* is the pencil $\mathcal{L}_Q$ from which we remove one line, i.e., it is $\mathcal{L}_Q \setminus \{\ell_2\}$ for some $\ell_2 \in \mathcal{L}_Q$. With the above notation and discussion, it is the set

$$\left\{\ell_{1,t} : (a_1 x + b_1 y + c_1) + t(a_2 x + b_2 y + c_2) = 0 : t \in \mathbb{R}\right\}.$$

The fact that we use one parameter instead of two to describe almost all lines passing through $Q$ simplifies calculations.

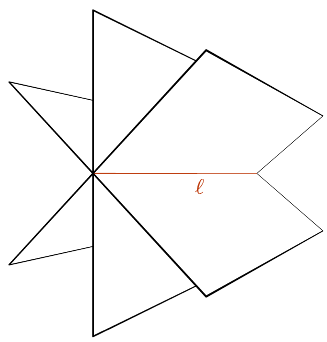
**Definition C.4.** Let $\mathbf{v} \in \mathbb{V}^2$. The set $\mathcal{L}_{\mathbf{v}}$ of all lines in $\mathbb{A}^2$ with direction vector $\mathbf{v}$ is called an *improper pencil of lines*, and $\mathbf{v}$ is called a *direction vector* of the pencil $\mathcal{L}_{\mathbf{v}}$.




The connection between pencils of lines and improper pencils of lines is best understood through projective geometry, where the improper pencil of lines is the set of all lines intersecting at the same point at infinity.

## C.2 Pencil of planes in $\mathbb{A}^3$

**Definition C.5.** Let $\ell \subseteq \mathbb{A}^3$ be a line. The set $\Pi_\ell$ of all planes in $\mathbb{A}^3$ containing $\ell$ is called a *pencil of planes*, and $\ell$ is called the *axis* (or *carrier line*) of the pencil $\Pi_\ell$.



**Proposition C.6.** If $\pi_1 : a_1x + b_1y + c_1z + d_1 = 0$ and $\pi_2 : a_2x + b_2y + c_2z + d_2 = 0$ are two distinct planes in the pencil $\Pi_\ell$, then $\Pi_\ell$ consists of planes having equations of the form

$$\pi_{\lambda,\mu} : \lambda(a_1x + b_1y + c_1z + d_1) + \mu(a_2x + b_2y + c_2z + d_2) = 0,$$

where $\lambda, \mu \in \mathbb{R}$ are not both zero.

Pencils of planes are useful when a line $\ell$ is given as the intersection of two planes (see Subsection 2.3.2) and one wants to find the equation of a plane containing $\ell$ and satisfying some other conditions. For example, one might require that it contains some point $P$ which does not belong to $\ell$ or that it is parallel to a given line.

As in the case of line pencils, there is redundancy in the two parameters $\lambda$ and $\mu$. If $\lambda \neq 0$, then one can divide the equation of $\pi_{\lambda,\mu}$ by $\lambda$ to obtain

$$\pi_{1,t} : (a_1x + b_1y + c_1z + d_1) + t(a_2x + b_2y + c_2z + d_2) = 0,$$

where $t = \frac{\mu}{\lambda} \in \mathbb{R}$. So $\pi_{1,\frac{\mu}{\lambda}}$ and $\pi_{\lambda,\mu}$ are in fact the same plane.

**Definition C.7.** A *reduced pencil* is the pencil $\Pi_\ell$ from which *we remove one plane*, i.e., it is $\Pi_\ell \setminus \{\pi_2\}$ for some $\pi_2 \in \Pi_\ell$. With the above notation and discussion, it is the set

$$\left\{\pi_{1,t} : (a_1x + b_1y + c_1z + d_1) + t(a_2x + b_2y + c_2z + d_2) = 0 : t \in \mathbb{R}\right\}.$$

The fact that we use one parameter instead of two to describe almost all planes containing $\ell$ simplifies calculations.

**Definition C.8.** Let $\mathbb{W} \subseteq \mathbb{V}^3$ be a vector subspace of dimension 2. The set $\Pi_{\mathbb{W}}$ of all planes in $\mathbb{A}^3$ which admit $\mathbb{W}$ as direction space is called an *improper pencil of planes*, and $\mathbb{W}$ is called the vector subspace associated with the pencil $\Pi_{\mathbb{W}}$.

The connection between pencils of planes and improper pencils of planes is best understood through projective geometry, where we can think of the improper pencil of planes as the set of all planes intersecting along a line at infinity.

APPENDIX D

# Some classical theorems in affine geometry

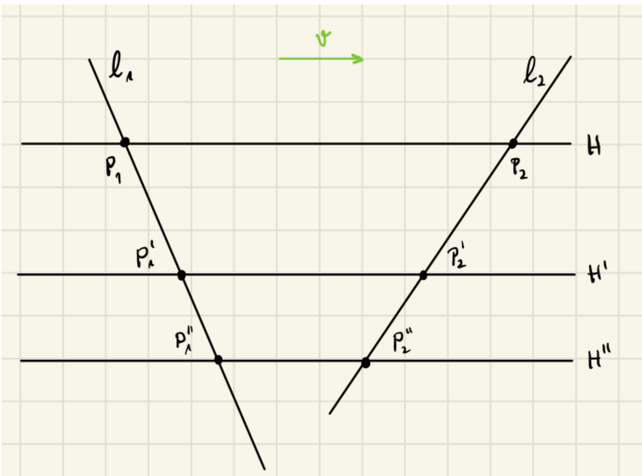
**Theorem D.1** (Thales' intercept theorem)**.** Let $H$, $H'$, and $H''$ be three distinct parallel lines in the affine plane $\mathbb{A}^2$. Let $\ell_1$ and $\ell_2$ be two lines not parallel to $H$, $H'$, and $H''$. For $i = 1, 2$, let

$$\begin{aligned} P_i &= \ell_i \cap H \\ P_i' &= \ell_i \cap H' \\ P_i'' &= \ell_i \cap H'' \end{aligned}$$

and let $k_1, k_2$ be scalars such that

$$\overrightarrow{P_iP_i''} = k_i\,\overrightarrow{P_iP_i'}.$$

Then, $k_1 = k_2$.

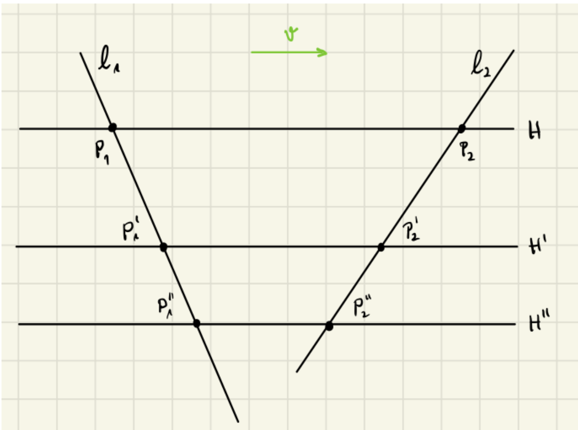


**Theorem D.2** (Converse of Thales' intercept theorem)**.** Let $\ell_1$ and $\ell_2$ be two lines in the affine plane $\mathbb{A}^2$. For $i = 1, 2$, let $P_i, P_i'$, and $P_i''$ be three distinct points on $\ell_i$. Let $H$ be the line passing through $P_1$ and $P_2$, and let $H'$ be the line passing through $P_1'$ and $P_2'$.

Suppose that $H$ is parallel to $H'$, and let $k$ be a scalar such that for $i = 1, 2$,

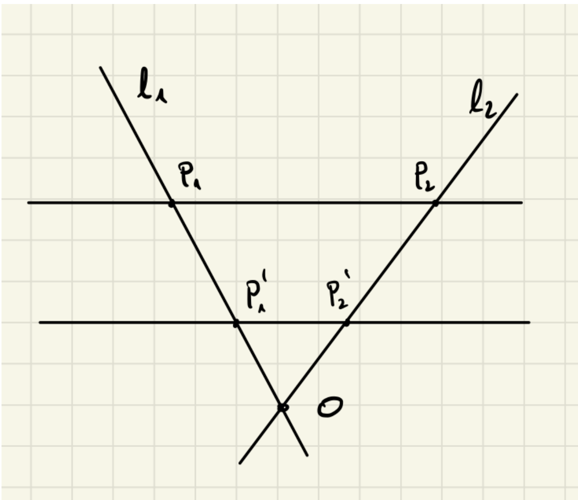
$$\overrightarrow{P_iP_i''} = k\,\overrightarrow{P_iP_i'}.$$

Then, the line $H''$ passing through $P_1''$ and $P_2''$ is parallel to $H$ and $H'$.

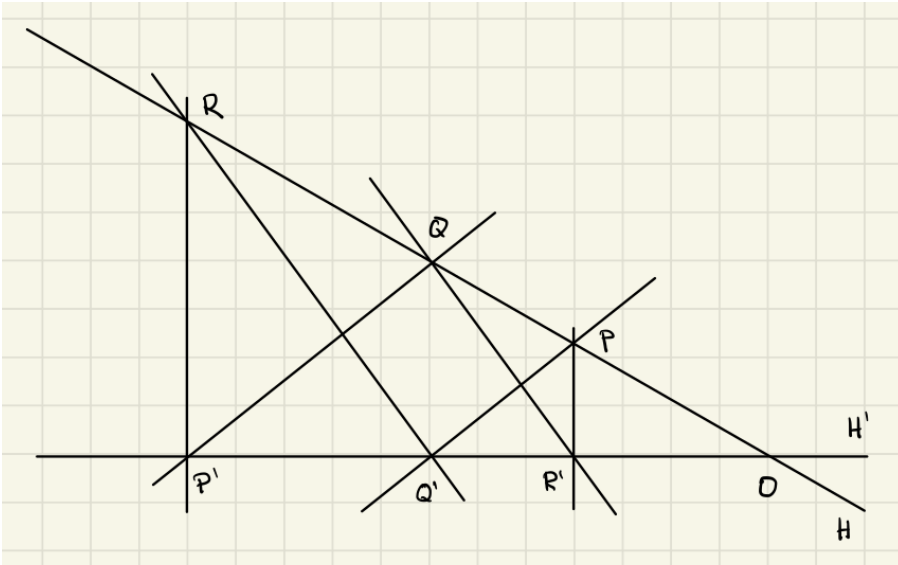



**Theorem D.3** (Thales' theorem for triangles)**.** Let $\ell_1$ and $\ell_2$ be two lines in the affine plane $\mathbb{A}^2$ intersecting at a point $O$. For $i = 1, 2$, let $P_i$ and $P_i'$ be points on $\ell_i$ distinct from $O$. Then the line $P_1P_2$ is parallel to the line $P_1'P_2'$ if and only if there exists a scalar $k$ such that

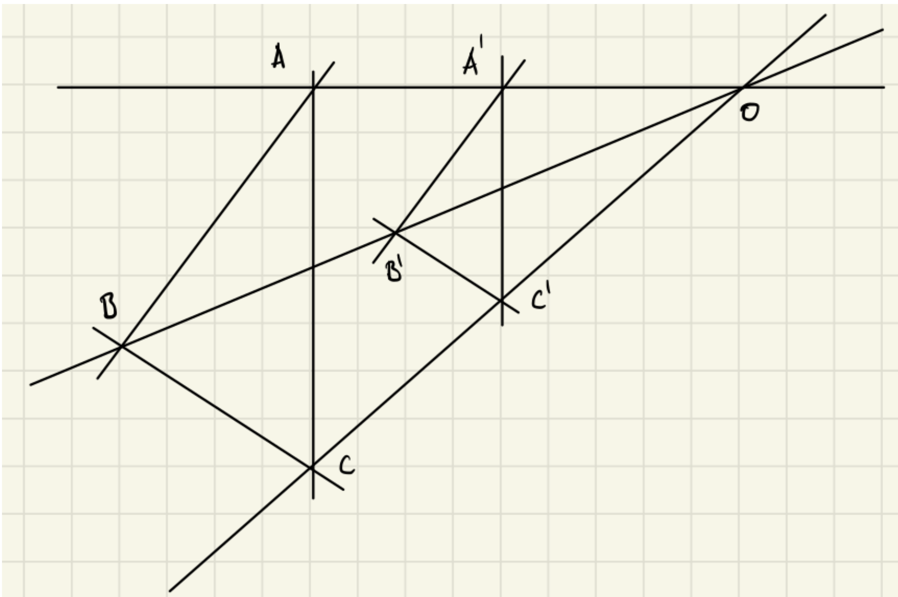
$$\overrightarrow{OP_i'} = k\,\overrightarrow{OP_i} \quad \text{for } i = 1, 2.$$



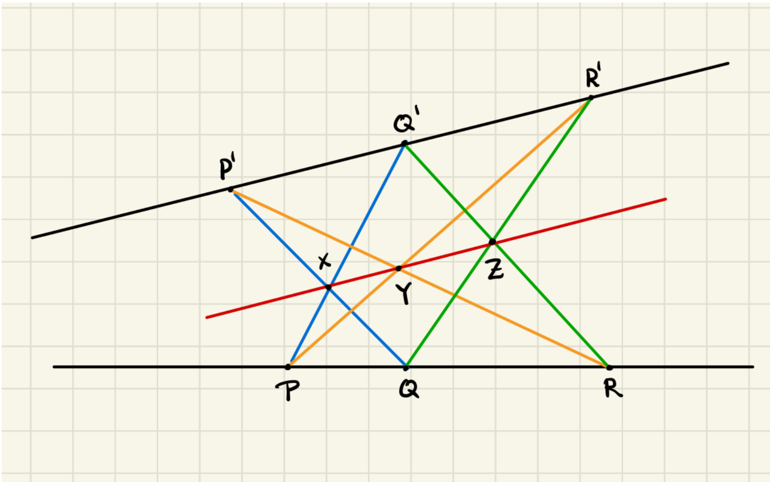
**Theorem D.4** (Pappus' affine theorem)**.** Let $H$ and $H'$ be two distinct lines in the affine plane $\mathbb{A}^2$. Let $P, Q, R \in H$ and $P', Q', R' \in H'$ be distinct points, none of which lies at the intersection $H \cap H'$. If $PQ' \parallel P'Q$ and $QR' \parallel Q'R$, then $PR' \parallel P'R$.




**Theorem D.5** (Desargues' theorem)**.** Let $A, B, C, A', B', C' \in \mathbb{A}^2$ be points such that no three are collinear, and such that $AB \parallel A'B'$, $BC \parallel B'C'$, and $AC \parallel A'C'$. Then the three lines $AA'$, $BB'$, and $CC'$ are either parallel or have a point in common.



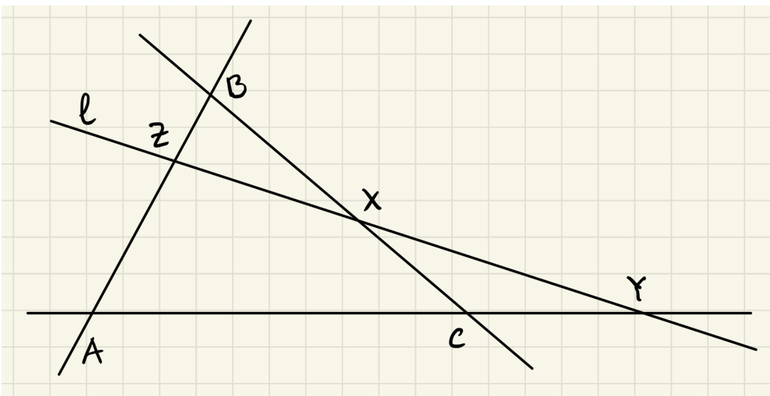
**Theorem D.6** (Pappus' hexagon theorem)**.** Let $H$ and $H'$ be two distinct lines in the affine plane $\mathbb{A}^2$. Let $P, Q, R \in H$ and $P', Q', R' \in H'$ be distinct points, none of which lies at the intersection $H \cap H'$. Assume that $PQ' \cap P'Q = \{X\}$, $PR' \cap P'R = \{Y\}$, and $QR' \cap Q'R = \{Z\}$. Then the points $X, Y, Z$ are collinear.




**Theorem D.7** (Newton-Gauss line)**.** A complete quadrilateral is the configuration of four lines, no three of which pass through the same point. Let $O, A, B, C, B', A'$ be the intersection points of such lines, i.e., the vertices of the complete quadrilateral, such that $A$ lies between $O$ and $B$ and such that $A'$ lies between $O$ and $B'$. The diagonals of this quadrilateral are the segments $[OC]$, $[AA']$, and $[BB']$. The midpoints of the diagonals of a complete quadrilateral are collinear.

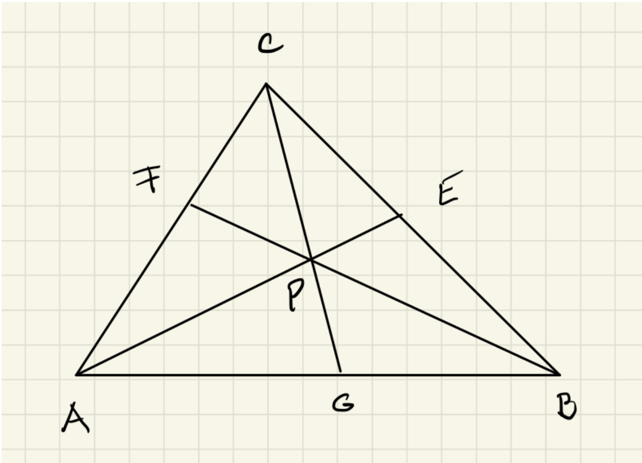
**Theorem D.8** (Menelaus' theorem)**.** Let $ABC$ be a triangle and $\ell$ a line which does not pass through the vertices of the triangle. Let $X, Y, Z$ be points on the lines $BC$, $CA$, and $AB$, respectively, and consider the signed ratios $\alpha = BX : XC$, $\beta = CY : YA$, and $\gamma = AZ : ZB$. Then $X, Y, Z$ are collinear if and only if

$$\alpha \cdot \beta \cdot \gamma = -1.$$

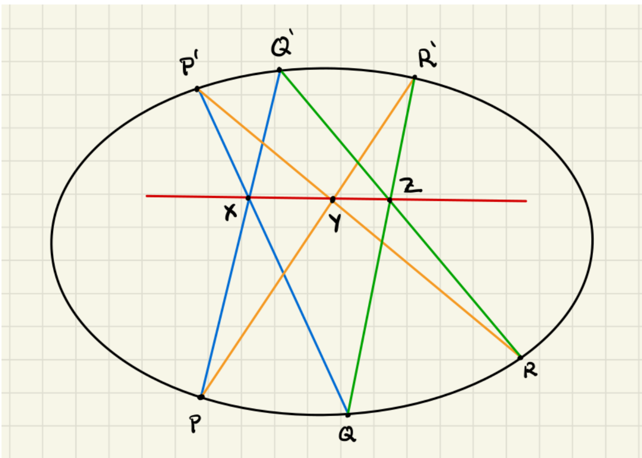


**Theorem D.9** (Ceva's theorem)**.** Let $ABC$ be a triangle. Let $E \in BC$, $F \in CA$, and $G \in AB$ and consider the signed ratios $\alpha = BE : EC$, $\beta = CF : FA$, and $\gamma = AG : GB$. The three lines $AE$, $BF$, and $CG$ are concurrent if and only if

$$\alpha \cdot \beta \cdot \gamma = 1.$$




**Theorem D.10** (Pascal's hexagrammum mysticum theorem)**.** If all six vertices of a hexagon lie on a conic section and the three pairs of opposite sides intersect, then the three points of intersection are collinear.



# APPENDIX E

## Eigenvalues and Eigenvectors

**Definition E.1.** Let $\mathbb{V}$ be an $\mathbb{R}$-vector space and let $\phi : \mathbb{V} \to \mathbb{V}$ be a linear map. A non-zero vector $\mathbf{v} \in \mathbb{V}$ is called an *eigenvector of $\phi$* if there is a scalar $\lambda \in \mathbb{R}$ such that $\phi(\mathbf{v}) = \lambda\mathbf{v}$. The scalar $\lambda$ is then called *the eigenvalue associated to the eigenvector* $\mathbf{v}$. The set of eigenvalues of $\phi$ is called the *spectrum of $\phi$*.

For $A \in \mathrm{Mat}_{n\times n}(\mathbb{R})$, an *eigenvector of $A$* is an eigenvector $\mathbf{v} \in \mathbb{R}^n$ for the map $\phi_A : \mathbb{R}^n \to \mathbb{R}^n$ defined by $A$ via $\phi_A(\mathbf{x}) = A\mathbf{x}$, and an *eigenvalue* of $A$ is an eigenvalue for $\phi_A$.

- If $\phi = \mathrm{Id}_{\mathbb{V}}$, then every non-zero vector $\mathbf{v}$ is an eigenvector of $\phi$ with eigenvalue $\lambda = 1$.
- Every non-zero vector in $\ker(\phi)$ is an eigenvector for $\phi$ with eigenvalue $\lambda = 0$.

**Proposition E.2.** The eigenvalue associated to an eigenvector is unique.

**Proposition E.3.** If $\mathbf{v}_1, \mathbf{v}_2 \in \mathbb{V}$ are eigenvectors with the same eigenvalue $\lambda$, then for every $c_1, c_2 \in \mathbb{R}$, the vector $c_1\mathbf{v}_1 + c_2\mathbf{v}_2$, if it is non-zero, is also an eigenvector with eigenvalue $\lambda$.

**Definition E.4.** From the above proposition, it follows that for each $\lambda \in \mathbb{R}$, the set

$$\mathbb{V}_\lambda(\phi) = \{\mathbf{v} \in \mathbb{V} : \mathbf{v} \text{ is an eigenvector of } \phi \text{ with eigenvalue } \lambda\} \cup \{0\}$$

is a vector subspace of $\mathbb{V}$, called the *eigenspace for the eigenvalue $\lambda$*. For a matrix $A \in \mathrm{Mat}_{n\times n}(\mathbb{R})$, the *eigenspace for the eigenvalue* $\lambda$ is defined to be the subspace $\mathbb{V}_\lambda(A) := \mathbb{V}_\lambda(\phi_A)$ in $\mathbb{R}^n$.

**Proposition E.5.** If $\mathbf{v}_1, \ldots, \mathbf{v}_k \in \mathbb{V}$ are eigenvectors with eigenvalues $\lambda_1, \ldots, \lambda_k$, respectively, and these $\lambda_i$ are pairwise distinct, then $\mathbf{v}_1, \ldots, \mathbf{v}_k$ are linearly independent.

**Proposition E.6.** If every $\mathbf{v} \in \mathbb{V} \setminus \{0\}$ is an eigenvector of $\phi$, then there exists $\lambda \in \mathbb{R}$ such that $\phi = \lambda\,\mathrm{Id}_{\mathbb{V}}$.

## E.1 Characteristic polynomial

In order to find the eigenvalues of a linear map $\mathbb{V} \to \mathbb{V}$, one uses the characteristic polynomial.

**Proposition E.7.** Let $\mathbb{V}$ be a finite-dimensional vector space and let $\phi : \mathbb{V} \to \mathbb{V}$ be a linear map. A scalar $\lambda \in \mathbb{R}$ is an eigenvalue of $\phi$ if and only if the map

$$\phi - \lambda \,\mathrm{Id}_{\mathbb{V}} : \mathbb{V} \to \mathbb{V} \quad \text{defined by} \quad (\phi - \lambda \,\mathrm{Id}_{\mathbb{V}})(\mathbf{v}) = \phi(\mathbf{v}) - \lambda \mathbf{v}$$

is *not* bijective, that is, if and only if $\det(\phi - \lambda \,\mathrm{Id}_{\mathbb{V}}) = 0$.

- Let $\mathcal{B} = (\mathbf{e}_1, \ldots, \mathbf{e}_n)$ be a basis of $\mathbb{V}$. The matrix associated to the map $\lambda \,\mathrm{Id}_{\mathbb{V}}$ is

$$[\lambda \,\mathrm{Id}_{\mathbb{V}}]_{\mathcal{B}} = \begin{bmatrix} \lambda & 0 & \ldots & 0 \\ 0 & \lambda & \ldots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \ldots & \lambda \end{bmatrix}$$

  and if $A = (a_{ij}) = [\phi]_{\mathcal{B}}$ then

$$[\phi - \lambda \,\mathrm{Id}_{\mathbb{V}}]_{\mathcal{B}} = \begin{bmatrix} a_{11} - \lambda & a_{12} & \ldots & a_{1n} \\ a_{21} & a_{22} - \lambda & \ldots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \ldots & a_{nn} - \lambda \end{bmatrix}.$$

**Definition E.8.** Let $A \in \mathrm{Mat}_{n \times n}(\mathbb{R})$. The determinant

$$P_A(T) := \det(A - T \,\mathrm{Id}_n) = \begin{vmatrix} a_{11} - T & a_{12} & \ldots & a_{1n} \\ a_{21} & a_{22} - T & \ldots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \ldots & a_{nn} - T \end{vmatrix}$$

is a polynomial of degree $n$ in $T$, called *the characteristic polynomial of* $A$. If $\phi : \mathbb{V} \to \mathbb{V}$ is linear and $\mathcal{B} = (\mathbf{e}_1, \ldots, \mathbf{e}_n)$ is a basis of $\mathbb{V}$, then the *characteristic polynomial of* $\phi$ is $P_\phi := P_{[\phi]_{\mathcal{B}}}$.

**Proposition E.9.** The definition of $P_\phi$ is independent of the basis.

**Corollary E.10.** Let $\mathbb{V}$ be a vector space of dimension $n$, and let $\phi : \mathbb{V} \to \mathbb{V}$ be linear. Then $\lambda \in \mathbb{R}$ is an eigenvalue of $\phi$ if and only if $\lambda$ is a root of the polynomial $P_\phi$. In particular, $\phi$ has at most $n$ eigenvalues.

**Proposition E.11.** Let $\mathbb{V}$ be a finite-dimensional vector space. A linear map $\phi : \mathbb{V} \to \mathbb{V}$ is diagonalizable if and only if there is a basis of $\mathbb{V}$ consisting entirely of eigenvectors of $\phi$.

**Theorem E.12.** Let $\mathbb{V}$ be a real vector space of dimension $n$, and let $\phi : \mathbb{V} \to \mathbb{V}$ be linear. If $\{\lambda_1, \ldots, \lambda_k\} \subseteq \mathbb{R}$ is the spectrum of $\phi$, then

$$\dim(V_{\lambda_1}(\phi)) + \cdots + \dim(V_{\lambda_k}(\phi)) \leq n$$

with equality if and only if $\phi$ is diagonalizable.

**Corollary E.13.** If $\dim(\mathbb{V}) = n$ and $\phi$ has $n$ distinct eigenvalues, then it is diagonalizable.

- We have a practical method for finding eigenvalues and eigenvectors.
- Suppose we are given $A \in \mathrm{Mat}_{n\times n}(\mathbb{R})$.
  1. Calculate the characteristic polynomial $P_A$.
  2. Find the eigenvalues of $A$ by calculating the roots of $P_A$ which lie in $\mathbb{R}$.
  3. For each eigenvalue $\lambda \in \mathbb{R}$, the homogeneous system of $n$ equations in $n$ unknowns:
  $$(A - \lambda\,\mathrm{Id}_n)\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$$
  has rank $r < n$ and therefore has nontrivial solutions. The space of solutions is the eigenspace $\mathbb{V}_\lambda(A)$.
- If the sum of the dimensions of the eigenspaces found by letting $\lambda$ vary over the roots of $P_A$ is equal to $n$, then $A$ is diagonalizable (by Theorem E.12).
- A linear map $\mathbb{V} \to \mathbb{V}$ need not have any eigenvalues nor eigenvectors.
- If $\dim(\mathbb{V})$ is odd, then the characteristic polynomial has odd degree and so has at least one real root. Thus, every linear map $\mathbb{V} \to \mathbb{V}$ on an odd-dimensional real vector space has at least one eigenvalue, and so at least one eigenvector.
- If we replace $\mathbb{R}$ with $\mathbb{C}$, by the Fundamental Theorem of Algebra, $P_A$ has roots in $\mathbb{C}$. Thus, every linear map $\mathbb{V} \to \mathbb{V}$ on a finite-dimensional complex vector space has at least one eigenvalue, and so at least one eigenvector. Of course, this does not mean that the linear map is diagonalizable.

**Definition E.14.** Let $\mathbb{V}$ be a finite-dimensional $\mathbb{R}$-vector space and let $\phi : \mathbb{V} \to \mathbb{V}$ be a linear map admitting $\lambda$ as an eigenvalue. The number $\dim(\mathbb{V}_\lambda(\phi))$ is called the *geometric multiplicity of $\lambda$ for $\phi$*. The *algebraic multiplicity of $\lambda$ for $\phi$* is instead the multiplicity of $\lambda$ as a root of the characteristic polynomial $P_\phi$; this is denoted by $h_\phi(\lambda)$.

**Proposition E.15.** For any linear map $\phi : \mathbb{V} \to \mathbb{V}$ and $\lambda \in \mathbb{R}$, one has

$$\dim(\mathbb{V}_\lambda(\phi)) \leq h_\phi(\lambda),$$

that is, the geometric multiplicity is not larger than the algebraic multiplicity.

APPENDIX F

# Bilinear forms and symmetric matrices

The purpose of this appendix is threefold:

1. We prove Sylvester's theorem (Theorem F.7) which shows that the properties in Proposition 3.10 define scalar products (Corollary F.10).
2. From the proof of Sylvester's theorem one obtains a simple algorithm for the affine classification of quadrics (see Chapter 9).
3. We prove the Spectral Theorem (Theorem F.14) which is used in the isometric classification of quadratic surfaces (see Chapter 9).

Bilinear forms are the natural context for the proofs of both theorems.

## F.1 Affine diagonalization

Throughout we let $\mathbb{V}$ denote a finite-dimensional real vector space. The proofs for the statements in this appendix can be found in [20, Chapter 15 and 16].

**Definition F.1.** A map $\phi : \mathbb{V} \times \mathbb{V} \to \mathbb{R}$ is called a *bilinear form* if it is linear in both arguments, i.e.,

$$\phi(a\mathbf{u} + b\mathbf{v}, \mathbf{w}) = a\phi(\mathbf{u}, \mathbf{w}) + b\phi(\mathbf{v}, \mathbf{w}) \quad \text{and} \quad \phi(\mathbf{u}, a\mathbf{v} + b\mathbf{w}) = a\phi(\mathbf{u}, \mathbf{v}) + b\phi(\mathbf{u}, \mathbf{w}). \tag{F.1}$$

for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{V}$ and all $a, b \in \mathbb{R}$.

With respect to a basis $\mathcal{B} = (\mathbf{e}_1, \ldots, \mathbf{e}_n)$ of $\mathbb{V}$, the values of a bilinear form $\phi$ are determined by its values on the basis vectors. These values are the entries in the *Gram matrix* $\mathrm{G}_{\mathcal{B}}(\phi)$ *of* $\phi$ *relative to* $\mathcal{B}$, which is defined by $\mathrm{G}_{\mathcal{B}}(\phi) = \phi(\mathbf{e}_i, \mathbf{e}_j)$. This matrix determines $\phi$ since

$$\phi(\mathbf{v}, \mathbf{w}) = [\mathbf{v}]_{\mathcal{B}}^T \cdot \mathrm{G}_{\mathcal{B}}(\phi) \cdot [\mathbf{w}]_{\mathcal{B}}.$$

Moreover, if $\mathcal{B}'$ is another basis, then

$$G_{\mathcal{B}'}(\phi) = M^T_{\mathcal{B},\mathcal{B}'} \cdot G_{\mathcal{B}}(\phi) \cdot M_{\mathcal{B},\mathcal{B}'}. \tag{F.2}$$

In particular, the rank of $G_{\mathcal{B}}(\phi)$ does not depend on $\mathcal{B}$. It depends only on $\phi$. We denote it by $\mathrm{rank}(\phi)$ and call it *the rank of* $\phi$.

**Definition F.2.** A bilinear form $\phi : \mathbb{V}^n \times \mathbb{V}^n \to \mathbb{R}$ is called *symmetric* if the Gram matrix $G_{\mathcal{B}}(\phi)$ is a symmetric matrix with respect to a basis $\mathcal{B}$. It follows from (F.2) that this definition does not depend on the basis $\mathcal{B}$.

**Definition F.3.** Let $\phi$ be a bilinear form on the vector space $\mathbb{V}$. The *quadratic form associated to* $\phi$ is the map

$$q_\phi : \mathbb{V} \to \mathbb{R} \quad \text{defined by} \quad q_\phi(\mathbf{v}) = \phi(\mathbf{v}, \mathbf{v}).$$

**Proposition F.4.** Let $\phi$ be a symmetric bilinear form on the vector space $\mathbb{V}$. The quadratic form $q_\phi$ associated to $\phi$ satisfies

$$\begin{array}{rcl} q_\phi(\lambda \mathbf{v}) & = & \lambda^2 q_\phi(\mathbf{v}) \\ 2\phi(\mathbf{v}, \mathbf{w}) & = & q(\mathbf{v}+\mathbf{w}) - q(\mathbf{v}) - q(\mathbf{w}) \end{array}$$

for every $\lambda \in \mathbb{R}$ and every $\mathbf{v}, \mathbf{w} \in \mathbb{V}$.

*Proof.* The first property is an immediate consequence of (F.1). Further,

$$\begin{array}{rcl} q_\phi(\mathbf{v}+\mathbf{w}) - q_\phi(\mathbf{v}) - q_\phi(\mathbf{w}) & = & \phi(\mathbf{v}+\mathbf{w}, \mathbf{v}+\mathbf{w}) - \phi(\mathbf{v}, \mathbf{v}) - \phi(\mathbf{w}, \mathbf{w}) \\ & = & \phi(\mathbf{v}, \mathbf{w}) + \phi(\mathbf{w}, \mathbf{v}) \\ & = & 2\phi(\mathbf{v}, \mathbf{w}). \end{array}$$

□

**Remark.** In particular, Proposition F.4 implies that the quadratic form $q_\phi$ determines uniquely the symmetric bilinear form $\phi$, i.e., the correspondence $\phi \leftrightarrow q_\phi$ is a bijection between symmetric bilinear forms and quadratic forms.

**Definition F.5.** Let $\phi$ be a symmetric bilinear form on the vector space $\mathbb{V}$. A *diagonalizing basis* for $\phi$ is a basis $\mathcal{B}$ of $\mathbb{V}$ such that $G_{\mathcal{B}}(\phi)$ is a diagonal matrix.

**Theorem F.6.** Let $\phi$ be a symmetric bilinear form on the vector space $\mathbb{V}$. Then there exists a diagonalizing basis for $\phi$.

*Proof.* We take the proof from [20, p.232]. The proof is 'by completing the squares' and induction on $n$. If $n = 1$ there is nothing to prove. Suppose therefore that $n \geq 2$, and that every symmetric bilinear form on a space of dimension less than $n$ has a diagonalizing basis. Choose a basis $\mathcal{B} = (\mathbf{b}_1, \dots, \mathbf{b}_n)$ of $\mathbb{V}$. If $\phi$ is the zero form, then $\mathcal{B}$ is diagonalizing and there is nothing to prove. Otherwise we can obtain from $\mathcal{B}$ a second basis $\mathcal{B}' = (\mathbf{c}_1, \dots, \mathbf{c}_n)$ for which $\phi(\mathbf{c}_1, \mathbf{c}_1) \neq 0$. Indeed, if there is some $i$ for which $\phi(\mathbf{b}_i, \mathbf{b}_i) \neq 0$ then it suffices to exchange $\mathbf{b}_1$ and $\mathbf{b}_i$. If, on the other hand, $\phi(\mathbf{b}_i, \mathbf{b}_i) = 0$ for all $i$, then there are $i \neq j$ for which $\phi(\mathbf{b}_i, \mathbf{b}_j) \neq 0$ (otherwise $\phi$ is the zero form), and again we can exchange these with $\mathbf{b}_1$ and $\mathbf{b}_2$ so that $\phi(\mathbf{b}_1, \mathbf{b}_2) \neq 0$. The new basis

$$\mathcal{B}' = (\mathbf{b}_1 + \mathbf{b}_2, \mathbf{b}_2, \dots, \mathbf{b}_n)$$

has the required property. In the basis $\mathcal{B}'$, the quadratic form $q_\phi$ associated to $\phi$ has the form

$$q(\mathbf{v}(y_1,\dots,y_n)) = h_{11}y_1^2 + 2\sum_{i=2}^{n} h_{1i}y_1y_i + \sum_{i,j=2}^{n} h_{ij}y_iy_j,$$

where $h_{ij} = \phi(\mathbf{c}_i,\mathbf{c}_j)$. Since $h_{11} = \phi(\mathbf{c}_1,\mathbf{c}_1) \neq 0$ we can rewrite the above equation as

$$q(\mathbf{v}(y_1,\dots,y_n)) = h_{11}\left(y_1 + \sum_{i=2}^{n} h_{11}^{-1}h_{1i}y_i\right)^2 + (\text{terms not involving } y_1).$$

We now change coordinates as follows

$$z_1 = y_1 + \sum_{i=2}^{n} h_{11}^{-1}h_{1i}y_i, \quad z_2 = y_2,\dots,z_n = y_n,$$

which corresponds to a change of basis from $\mathcal{B}'$ to $\mathcal{B}'' = (\mathbf{d}_1,\dots,\mathbf{d}_n)$ given by $\mathbf{d}_1 = \mathbf{c}_1$, and for $i > 1$, $\mathbf{d}_i = \mathbf{c}_i - h_{11}^{-1}h_{1i}\mathbf{c}_1$. In these coordinates, $q_\phi$ has the form

$$q(\mathbf{v}(y_1,\dots,y_n)) = h_{11}z_1^2 + q'(z_2,\dots,z_n),$$

where $q'$ is a homogeneous polynomial of degree 2 in $z_2,\dots,z_n$, and so defines a quadratic form on the space $\langle \mathbf{d}_2,\dots,\mathbf{d}_n\rangle$. By the inductive hypothesis, $\langle \mathbf{d}_2,\dots,\mathbf{d}_n\rangle$ has a basis $(\mathbf{e}_2,\dots,\mathbf{e}_n)$ which diagonalizes $q'$. Thus, the basis $(\mathbf{d}_1,\mathbf{e}_2,\dots,\mathbf{e}_n)$ is diagonalizing for $q$. □

**Theorem F.7.** (Sylvester's[1] Theorem) Let $\phi$ be a symmetric bilinear form of rank $r$ on the vector space $\mathbb{V}$. Then there is an integer $p$ depending only on $\phi$, and a basis $\mathcal{B}$ of $\mathbb{V}$ such that the Gram matrix $\mathrm{G}_{\mathcal{B}}(\phi)$ has the form

$$\begin{bmatrix} I_p & 0 & 0 \\ 0 & -I_{r-p} & 0 \\ 0 & 0 & 0 \end{bmatrix} \tag{F.3}$$

where 0 stands for zero matrices of appropriate sizes.

*Proof.* We take the proof from [20, p.232]. By Theorem F.6 there is a basis $\mathcal{B} = (\mathbf{b}_1,\dots,\mathbf{b}_n)$ for which

$$q_\phi(\mathbf{v}) = b_{11}y_1^2 + \cdots + b_{nn}y_n^2.$$

for each $\mathbf{v} = y_1\mathbf{b}_1 + \cdots + y_n\mathbf{b}_n \in \mathbb{V}$. The number of non-zero coefficients $b_{ii}$ is equal to the rank of the quadratic form $q_\phi$, and so, depends only on $\phi$. After possibly reordering the basis, we can suppose that the first $p$ coefficients are positive, the next $r-p$ are negative, and the remaining $n-r$ are zero. Then one has

$$b_{11} = \beta_1^2,\dots,b_{1p} = \beta_p^2, \quad b_{p+1,p+1} = -\beta_{p+1}^2,\dots,b_{r,r} = -\beta_r^2, \quad b_{r+1,r+1} = \cdots = b_{n,n} = 0.$$

[1] 1814 – 1897

for appropriate $\beta_1,\dots,\beta_r \in \mathbb{R}$, which can be taken to be positive. Then, with respect to the basis

$$\mathbf{c}_1 = \mathbf{b}_1/\beta_1,\dots,\mathbf{c}_r = \mathbf{b}_r/\beta_r, \quad \mathbf{c}_{r+1} = \mathbf{b}_{r+1},\dots,\mathbf{c}_n = \mathbf{b}_n,$$

the matrix of $\phi$ is that given in (F.3), and so the quadratic form $q_\phi$ is

$$q_\phi(\mathbf{v}(x_1,\dots,x_n)) = x_1^2+\dots+x_p^2-x_{p+1}^2-\dots-x_r^2.$$

for each $\mathbf{v} = x_1\mathbf{c}_1+\dots+x_n\mathbf{c}_n \in \mathbb{V}$.

There remains only to prove that $p$ depends only on $\phi$ and not on the particular basis $\mathcal{B}$. Suppose that with respect to another basis $\mathcal{B}' = (\mathbf{c}_1,\dots,\mathbf{c}_n)$, $q_\phi$ is

$$q_\phi(\mathbf{v}) = z_1^2+\dots+z_t^2-z_{t+1}^2-\dots-z_r^2,$$

for each $\mathbf{v} = z_1\mathbf{c}_1+\dots+z_n\mathbf{c}_n$, for some integer $t \le r$. We need to show that $t = p$. If $p \neq t$ we can suppose that $t < p$. Consider the subspaces of $\mathbb{V}$:

$$S = \langle \mathbf{b}_1,\dots,\mathbf{b}_p\rangle \quad \text{and} \quad T = \langle \mathbf{c}_{t+1},\dots,\mathbf{c}_n\rangle.$$

Since $\dim(S)+\dim(T) > n$ it follows from Grassmann's formula that $S\cap T \neq \{0\}$, and so there is a $\mathbf{v}\in S\cap T$ with $\mathbf{v}\neq 0$. Then,

$$\mathbf{v} = x_1\mathbf{b}_1+\dots+x_p\mathbf{b}_p = z_{t+1}\mathbf{c}_{t+1}+\dots+z_n\mathbf{c}_n.$$

Since $\mathbf{v}\neq 0$ it follows, using the expression of $q_\phi$ relative to $\mathcal{B}$, that

$$q_\phi(\mathbf{v}) = x_1^2+\dots+x_p^2 > 0$$

and, using the expression of $q_\phi$ relative to $\mathcal{B}'$, we have

$$q_\phi(\mathbf{v}) = -z_{t+1}^2-\dots-z_r^2 < 0.$$

This is clearly a contradiction, so $t = p$. □

**Theorem F.8.** (Sylvester's Theorem - matrix formulation) Let $A$ be a symmetric matrix of rank $r$. Then there is an integer $p$ depending only on $A$, and an invertible matrix $P$ such that

$$P^TAP = \begin{bmatrix} I_p & 0 & 0\\ 0 & -I_{r-p} & 0\\ 0 & 0 & 0\end{bmatrix}$$

where 0 stands for zero matrices of appropriate sizes.

*Proof.* Let $\phi$ be the symmetric bilinear form associated to the matrix $A$ with respect to some basis $\mathcal{B}'$ of $\mathbb{V}$. Then $\mathrm{G}_{\mathcal{B}'}(\phi) = A$. By Theorem F.7 there is a basis $\mathcal{B}$ and an integer $p$ depending only on $\phi$ such that

$$P^TAP = \mathrm{M}_{\mathcal{B}',\mathcal{B}}^T\mathrm{G}_{\mathcal{B}'}(\phi)\mathrm{M}_{\mathcal{B}',\mathcal{B}} = \mathrm{G}_{\mathcal{B}}(\phi) = \begin{bmatrix} I_p & 0 & 0\\ 0 & -I_{r-p} & 0\\ 0 & 0 & 0\end{bmatrix}.$$

where $P = \mathrm{M}_{\mathcal{B}',\mathcal{B}}$. Since $p$ depends only on $\phi$, it does not depend on $\mathcal{B}$ and $\mathcal{B}'$, i.e., it depends only on $A$ and the claim follows. □

**Corollary F.9.** Let $\phi$ be a positive definite symmetric bilinear form on an $n$-dimensional vector space $\mathbb{V}$. There is a basis $\mathcal{B}$ of $\mathbb{V}$ such that $\mathrm{G}_{\mathcal{B}}(\phi) = I_n$.

*Proof.* By Theorem F.7 there is a basis $\mathcal{B} = (\mathbf{b}_1, \dots, \mathbf{b}_n)$ such that the Gram matrix of $\phi$ has the form (F.3). Since $\phi$ is positive definite, i.e., $\phi(\mathbf{v}, \mathbf{v}) > 0$ for all non-zero vectors $\mathbf{v}$, its Gram matrix must equal the identity matrix $I_n$ (otherwise $\langle \mathbf{b}_n, \mathbf{b}_n \rangle$ equals 0 or $-1$, a contradiction). □

**Corollary F.10.** Assume that the dimension of the space $\mathbb{V}$ of geometric vectors is $n$ and assume that a unit segment has been chosen. Then, the properties in Proposition 3.10 define the scalar product. More concretely, there is a unique positive definite symmetric bilinear form $\phi$ which recognizes right angles and unit lengths.

*Proof.* The first three properties in Proposition 3.10, assert that the scalar product is a positive definite symmetric bilinear form on the space $\mathbb{V}^n$ of geometric vectors. By Corollary F.9 there is a basis such that the Gram matrix of the scalar product equals $I_n$. Therefore, by property (SP4), $\mathcal{B}$ is an orthonormal basis.

Let $\phi$ be another positive definite symmetric bilinear form. Since $\phi$ is symmetric, the Gram matrix $\mathrm{G}_{\mathcal{B}}(\phi)$ is symmetric. If $\phi$ satisfies property (SP4), i.e., if it recognizes right angles and unit lengths, then we must have $\mathrm{G}_{\mathcal{B}}(\phi) = I_n$ since $\mathcal{B}$ is orthonormal. Hence $\phi$ is the scalar product. □

## F.2 Isometric diagonalization

In essence, the spectral theorem says that a symmetric linear map or a symmetric bilinear form can be diagonalized with orthogonal transformations. In order to make this statement precise, we first put side by side some facts about linear maps and bilinear forms, and make precise what we mean by a 'symmetric' linear map. Symmetric bilinear forms were discussed in the previous section F.1.

Given a basis $\mathcal{B}$ of $\mathbb{V}^n$, an $n \times n$-matrix $M$ with real entries gives rise to the linear map

$$\phi : \mathbb{V}^n \to \mathbb{V}^n, \quad \text{defined by} \quad \phi(\mathbf{v}) = M \cdot [\mathbf{v}]_{\mathcal{B}}$$

and to the bilinear form

$$\psi : \mathbb{V}^n \times \mathbb{V}^n \to \mathbb{R}, \quad \text{defined by} \quad \psi(\mathbf{v}, \mathbf{w}) = [\mathbf{v}]_{\mathcal{B}}^T \cdot M \cdot [\mathbf{w}]_{\mathcal{B}}.$$

We say that $\phi$ is the *linear map associated to $M$ in the basis $\mathcal{B}$* and $\psi$ is the *bilinear form associated to $M$ in the basis $\mathcal{B}$*. The other way around, given a linear map $\phi : \mathbb{V}^n \to \mathbb{V}^n$, it has an associated matrix $\mathrm{M}_{\mathcal{B},\mathcal{B}}(\phi)$ with respect to the basis $\mathcal{B}$. Similarly, given a bilinear form $\psi : \mathbb{V}^n \times \mathbb{V}^n \to \mathbb{R}$ we associate to it the Gram matrix $\mathrm{G}_{\mathcal{B}}(\psi)$ (see Section F.1). Then

$$\phi(\mathbf{v}) = \mathrm{M}_{\mathcal{B},\mathcal{B}}(\phi) \cdot [\mathbf{v}]_{\mathcal{B}} \quad \text{and} \quad \psi(\mathbf{v}, \mathbf{w}) = [\mathbf{v}]_{\mathcal{B}}^T \cdot \mathrm{G}_{\mathcal{B}}(\psi) \cdot [\mathbf{w}]_{\mathcal{B}}.$$

If we change the basis from $\mathcal{B}$ to $\mathcal{B}'$, it is an exercise in linear algebra to show that

$$\mathrm{M}_{\mathcal{B}',\mathcal{B}'}(\phi) = \mathrm{M}_{\mathcal{B},\mathcal{B}'}^{-1} \cdot \mathrm{M}_{\mathcal{B},\mathcal{B}}(\phi) \cdot \mathrm{M}_{\mathcal{B},\mathcal{B}'} \quad \text{and} \quad \mathrm{G}_{\mathcal{B}'}(\psi) = \mathrm{M}_{\mathcal{B},\mathcal{B}'}^T \cdot \mathrm{G}_{\mathcal{B}}(\psi) \cdot \mathrm{M}_{\mathcal{B},\mathcal{B}'} \tag{F.4}$$

where $\mathrm{M}_{\mathcal{B},\mathcal{B}'}$ is the change-of-basis matrix from $\mathcal{B}'$ to $\mathcal{B}$.

In this setting we add the following assumptions: we consider the scalar product $\langle \_, \_ \rangle$ on $\mathbb{V}^n$, we let $\mathcal{B}$ be an orthonormal basis and we assume that $M$ is a symmetric matrix, i.e., $M = M^T$. Then, we let $\phi$ be the linear map associated to $M$ in the basis $\mathcal{B}$ and we let $\psi$ be the bilinear form associated to $M$ in the basis $\mathcal{B}$. Then, since the Gram matrix of the scalar product is $I_n$ we have

$$\psi(\mathbf{v},\mathbf{w}) = [\mathbf{v}]_{\mathcal{B}}^T \cdot M \cdot [\mathbf{w}]_{\mathcal{B}} = \langle \mathbf{v}, M \cdot [\mathbf{w}]_{\mathcal{B}} \rangle = \langle \mathbf{v}, \phi(\mathbf{w}) \rangle.$$

Since $M$ is symmetric, we also have

$$[\mathbf{v}]_{\mathcal{B}}^T \cdot M \cdot [\mathbf{w}]_{\mathcal{B}} = (M^T \cdot [\mathbf{v}]_{\mathcal{B}})^T \cdot [\mathbf{w}]_{\mathcal{B}} = (M \cdot [\mathbf{v}]_{\mathcal{B}})^T \cdot [\mathbf{w}]_{\mathcal{B}}$$

and therefore

$$\langle \phi(\mathbf{v}), \mathbf{w} \rangle = \psi(\mathbf{v},\mathbf{w}) = \langle \mathbf{v}, \phi(\mathbf{w}) \rangle. \tag{F.5}$$

**Definition F.11.** A linear map $\phi : \mathbb{V}^n \to \mathbb{V}^n$ is called *symmetric (relative to the scalar product)* if (F.5) holds for all $\mathbf{v}, \mathbf{w} \in \mathbb{V}^n$.

The proof of the Spectral Theorem uses the concept of orthogonal complement to a vector.

**Definition F.12.** Let $\mathbf{v} \in \mathbb{V}^n$. The *orthogonal complement*, denoted by $\mathbf{v}^\perp$, is the set of all vectors in $\mathbb{V}^n$ which are orthogonal to $\mathbf{v}$. So

$$\mathbf{v}^\perp = \Big\{ \mathbf{w} \in \mathbb{V}^n : \langle \mathbf{v}, \mathbf{w} \rangle = 0 \Big\}.$$

Since the scalar product is bilinear, the map $f_{\mathbf{v}} : \mathbb{V}^n \to \mathbb{R}$ defined by $f(\mathbf{w}) = \langle \mathbf{v}, \mathbf{w} \rangle$ is linear and we notice that $\mathbf{v}^\perp = \ker(f_{\mathbf{v}})$. Thus, if $\mathbf{v}$ is non-zero, $\mathbf{v}^\perp$ is an $(n-1)$-dimensional vector subspace of $\mathbb{V}^n$.

The proof of the Spectral Theorem also uses the following linear algebra fact.

**Lemma F.13.** The characteristic polynomial of a symmetric matrix $M \in \mathrm{Mat}_{n\times n}(\mathbb{R})$ has only real roots. Equivalently, the eigenvalues of a symmetric linear map are all real.

*Proof.* The equivalence of the two statements is obvious, by considering the linear map associated to $M$. We take the proof of the first claim from [20, p.232].

We can consider $M$ as a matrix over $\mathbb{C}$ (that is, with complex entries), and so we may view $\phi$ as a linear map $\mathbb{C}^n \to \mathbb{C}^n$. Let $\lambda \in \mathbb{C}$ be a root of the characteristic polynomial of $M$, and let $\mathbf{x}(x_1, \dots, x_n) \in \mathbb{C}^n$ be a corresponding eigenvector. Then

$$M\mathbf{x} = \lambda \mathbf{x}.$$

Taking the complex conjugates of both sides gives

$$M\overline{\mathbf{x}} = \overline{\lambda}\overline{\mathbf{x}}.$$

Consider the scalar $\overline{\mathbf{x}}^T M\mathbf{x}$. Writing it in two different ways using the above equations gives

$$\overline{\mathbf{x}}^T M\mathbf{x} = \overline{\mathbf{x}}^T (M\mathbf{x}) = \overline{\mathbf{x}}^T (\lambda \mathbf{x}) = \lambda \overline{\mathbf{x}}^T \mathbf{x} \tag{F.6}$$

$$\overline{\mathbf{x}}^T M\mathbf{x} = (\overline{\mathbf{x}}^T M)\mathbf{x} = (M\overline{\mathbf{x}})^T \mathbf{x} = (\overline{\lambda}\overline{\mathbf{x}})^T \mathbf{x} = \overline{\lambda}\overline{\mathbf{x}}^T \mathbf{x} \tag{F.7}$$

Note that

$$\overline{\mathbf{x}}^T \mathbf{x} = \overline{x_1} x_1 + \dots \overline{x_n} x_n$$

is a strictly positive real number, since $\mathbf{x} \neq 0$. We can therefore deduce from (F.6) and (F.7) that $\overline{\lambda} = \lambda$, that is that $\lambda$ is real. $\square$

**Theorem F.14** (Spectral Theorem)**.** Let $\phi : \mathbb{V}^n \to \mathbb{V}^n$ be a symmetric linear map. Then, there exists an orthonormal basis $\mathcal{B}$ such that $\mathrm{M}_{\mathcal{B},\mathcal{B}}(\phi)$ is a diagonal matrix.

Equivalently, let $\psi : \mathbb{V}^n \times \mathbb{V}^n \to \mathbb{R}$ be a symmetric bilinear form. Then, there exists an orthonormal basis $\mathcal{B}$ such that $\mathrm{G}_{\mathcal{B}}(\psi)$ is a diagonal matrix.

*Proof.* We take the proof of the first claim from [20, p.232]. The proof is by induction on $n = \dim(\mathbb{V}^n)$. If $n = 1$ there is nothing to prove. Suppose therefore that $n \geq 2$, and that the theorem holds for spaces of dimension $n-1$. Since $\phi$ is symmetric, it has only real eigenvalues, by Lemma F.13. Thus $\phi$ has an eigenvalue $\lambda$; let $\mathbf{e}_1$ be a corresponding eigenvector, which we can take to be of length 1. Let $\mathbb{U} = \mathbf{e}_1^{\perp}$, the orthogonal complement to $\mathbf{e}_1$. For each $\mathbf{u} \in \mathbb{U}$,

$$\langle \phi(\mathbf{u}), \mathbf{e}_1 \rangle = \langle \mathbf{u}, \phi(\mathbf{e}_1) \rangle = \langle \mathbf{u}, \lambda \mathbf{e}_1 \rangle = \lambda \langle \mathbf{u}, \mathbf{e}_1 \rangle = \lambda \cdot 0 = 0,$$

and so $\phi(\mathbf{u}) \in \mathbb{U}$, that is $\phi$ induces a map $\phi_{\mathbb{U}} : \mathbb{U} \to \mathbb{U}$. Since $\phi_{\mathbb{U}}(\mathbf{u}) = \phi(\mathbf{u})$ for every $\mathbf{u} \in \mathbb{U}$, the map $\phi$ is a symmetric linear map. By the inductive hypothesis, $\mathbb{U}$ has an orthonormal basis $(\mathbf{e}_2, \ldots, \mathbf{e}_n)$ which diagonalizes $\phi_{\mathbb{U}}$. Thus $\mathcal{B} = (\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n)$ is an orthonormal basis of $\mathbb{V}$ which diagonalizes $\phi$.

For the second claim we use (F.5). Let $\phi$ be the symmetric linear map associated to the Gram matrix $\mathrm{G}_{\mathcal{B}'}(\psi)$ of $\psi$ for some basis $\mathcal{B}'$. Let $\mathcal{B} = (\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n)$ be the diagonalizing orthonormal basis for $\phi$ obtained in the first part of the proof. By construction, it consists of eigenvectors of $\phi$. Then, by (F.5), we have

$$\psi(\mathbf{e}_i, \mathbf{e}_j) = \langle \phi(\mathbf{e}_i), \mathbf{e}_j \rangle = \langle \lambda_i \mathbf{e}_i, \mathbf{e}_j \rangle = \lambda_i \langle \mathbf{e}_i, \mathbf{e}_j \rangle = \begin{cases} \lambda_i & \text{if } i = j \\ 0 & \text{if } i \neq j. \end{cases}$$

hence $\mathcal{B}$ is a diagonalizing basis for $\psi$. □

**Theorem F.15** (Spectral Theorem - matrix formulation)**.** Let $M \in \mathrm{Mat}_{n\times n}(\mathbb{R})$ be a symmetric matrix. There exists an orthogonal matrix $P$ such that

$$P^{-1}MP = P^TMP = \begin{bmatrix} \lambda_1 & 0 & \ldots & 0 \\ 0 & \lambda_2 & \ldots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \ldots & \lambda_n \end{bmatrix}$$

where $\lambda_1, \ldots, \lambda_n$ are the eigenvalues of $M$.

*Proof.* Let $\mathcal{B}'$ be an orthonormal basis of $\mathbb{V}^n$. Let $\phi$ be the linear map associated to $M$ in the basis $\mathcal{B}'$. By the proof of Theorem F.14, there is an orthonormal basis $\mathcal{B}$ such that

$$\mathrm{M}_{\mathcal{B},\mathcal{B}}(\phi) = \begin{bmatrix} \lambda_1 & 0 & \ldots & 0 \\ 0 & \lambda_2 & \ldots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \ldots & \lambda_n \end{bmatrix}$$

where $\lambda_1, \ldots, \lambda_n$ are the eigenvalues of $\phi$, and therefore of $M$. Then we may choose $P = \mathrm{M}_{\mathcal{B}',\mathcal{B}}$ since

$$P^{-1}MP = \mathrm{M}_{\mathcal{B}',\mathcal{B}}^{-1}\mathrm{M}_{\mathcal{B}',\mathcal{B}'}(\phi)\mathrm{M}_{\mathcal{B}',\mathcal{B}} = \mathrm{M}_{\mathcal{B},\mathcal{B}}(\phi)$$

by (F.4). Moreover, since $\mathcal{B}$ and $\mathcal{B}'$ are both orthogonal, it follows that $P$ is an orthogonal matrix, i.e., $P^{-1} = P^T$ and the proof is finished. Indeed, since $\mathcal{B}'$ is orthonormal and since $P = \mathrm{M}_{\mathcal{B}',\mathcal{B}}$ is the matrix with columns consisting of the components of the basis vectors in $\mathcal{B}$ (which is an orthonormal basis), we have $P^T \cdot P = I_n$, hence $P^{-1} = P^T$. $\square$

# APPENDIX G

## Trigonometric functions

Let $\mathbf{a}$ and $\mathbf{b}$ be two vectors in $\mathbb{V}^2$. In Section 3.1 we noticed that there is a bijection between the set $\mathbb{W}$ of unoriented angles $\measuredangle(\mathbf{a}, \mathbf{b})$ and the semicircle $\mathbb{S}^{\frac{1}{2}}$ as well as a bijection between the set $\mathbb{W}_{\text{or}}$ of oriented angles $\measuredangle_{\text{or}}(\mathbf{a}, \mathbf{b})$ and the unit circle $\mathbb{S}^1$. It is clearly uncomfortable to do calculations in this setting. For a more comfortable manipulation, one attaches numbers to angles (we parametrize angles). The standard way to do this is to define a measure of an angle, a numerical value which we may use instead of describing the angle as two rays emanating from a point. The standard measure is that of radians, which can be introduced as follows

**Definition G.1.** Let $\mathbf{a} = \overrightarrow{OA}$ and $\mathbf{b} = \overrightarrow{OB}$ be two non-zero vectors in $\mathbb{V}^2$. The *interior of the angle* $\measuredangle AOB$ is the set of points $P$ with the property that there are scalars $\alpha, \beta \geq 0$ such that $\overrightarrow{OP} = \alpha\mathbf{a} + \beta\mathbf{b}$. The *sector* $\mathcal{S}(AOB)$ is the set of points $P$ in the interior of the angle and in the interior of the circle of radius 1 centered in $O$, i.e. $d(O, P) \leq 1$.

The *measure of the angle* $\measuredangle(\mathbf{a}, \mathbf{b})$, denoted by $m(\measuredangle(\mathbf{a}, \mathbf{b}))$, is

$$m(\measuredangle(\mathbf{a}, \mathbf{b})) = \begin{cases} 2 \cdot \text{Area}(\mathcal{S}(AOB)) & \text{if } \mathbf{a} \text{ and } \mathbf{b} \text{ are linearly independent} \\ 0 & \text{if } \mathbf{a} \text{ and } \mathbf{b} \text{ have the same direction} \\ \pi & \text{if } \mathbf{a} \text{ and } \mathbf{b} \text{ have opposite direction} \end{cases}$$

where $\pi$ is by definition the area of a unit circle. The definition does not depend on the representatives.

**Theorem G.2.** The properties of the area function translate into properties of angles. For two non-zero vectors $\mathbf{a}$ and $\mathbf{b}$ we have:

1. $0 \leq m(\measuredangle(\mathbf{a}, \mathbf{b})) \leq \pi$.
2. If $\mathbf{v}$ is between $\mathbf{a}$ and $\mathbf{b}$, then $m(\measuredangle(\mathbf{a}, \mathbf{v})) + m(\measuredangle(\mathbf{v}, \mathbf{b})) = m(\measuredangle(\mathbf{a}, \mathbf{b}))$.
3. For any $\theta \in [0, \pi]$ there is a vector $\mathbf{c} \in \mathbb{V}$ such that $m(\measuredangle(\mathbf{a}, \mathbf{c})) = \theta$.

4. $m(\measuredangle(\mathbf{a},\mathbf{b})) = m(\measuredangle(\mathbf{c},\mathbf{d}))$ if and only if $\frac{\langle \mathbf{a},\mathbf{b}\rangle}{|\mathbf{a}|\cdot|\mathbf{b}|} = \frac{\langle \mathbf{c},\mathbf{d}\rangle}{|\mathbf{c}|\cdot|\mathbf{d}|}$.
5. $m(\measuredangle(\mathbf{a},\mathbf{b})) = m(\measuredangle(\mathbf{b},\mathbf{a}))$.
6. $m(\measuredangle(\mathbf{a},\mathbf{b})) + m(\measuredangle(\mathbf{b},-\mathbf{a})) = \pi$.
7. $m(\measuredangle(\mathbf{a},\mathbf{b})) = 0$ if and only if $\mathbf{b}$ and $\mathbf{a}$ have the same direction.
8. If $\mathbf{a}\perp\mathbf{b}$, then $m(\measuredangle(\mathbf{a},\mathbf{b})) = \frac{\pi}{2}$.
9. If $\langle \mathbf{a},\mathbf{b}\rangle > 0$, then $0 \le m(\measuredangle(\mathbf{a},\mathbf{b})) < \frac{\pi}{2}$ and we say that the angle is *acute*.
10. If $\langle \mathbf{a},\mathbf{b}\rangle < 0$, then $\frac{\pi}{2} < m(\measuredangle(\mathbf{a},\mathbf{b})) \le \pi$ and we say that the angle is *obtuse*.
11. In particular, $\mathbf{a}\perp\mathbf{b}$ if and only if $m(\measuredangle(\mathbf{a},\mathbf{b})) = \frac{\pi}{2}$.

**Definition G.3.** The *measure of the angle* $\measuredangle_{\text{or}}(\mathbf{a},\mathbf{b})$, denoted by $m(\measuredangle_{\text{or}}(\mathbf{a},\mathbf{b}))$, is

$$m(\measuredangle_{\text{or}}(\mathbf{a},\mathbf{b})) = \begin{cases} m(\measuredangle(\mathbf{a},\mathbf{b})) & \text{if } [\mathbf{a},\mathbf{b}] \ge 0 \\ -m(\measuredangle(\mathbf{a},\mathbf{b})) & \text{if } [\mathbf{a},\mathbf{b}] < 0 \end{cases}$$

**Proposition G.4.** The measure of an oriented angle satisfies the following properties:

1. $-\pi < m(\measuredangle_{\text{or}}(\mathbf{a},\mathbf{b})) \le \pi$.
2. $0 < m(\measuredangle_{\text{or}}(\mathbf{a},\mathbf{b})) < \pi$ if and only if $(\mathbf{a},\mathbf{b})$ is right oriented.
3. $m(\measuredangle_{\text{or}}(\mathbf{a},\mathbf{J}(\mathbf{a}))) = \frac{\pi}{2}$.
4. $m(\measuredangle_{\text{or}}(\mathbf{a},\mathbf{b})) = -m(\measuredangle_{\text{or}}(\mathbf{b},\mathbf{a}))$.

These definitions of measures of angles give the standard bijections

$$\mathbb{W} \leftrightarrow [0,\pi] \quad \text{and} \quad \mathbb{W}_{\text{or}} \leftrightarrow (-\pi,\pi]. \tag{G.1}$$

This allows us to view sin, cos, tan as functions $(-\pi,\pi] \to \mathbb{R}$. Notice also that if $\theta$ is an oriented angle, then the sign is no longer necessarily positive and the sine function changes the sign while the cosine function does not, i.e., we have the sign rules that we are familiar with:

$$\sin(-\theta) = -\sin(\theta) \quad \text{and} \quad \cos(-\theta) = \cos(\theta).$$

In what follows, we work out the main properties of the trigonometric functions. The bijections (G.1) allow us to simplify notation: we write $\measuredangle_{\text{or}}(\mathbf{a},\mathbf{b})$ instead of $m(\measuredangle_{\text{or}}(\mathbf{a},\mathbf{b}))$.

**Proposition G.5.** If $\mathbf{a}$ and $\mathbf{b}$ are two non-zero vectors in the oriented Euclidean plane $\mathbb{E}^2$, then

$$\cos(\theta) = \frac{\langle \mathbf{a},\mathbf{b}\rangle}{|\mathbf{a}|\cdot|\mathbf{b}|} \quad \text{and} \quad \sin(\theta) = \frac{[\mathbf{a},\mathbf{b}]}{|\mathbf{a}|\cdot|\mathbf{b}|}$$

where $\theta = \measuredangle_{\text{or}}(\mathbf{a},\mathbf{b})$.

**Proposition G.6.** If $\mathbf{a}$ is a non-zero vector in the oriented Euclidean plane $\mathbb{E}^2$, and if $c^2 + s^2 = 1$, then there is a vector $\mathbf{b}$ such that

$$\cos(\measuredangle_{\mathrm{or}}(\mathbf{a},\mathbf{b})) = c \quad \text{and} \quad \sin(\measuredangle_{\mathrm{or}}(\mathbf{a},\mathbf{b})) = s$$

and all such vectors are proportional.

Instead of viewing $\measuredangle_{\mathrm{or}}(\mathbf{a},\mathbf{b})$ as a value in $(-\pi,\pi]$, it is more convenient to view it as an equivalence class of $\mathbb{R}$ modulo $2\pi$, i.e., $\theta \in \mathbb{R}$ represents the angle $\measuredangle_{\mathrm{or}}(\mathbf{a},\mathbf{b})$ if $\theta = \measuredangle_{\mathrm{or}}(\mathbf{a},\mathbf{b}) + 2k\pi$ for some integer $k$. With this convention, we have:

**Proposition G.7.** If $\mathbf{a},\mathbf{b},\mathbf{c}$ are three non-zero vectors in the oriented Euclidean plane $\mathbb{E}^2$, then

$$\measuredangle_{\mathrm{or}}(\mathbf{a},\mathbf{b}) = \measuredangle_{\mathrm{or}}(\mathbf{a},\mathbf{c}) + \measuredangle_{\mathrm{or}}(\mathbf{c},\mathbf{b}) \mod 2\pi.$$

**Theorem G.8.** For all $\alpha,\beta \in \mathbb{R}$, we have

$$\cos(\alpha+\beta) = \cos(\alpha)\cos(\beta) - \sin(\alpha)\sin(\beta)$$
$$\sin(\alpha+\beta) = \sin(\alpha)\cos(\beta) + \cos(\alpha)\sin(\beta).$$

**Proposition G.9.** The following half-angle formulas hold true:

$$\begin{aligned} \tan\left(\tfrac{\alpha}{2}\right) &= \tfrac{\sin(\alpha)}{1+\cos(\alpha)} \\ 2\cos^2\left(\tfrac{\alpha}{2}\right) &= 1+\cos\alpha \\ 2\sin^2\left(\tfrac{\alpha}{2}\right) &= 1-\cos\alpha \end{aligned}$$

**Proposition G.10.** For all $\frac{\pi}{2} > \theta \geq 0$, we have

$$\sin(\theta) \leq \theta \leq \tan(\theta).$$

**Corollary G.11.** For all $\frac{\pi}{2} > \theta \geq 0$, we have

$$0 \leq 1-\cos(\theta) \leq \theta^2.$$

**Proposition G.12.** We have the following limits:

$$\lim_{\theta\to 0}\left(\frac{1-\cos(\theta)}{\theta}\right) = 0 \quad \text{and} \quad \lim_{\theta\to 0}\left(\frac{\sin(\theta)}{\theta}\right) = 1.$$

**Corollary G.13.** The sine and cosine functions are differentiable, and the derivatives are:

$$\cos'(\theta) = -\sin(\theta) \quad \text{and} \quad \sin'(\theta) = \cos(\theta).$$

# APPENDIX H

## Quaternions and rotations

### H.1 Algebraic considerations

Some of the aspects considered here are also covered in [**?**, Section 4.4].

**Definition H.1.** Denote the standard basis of $\mathbb{R}^4$ by $1$, $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$ and consider the bilinear map

$$\_\cdot\_ : \mathbb{R}^4 \times \mathbb{R}^4 \to \mathbb{R}^4$$

given on the basis vectors by

$$\begin{array}{c|cccc} & 1 & \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \hline 1 & 1 & \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \mathbf{i} & \mathbf{i} & -1 & \mathbf{k} & -\mathbf{j} \\ \mathbf{j} & \mathbf{j} & -\mathbf{k} & -1 & \mathbf{i} \\ \mathbf{k} & \mathbf{k} & \mathbf{j} & -\mathbf{i} & -1 \end{array}.$$

We denote $\mathbb{R}^4$ with the above multiplication by $\mathbb{H}$. The elements of $\mathbb{H}$ are called *quaternions*. The product is the *Hamilton product*.

**Remark.** From the definition, we observe that

1. The multiplication map on arbitrary quaternions $p = a_1 + b_1\mathbf{i} + c_1\mathbf{j} + d_1\mathbf{k}$ and $q = a_2 + b_2\mathbf{i} + c_2\mathbf{j} + d_2\mathbf{k}$ is
$$\begin{aligned} pq &= (a_1a_2 - b_1b_2 - c_1c_2 - d_1d_2) + (a_1b_2 + a_2b_1 + c_1d_2 - c_2d_1)\mathbf{i} \\ &+ (a_1c_2 + a_2c_1 - b_1d_2 + b_2d_1)\mathbf{j} + (a_1d_2 + a_2d_1 + b_1c_2 - b_2c_1)\mathbf{k} \end{aligned} \tag{H.1}$$

2. Direct calculations show that $\mathbb{H}$ is an algebra, usually called *the quaternion algebra*.

3. $\mathbb{H}$ is not commutative: $\mathbf{i}\cdot\mathbf{j}=\mathbf{k}=-\mathbf{j}\cdot\mathbf{i}$.
4. $\mathbb{R}\cdot 1$ is a subfield of $\mathbb{H}$, so we just write $\mathbb{R}$ for it.
5. $\mathbb{C}=\mathbb{R}\cdot 1+\mathbb{R}\cdot\mathbf{i}$ is a subfield of $\mathbb{H}$.

**Definition H.2.** For a quaternion $q=a+b\mathbf{i}+c\mathbf{j}+d\mathbf{k}\in\mathbb{H}$, $a$ is the *real part* $\mathfrak{Re}(q)$ *of* $q$ and $b\mathbf{i}+c\mathbf{j}+d\mathbf{k}$ is the *imaginary part* $\mathfrak{Im}(q)$ *of* $q$. We say that $q$ is *real* if it equals its real part. We say that $q$ is *purely imaginary* if it equals its imaginary part.

**Proposition H.3.** A quaternion is real if and only if it commutes with all quaternions, i.e., the center of $\mathbb{H}$ is $\mathbb{R}$.

**Proposition H.4.** A quaternion is purely imaginary if and only if its square is real and non-positive.

**Definition H.5.** For a quaternion $q=a+b\mathbf{i}+c\mathbf{j}+d\mathbf{k}\in\mathbb{H}$, the *conjugate of* $q$ is

$$\overline{q}=a-b\mathbf{i}-c\mathbf{j}-d\mathbf{k}=\mathfrak{Re}(q)-\mathfrak{Im}(q)\in\mathbb{H}.$$

**Proposition H.6.** For $p,q\in\mathbb{H}$ and $a\in\mathbb{R}$, we have

1. $\overline{p+q}=\overline{p}+\overline{q}$
2. $\overline{ap}=a\overline{p}$
3. $\overline{\overline{p}}=p$
4. $\overline{p\cdot q}=\overline{q}\cdot\overline{p}$
5. $p\in\mathbb{R}\Leftrightarrow\overline{p}=p$
6. $p$ is purely imaginary $\Leftrightarrow\overline{p}=-p$
7. $\mathfrak{Re}(p)=\frac{1}{2}(p+\overline{p})$
8. $\mathfrak{Im}(p)=\frac{1}{2}(p-\overline{p})$

## H.2 Geometric considerations

By construction, $\mathbb{H}$ is $\mathbb{R}^4$ as a real vector space, so we may view it as a 4-dimensional real affine space. If, in addition, we consider the 4-dimensional Euclidean structure, we may identify $\mathbb{H}$ with $\mathbb{E}^4$. In particular, we may consider the standard scalar product $\langle\_,\_\rangle:\mathbb{H}\times\mathbb{H}\to\mathbb{H}\cong\mathbb{R}^4$.

**Proposition H.7** (Compare this with the similar statements for $\mathbb{C}\cong\mathbb{E}^2$)**.** For $p,q\in\mathbb{H}$, we have

1. $\langle p,q\rangle=\frac{1}{2}(\overline{p}q+\overline{q}p)$
2. $\langle p,p\rangle=\overline{p}p$
3. $|p|=\sqrt{\overline{p}p}$

If, in addition, $p$ and $q$ are purely imaginary, we have

4. $\langle p,q\rangle = -\frac{1}{2}(pq+qp) = -\mathfrak{Re}(pq)$
5. $\langle p,p\rangle = -p^2$
6. $|p| = \sqrt{-p^2}$
7. $\langle p,q\rangle = 0 \Leftrightarrow pq = -qp$.

**Definition H.8.** With our identification, quaternions are vectors in $\mathbb{V}^4 \cong D(\mathbb{H}) \cong \mathbb{H}$, and the *norm* $|q|$ of a quaternion $q$ equals $(\overline{q}\,q)^{\frac{1}{2}}$. If $|q| = 1$, we say that $q$ is a *unit quaternion*.

**Proposition H.9.** For any $p,q \in \mathbb{H}$, we have

$$|pq| = |p|\cdot|q|.$$

In particular, left and right multiplication by unit quaternions are isometries.

**Proposition H.10.** $\mathbb{H}$ is a skew field. The inverse of $q \in \mathbb{H}\setminus\{0\}$ is

$$q^{-1} = \frac{\overline{q}}{|q|^2}.$$

We identified $\mathbb{H}$ with $\mathbb{E}^4$. Next, we view $\mathbb{E}^3$ as a subspace of $\mathbb{H}$ by identifying it with the purely imaginary quaternions $\mathfrak{Im}(\mathbb{H}) = \mathbb{R}\mathbf{i} + \mathbb{R}\mathbf{j} + \mathbb{R}\mathbf{k}$.

**Proposition H.11.** Let $q_1, q_2$ be two quaternions with $a_i = \mathfrak{Re}\, q_i$, $v_i = \mathfrak{Im}\, q_i$. Making use of the scalar product and the cross product in $\mathbb{E}^3$, we have

$$q_1 q_2 = (a_1+v_1)(a_2+v_2) = a_1 a_2 - \langle v_1, v_2\rangle + a_2 v_1 + a_1 v_2 + v_1 \times v_2. \tag{H.2}$$

**Proposition H.12.** Let $v = v_i\mathbf{i} + v_j\mathbf{j} + v_k\mathbf{k} \in D(\mathbb{E}^3) \cong \mathfrak{Im}(\mathbb{H})$ be a unit quaternion and $p \in \mathbb{E}^3 \cong \mathfrak{Im}(\mathbb{H})$ a point. The rotation of $p$ around the axis $\mathbb{R}v$ by an angle $\theta$ is given by

$$p' = qpq^{-1}$$

where

$$q = \cos\left(\frac{\theta}{2}\right) + \sin\left(\frac{\theta}{2}\right)v.$$

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